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From the Editor Desk

Biospectra is an official publication of MSET-ICCB, Ranchi. On behalf of the Editorial Board of the Journal Biospectra and my Co-Editors, I am glad to present the Volume 17, Issue 1 of the Journal. The publication of the Journal started in September, 2006. With the great effort of Former Editor-in-Chief, Late Prof. (Dr.) Mahendra Prasad the Journal is indexed in Clarivate Analytics as Zoological Records. The Journal has a Scientific Journal Impact Factor (SJIF-2022) of 8.379. The Journal is also listed in UGC-CARE Group II through Web of Science (Journals indexed in globally recognized databases). We are putting our effort to list the Journal in UGC-CARE Group I & also preparing for obtaining NAAS rating. As we had NAAS rating of 2.3 in the list of "Rating of Scientific Journals 2012 (Effective from Jan 1, 2013)". From the last issue onwards the Academic Council of MSET-ICCB & Executive Committee of Biospectra has nominated me as the Editor-in-Chief of Biospectra. As I have stated in earlier issue that the papers are published only after checking plagiarism and careful selection through a double-blind peer-review process.

I am happy to state that we have worked hard on our words and the policies of publication. We have connected with several new reviewers from all over India and abroad also.

I extend my great honour and sincerest thanks to all the contributors of the articles, members of trust as well as editorial board and innumerable man power support in bringing out BIOSPECTRA, Volume 17, Issue 1, March, 2022, on ground for the enlightenment of people of Life Sciences.

31.03.2022

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Antagonistic activity of *Lactobacilli* derived from raw milk

Shwet Nisha*, Kumar Ramashankar & Rashmi Kumari
University Department of Zoology, B.N.M. University, Madhepura, Bihar, India

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Abstract- Altogether 15 milk samples were collected from local dairy firm of Madhepura district and *Lactobacilli* were isolated. Altogether 3 isolates were obtained, LM1, LM2 and LM3 which were identified as *Lactobacillus casei*, *Lactobacillus brevis* and *Lactobacillus fermentum* respectively. The identification were made on the basis of their morphological, biochemical and sugar test as prescribed in Bergey's manual of systematic Bacteriology (second edition-2001). The antagonistic study of isolated strain was made against four pathogenic bacteria *Pseudomonas aeruginosa*, *Escherichia coli*, *Staphylococcus aureus* and *Proteus sp.*. The inhibition zone against *Pseudomonas aeruginosa, Escherichia coli* and *Proteus sp.*was observed maximum by *Lactobacillus brevis* and against *Staphylococcus aureus*, the maximum inhibition zone was observed by *Lactobacillus fermentum*.

Key words: Niche, Niche Relaxation, Correlation, Amoeba, Copepod

INTRODUCTION

Raw milk is natural growth medium for micro-organism. An integral part of raw milk microflora is species of *Lactobacilli* as *Lactobacillus casei*, *L. plantarum*, *L. rhamnosus*, *L. curvatus*, *L. bravis*, *L. fermentum* and some other Lactic acid bacteria like *Leuconostoc lactis*.

Lactobacilli are rod shaped gram positive, Catalase negative, non-pathogenic bacteria present in milk and milk products. Several *Lactobacillus* strains are used as starter culture for the manufacture of fermented food.[1] These bacteria play a vital role in our daily life for fermentation, preservation, production and of wholesome food.

Lactic acid bacteria produced various compounds as organic acids, diacetyls, hydrogen peroxide, Vitamins and anti-microbial substance, bacteriocin. Antimicrobial activity of *Lactobacillus* against *E. coli*, *S. aureus* and *Solmonella* was reported by several workers.[2-5]

Lactobacillus casei have a wide range of pH and temperature and complement the growth of *L. acidophillus*, a producer of enzyme amylase. *L. sporogenesis* supports the growth of beneficial bacteria and help to maintain a healthy balance of microflora in the intestinal environment. A lot of LAB has been noted to have nutritional benefits, improve Lactose utilization, and have anti-cholesterol, anti-carcinogenic activity. Intestinal Lactic acid bacteria are considered to have health benefits for human being and introduced as probiotic including *L. rhamnosus*, *L. casei* and *L. johnsonii*.[6]

MATERIAL & METHOD

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Milk samples were collected from local diary firm of Madhepura in 500ml sterilized bottles and brought to the laboratory in ice boxes. From each sample, 1ml of milk was mixed in 10ml distilled water and serial dilution was made up to 10^{-5}. From each sample, 0.1ml inoculum was inoculated in MRS medium and incubated at 30°C temperature for 48 hours. Colonies were purified by sub-culturing in same medium. From each plate, colonies were examined for morphological study, gram staining, biochemical test and sugar test. On the basis of biochemical test and sugar test, isolates were categorised in following three categories: LM1, LM2 and LM3. The strains were identified on the basis of their morphological character, biochemical and sugar test as prescribed in Bergey's manual of systematic Bacteriology (second edition-2001).

Four pathogens *Pseudomonas aeruginosa*, *Escherichia coli*, *Staphylococcus aureus* and *Proteus sp.*were obtained from local pathological lab. All test pathogenic bacteria were cultured in Nutrient broth.

Antagonistic study:

For the antagonistic study of isolated strains of *Lactobacilli* against pathogenic bacteria was performed by well diffusion method.

Petri dishes containing 20ml of Muller Hinton agar were prepared and inoculated with 0.1ml pathogenic culture. The plates were allowed to solidify and stored in refrigerator for 2 hours. Four wells were made in culture plate and filled using different concentration (0.01ml, 0.02ml, 0.03ml and 0.04ml) of cell free filtrate of each isolate. Plates were incubated at 37°C for 24 hours. The diameter of inhibition zone was measured using Callipers.

Table 1- Biochemical test

Sl. No.	Isolate	Gram staining	Indole	MR	VP	Catalase	Oxidase	Nitrate reducation
1	LM1	+ve	-ve	-ve	-ve	-ve	-ve	+ve
2	LM2	+ve	-ve	-ve	-ve	-ve	-ve	+ve
3	LM3	+ve	-ve	-ve	-ve	-ve	-ve	+ve

Table 3- Antagonistic activity of selected isolates against pathogens

Sl. No.	Pathogens	LM1	LM2	LM3
1	*Pseudomonas aeruginosa*	7mm	13mm	6.5mm
2	*Escherichia coli*	7.5mm	13.5mm	9.5mm
3	*Staphylococcus aureus*	9.5mm	10mm	12.5mm
4	*Proteus sp.*	8mm	14mm	11mm

RESULT

Altogether 15 samples were tested in which 3 isolates were obtained which were assigned as LM1, LM2 and LM3. All isolates were gram positive while Indole, Catalase and Oxidase show negative. In sugar test, LM1 was negative in Glucose and sorbitol and positive in all other tested sugars. LM2 was positive for Xylose and negative for all other sugars. LM3 was negative in Rhamonose and positive in all other test sugars. On the basis of biochemical test and sugar fermentation test, LM1 was identified as *Lactobacillus casei*, LM2 as *Lactobacillus brevis* and LM3 as *Lactobacillus fermentum*.

The result is tabulated in Table no. 01 and 02. All three isolates strains inhibited the growth of all four pathogenic bacteria under test. The inhibition zone varies for different strains for different pathogenic bacteria. The maximum inhibition zone against *Pseudomonas aeruginosa* was of LM2 (13mm) and minimum of LM3 (6.5mm), against *E. coli* the highest inhibition zone was recorded for LM2 (13.5mm) and minimum for LM1 (7.5mm). The maximum inhibition zone against *S. aureus* was recorded for LM3 (12.5mm) and minimum for LM1 (9.5mm) and Inhibition zone against *Proteus* was maximum for LM2 (14mm) and minimum for LM1 (8mm). The result is tabulated in Table no. 03.

CONCLUSION

Altogether 15 milk samples were collected from local area of Madhepura district. *Lactobacilli* from each milk were isolated by culturing in MRS medium, Gram staining, Biochemical test and morphological study of bacteria from

Table 2- Sugar fermentation test

Sugar test	LM1	LM2	LM3
Xylose	+ve	+ve	+ve
Cellubiose	+ve	-ve	+ve
Glucose	-ve	-ve	+ve
Mannitol	+ve	-ve	+ve
Ramonose	+ve	-ve	-ve
Raffinose	+ve	-ve	+ve
Ribose	+ve	-ve	+ve
Sorbitol	-ve	-ve	+ve
Trehalose	+ve	-ve	-ve
Fructose	+ve	-ve	+ve

all samples were performed. Three different isolates were obtained from all samples. On the basis of their biochemical study and sugar test, the isolates were identified as *Lactobacillus casei*, *Lactobacillus brevis* and *Lactobacillus fermentum*. The antagonistic study of these *Lactobacilli* species were performed against four pathogenic bacteria- *Pseudomonas aeruginosa, Escherichia coli, Staphylococcus aureus* and *Proteus* sp.. The antagonistic study was performed by well diffusion method. The inhibition zone against *Pseudomonas aeruginosa, Escherichia coli* and *Proteus sp.* was observed maximum by *Lactobacillus brevis* and against *Staphylococcus aureus*, the maximum inhibition zone was observed by *Lactobacillus fermentum*.

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Habenaria gibsonii var. *foetida* Blatt. & McCann (Orchidaceae): New distribution recorded from Sitamata Wildlife Sanctuary, Rajasthan, India

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Abstract- Orchid species *Habenaria gibsonii* var. *foetida* Blatt. & McCann (Orchidaceae) is reported from Sitamata Wildlife Sanctuary of Rajasthan, India for the first time along with its description, illustration, photographs and location in Rajasthan.

Key words: *Habenaria gibsonii* var. *foetida*, Orchidaceae, Sitamata Wildlife Sanctuary, Description, Report

INTRODUCTION

During the present investigation and characterization of orchid plants of Rajasthan, an interesting Orchid species was collected from Sitamata Wildlife Sanctuary, Rajasthan. After critical studies of the collected sample, this orchid was identified as *Habenaria gibsonii* var. *foetida*. Almost all floristic research work has been examined critically but *Habenaria gibsonii* var. *foetida* Blatt. & McCann has not been reported from Sitamata Wildlife Sanctuary, Rajasthan to date.[1-15]

Description:

Habenaria gibsonii var. *foetida* Blatt. & McCann, J. Bombay Nat. Hist. Soc. 16 (1932); Tiwari & Maheshwari, Orch. Madh. Pra.: 435, (1963); Misra, Orch. Ori.: 186 (2004).

Terrestrial, 30-40 cm high, Tuber single, spheroid or oblong ellipsoid, vertically below the stem roots few. Stem cylindrical. Leaves 5-8, dark green above and pale below, alternate, sessile, elliptic-lanceolate, acute, slightly undulate, base narrow, sheathing below. Inflorescence erect, 14-20 cm long, peduncle cylindrical, with large obovate lanceolate, acuminate, sub-sheathing sterile bracts subtending raceme. Raceme lax, not compact, up to 10-12 flowers. Bracts longer than ovary, upper ones smaller, ovate-lanceolate, subulate, 5-veined, green. Flower foul scented, greenish-white. Pedicel 1-2 cm long. Sepals unequal, concave, dorsal erect, hooded, broadly ovate, sub-obtuse; laterals longer, spreading to the base, oblong-lanceolate, rounded, falcate. Petals bipartite nearly to the base, segments linear, tapering upward, acute, upper strongly sickle-shaped, lower segment almost at a right angle to the upper segment, strongly incurved upward. Lip 3-partite nearly to the base with subequal segments, midsegment decurved; laterals forming acute angles with the middle, reflexed, decurved upwards. Spur shorter and close to the ovary, clavate with an obtuse apex. Column erect, more or less oblong. Stigmatic processes stretched forward, slightly divergent, linear narrowing the apices.

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Rostellum broad and triangular, rounded. Anther locules waxy yellow, parallel, broad. Pollinia obliquely obovate, apex bilobed. Ovary 1-2 cm long, slightly curved at the apex, ribbed, green.

Flowering: August, **Fruiting**: September – October.

Ecological notes: This species grows underneath of mixed type forest with tree species like *Terminalia tomentosa, Madhuca indica, Mangifera indica, Terminalia bellirica, Anogeissus pendula, Ficus benghalensis* and *Butea monosperma* etc. in humus-rich soil. The plant grows with herbaceous plants like *Desmodium* sp., *Diocscorea* sp., *Chlorophytum* sp. and other associated herbs.

Diagnostic features: *H. gibsonii* var. *foetida* can be differentiated in the field on the basis of foetid smelling flowers.

Distribution in India: Jharkhand, Madhya Pradesh, Chhattisgarh, Orissa, Maharashtra, Gujarat and Rajasthan

The Genus *Habenaria* is a large genus with around 600 species widely distributed all over the world. Within the Rajasthan, only six species are reported *vis. Habenaria digitata, Habenaria frucifera, Habenaria longicorniculata, Habenaria marginata, Habenaria plantaginea* and *Habenaria gibsonii.* Earlier *H. gibsonii* var. *foetida* Blatt. & McCann has been reported from Jharkhand, Madhya Pradesh, Chhattisgarh, Orissa, Maharashtra, Gujarat and Rajasthan (Sirohi District)[15-17]. This species is common in the Sitamata Wildlife Sanctuary and can be easily traced out during the flowering time. The previous record of this plant was only from Mount Abu Wildlife Sanctuary of Sirohi District. This is the first time it has been discovered in the Sitamata Wildlife Sanctuary, Rajasthan[15].





Fig. 1 *Habenaria gibsonii* var. *foetida* Blatt. & McCann: A. Whole plant in habitat; B. Plant Inflorescence

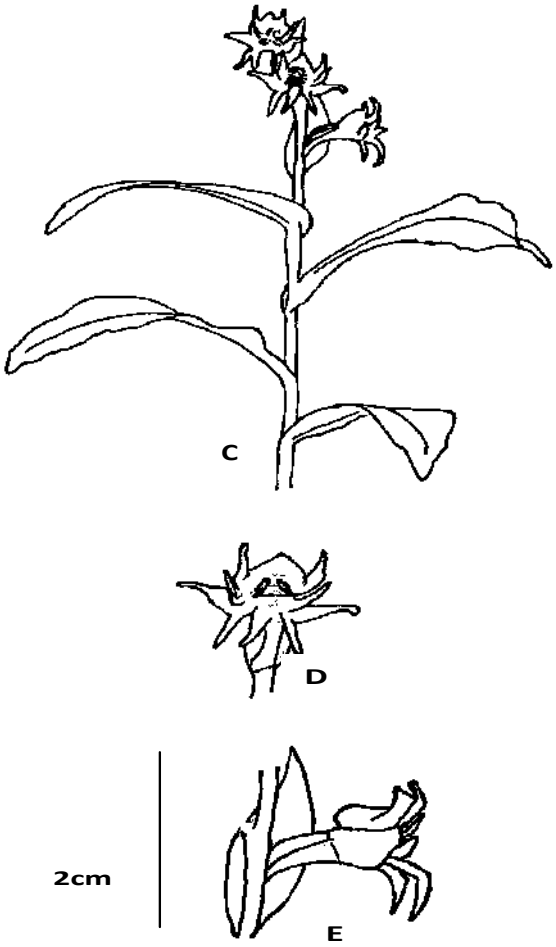


Fig. 2: *Habenaria gibsonii* var. *foetida* Blatt. & McCann: Illustration: C. Plant with inflorescence D. Close up of single flower E. lateral view of flower A

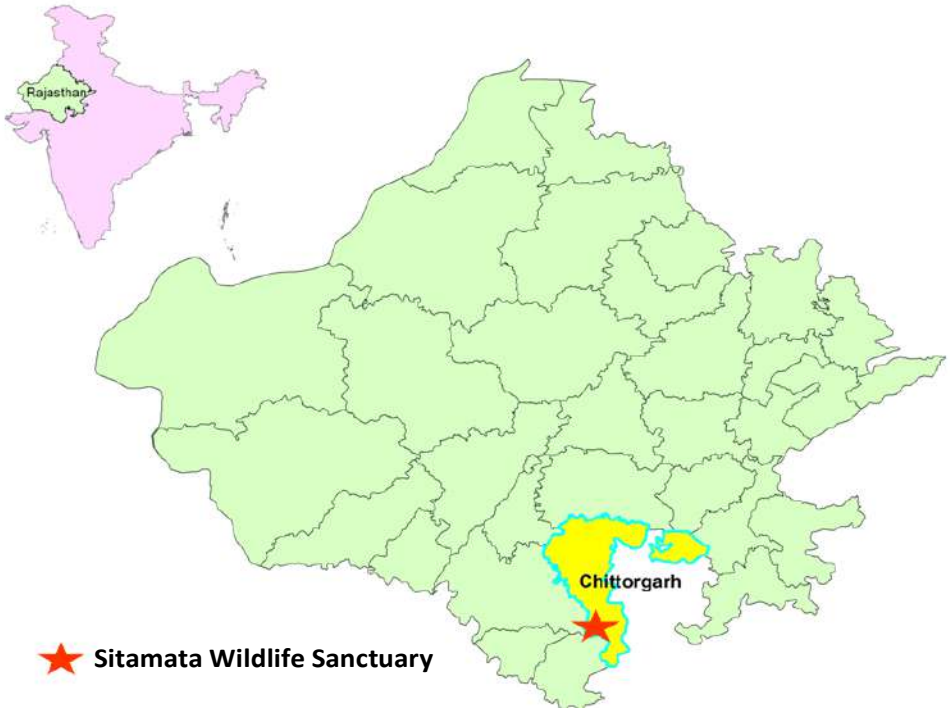


Figure 3: Location of *Habenaria gibsonii* var. *foetida* Blatt. & McCann in Sitamata Wildlife Sanctuary

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Birds diversity and feeding guild in and around Abu road and Mount Abu Municipal area, Rajasthan, India

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Abstract- Birds are well-known bioindicator species and play a vital role in both natural and man-made ecosystems. Garbage dumps provide a variety of food resources for a large number of bird species, which belongs to different feeding guilds. During study, from July 2019 to December 2021, 68 avian species were found feed on household waste, garbage and dump sites in the Abu Road and Mount Abu areas. Out of 68 species, 65 species were categorized as least concerned, two species (*Threskiornis melanocephalus* and *Mycteria leucocephala*) near threatened and one species (*Ciconia episcopus episcopus*) was classified as vulnerable. During study, maximum relative diversity was found in two families: Columbidae and Muscicapidae (RDi= 8.82), followed by three families: Accipitridae, Ardeidae and Phasianidae (RDi= 7.35) and five families, Corvidae, Leiothrichidae, Sturnidae, Threskiornithidae and Turdidae (RDi= 4.41) and the remaining eight families: Ciconiidae, Cisticolidae, Coraciidae, Cuculidae, Dicruridae, Hirundinidae, Pyncnotidae, and Rallidae represent (RDi= 2.94) relative diversity. The remaining ten families (Charadriidae, Falconidae, Laniidae, Leiothrichidae, Motacillidae, Muscicapidae, Passeridae, Psittacidae, Rhipiduridae, Strigidae, and Upupidae) each had only one species and relative diversity (RDi = 1.47) was observed in the study area. During study, most birds belonged to the insectivore (22 species), followed by the omnivore (21), carnivore (17), granivores (7) and frugivores (*Psittacula krameri parvirosnis*) feeding guild. The highest average number of individuals belonged to the omnivore feeding guild, followed by garnivores, insectivore, carnivore and frugivore.

Key words: Birds, Garbage dump site, feeding guild, Mount Abu and Abu road

INTRODUCTION

Excessive growth of the human population is largely responsible for the expansion of urban areas. As a result, between 2000 and 2030, global urban areas will increase three fold, transforming large natural area into urban landscape.[1] Due to the rapid loss of wildlife and their habitats, urbanisation now seems like a challenging ecosystem for sustaining biotic communities and diversity.[2] Very limited flora and fauna are able to be sustained in urban ecosystems. Due to the rapid loss of natural habitats for birds, they adapt to anthropogenic modified habitats such as artificial feeding sites like dumpsites, sewage ponds and garbage sites.[3] These man-made sites have been found to provide unlimited food resources for birds and are one of the factors that attract birds to the sewage stabilisation pond and dumping sites area.[3] In urban areas, solid waste management is the main issue for responsive development in developing countries.

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It is essential to recognise that both the economy and population prosperity have an influence on solid-waste generation rates.[4]

Most developing countries are facing various problems, like rapid population growth and urbanization-related issues. The environment has been subjected to a great deal of stress as a result of increased urbanisation and growing economies. As the world becomes more industrialised, people are infesting the environment with new and complex chemicals, with no frequent bio-assessment of their toxicity.[4] As a result, waste management has emerged as one of the most pressing concerns for authorities in developing countries rapidly expanding cities.[5] Throughout history, humans have been able to dispose of biological waste such as bones, skins, fire, ash, stools, shells and other materials.[6] These wastes are effective because they are generally biodegradable and low in density, as well as because they use simple waste management technologies.[7]

As the human population and urbanisation continue to grow, effective solid waste management has become a main issue in urban areas.[8] Tons of municipal solid waste, mostly non-hazardous garbage, trash and garbage from homes, institutions and industrial facilities, end up in landfills. Construction of dumpsites not only reduces wildlife habitat, but also increases human-wildlife conflicts and interactions.[9-11] Despite the fact that improper waste management can have serious consequences for biodiversity and the environment, including disease-carrying organisms, poisonous gases and pollution, waste sites can serve as a resource for birds, which adapt to the area based on food availability.[12]

Toxins, pathogenic bacteria and poor food quality are now recognized as major contributors to health problems, individual morality and ultimately biodiversity decline at the local and regional levels.[13-16] Waste disposal sites are important for the attraction of various wildlife species. Many wildlife species, particularly opportunistic scavengers, use them as feeding grounds, including invertebrates and vertebrates, especially mammals and birds.[17,18] Food obtained from dump sites has a multitude of benefits, including that waste disposal sites have a number of advantages, including increased survival rates and body condition, improved reproductive performance, reduced feeding time costs and also reduced risk of predation, migration and extinction of species, especially threatened species.[19] Reduced food availability in waste disposal sites, on the other hand, may have an adverse effect on wildlife, such as shifting of diet and feeding habitat and dependency on anthropogenic derived food.[20]

Improper waste management has the potential to have serious environmental and biodiversity consequences. Pollutants and toxicant materials influence atmospheric conditions such as water, air and land. Other side waste disposal sites, on the other hand, provide several benefits and suitable feeding grounds for wildlife, particularly birds. Garbage dumps provide food for a variety of bird species from various feeding guilds.[21-26] Several studies have been made that garbage dumps have had a significant impact on the population, abundance and distribution of the cattle egret (*Bubulcus ibis*) and white stork (*Ciconia ciconia*) in the Iberian Peninsula region.[27,28] Garbage dumps are important not only during the migration season, but also during the breeding and wintering seasons. Several species of carnivores and scavenger birds like black kites, red kites, griffon vultures, crows, ravens and magpies utilize garbage dump areas as feeding stations and roosting grounds.[27-29]

Migratory birds also feed on garbage dump sites and several investigations also found positive results because garbage dump sites provide food material during the long path of migration and also enhance the survival rate during stressful and mortality prone periods. Due to sufficient food availability around dump sites, some bird species drastically reduce migration routes in some cases, like the white stork in Spain.[24,27] Garbage dump sites also provide new wintering ground habitat for resident and migratory avian species.[27,30] A more densely forested area can support a greater number of individuals or species of birds.[31] Additionally, the presence of water and trees near the dumping site may provide additional ecological services to birds. Although birds primarily use waste disposal sites for feeding, the surrounding habitats are also utilized for other ecological processes and requirements.[32]

Birds are a well-known group of organisms; they can survive in a wide variety of habitats, including natural and man-made ecosystems. They perform vital ecological and functional roles such as bio-indicators of ecosystems, pollination, nutrient recycling, seed dispersal and maintaining ecological balance in the food chain and food web.[33] According to Sethy *et al.,* (2015)[34] assessing bird

communities is an important part of biodiversity conservation and identifying the need for conservation actions towards ecosystems and natural resources. It is also essential to understand the diversity and composition of avian communities in order to assess the health status of local and regional landscapes for bird conservation.

In order to present study was conducted on avian diversity and feeding guilds at garbage dump sites, household and municipal waste in Mount Abu and Abu Road area of Sirohi district.

MATERIALS & METHODS

Data collection started from July, 2019 to December, 2021 in the Mount Abu and Abu Road area. Data were collected using point count and direct observation methods in the vicinity of hotels, houses and colonies where household and municipal waste were dumped in the vicinity of road sites where household and garbage were dumped by locals and tourists. At-least minimum 15 minutes observations were done at each site and observations were supported with the help of Nikon 8x40 binocular and Nikon P1000 camera. Birds were identified with help of standard filed guides by Grimmet *et al.,* (2011)[35] and Vyas, (2013).[36]

RESULT & DISSCUSION

Birds are well-known organisms that play a role in both natural and man-made ecosystems. They play an important role in ecological processes and functions such as serving as bio-indicators of ecosystem, assisting in the reduction of disposable waste and regulating insect and pest populations in agricultural landscapes.[33] Interspecific and intraspecific competitions were also observed between two similar food habitats for birds.[37] A large number of house crows and brahminy kites congregate during the daytime when waste is transported from the slaughterhouse to garbage dump sites. Due to similar food habitats, interspecific competitions are frequently observed between brahminy kites and house crows at dump sites.[38]

Similar observations we were also observed between house crow, jungle crow and black kites during the foraging, feeding and roosting time at garbage site in study area. We also observed interspecific completion for food in Asian pied starling, Common Myna and Brahmin starling. These birds were most abundant and more preferred feeding on garbage. During study, we observed 68 avian species on garbage and dump sites in the Abu Road and Mount Abu areas and they belonged to 28 families of birds. According to the Birdlife International (2021)[39], 65 species were least concerned, two species (*Threskiornis melanocephahus* and *Mycteria leucocephala*) belonged to two near-threatened categories and one species *Ciconia episcopus episcopus* belongs to vulnerable categories (Table -1).

The relative diversity of avian families was calculated with the help of the following formula.[40]

$$\text{RDi (Relative diversity)} = \frac{\text{Number of avian species in the family}}{\text{Total number of species}} \times 100$$

Relative diversity was highest in two families Columbidae and Muscicapidae (RDi = 8.82), followed by three families each of Accipitridae, Ardeidae, and Phasianidae (RDi = 7.35), five families Corvidae, Leiothrichidae, Sturnidae, Threskiornithidae and Turdidae (RDi = 4.41), and the remaining ten families (Charadriidae, Falconidae, Laniidae, Leiothrichidae, Motacillidae, Muscicapidae, Passeridae, Psittacidae, Rhipiduridae, Strigidae and Upupidae) each have only one species, and relative diversity was observed (RDi = 1.47) (Table 1).

Maximum birds species were belongs to Columbidae and Muscicapidae (6 species), followed by three families Accipitridae, Ardeidae, and Phasianidae were represent (5 species), five families Corvidae, Leiothrichidae, Sturnidae, Threskiornithidae and Turdidae (3 species), and the remaining ten families (Charadriidae, Falconidae, Laniidae, Leiothrichidae, Motacillidae, Muscicapidae, Passeridae, Psittacidae, Rhipiduridae, Strigidae and Upupidae) each have only one species (Table 1, Graph 1). Birds were also categorized on the basis of their feeding guilds and food choices.[41,42]

During present study birds were categorized into five feeding guilds namely Insectivore, Granivores, Frugivore, Carnivore and Omnivore according to their food preference. During the study, the most bird species belonged to the Insectivore (22) feeding guild, followed by the Omnivore (21), Carnivore (17), Granivores (7) and the Frugivore (*Psittacula krameri parvirosnis*) feeding guild. The average number of individuals was highest for individuals belonging to the Omnivore feeding guild, followed by Granivores, Insectivore and Carnivore, and lowest individual belonged to the Frugivore feeding guild (Table 2; Garph-2).

Dead animals dumped in landfills may have died as a result of infections that could have killed other birds, such as avian influenza, which is frequently observed in the Middle East.[43] Because infectious diseases, such as avian influenza and the West Nile virus will emerge more frequently in new areas as a result of climate change[44], wild birds may be able to transmit diseases along migratory routes.[45] Various wildlife species can now be found in and around urban areas, especially in developing countries due to lack of species and habitat management, outside their natural habitats such as agricultural and cultivated land and garbage dump sites.[46]

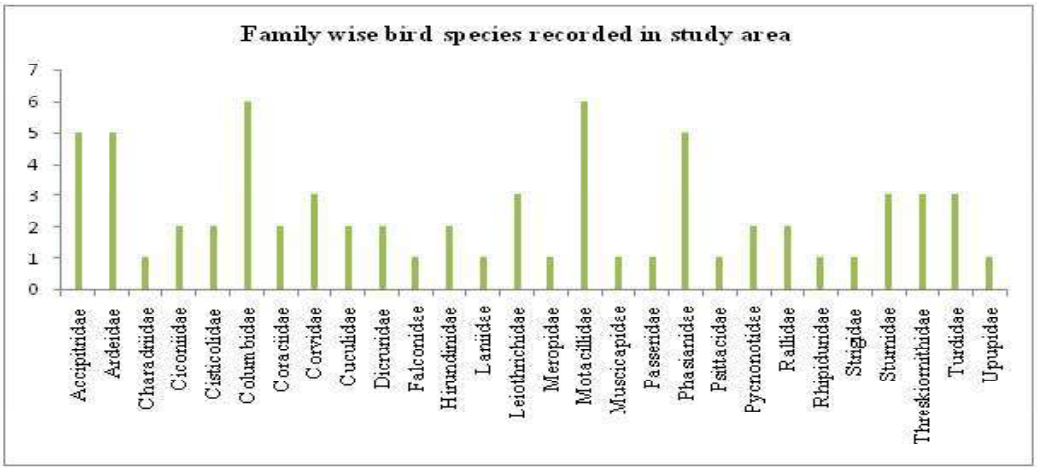
Table 1- Birds species recorded on garbage dump site in Abu road and Mount Abu

Sl. No.	Common name	Zoological name	Family	Feeding guild	IUCN status
1	Ashy prinia	*Prinia socialis stewarti*	Cisticolidae	Insectivore	LC
2	Asian koel	*Eudynamys scolopaceus scolopaceus*	Cuculidae	Omnivore	LC
3	Asian pied Starling	*Gracupica contracontra*	Sturnidae	Omnivore	LC
4	Bay bake shrike	*Lanius vittatus nargianus*	Laniidae	Insectivore	LC
5	Black drogue	*Dicrurus macrocercus minor*	Dicruridae	Insectivore	LC
6	Black kite	*Milvus migrans govinda*	Accipitridae	Carnivore	LC
7	Black Redstart	*Phoenicurus ochruros rufiventris*	Muscicapidae	Insectivore	LC
8	Black-headed white ibis	*Threskiornis melanocephalus*	Threskiornithidae	Carnivore	NT
9	Black-shouldered kite	*Elanus caeruleus*	Accipitridae	Carnivore	LC
10	Brahmin starling	*Sturnus pagodarum*	Sturnidae	Omnivore	LC
11	Brown Rock Chat	*Cercomela fusca*	Turdidae	Insectivore	LC
12	Cattle egret	*Bubulcus ibis*	Ardeidae	Insectivore	LC
13	Common Babbler	*Turdoides caudata*	Leiothrichidae	Omnivore	LC
14	Common hoopoe	*Upupa epops*	Upupidae	Insectivore	LC
15	Common Kestrel	*Falco tinnunculus*	Falconidae	Carnivore	LC
16	Common moorhen	*Gallinula chloropus*	Rallidae	Omnivore	LC
17	Common Myna	*Acridotheres tristis*	Sturnidae	Omnivore	LC
18	Common Swallow	*Hirundo rustica*	Hirundinidae	Insectivore	LC
19	Crested-serpent eagle	*Spilornis cheela*	Accipitridae	Carnivore	LC
20	Eurasian collared Dove	*Streptopelia decaocto decaocto*	Columbidae	Granivores	LC
21	European Roller	*Coracias garrulus*	Coraciidae	Carnivore	LC
22	Glossy ibis	*Plegadis falcinellus*	Threskiornithidae	Carnivore	LC
23	Greater cukaal	*Centropus sinensis sinensis*	Cuculidae	Carnivore	LC
24	Green bee- eater	*Merops orientalis*	Meropidae	Insectivore	LC
25	Grey Heron	*Ardea cinerea cinerea*	Ardeidae	Carnivore	LC
26	Grey jungle fowl	*Gallus sonneratii*	Phasianidae	Omnivore	LC
27	Grey Wagtail	*Motacilla cinerea*	Motacillidae	Insectivore	LC
28	House crow	*Corvus splendens splendens*	Corvidae	Omnivore	LC
29	House sparrow	*Passer domesticus indicus*	Passeridae	Omnivore	LC
30	Indian Black ibis	*Pseudibis papillosa*	Threskiornithidae	Omnivore	LC
31	Indian Grey Francolin	*Francolinus pondicerianus*	Phasianidae	Omnivore	LC

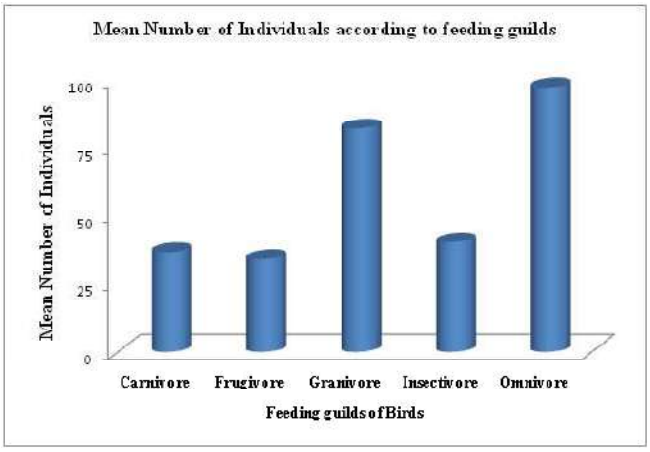
32	Indian Peafowl	*Pavo cristatus*	Phasianidae	Omnivore	LC
33	Indian pond heron	*Ardeola grayii*	Ardeidae	Carnivore	LC
34	Indian Robin	*Saxicoloides fulicatus cambaiensis*	Turdidae	Insectivore	LC
35	Indian roller	*Coracias benghalensis*	Coraciidae	Insectivore	LC
36	Intermediate egret	*Egretta intermedia*	Ardeidae	Carnivore	LC
37	Jungle Babbler	*Turdoides striata*	Leiothrichidae	Omnivore	LC
38	Jungle crow	*Corvus macrorhynchos*	Corvidae	Omnivore	LC
39	Large grey babbler	*Turdoides malcolmi*	Leiothrichidae	Omnivore	LC
40	Large Pied wagtail	*Motacilla maderaspatensis*	Motacillidae	Insectivore	LC
41	Laughing Dove	*Streptopelia senegalensis cambayensis*	Columbidae	Granivores	LC
42	Little egret	*Egretta garzetta garzetta*	Ardeidae	Carnivore	LC
43	Oriental magpie-robin	*Copsychus saularis saularis*	Turdidae	Insectivore	LC
44	Oriental turtle dove	*Streptopelia orientalis*	Columbidae	Granivores	LC
45	Paddy field Pipit	*Anthus rufulus waitei*	Motacillidae	Insectivore	LC
46	Painted Stork	*Mycteria leucocephala*	Ciconiidae	Carnivore	NT
47	Plain Prinia	*Prinia inornata inornata*	Cisticolidae	Insectivore	LC
48	Rain Quill	*Coturnix coromandelica*	Phasianidae	Granivores	LC
49	Red collared Dove	*Streptopelia tranquebarica*	Columbidae	Granivores	LC
50	Red Spur- fowl	*Galloperdix spadicea*	Phasianidae	Omnivore	LC
51	Red vented bulbul	*Pycnonotus cafer*	Pycnonotidae	Omnivore	LC
52	Red wattled lapwing	*Vanellus indicus indicus*	Charadriidae	Insectivore	LC
53	Rock pigeon	*Columba livia intermedia*	Columbidae	Granivores	LC
54	Rose-ring parakeet	*Psittacula krameri parvirosnis*	Psittacidae	Frugivore	LC
55	Rufous Tree pie	*Dendrocitta vagabunda*	Corvidae	Omnivore	LC
56	Shikra	*Accipiter badius*	Accipitridae	Carnivore	LC
57	Spotted dove	*Streptopelia chinensis chinensis*	Columbidae	Granivores	LC
58	Spotted Owlet	*Athenebrama indica*	Strigidae	Carnivore	LC
59	Tree pipit	*Anthus trivialisha ringtoni*	Motacillidae	Omnivore	LC
60	White breasted water hen	*Amaurornis phoenicurus*	Rallidae	Omnivore	LC
61	White browned fantail	*Rhipidura aureola compressirostris*	Rhipiduridae	Insectivore	LC
62	White eared bulbul	*Pycnonotus leucotis*	Pycnonotidae	Omnivore	LC
63	White wagtail	*Motacilla alba*	Motacillidae	Insectivore	LC
64	White-bellied Drogue	*Dicruru scaerulescen scaerulescens*	Dicruridae	Insectivore	LC
65	White-throated kingfisher	*Halcyon smyrnensis smyrnensis*	Alcedinidae	Carnivore	LC
66	Wire tailed swallow	*Hirundo smithii*	Hirundinidae	Insectivore	LC
67	Asian Wolly-necked Stork	*Ciconia episcopus episcopus*	Ciconiidae	Carnivore	V
68	Yellow Wagtail	*Motacilla flava*	Motacillidae	Insectivore	LC

Table 2- Feeding guilds and Mean number of birds individual observed during the study

Feeding guild	N	Mean Number of Individuals	Std. Deviation	Std. Error	Minimum	Maximum	p value
Carnivore	17	36.47	29.313	7.109	3	134	0.082
Frugivore	1	34	----	----	34	34	
Granivores	7	82.29	121.46	45.908	13	356	
Insectivore	22	40.14	24.445	5.212	17	121	
Omnivore	21	97.24	113.194	24.701	2	403	
Total	**68**	**61.1**	**79.396**	**9.628**	**2**	**403**	



Graph 1- Family wise bird's species recorded in study area



Graph 2- Mean Number of Individuals according feeding guilds of birds



Fig. 1- Jungle crow finds food material inside plastic waste



Fig. 2- Red wattled lapwing foraging at dump site



Fig. 3- Household dump site in Mount Abu



Fig. 4- Red Spur-fowl foraging and feeding on Garbage

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Study of the effect of neem leaves powder and orange peels powder on weevil growing on maize grains

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Abstract- A laboratory study with the purpose of investigating the effects of neem leaf and orange peels powder against *Sitophilus zeamais*, (Coleoptera: Curculionidae) was conducted at Department of Botany, J. P. University, Chapra, Saran. Three sets of testing were set including treatment of seeds with two different plant powder extracts such as Orange peels powder and neem leaf powder and a control where no plant powder were applied to the grains. The dosage of powder, 5g, 10g, 15g, 20g, 25g, and 30g were applied to 500 grains included weevil mortality, survival and grain damage. In view of the percentage of mortality, the toxicity of powder's increased gradually with increasing concentration. The dead adult weevils were counted within 6 weeks after the maize was treated by neem leaf powder and orange peels powder. The results from the study showed that the highest percentage weevil mortality was recorded for neem followed by Citrus. The lowest and highest mortality rate have been seen of 5g & 30g for neem extract of 24 & 28 respectively, where of these value was noticed of 4 & 8 at 30g & 5g respectively for Citrus extracts. The percentage calculated 80 & 93.3% of 5g & 30g of neem extract whereas 73.3 % & 86.6% for Citrus extract respectively.

Key words: Neem Leaf, Orange Peel, *Sitophilus*, Maize grain.

INTRODUCTION

Maize (*Zea mays* L.) is a cereal grain belonging to the family of Gramineae. It is a cereal grain linked to wheat, rice, oat and barley. In terms of world grain manufacture, it is ranked second next to wheat.[1] Maize can flourish in diverse climattic condition and is grown-up by small holder families in many countries of the world. In India maize is habitually grown during the Kharif season with high temperature and low and surplus rains, creating adverse condition for changing length of time during the two growing seasons (that is from March to June and from mid of July to September). It is measured as versatile plant with many uses. First of all, it is one of the major sources of food for both human and animals. Distant from this, it is being processed into various food and industrial products together with starches, sweeteners, oil, beverages, industrial alcohol and firewood ethanol. Maize is a significant staple food for human being and animals. According to Addo-Quaye *et al.* (1993)[2], it has an average composition of 13.2% protein, 60% starch, 1.2% sugar and 2.5% crude fibre. It is principally an energy giving food because of its high starch content. The grain in also used to feed monogastric such as pigs and

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poultry as the main basis of energy whereas the plant also serves as fodder for ruminants.[2] The starch is used in creation ice cream, alcohol, beverages, cosmetics and adhesive, cooking oil, salad oil, glycerine, soap and pharmaceuticals can as well be made from maize.[3]

Post harvest losses occur between harvest and the moment of human utilization. They include on-farm losses, such as when grain is threshed, winnowed, and dried, as well as losses next to the chain through carrying, storage and processing. Harvesting is done by defiance the ear from the stalk using the hand with both cutlass and sickle. This is done through natural draying using Sun draying or non-natural draying like photovoltaic solar energy.[4] Maize weevil (*Sitophilus zeamais*) is a major pest that attack stored maize grains in the tropics and temperate regions of the world.[5,6] In spite of the mentioned uses of maize, it is attacked by different storage pests. Maize weevil (*Sitophilus zeamais*) is a major pest that attack stored maize grains in the tropic and temperate regions of the world.[5,6] The attack may begin in the adult crop when the Moisture Content (MC) of the grain had fallen to 18-20%.[7] The pest as well infests other stored cereal grains as substitute hosts. Famous among its secondary hosts is wheat, rice, sorghum and barley both in field before harvest and in storage. This pest is an internal feeder and causes significant loss to cereals affecting the quantity, as well as quality of stored cereal grains through storage.[8]

Such plant materials include powders from parts of the neem tree (*Azadirachta indica*). Mixing dried neem leaves with grain in storage is classic example of natural produce use that has been practiced by farmers in many countries for many years. The main active insecticidal ingredients are present mainly in the seeds, leaves, and other parts of the neem.[9] Its various plant parts have been habitually used to control domestic insect pests in stored grains, crop, in human and cattle medicine. Hence, the objective of this laboratory experiment was to determine the effect of ground neem leaf and orange peels against maize weevil and storage of neem leaf and orange peels on its insecticides movement. Many studies reported its effective use in administration insect pest. Plant based necessary oil like citrus and neem have stepped in market place and is best suited for organic food invention in developing and industrial Countries. Azadirachtin, extracted from neem leaf, act on target organism as antifidant growth regulator. For example, plant powders from neem and citrus had strong insecticidal activity on *Sitophilus oryzae* and *S. granaries, S. zeamais*.[10] The study proves the insecticidal activity of plant powder of neem leaf and orange peels against *S. zeamais* in addition; the current study was designed to investigate the suitability of these plant powders to protect stored maize grains.

MATERIALS & METHODS

Experimental Site: - An experiment was conducted at the laboratory at Department of Botany Jai Prakash University, Chapra. Which is located between latitude 180° (90° at the North and 90° South) and 360° of longitude (+180° eastward and -180° westward). The distance between each degree of latitude is about 69 miles (110 kilometers).

Insect Culture:- The adult *S. zeamais* used for the experiment were cultured in a plastic container under ambient laboratory temperature of 34.5°C. The fresh maize grains were purchased from Mashrak Saran.

Experimental details:- Adult *S. zeamais* used for the experiments were obtained from already infected private warehouses, Mashrak Saran. Plant protection Orange peels were collected from an orange juice seller in Chapra. Neem leaves were obtained from the B.S.M. Degree College Campus in Mashrak.

An amount of five hundred (500) of maize grains per treatment was mixed independently with tested powders. 500 (five hundred) grains of maize were placed is small plastic container.

Two plant powder extract were applied for three weeks to see the efficacy of the powders. Different amount of the powder, thus 5g, 10g, 15g, 20g, 25g, and 30g were measured using an electronic balance scale and were separately kept in small plastic containers. The containers were covered with a fine mesh muslin cloth held in place with rubber bands for exposure to air. Thirty (30) adult *S. zeamais* were introduced into each container and shaken well, so that the powder gets mixed over the weevils. After mixing it is covered with muslin cloth and tied with rubber bands. The grains in the controls contain no plant powder. The setup was kept for 6 weeks and the observations were made weekly.

RESULTS

The safety of stored products by the use of plant materials is a general practice among smallholder farms in India. The plant powder showed different levels of bioactivity against adult *S. zeamais* during the 6 weeks experimental periods (Table-1). The toxicity of powder improved slowly with increasing amount of extract. Neem leaf powder resulted in weevil mortality up to lowest and highest mortality rate been seen in 5g & 30g for neem extract of 24 & 28 in numbers respectively. Where these value was noticed of 22 & 26 at 30g &5g respectively for citrus extracts. The percentage alteration has also been calculated for neem and citrus extracts. The maximum percentage alteration has been calculated is 20 & 6.6 of 5g & 30g for neem extract whereas 16.6 & 3.3 for citrus extract respectively.

Table 1- Percentage weevil mortality of maize grain treated with different dose of plant powder extracts, during summer season.

Total no-of grain taken	Control		Dose (gram)	Experimental					
				Neem extract			Citrus extract		
	Total no-of insect taken	Mortality		Mortality	Differences	% Alteration	Mortality	Differences	% Alteration
500	30	4	5g	24	6	20	22	8	26.6
500	30	4	10g	25	5	16.6	24	6	20
500	30	4	15g	25	5	16.6	24	6	20
500	30	3	20g	26	4	13.3	25	5	16.6
500	30	3	25g	26	2	10	26	4	13.3
500	30	3	30g	28	2	6.6	26	4	13.3

In this study:- The percentage of weevil of maize grain treated with different dose of plant powder extract during summer season have been described in table- 2 & percentage by fig- 2. Dose of neem and citrus extract (5g, 10g, 15g, 20g, 25g and 30g) have been applied on 500 maize grains, inoculated e^- 30 insects.

The lowest and highest mortality rate been seen of 5 & 30g for neem extract of 24 & 28 respectively. Whereas these value was noticed of 4 & 8 at 30g & 5g respectively for citrus extracts. The percentage alteration has after calculated for neem and citrus extracts.

The maximum percentage alteration has been calculated of 20 & 6.6 of 5 & 30g for neem extract whereas 26.6 & 13.3 for citrus extract respectively.

Table 2-Percentage weevil survival of maize grain treated with different dose of plant powder extracts, during summer season.

Total no-of grain taken	Control		Dose (gram)	Experimental					
				Neem extract			Citrus extract		
	Total no-of insect taken	Survival		Survival	Differences	% Alteration	Survival	Differences	% Alteration
500	30	26	5g	6	24	80	8	22	73.3
500	30	26	10g	5	25	83.3	6	24	80
500	30	26	15g	5	25	83.3	6	24	80
500	30	27	20g	4	26	86.6	5	25	83.3
500	30	27	25g	3	27	90	4	26	86.6
500	30	27	30g	2	28	93.3	4	26	86.6

In current study:- The percentage of weevil survival of maize grain treated with six doses of plant powder extract during summer season have been described in table- 2 & percentage by fig- 2. Dose of neem and citrus extract (5 g,10 g.....30 g) have been applied on 500 maize grain, inoculated e^- 30 insects.

The lowest and highest survival rate been seen of 5 & 30 g for neem extract of 6 & 2 respectively. Where of

these value was noticed of 26 & 22 at 30 & 5g respectively for citrus of extracts.

The percentage alteration has after calculated for neem and citrus extracts. The maximum percentage alteration has been calculated of 80 & 93.3 of 5 & 30 g for neem extract therefore 73.3 & 86.6 for citrus extract respectively.

Table 3- Showing damage of maize grain after application of plant extract (N+C) during summer season.

Total no-of grain taken	Control		Dose (gram)	Experimental					
	Total no-of insect taken	Damage		Neem extract			Citrus extract		
				Damage	Differences	% Alteration	Damage	Differences	% Alteration
500	30	30	5g	8	492	98.4	10	490	98
500	30	30	10g	8	492	98.4	10	490	98
500	30	30	15g	6	494	98.8	8	492	98.4
500	30	32	20g	5	495	99	7	493	98.6
500	30	32	25g	5	495	99	6	494	98.8
500	30	33	30g	3	497	99.4	5	495	99

In this survey: -The percentage damage of maize grain treated with different doses of plant powder extract during summer season have been described in table- 2 & percentage by fig-2. Dose of neem and citrus extract (5g, 10g, 15g…….30g) have been applied on 500 maize grain, inoculated e^- 30 insects.

The lowest and highest damage rate has been seen at 5 and 30g neem extract of 8 & 3 respectively. Whereas value was noticed of 490 & 495 at 30 & 5g respectively for citrus of extract.

The percentage alteration has after calculated for neem and citrus extract. The maximum percentage alteration has been calculated of 98.4 & 99.4 of 5 & 30g for neem extract therefore 98 & 99 for citrus extract respectively.

DISCUSSION

Results from the current study shows that the uses of the said botanical powders have the great potential in protection of the maize grains against *S. zeamais*.

Comparing the Neem leaf powder and Orange peels powder treatment, the highest mortality rate was recorded at the neem leaf powder (93.3). However, this high amount recorded at the neem leaf powder is by below than the considered the minimum effective concentration. According to Collingwood and Marchart (1971)[11] the concentration that causes at least 95% mortality is considered the minimum effective concentration. Comparing the findings of the present study with other findings, the present study is in line with the findings of lleke and Oni (2011)[12] and Khaliq *et al.* (2014)[13]. From their study on the toxicity of some plant powders to maize weevil, *S. zeamais* Motschulsky (Coleoptera: Curculiondae) and on stored wheat grains (*Triticum aestivum*), lleke and Oni (2011)[12] found mortality of adult *S. zeamais* that increase with concentration from 2.5-25% w/w. Moreover, Khaliq *et al.* (2014)[13] found at concentration of 5% w/w maize weevil mortality of 35.81% from *Glycyrrhiza glabra* and mortality rate of 35.55% due to 24 h exposure to the same plant.

Weevil Mortality:- Weevil mortality was high in grains treated with the plant extract compared to the control. Maize grains treated with neem leaf powder recorded high weevil mortality than grain treated by citrus peel powders. The control treatment had more weevils surviving than dead as was expected. Higher plant dosage on the other hand were effective and controlling the weevils than lower dosages (5g). It was observed from the experiment that high dosages of the plant extract resulted in high weevil mortality. Trends seen in weevil mortality were inversely correlated with weevil survival as high weevil mortality resulted from low weevil survival and vice-versa. It was also observed that increasing the dosage of neem leaf powder from 30g and completely killed the weevils. Citrus peel powder was effective at 25g and above of the powder applied. Hence it can be said that the higher dosage applied, the more effective it becomes, thus the higher mortality seen in the weevil population.

Results are in harmony with of several previous workers, who found toxicity effect of plant material on stored grain pests. The National Research Council

(1992)[14] showed that azadirachtin, the active component of Neem disrupts the metamorphosis of insect larvae. Akou-Edi (1984)[15] indicated that neem leaf powder and oils were effective in repelling and killing maize weevils. Sharaby (1988)[16] also reported that orange peel oil have toxic effect on weevils due d-limonen. Onu and Sulyman, (1997)[17] pointed out that plant volatile essential oils of some Citrus species had insecticidal properties against stored grains insect pest.

Weevil Survival:- The current study showed that the maize grains treated with the plant extracted powder recorded low weevil survival. Out of the two plant extract used maize grain treated with neem leaf powder had less weevils surviving compared to that leaf treated with citrus peel powders. The control treatment as was expected had almost all the weevils surviving since there was no application of the plant powder extract. On the other hand, plant powder dosage 15g and above was more effective in controlling the weevils than lower dosage (5g & 10g). Appert, (1987)[4] reported that the leaves of neem tree garlic, dry peel of citrus, are used for storing maize by farmers who believe that these plants could deter storage insects. Sowunmi and Akinusi (1983)[18] indicated that maize seeds could be protected from attack by weevils using 1% & 2% & neem kernel powder.

Grain Damage:- In conditions of grain damage, minimal damage from weevils was recorded in the grain treated with the two plant powder extracts. Grain damage was very low in seeds treated neem leaves powder. The low values recorded can be attributed to the high mortality of the weevils to the plant extract application since low numbers of weevils are ineffective in causing substantial damage in storage. The lowest and highest damage rate has been seen at 5g, & 30g neem extract of 8, & 3 respectively. Whereas value was noticed of 488 & 495 at 5g & 30g respectively for citrus of extract. The percentage alteration has after calculated for neem and citrus extract. The maximum percentage alteration has been calculated at 98 & 99.4 of 5g & 30g for neem extract while 97.6 & 99 for citrus extract respectively.

Kraus *et al.* (1981)[19] also reported that neem extracts have anti-feeding and growth disrupting properties.The orange peel oil was toxic to *Culex pipiens*.[20] Odeyemi *et al*. (2000)[21] indicated that application of citrus peel powder caused mortality of weevils.

CONCLUSION

The results showed that plant powder extracts were more effective at higher dosage of 25g & 30g. It was also observed that the quality of the maize grains was better in grains stored with neem leaf powder and orange peel powders. The two plant powder indicates that higher dosage is more efficient in management of pests. Again, when the plant powders were applied to maize grains in storage, there were important reductions in maize damage.

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Composition and abundance of zooplankton in freshwater perennial ponds of Madhepura district, North Bihar

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Abstract- The study was conducted to explore the zooplankton composition and abundance in freshwater perennial ponds, locally named as Puraini Pokhar (Pond 1) and Baghmara Talab (Pond 2) located at Madhepura district, North Bihar. Water and zooplankton samples were collected monthly from March, 2019 to February, 2020. The investigation on physico-chemical characteristics at both the ponds revealed its alkaline nature, suitable for aquaculture practices. Four groups of zooplankton viz. Rotifera, Cladocera, Copepoda and Ostracoda comprising of total 26 species were identified.Of these 23 species consisting 13 species of Rotifera followed by Cladocera 5 species, Copepoda 4 species and Ostracoda 1species were recorded from Pond 1, though, 17 species comprising 11 species of Rotifera followed by Cladocera and Copepoda 3 species each were identified from Pond 2. Zooplankton abundance ranged from 82 ind./l to 318 ind./l in Pond-1 and from 54 ind./l and 376ind./l in Pond-2. Zooplankton density was minimum during monsoon and maximum in summer. Percentage compositions showed dominance of Rotifera contributing 55.56% followed by Copepoda 22.53%, Cladocera 20.73% and Ostracoda1.18% in total zooplankton abundance in Pond 1, while, Rotifera contributing 57.75%, followed by Copepoda 23.04% and Cladocera 19.86% in Pond 2. Most dominant rotifers were *Brachionus angularis*, *Brachionus caudatus*, *Brachionus falcatus*, *Filinia longiseta* and *Keratella tropica* in the studied ponds. No important differences were observed in abundance of Cladocera and Copepoda. Among cladoceran group, *Daphnia* sp. were found most dominant followed by *Moina dubia*. Copepoda was dominated by *Nauplii.* Sørensen's similarity index was applied to assess similarity in the species composition among the ponds. Index values showed highest similarity in copepods species 85.71% followed by rotifers 75% and cladoceran 50%. The distribution and density of zooplankton was influenced by physical and chemical factors of the pond environment were discussed.

Key words: Zooplankton, abundance, composition, perennial ponds, physico-chemical factors

INTRODUCTION

Zooplanktons are most common heterogeneous assemblage of free floating minute animal forms found in all aquatic ecosystems. Zooplankton comprises of various taxonomic groups belonging to 4 major groups, Rotifera, Cladocera, Copepoda and Ostracoda often dominate entire consumer communities. Zooplanktons are most important biotic components influencing all the functional aspects of aquatic ecosystem, such as food chain, food web, energy flow and cycling of materials.[1] They are highly sensitive to physical and chemical changes in water. Zooplankton occupies the secondary level in the food chain and mediate transformation of the food energy synthesized by the phytoplankton to the higher trophic level.[2] They are

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important food for fish and many invertebrate predatorsand graze heavily on algae.Carp fish fry feed mostly on zooplankton.[3] They play an important role in indicating the trophic condition of aquatic system.[4] Zooplankton composition and abundance of zooplankton is affected by several environmental and biological factors and also limited by the various physico-chemical parameters.[5,6] There is, however, no information on zooplankton communities found in the water bodies of this region, particularly in Madhepura district. Thus, the present investigation was made to determine physico-chemical condition and zooplankton composition and abundance as well as community similarities in two perennial ponds of Madhepura district, North Bihar.

MATERIALS & METHODS

Study area

The present study was carried out on freshwater perennial ponds namely Puraini pokhar (Pond-1) and Baghmara talab (Pond-2) of Madhepura district, North Bihar (Fig 1). Pond-1 is a rectangular shape water body spread over an area of 2.79 acre located at latitude 25°35'53" N and longitude 86°59'31"E. Fishery of pond is being managed by local fishermen. Pond-2 is an irregular shape surrounded by farming lands spread over an area of 2.2 acre situated at latitude 25° 34' 26" N and longitude 87° 0' 34" E. No organized fishing activity is being carried out in this pond.

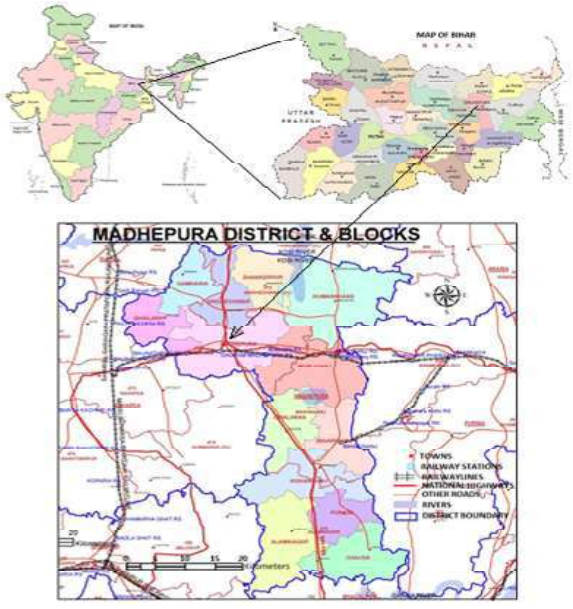


Fig. 1: Geographical location of Madhepura district, North Bihar

Water quality monitoring

The physico-chemical parameters of water were analyzed monthly from March 2019 to February 2020. Water samples were collected in 2-liter polyethylene containers between 08 am to 10am. Temperature was recorded using mercury thermometer graduated up to 50°C. Transparency was measured by Secchi disc (20cm diameter). The parameters, such as temperature, Secchi depth, pH, dissolved oxygen, free carbon dioxide, carbonate and bicarbonate alkalinity were recorded at sampling site and parameters like conductivity, calcium, magnesium, chloride, nitrate-nitrogen, phosphate-phosphorus and BOD were analyzed in the laboratory. Physico-chemical parameters were analyzed following standard methods.[7]

Zooplankton sampling

Zooplankton samples were collected using plankton net made up of bolting cloth, No. 25 (Mesh size approx. 56m). For zooplankton collection 50 liter of sub-surface pond waters were filtered through the net from the ponds. Samples were then transferred to specimen bottles and preserved by adding 5% formalin on site. Identification and analysis of the zooplankton was conducted in laboratory under a binocular compound microscope with different magnifications. Quantitative analysis of zooplankton was done using Sedgewick-Rafter plankton counting chamber. Analysis involved transfer of 1 ml sub-sample from each of the samples to the Sedgewick-Rafter chamber using pipette. The zooplanktons were identified up to genus and species level with the help of standard taxonomic references,[1,8-10] and counts were expressed numerically per liter of water of the pond. Community similarities were analyzed using Sørensen's similarity index.[11]

RESULTS & DISCUSSION

Physico-chemical characteristics

The range and mean with standard error (SE) of physico-chemical variables of water are shown in Table 1. Seasonal variations in water temperature might be related to the weather conditions. No important differences were observed between the water temperature ranged between 17.1°C and 30.9° C in Pond-1 and between 18.5° C and 31.2° C in Pond-2. Maximum water temperature was recorded in summer might be due to longest sunshine hours, higher atmospheric temperature

Table 1: Physico-chemical parameters of Pond 1 and Pond 2 showing the ranges, mean averages and Standard Error (±SE)

Parameters	Pond-1		Pond-2	
	Min. - Max.	Min. - Max.	Min. - Max.	Min. - Max.
Water temperature ^{o}C	17.1 - 30.9	26.43 ± 0.82	18.5 - 31.2	26.65 ± 0.76
Secchi disc depth cm	19.3 - 55.3	36.42 ± 2.24	11.2 - 36.7	23.83 ± 1.41
EC µmohs/cm	91.2 - 221.7	164.11 ± 7.47	183.6 - 412.3	286.33 ± 13.53
pH	6.9 - 8.3	7.6 ± 0.07	6.8 - 8.1	7.5 ± 0.06
Dissolved oxygen mg.l^{-1}	5.2 - 8.2	6.69 ± 0.19	5.1 - 8.0	6.33 ± 0.17
Free carbon dioxide mg.l^{-1}	2.1 - 7.6	5.6 ± 0.71	5.4 - 10.1	7.9 ± 1.07
Carbonate alkalinity mg.l^{-1}	6.8 - 13.7	9.8 ± 0.44	11.4 - 26.8	18.98±0.88
Bicarbonate alkalinity mg.l^{-1}	98.6 - 253.2	161.31 ± 7.73	174.2 - 311.3	214.4 ± 7.78
Calcium mg.l^{-1}	17.3 - 33.6	25.13 ± 0.87	23.2 - 45.3	33.13 ± 1.19
Magnesium mg.l^{-1}	6.2 - 15.4	9.52 ± 0.45	4.8 - 21.5	14.7 ± 0.94
Chloride mg.l^{-1}	9.7 - 36.9	32.08 ± 1.65	14.9 - 54.5	34.35±2.0
Nitrate-nitrogen mg.l^{-1}	0.298 - 0.736	0.782 ± 0.03	0.289 - 0.845	0.538 ± 0.04
Phosphate-phosphorus mg.l^{-1}	0.049 - 0.352	0.160 ± 0.02	0.131 - 0.681	0.334 ± 0.03
BOD mg.l^{-1}	2.1 - 5.9	3.80 ± 0.21	3.5 - 8.7	6.08 ± 0.28

and decreased water level, while minimum during winter could be due to shorter sunshine hours and low temperature. Similar trends were reported by several workers.[12,13] Secchi disc depths recorded showed significant difference, higher in Pond-1(36.42 ± 2.24 cm) than in pond-2 (23.83± 1.41cm). Data reflects that light penetrated much deep in water column in Pond-1, however, light penetration in water was blocked by suspended particles thus reducing water clarity in Pond-2. Conductivity is an important parameter to evaluate water quality.[14] The value of conductivity was higher in pond-2 (286.33±13.53 µmohs/cm) than in Pond-1 (164.1±7.47 µmohs/cm). pH ranged between slight acidic and alkaline, varied between 6.9 and 8.3 in Pond-1 and between 6.8 and 8.1 in Pond-2. The pH value was lower during monsoon probably due to addition of the rainwater. The amount of dissolved oxygen was slight higher in pond-1 (7.6±0.07) than in Pond-2 (7.5±0.06). Dissolved oxygen was observed maximum during winter probably due to high rate of photosynthetic activities due to greater phytoplankton density and low temperature holds more oxygen, while minimum in monsoon could be due to high decompositions rate and high rate of respiration by the aquatic biotaconsume more oxygen. Free carbon dioxide was recorded only in monsoon could be due to higher decompositions rate released more carbon dioxide in water. Carbonates was recorded when the free carbon dioxide was absent. Carbonate and bicarbonate along with free carbon dioxide form an equilibrium system in water and acts as buffer system not allowing pH to fluctuate more. Higher value of calcium and magnesium was observed in Pond-2 might be due to discharge of domestic sewage and lower in Pond-1 could be due to its utilization by aquatic organisms. Chloride is good indicators of organic pollution.[15] The higher amount of chloride was recoded in Pond-2 (34.35±2.0 mg/l) than in Pond-1(32.08±1.65 mg/l). Maximum chloride was noticed in summer probably due to decreased water level concentrate content, while minimum during monsoon could be due to increased water level dilutes content.[16,17] The higher amount of nitrate-nitrogen was recorded in Pond-1 (0.782±0.03 mg/l) than in Pond-2 (0.538±0.04 mg/l). Nitrate value was maximum during summer might be due to decreased water level concentrate content and minimum during winter could be due to its utilization by plant and algae. The amount of phosphate-phosphorus was higher in Pond-2 (0.334±0.03 mg/l) than in Pond-1 (0.160±0.02 mg/l) could be due to discharge of domestic sewage and wastes from nearby cultivated lands increasing content. Phosphate-phosphorus was maximum in summer could be due to higher temperature increasing biological degradations of organic materials released more phosphate in water. BOD value was higherin Pond-2 (6.08±0.28 mg/l) indicated the higher loads of organic matters than in Pond-1 (3.80±0.21 mg/l). BOD value was maximum in summer might be due to increased microbial activities

consume more oxygen and minimum during winter could be due to reduced decomposition rates and decreased temperature. Physico-chemical parameters of pond water measured were found within suitable range for zooplankton production.

Analysis of zooplankton

The study recorded 26 species of zooplankton belonging to four important groups, viz., Rotifera, Claocera, Copepoda and Ostracoda. Systematic list of recorded zooplankton has been presented in Table 2, and abundance and percentage distribution shown in Figure 2 to 4. Out of 26 species identified, 23 species were reported from Pond-1, of these, 13 species were Rotifera followed by 5 Cladocera, 4 Copepoda and 1 Ostracoda, while 17 species recorded from Pond-2, of these 11 species were Rotifera followed by 3 Cladocera and 3 Copepoda. Zooplankton abundance ranged between 82 individual/l and 318 individual/l in Pond-1, whereas between 54 individual/l and 376 individual/l in Pond-2. Zooplankton abundance followed a bimodal annual variation, with major peak during summer, after following decline in monsoon attained minor peak during winter (Fig 2). Zooplankton density was highest during summer might be due to availability ofrich food in form of algae, bacteria and detritus.[18,19] Zooplankton density appears to decline in monsoon probably linked with dilution factors as addition of rainwater dilutes population density.[20]

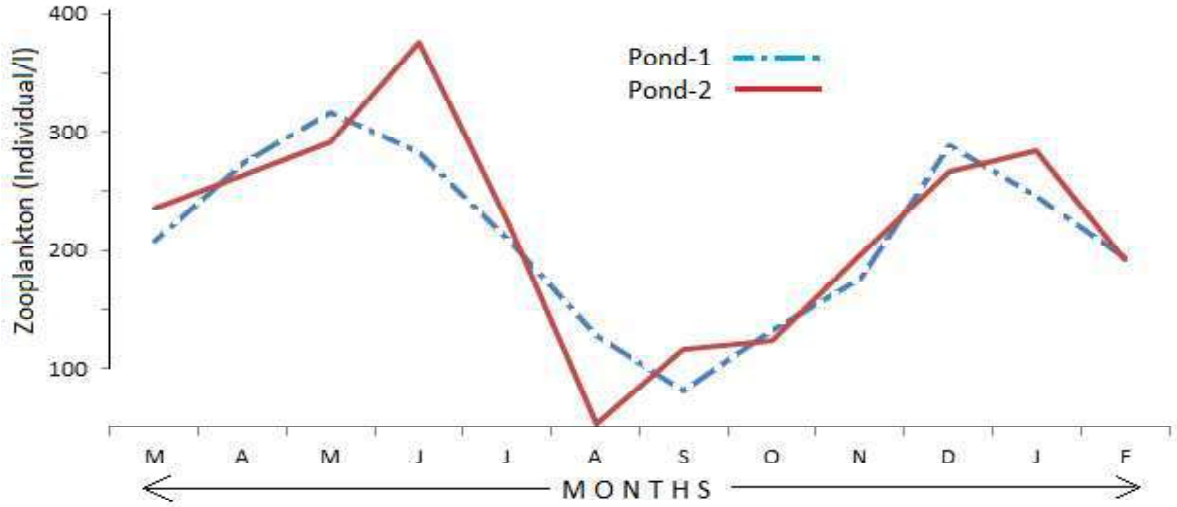


Fig. 2: Monthly variation of total zooplankton population in the studied ponds.

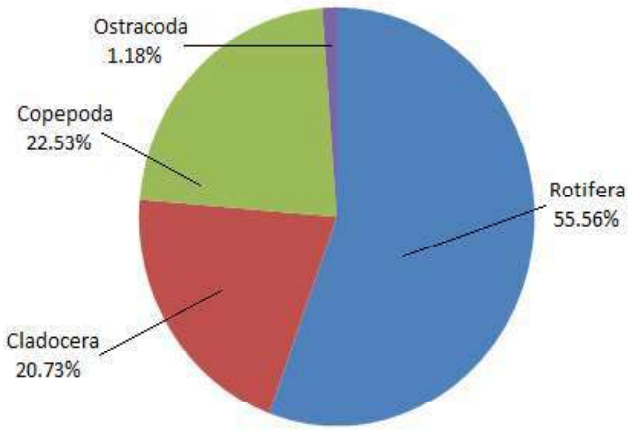


Fig. 3: Percentage distribution of zooplankton in Pond-1.

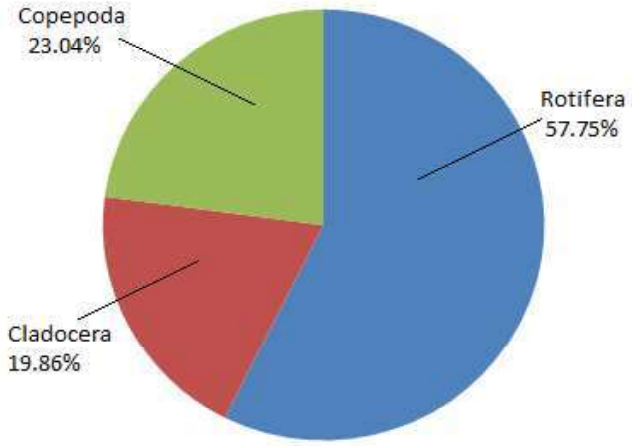


Fig. 4: Percentage distribution of zooplankton in Pond-2.

Table 2: Systematics and occurrence of zooplankton species in ponds.

Zooplankton	Pond-1	Pond-2
Phylum: Rotifera		
Subclass: Eurotatoria		
Superorder: Monogononta		
Order: Ploima		
Family: Brachionidae		
Brachionus angularis (Gosse, 1851)	+	+
Brachionus calyciflorus (Wierzejski, 1891)	+	+
Brachionus caudatus (Borrois & Daday, 1894)	+	+
Brachionus falcatus (Zacharias, 1898)	+	+
Brachionus forficula (Wierzejski, 1891)	+	+
Keratella tropica (Apstein, 1907)	+	+
Keratella cochlearis (Gosse, 1851)	+	
Keratella lenzi (Hauer, 1938)	-	+
Family: Lecanidae		
Lecane sp.	+	-
Monostyla sp.	+	+
Family: Filinidae		
Filinia longiseta (Ehrenbergh, 1834)	+	+
Filinia terminalis (Plate, 1886)	+	-
Family: Asplanchnidae		
Asplanchna sp.	+	+
Family: Synchaetidae		
Polyarthra sp.	+	-
Family: Testudinellidae		
Testudinella sp.	-	+
Super class: Crustacea		
Class: Branchiopoda		
Super order: Cladocera		
Family: Bosminidae		
Bosmina longirostris (O.F. Muller, 1776)	+	-
Family: Daphania		
Ceriodaphnia sp.	+	-
Daphnia carinata (King, 1853)	+	+
Daphnia lumholtzi (Sars, 1885)	+	+
Family: Moinidae		
Moina dubia (Guerne & Richard, 1892)	+	-
Moina brachiate (Jurine, 1820)	-	+
Class: Copepoda		
Family: Cyclopoidae		
Cyclops sp.	+	+
Diaptomus sp.	+	+
Mesocyclops	+	-
Nauplii larvae	+	+
Subclass: Ostracoda		
Cypris sp.	+	-
Total	**23**	**17**

Rotifera was the most abundant group of zooplankton contributing 55.56% in total zooplankton abundance in Pond-1 followed by Copepoda 22.53%, Cladocera 20.73% and Ostracoda 1.18%, while in Pond-2, Rotifera contributing 57.75% followed by Copepoda 23.04% and Cladocera 19.86% (Fig. 3 and 4). Rotifera abundance was dominated by *Brachionus angularis*, *Brachionus caudatus*, *Brachionus falcatus*, *Filinia longiseta* and *Keratella tropica*. Dominance of Rotifera over Cladocera and Copepoda in terms of species number was reported by various studies.[21,22] *Brachionus* species dominated rotifers reflect that ponds investigated were approaching towards eutrophication.[23,24]

Copepoda was second most abundant group consists of *Cyclops* sp., *Diaptomus* sp., *Mesocyclops* sp. and *nauplii* larvae. Population density of Copepoda was minimum in monsoon and maximum during summer. Similar trend was recorded by various workers.[25,26] The occurrence of *Nauplii* throughout the study period showed an active continuous reproductive phase of the cyclopoids.

Cladocera was the next dominant group of zooplankton comprised of *Ceriodaphnia* sp., *Bosmina longirostris*, *Daphnia carinata* and *Daphnia lumholtzi*. After attaining highest density in summer cladocerans reached least abundance in monsoon and then increased during winter probably due to high pH and rich phytoplankton. Cladoceran density was highest during summer could be due to availability of abundant food. *Cerio daphnia* sp. and *Moina dubia* were only found in Pond-1 and *Moina brachiate* in Pond-2 (Table 2).

The group Ostracoda was found comparatively least in number represented by a single species, *Cypris* sp. contributing insignificantly to total abundance of zooplankton.

Community similarities

Sørensen's index[11] used to know compositional similarity between zooplankton communities of two ponds. The similarity index value of zooplankton and their different groups is presented in Table 3. Zooplankton communities of Pond-1 and 2 registered 70% similarity (*vide* Sørensen's similarity index) indicating more homogeneity in the species composition. Results of Sørensen's index showed 85.71% similarity, in the species composition of Copepoda followed by Rotifera 75% and Cladocera 50% (Table 3). The index value was relatively much higher indicating more similarity between two communities of zooplankton could be attributed to greater homogeneity in environmental condition, as studied ponds are located in same climate and similar geographic location. However, it is rather difficult to account that environmental factor alone responsible for high similarity in zooplankton communities, because the other factors, like high pH, conductivity and nutrients were also accountable for greater similarity in species compositions. The results of similarity index analysis indicate minor differences between two zooplankton communities.

Table 3: Sørensen's similarity index values between zooplankton communities

Zooplankton	Co-efficient of similarity (%)	Dissimilarity (%)
Rotifera	75.00	25.00
Cladocera	50.00	50.00
Copepoda	85.71	14.29
Total zooplankton	70.00	30.00

CONCLUSION

The results of present study showed that zooplankton composition and abundance are not same in perennial ponds. No single factor is responsible for this variability. Zooplankton communities were comprised of mainly Rotifera, Cladocera, Copepoda and Ostracoda and their occurrence and abundance indicates water bodies studied were productive. Zooplankton contributes significantly to the secondary production as they forms important food constituent of fishes and many other aquatic animals thus play major role in flow of energy in aquatic ecosystem. Results showed that most of the physico-chemical parameters were within the optimum level thus water is in the ideal state and is conducive for growth and reproduction of zooplankton, ultimately fish production. The present information on zooplankton abundance and composition would provide a valuable tool for further study and monitoring of aquatic ecosystem.

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Role of nutmeg in minimizing hepatotoxicity due to arsenic in albino rats

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Abstract- Arsenic is pervasive element in the environment and cause serious health problem in most of the countries. In India west Bengal, Bihar and Jharkhand found arsenic contamination in underground water. It causes serious health problem in human such as cancer in liver, kidney skin etc. It became severe in low-income nation such as Bangladesh. Arsenic out-turn into liver injury. In present study was conducted to evaluate the protective role of nutmeg (Myristica fragrans) against arsenic induced toxicity in albino rats. Albino rats were divided into three groups. Group A were control rat, group B received an acute dose of arsenic (8mg/kg bw) orally, group C received an acute dose of arsenic followed by daily administration of nutmeg (300mg/kg bw) orally. Arsenic treatment leads to increase in the liver injury. Biochemical analyses of treatment group show Arsenic induced rats had significantly (p\0.0001) found the changed serum levels of Serum Glutamic-Pyruvic Transaminase (SGPT), Serum Glutamic-Oxaloacetic Transaminase (SGOT), Alkaline Phosphatase (ALP), total bilirubin, urea, uric acid, creatinine and albumin and its degenerative effect; But, after the administration of M. fragrans extract, there was significant ($p<0.0001$) restoration observed in these liver and kidney function. Nutmeg become significant protection in the palliation of arsenic induced liver injury. The aim of the present review is to introduce *Myristica fragrans* as a potent medicinal property by highlighting its pharmacological and clinical application.

Key words: Arsenic, Hepato-Renal, *Myristica fragrans*, Medicinal Property

INTRODUCTION

Arsenic is a highly toxic metalloid, which is distributed in the natural and anthropogenic sources.[1] It became an important health concern. Arsenic in the bedrock or soil readily dissolves in surrounding and ground water and a high concentration of arsenic i.e above the WHO guideline level of 10µg L-1. Diseases associated with arsenic were pandemic in many countries.[2] Skin, liver, lung, and bladder cancers are caused by chronic exposure to arsenic-contaminated water and food.[3-7]

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Due to growing human activities such as mining, smelting, and pesticide manufacture, the risk of arsenic compounds has been elevated, resulting in arsenic accumulation in the soil.[8] Arsenic induces lipid peroxidation and thus act as a prooxidant and bring about protein and enzymeoxidation,glutathione depletion, DNA oxidation and DNA adducts.[9] Arsenic become serious problem to human life.

Plants and its products are being used as medicine since long year. There are large number of phytochemical property is found in plant food product.[10] *Myristica fragrans* is commonly known as "nutmeg". It produces

two spiece: mace and nutmeg. It belongs to the family myristicacea in the order magonliales. It possesses various therapeutic properties. For a long time *M.fragrans* has been used as a folklore medicine for treating diarrhea, mouth, sore and insomnia.[11] Nutmeg crude extract has been shown to have chemopreventive and anti-Helicobacter pylori properties.[12] It also shows hepatoprotective activities. Nutmeg contains numerous constituents, such as essential oil (monoterpenes, phenylpropanoids, myristicin, elemicin, safrole, eugenol, methyl isoeugenol), lignans and neolignans, diphenylalkanes, phenylpropanediols, steroids and cyclobutanones.[13] Cytotoxic and apotoxic effects of Myristicine have been reported such that cell viability was reduced by exposure to Myristicine in a dose dependent manner.[14]

The liver is an intra-abdominal organ (belonging to the gastrointestinal tract) that plays an important role in the detoxification and excretion of many endogenous and exogenous substances. The liver is a natural chemical factory that helps metabolism and detoxifies complex molecules. It neutralizes toxins and produces bile which helps in the digestion of fats and the elimination of toxins from the intestines.[15] The liver plays an important role in breaking down fat, converting glucose into glycogen, and maintaining proper blood sugar levels. Nutmeg is commonly used at home for cooking, flavouring, and in medicine, which is believed to have intermediate histological effects also on the kidney distorting the cyto-architecture of the renal corpuscle[16] at higher dose (500mg/kg) as aphrodisiac dose and psychoactive agent in male Wister rats. Nutmeg allergies are extremely rare, and it is a very safe spice.

The aim of the present study is to introduce *M. fragrans* as a potent medicinal plant by highlighting its traditional applications as well as the recent findings for novel pharmacological and clinical applications. Nutmeg is often used as a spice in a variety of meals, as well as in teas and soft drinks, as well as in milk and alcohol. the possible effects of nutmeg on the microscopic architecture of the liver would be assessed in the present studies.

METHOD & METHODOLOGY

Chemical :- Chemical Arsenic was used as sodium (meta) arsenite (90%) manufactured by Sigma-Aldrich, USA (CAS Number: 7784-46-5; S7400-100G), Lot# SLBH5736V, P Code 1001683292 was purchased from the licensed and approved scientific store of Patna, Bihar, India.

Preparation of *Myristica fragrans* seed Ethanolic Extract :- In the present study, the nutmeg from the nearby local market of sufficient amount were bought, washed with distilled water to remove dirt and soil, dried in shade and finally crushed to powder. The powdered material was soaked with absolute ethanol approximately 200 ml ethanol. The material was then left for 48 hrs, the material was put in the water bath at 65^0C and pressure was 3000-4000 psion a rotary evaporater. Now ethanolic extract was stored in fresh container. For administration 10 ml dose was prepared before administration.

Animal :- Twenty-four female healthy Charles Foster rats, were randomly selected of 8 weeks old weighing 160 g to 180 g for this study. The animals were procured from the animal house of Mahavir Cancer Sansthan and Research Centre, Patna, India (CPCSEA Reg-No. 1129/bc/07/CPCSEA). The experimental work was approved by the Institutional Animal Ethics Committee (IAEC) with IAEC No. 2020/1B-27/08/20. All animal trials followed the rules set forth by the Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), based in New Delhi, India.

Experimental Design :- Rats were randomly distributed into control and treated groups:

Group I: (n = 6) Control. Group II: (Arsenic treated)-18 rats were orally induced with arsenic at the dose of 8 mg/kg body weight/day for 90 days. After 90 days of treatment with arsenic, group II rats were further divided into three sub groups (n = 6 each).

Subgroup I: (Arsenic-treated rats)-Rats were euthanized to measure the level of arsenic toxicity.

SubgroupII: (Arsenic pretreated control subgroup)-Rats were left untreated for the next 90 days to allow for auto recovery.

Subgroup III: (*M.fragrans* administered sub group)-Rats were orally administered with *M.fragrans* ethanolic rhizome extract at the dose of 300 mg/kg body weight/day for further next 90 days.

Biochemical Assay :-Biochemical analysis was performed through the serum by standard kit process (Coral crest) on (UV- Vis) spectrophotometer (UV-10, Thermo Scientific, USA). In liver function tests (LFT), Serum glutamate pyruvate transaminase (SGPT) and

serum glutamate oxaloacetate transaminase (SGOT) were measured using the Reitman and Frankel method, alkaline phosphatase (ALP) using the the Kind and King method, and total bilirubin activity using the Jendrassik and Grofs method.

RESULT

When compared to the control group, the arsenic-treated group had a substantial (p0.0001) rise in serum levels of SGPT, SGOT, ALP, and total bilirubin. The presence of an elevated hepatic serum marker implies a problem with liver function.

However, when arsenic-treated rats were left without medication for another 90 days for auto repair, serum levels of SGPT, SGOT, total bilirubin, and ALP were significantly lower (p0.0001) than when arsenic-treated rats were left without treatment. After the *M.fragrans* administration upon arsenic treated rats, there were significant (p\0.0001) decrease in the serum levels of SGPT, SGOT, ALP and total bilirubin in comparison to the arsenic treated control rats, denoting the protective effect of the plant extract against arsenic induced hepatotoxicity (Figs. 1, 2, and 3).

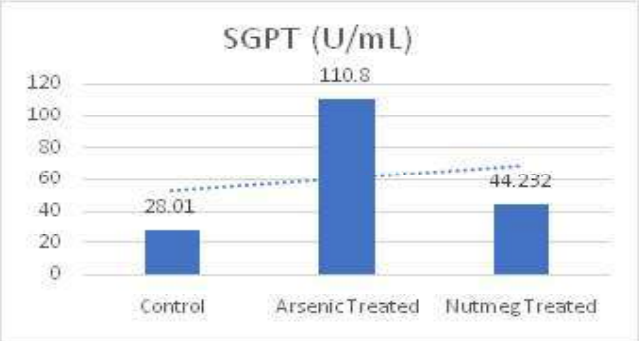


Fig. 1 - Comparative level of SGPT in various groups of rats(n=6)

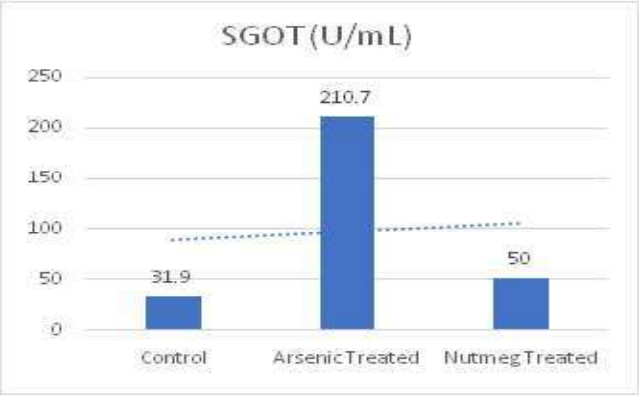


Fig. 2 - Comparative level of SGOT in various groups of rats (n=6)

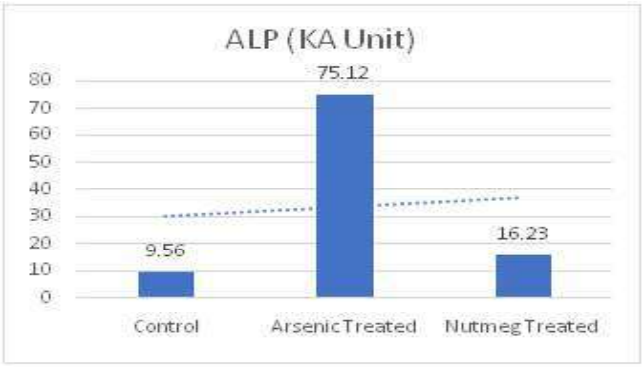


Fig. 3 - Comparative level of ALP in various groups of rats (n=6)

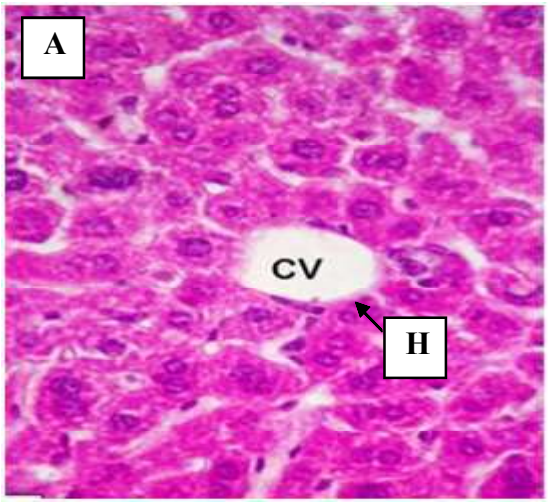
DISCUSSIONS

Now a day's no of people from different countries like India, Bangladesh and West Bengal suffers from chronic and acute toxicity. The chronic exposure to arsenic causes various types of carcinogenic and non-carcinogenic health effects. Arsenic has been found to affect a few downstream signalling cascades, as well as cell development, proliferation, and death. Prolonged consumption of arsenic contamination liquids such as flowle's solution has been known to cause similar hepatic lesions. In the present study, arsenic treated rats showed significantly increased in the levels of SGPT, SGOT, ALP and total Bilirubrin. It causes impairment in liver function and its property as the human subjected to long term consumption of arsenic contaminated water was needed for generation of hepatic fibrosis. After 6 months of arsenic exposure, there is elevation of hepatic arsenic, it confirms arsenic accumulation in the liver.

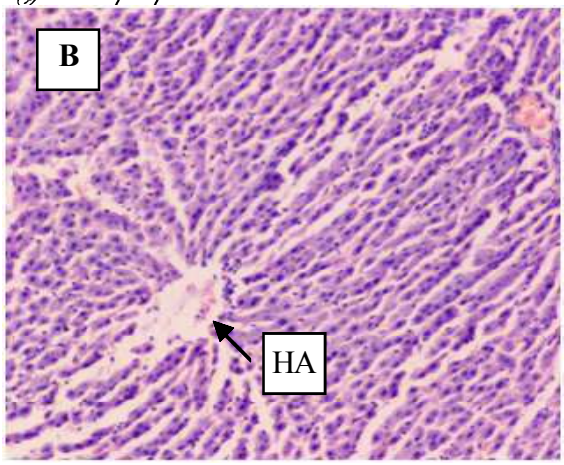
Gora *et al.* (2015)[17] evaluated that high dose of arsenic cause acute liver damage and hepatocyte necrosis, causing hepatocyte enzymes to leak into the blood. In addition, they observed that the degree of damage to hepatocytes was detected by the activity of antioxidant enzymes. In the present study it is observed that the relative weight of the rat liver was increased in arsenic treated in comparison to control rat. This observation was confirmed in the study by Shiguang and Beynem (2001)[18]., they also reported a significant increase in the liver weight in male rat feed 100mg/kg of arsenic for 2 week. Because the liver is a highly active metabolic site, it is a major site of arsenic poisoning and is involved in the methylation process using S-adenosylmethionine as the methyl donor by the arsenic methyl transferase enzymes GSH as an essential cofactor.[19]

Because the liver is the primary site of protein synthesis, arsenic-induced hepatic damage results in a decrease in the level of antioxidant enzymes with arsenic trioxide. Recent studies have clearly shown that arsenic compounds during metabolism in cell generate reactive oxygen species, hydroxy radical and hydrogen peroxide leading to oxidative stress. Liu *et al.* (2001)[20] suggested that enhanced production of free radicals and inhibition of antioxidant enzymes as possible to explain arsenic induced oxidative damage.

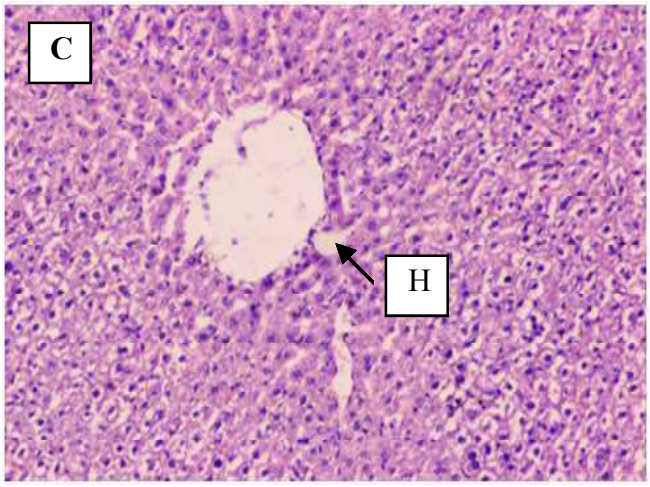
Arsenic induced toxicity arousing increased production of hydrogen peroxide. These effect cause formation of reactive oxygen species resulting in oxidative stress.[21] As shown in fig. 1, the activity of SGPT, SGOT, ALP, and total bilirubin in serum can be used to identify liver impairment. The increase in SGOT and SGPT in serum may be due to hepatocellular necrosis, which causes increase in the permeability of the cell membrane resulting in release of transaminases in the blood stream consequently, the hepatic serum biomarkers perturbations seem to be correlated with the liver histopathological study. The histological examination revealed severe abnormalities in the hepato-cellular architecture, including degraded hepatocytes, sinusoidal vacuolizations, and portal vein haemorrhages, which are consistent with prior studies.[22] Furthermore, serum levels of SGPT, SGOT, ALP, and total bilirubin were significantly reduced after *M.fragrans* was administered to arsenic-pretreated rats. *M.fragrans* also stabilised the hepatic membrane's functional integrity and histological alterations.



A: The normal architecture of the liver was visible in the control liver segment, with well-arranged H and CV X500.



B: Degeneration in the H, haemorrhages in the PV, and vacuolizations in the sinusoids X400 in an arsenic-treated liver segment.



C: The *M.fragrans* administration upon arsenic pretreated sub group liver sections showing significant restoration in the CV and H X500.

(CV- Central vein, H- hepatocytes, HA- Hepatic Artery)

Fig. 4 Microphotograph of liver sections stained with hematoxylin and eosin

CONCLUSION

The overall conclusion of the study is that arsenic intoxication causes severe liver damage. Furthermore, there was a considerable improvement in the liver's functioning after *M. fragrans* administration. Through suppression of lipid peroxidation, there was also considerable restoration at the cellular level. Hence, *M. fragrans* possesses phytoremedial properties against arsenic induced toxicity in rats, which in the future might be taken for the therapeutic development of antidote drug against arsenic poisoning in liver and kidney.

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Addition of three new species to the flora of Allahabad (Prayagraj), Uttar Pradesh, India

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Abstract- The author continuously visited to study area for the floristic survey which deals with three new species addition to the Flora of Allahabad, Uttar Pradesh. These three species are herbaceous in nature which is *Portulaca pilosa* L. (Portulacaceae), *Pentapetes phoenicea* L. (Malvaceae) and *Cyperus digitatus* Roxb. (Cyperaceae). It was first noticed between June to September 2021, and was identified as new addition from Allahabad (Prayagraj) District, Uttar Pradesh. Brief description provided with up to date citation, description, photographs, ecology and geographical distribution.

Key words: Addition, *Portulaca pilosa* L., *Pentapetes phoenicea* L., *Cyperus digitatus* Roxb., Prayagraj, Uttar Pradesh.

INTRODUCTION

The floristic documentation is mainly based on the field observations. These species are mostly occurring in waste places, moist and cultivated field which is distributed tropical and sub-tropical region of the world. There are 2 species of genus *Portulaca* L. (Portulacaceae), 7 genus of family Malvaceae and 25 species, 2 subspecies of genus *Cyperus* L. (Cyperaceae) out of 713 species belonging to 419 genus and 113 families of angiosperms by Mishra and Verma (1992)[1] made a latest complete floristic account of Allahabad district.

A large number of research works has been published from study area by various workers in a form of floristic accounts, ethnobotanical observations and ecological studies. The floristic work was first time undertaken by Srivastava (1938)[2], biological spectrum of Allahabad flora Srivastava (1944)[3] and a supplement to flora of Allahabad Srivastava (1949)[4]. Panigrahi and Arora (1962)[5], Panigrahi and Saran (1968)[6] also have been conducted township floristic work. After Botanical Survey of India, central circle, Allahabad undertook a project to bring out an illustrated manual of the flora of Allahabad town (Rajagopal 1965[7], Rajagopal and Panigrahi 1965[8], Rajagopal and Panigrahi 1966[9]). Sharma and Pandey (1984)[10] have published an enumerative account of exotic species in district Allahabad. So far, there is no authentic published record has been available from the study area of these three new species.

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MATERIALS & METHODS

During the intensive floristic survey, the author found that most of the parts of study area are abundance of grasslands and cultivated fields. The *Portulaca pilosa* L., *Pentapetes phoenicea* L. and *Cyperus digitatus* Roxb. recorded first time from Allahabad. These plants were collected from different localities of Prayagraj district in

between the month of May to December (2021). The permanent characters which are habit / habitat, flower color, fruiting, seeds as well as other key points and photograph along with the latitude and longitude were also recorded. The flowers were dissected out and plants were carefully identified with regional floras that is Flora of Upper Gangetic plains[11] and flora of Uttar Pradesh[12]. The specimens were selected and then were pressed and dryied. To avoid fungal infection specimens were poisoned with alcohol and mercuric chloride solution. The well prepared specimens pested on mounting board listed as per the Bentham and Hooker (1862-1883) system of classification. The specimens were stored in Duthie Herbarium, Department of Botany, University of Allahabad, Prayagraj.

Study area

In 2018 the name of the district was change to Prayagraj which was previously known as Allahabad. Prayagraj district having very rich diversity[13] which is part of Gangetic plains of India. It located at 25.45°N 81.84°E in the southern part of the Uttar Pradesh at an elevation of 98 meters (322 ft). It is sharing border with Rewa District (Madhya Pradesh) to the South, Jaunpur District to the East, Kaushambi District to the west, Mirzapur District to the East, Pratapgarh District to the North. It occupies an area of approximately 365 km^2.

RESULT & DISCUSSION

A field survey was conducted by the author in the study area to know the three new species which belong to the three different genus and families. Out of three species, two species belongs to dicot (Portulacaceae & Malvaceae) and one species belongs to monocot (Cyperaceae) but all were herbaceous in nature. The plant specimens were collected and correlate with the occurrence of various herbaceous species. Reporting of these new species indicate towards increase in the diversity of herbaceous flora of Prayagraj district. In this continuation *Portulaca pilosa* L. and *Cyperus digitatus* Roxb. is included as species and *Pentapetes phoenicea* L. is included as genus in Flora of Allahabad.

Taxonomic treatment

Portulaca pilosa L., Sp. Pl.; 1753; Vasudeva Rao in B.D. Sharma & Sanjappa, Fl. India 3:6. 1993; K.P.Singh & al, Fl. U.P. 1: 202. 2016.

Herb, erect, annual, much branched, up to 30 cm high, with branched woody roots. Leaves spiral, crowded at apices of the branches, subterete or linear lanceolate, 4-25 x 1-4 mm,; axillary hairs sparse. Flowers 2-6 in capituli. Sepals ovate, 2-5 x 1-4 mm, ecarinate. Petals 4-6, pink of reddish purple, obovate, 2.5-8 x 1.8-11 mm. Stamens 10-16; filaments 1-5 mm long. Style 2-8 mm long, 3-7 armed. Capsule yellow to green, globose, 2-3 mm across; operculum present, shining. Seeds dull or bluish, 0.4-0.7 mm across.

Flowering and Fruiting: July – October.

Type: LT: Herb. Linn. - 625.2. (LINN/A:625.2)

Occurs in sandy soil along the road side.

Voucher specimens: Dhanuha, Prayagraj, Oct. 20.2021, P. Kumar: 35525 (Duthie herbarium) (25° 22' 14.664”N, 81° 50' 25.5156” E)

Pentapetes phoenicea L., Sp. Pl.: 698. 1753; mast. in Hook.f., Fl., Fl. Brit. India 1: 371. 1874; Duthie, Fl. Gangatic Plain 1: 105.1903; R. Dutta in B.D. Sharma & Sanjappa, Fl. India 3: 443. 1993; K.P.Singh & al, Fl. U.P. 1: 259. 2016.

An erect herb or undershrub. Leaves 5-13 cm long, lanceolate or liner-lanceolate, crenate-serrate, glabrous obove, stellately hairy beneath especially on veins. Stipules subulate, linear. Flowers red. Capsules 1.6 cm long, sub-globose, bristly nearly half the length of the persistent calyx. Seeds 8-12, 2-seriate, angular, rough, brownish-black.

Flowering and Fruiting: August- November.

Type: LT: LINN-860.1.

In waste place and some time in gardens.

Voucher specimens: Ghoorpur, Prayagraj, Nov. 12. 2021, P. Kumar: 35527 (Duthie herbarium), (25° 19' 52.9428'', N 81° 50' 20.0364'' E).

Cyperus digitatus Roxb., Fl.Ind. 1: 209. 1820; C.B. Clarke in Hook. f., Fl. Brit. India 6: 618. 1893; Prain, Beng. Pl. 1145; Cooke, Fl. Bomb. 2: 873; Duthie, Fl. Gangatic Plain 3: 331. 1915.

A tall glabrous herb, 2 to 4 fit high. Rhizome short horizontal. Stem triquetrous above, smooth, strianth, covered at the base with coarse not fibrous usually dark leaf sheaths. Leaf-blades liner tapering above to an acuminate apex, variable in length and width, some time as long as stem, often shorter up to 0.5 in. broad, smooth, midrib well marked and keeled below, numerous lateral veins. Inflorencence a compound umble of spiks, primary

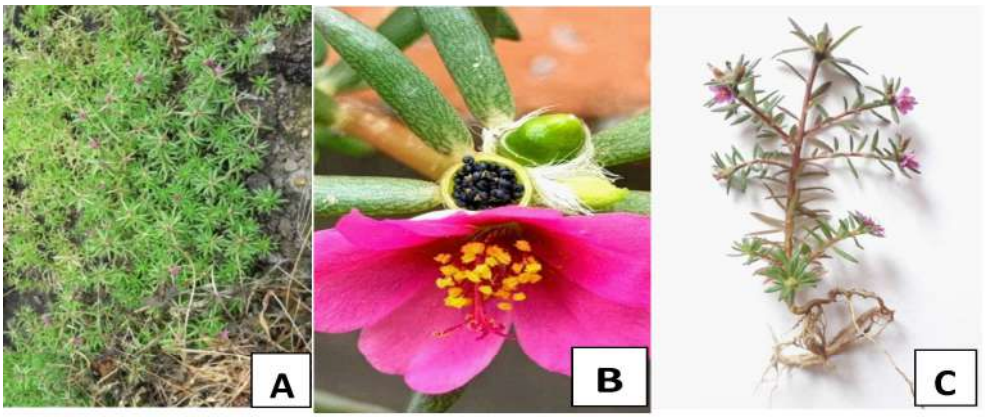
rays 0.5 to 7 ins. Long very unequal in same inflorescence; primary bract 6 to 8, leaf like often broade at the base, unequal in length and width the largest much longer than the rays; secondary bracts narrower. Spike 1 to 2 ins. Long composed of numerous spikelets which for the most part do not overlap but spread at right angles. Stamens 2. Style long; stigma 3, long exserted. Nuts yellowish brown, oblong-ellipsoid, trigonus, 1-1.3 mm long, epiculate.

Flowering and Fruiting: May – December.

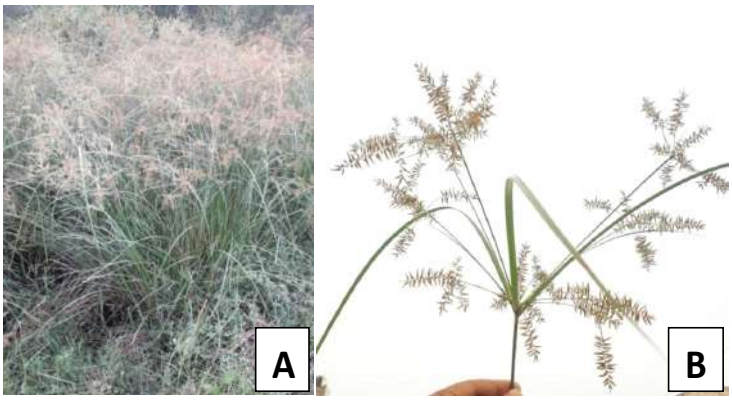
Type: "East India", Roxburgh; ex Hrb. Forsyth in Hrb. Bentham, (K!).

Common in moist and water logging places.

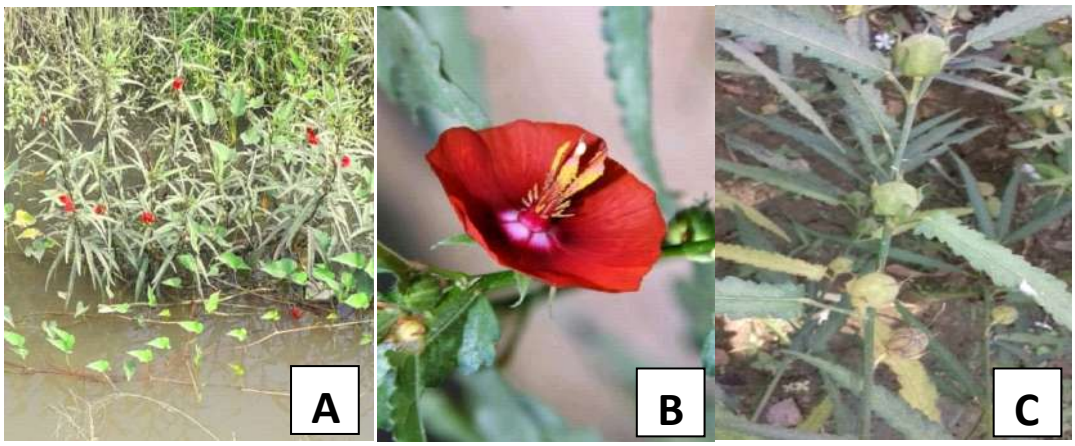
Voucher specimens: Dhanuha, Prayagraj, Nov.05.2021, P. Kumar: 35526 (Duthie herbarium), (25° 22' 14.664" N, 81° 50' 25.5156" E).



Portulaca pilosa L., **A**:- Habit & Habitat of plant, **B**:- Flower & capsule with ripen seeds, **C**:- A plant twig.



Cyperus digitatus Roxb. **A**:- Habit and Habitat, **B**:- Inflorescence



Pentapetes phoenicea L., **A:**-Habit & Habitat, **B:-**A flower, **C:**-Fruiting twig

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Chemical communication and methods of self-therapy in the honey bee

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Abstract- A critical area of physiology of the honey bee is chemical association among individuals and castes in the swarm which balance of its up rightness and limit. The uncommonly friendly interconnection of honeybee is interceded through pheromones. We here discuss the effect of both self-produced organ discharges and foraged hive item on colony health. Explicit honey bee health improving pursuit of honeybee items ought to plainly be recognized from the impact of unblemished nourishment guaranteeing the fundamental resistant ability of honey bee. We should see how the individual working bee with its incredible identification detects consolidates possibly accessible compounds.

Key words: Honey bee, physiology, chemical, colony, health

INTRODUCTION

Honey bee (*Apis mellifera* Linn.) colonies regularly have an enormous number of laborers coming to 15000 to 60,000 individuals of honey bees.[1] Such enormous numbers of honey bees require a lot of food to support their endurance and improvement. Adequate nutrition can support the advancement of honey bee colonies.[2]

Animals incorporating insects search food to get fitting supplements for their food which incorporate development, advancement and proliferation on account of social insects; it isn't simply to fulfill the necessity of people however the scavengers gather the nourishing assets likewise for different individuals living in the hive.[3]

Social resistance sums up all antiparasitic colony level components, not simply decreasing any parasite admission into the colonies yet additionally lessening the speed and transmission among people and colonies.[4] This came while a wondering, given the incredible intra-colonial circumstances for microbes and the evident requirement for control. Nonetheless, colony resistance is administered by individual invulnerable skill as well as by outside safe guard.[5]

Nutrient adjusting is a multiplex process where creatures coordinate taste prompts with post-ingestive data about food quality to acquire ideal nourishment.[6] How rapidly creatures learn signals related with suitable sustenance's. Grasshoppers are bound to figure out how to connect smells and tastes with protein in food assuming they have protein insufficient.[7]

Primary pheromones are proficient means for keeping with social concordance in the colony and their belongings are significant. These pheromones act by influencing the physiology of the beneficiaries with an ensuring change in their way of behaving.[8]

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CHEMICAL COMMUNICATION IN THE HONEYBEE

The most bountiful queen mandibular organ pheromone part of keto-2(E) decenoic corrosive (q-ODA) and two fragment parts, 4-hydroxy, 3-hydroxy phenyl ethanol (HVA) and methyl 1p-hydroxy benzoate (HOB) are comparatively communicated.[9] Discharge from the Dufour organ is position controlled.

Martin and Jones (2004)[10] saw that as C28-C38 esters are related with sovereigns and liquor eicosenol is related with non-laying working honeybee. The two esters and eicosenol are synthetically comparable compounds (Both are results of unsaturated fat biosynthesis). 3M2BA may be uniform to certain individuals of Africanized honeybees.[11]

Rhodes *et al.,* (2007)[12] recorded changes in constituent levels from head concentrates of sovereign with expanding age. Non-mated multiday old sovereign had higher normal degree of 9 HDA, 9 ODA and 10 HDA than mated multiday old sovereigns. These outcomes recommend these specific three constituents engage in sexual relations pheromones capacities in the honey bee 9 octa decenoic corrosive decyl decanote from the tarsal organ may likewise take part in the correspondence between genders.

Pheromones functions in the honey bee, 9-octadecenoic acid decyl decanoate from the tergal gland may also participate in communication between sexes. With expanding time of working honey bees, the size of mandibular organ and much 2HPT continuously increments.[13] At the point when the substances are infused into hive, they essentially increment the number of foragers leaving the hive.[14] This recommends that these mixtures might play a pheromonal role in worker prerequisite. An essential quality of eusocial life is a division of work.[15]

Working honeybees for the most part perform various errands in the home (*i.e.,* cell clearing, brood raising brush building, nectar raising for the queen and drones and so on) for the initial three weeks of grown up life and afterward adventure outside to gather food and safeguard the home when they age.[16-18]

In any case, divisions of work in honey bee colonies are unbending.[18, 19] On the connection among wide and working bees a synthetic sign on the outer layer of larvae called brood pheromone is significant.[20,21] A few parts are more every one of the ten individuals compounds show some releaser pheromones impact on adult honey bees.[22] The esters are available in various sums and extents as a component of standing and larvae age.[23]

METHODS OF SELF-THERAPY IN THE HONEY BEE

The additionally influence the retention of synthetic compounds from the climate can serve as pheromones and generally significant for honey bee health, structure a most proficient impediment against the connection to or entrance of the insect cuticle by microorganisms.[24,25] The structure an oily layer on the cuticle which is ceaselessly renewed by glandular emission and keeps organisms and microorganisms from entering the honeybee's body.[26-28] The cuticular hydrocarbon structure fluctuates during individual turn of events and is subject to both the climate and the individual health status.[26] The honey bees cuticular wax layer basically comprises of hydrocarbons, mono and polyester, free unsaturated fat and other polar the substances.[26]

Properties of bee wax have been exhaustively depicted in two monographs by the late.[29,30] Bee wax is discharged by the wax organ and is made out of a perplexing blend of alkanes, alkenes, hydrocarbon 14%, free unsaturated fat 12%, mono esters 35%, diesters 14%, hydroxy monoesters and a few minor constituents (Fatty alchols and hydroxyl diesters).[30,31] The most dynamic compound with upto half of dry weight is melittin. Its high antitoxin and natural action has been read up for a long time in human api-therapy.[31,32] The pollen foragers deposit gathered pollen plates straight for wardly into the capacity cell. The put away pollen is the changed to honey bee bread by in hive honey bees through tactic acid fermentation of pollen blended in with disgorged nectar honey and glandular emission.[2]

Polyfloral pollen further increments endurance rates for stone brood tainted honey bees, as displayed with in vivo tests utilizing honey bee larvae, in contract with monofloral pollen.[33] Protein and amino corrosive quality and amount are the principle measures in regards to the well-being advancing movement of pollen. For instance a new report showed that poly floral pollen isn't really better compared to a monofloral pollen diet of high protein content.[34]

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Solanum sisymbriifolium Lam. (Solanaceae): An extended distribution in Uttar Pradesh, India.

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Abstract- *Solanum sisymbriifolium* Lam. of family Solanaceae is a prickly invasive taxon in Uttar Pradesh and now naturalized in Uttar Pradesh. Still it is reported only from northern tarai region from Saharanpur and Gorakhpur districts of Uttar Pradesh and now extended to Sonbhadra and Ghazipur district of South-East area of Uttar Pradesh. Brief description is provided along with up to date citation, phenology, ecology and geographical distribution.

Key words: Extended distribution, *Solanum sisymbriifolium* Lam., invasive, Uttar Pradesh

INTRODUCTION

The overall vegetation of study area (Uttar Pradesh) is mixed dry deciduous and scrubby forest. It lies between altitude 23° 45' –24° 38' N and longitude 82° 30'–83°33' E. The average temperature of the area fluctuates from 32°C- 46°C in summer and 5°C-18°C in the winter. *Solanum sisymbriifolium* Lam. of family Solanaceae shows peculiar character like orange-red spine all over the body and large white calyx. It was first introduced in lower Bengal in 19th century and reported by D. Prain in (1896)[1]. After that subsequently in Assam (Kanjilal *et al.*, 1939)[2], Madras (Gamble, 1956)[3], Uttarakhand (Raizada, 1976)[4], Bihar (Mishra & Kumar, 1992)[5], Rajasthan (Yadav & Meena, 2007)[6], Maharashtra (Rahul B. Kamble *et al*., 2013)[7], Tripura (Saha & Dutta, 2013)[8], Gujarat (Tiwari & Ravi Kumar, 2014)[9], Jharkhand (Panda *et al*., 2014)[10], Delhi (Mishra, 2015)[11], Chhattisgarh (Singh, 2016)[12], Odisha (Sahu *et al.,* 2017)[13] and Jammu and Kashmir (Bhellum, 2019)[14]. Recently it is published as additions from Gorakhpur and Saharanpur by Srivastava *et al*. (2015)[15] and Mohammed *et al*., (2019)[16] respectively. During botanical exploration, the author has collected *Solanum sisymbriifolium* Lam. from Sonbhadra and Ghazipur district of South-East Uttar Pradesh. (Plate- 1)

The genus *Solanum* is represented by 42 species and 3 varieties distributed in different parts of India (Kalidass and Panda, 2019)[17] and 7 species has been reported from upper gangetic plain (Duthie, 1903-1929)[18] while *Solanum sisymbriifolium* Lam. has been published by Sahni & Naithani (1974)[19] from Rishikesh, District Dehra Dun. Floristic and ethno-botanical studies have been revealed that there is no published record of this species from South-East Uttar Pradesh. Therefore, the present finding of the species is more considerable with respect to the extended distribution in Uttar Pradesh. A brief taxonomic description along with up to date

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nomenclature, illustration and phenology are helpful for the identification of plant species. The herbarium specimen of this plant has been deposited in the Duthie Herbarium, Department of Botany, University of Allahabad, Prayagraj.

Description

Solanum sisymbriifolium Lam. Tabl. encycl. 2: 25 (1794).; Prain, J. Asiat. Soc. Bengal 65:542. 1896; J.S. Gamble, Fl. Pres. Madras Vol. 2 (Part 4), 658. 1956; Sahni & Naithani, in Acta Botanica India 2:151. 1974; Raizada, Suppl. Fl. Upper. Gan. Plain 176. 1976. *Solanum balbisii* Dunal – Sims, Bot. Mag. 52: t. 2568. 1825; vern.; Lal Bhatkatayia.

Annual, much spiny some time erect, perennial, under shrub, up to 1 m tall. Stem woody at the base, with sharp yellow prickles up to 1.5 cm long, erect, branched. Leaves alternate, simple, pinnately, ovate-oblong, petiolate; lamina 10-16 x 7-12 cm long, deeply pinnatifid with many spines along with the white veins on both surface and petioles 2- 5.5cm long. Inflorescence extra auxiliary raceme, 10-15 cm long, 9 -13 flowered, peduncle unbranched and 6-8 cm long covered with red spine. Flower white, hermaphrodite, actinomorphic and perianth is biseriate, dichlamydeous. Calyx yellowish green, sepals 5, gamosepalous, lanceolate, volvate astivation, covered with pinkish spines and some time persistent. Corolla white, petal 5, gamopetalous, lobed, rotate. Stamens 5, epipetalous, yellow, lanceolate, anther- lobes longer than filament. Overy and style glabrous, superior overy, bicarpillary, syncarpous and axile placentation. The fruit is berry bright red and spiny. (Plate- 2.)

Flowering and Fruiting: Throughout the year in the study area.

Ecology: This plant species commonly grows along roadsides and wastelands.

Distribution

Solanum sisymbriifolium Lam. is a native of Central and South America (Argentina, southern Brazil, Paraguay, Uruguay, Bolivia, and Colombia). The known introduced ranges of this species are North America, Europe (Hawkes & Edmonds, 1964)[20], Africa, Australasia, Asia (China, Taiwan, India), India (West Bengal, Tamil Nadu, Assam, Uttarakhand, Maharashtra, Bihar, Rajasthan, Jammu and Kashmir, Odisha, Tripura, Gujarat, Jharkhand, Karnataka, Delhi, Chhattisgarh and Uttar Pradesh).

Voucher specimen: India: Uttar Pradesh: Sonbhadra District, Renukut, Renukut railway station, Sept. 01. 2019, Prabhat Kumar, 35135 (Duthie). (N 24^0 12’ 11.88"; E 83^0 02’ 23.64") and Ghazipur District, Karamahari, Zamania, beside Karmanasa River, Sept. 23. 2020. Prabhat Kumar, 35225 (Duthie), (N 25^0 20’ 54. 68"; E 83^0 39’ 11.21").

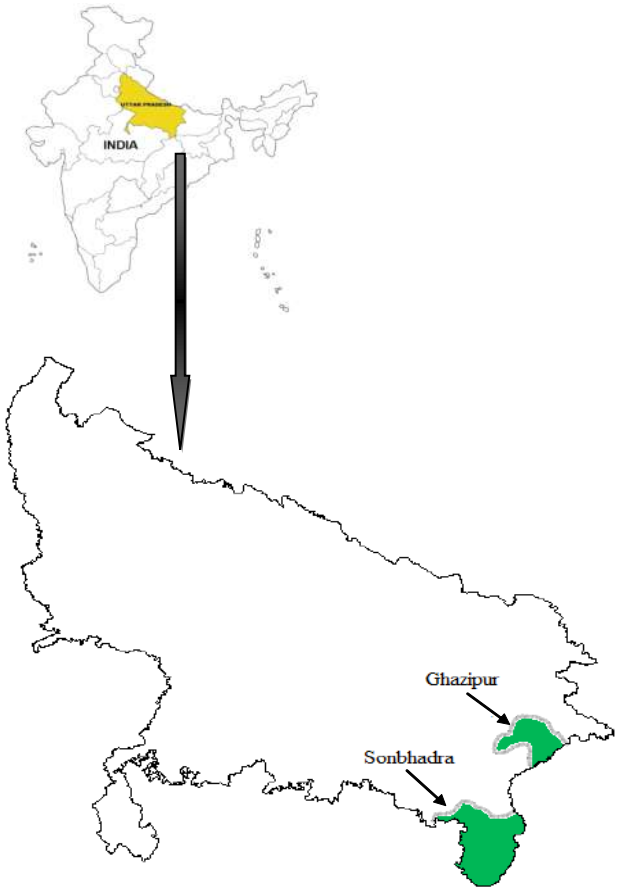


Plate 1. Map of Uttar Pradesh showing the collection site



Fig. A



Fig. B



Fig. C



Fig. D



Fig. E

Plate 2. *Solanum sisymbriifolium* Lam.: **Fig. A.** Plant habit and habitat; **Fig. B**. Inflorescence; **Fig. C.** Flower; **Fig. D**. Ripe berries (fruit) and prickles **Fig. E**. Fruit with seeds.

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Role of microorganisms in municipal solid waste management -A case study

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Abstract- Microorganisms play a vital roles in the maintenance of all the phenomenon whether manmade or natural. They makes survival easier for human beings. So microorganisms are used is in municipal waste management. The proper treatment of the municipal waste generated by humans in their day to day activities is a challenging task faced by government and environmental agencies. A vital way of successfully reducing this problem is by use of microorganisms. Thus, this paper contains many methods of microorganisms used in the management of municipal waste. And also their roles of used in the environment, such as in sewage, soil treatment & energy generation. It also discusses waste generation and its management methods, and some specific uses of microorganisms (bacteria, fungi, algae, virus and protozoa) in waste management. It examines by showing some recent activity in bio techniques waste management. A major challenge to all the countries world-wide is effective management of the waste generated. Waste can be categorized by many methods but by the classic classification is degradable waste, recyclable materials, inert waste, electrical and electronic, composite wastes, harmful waste and Toxic waste. Microbes can continue to exist in the extreme environmental can solve various types of problems of mankind. Municipal solid waste generation in India is about 42 million tons/year and in this microorganism plays an important role.

Key words: Microorganisms, waste management, composting, Fungi, Disposal.

INTRODUCTION

Microorganisms are very essential in nature where they have a variety of important functions. Many microbes are unique to specific environmental surrounding. Microbes play an important role in the naturally recycling of living materials. All substances that are naturally produced are biodegradable in nature, they are broken down by living organisms such as bacteria or fungi. Microorganisms have been used in finding solutions for many problems humans have faced in maintaining the environment such as they have played a beneficial role in human and animal health, genetic engineering, environmental protection, and municipal and industrial waste treatment. Microorganisms have enabled easy and affordable responses which would have been impossible via chemical or physical engineering methods in turn cost would have increased. Moreover, microbial methods have successfully been used to a large number of environmental issues, such as municipal waste management issues. In India with a population of 138Cr, Municipal waste management is a severe threat with various challenges to

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be faced. The municipal wastes in India is caused due to household, farming and commercial sources, which may be characterize into liquid, solid, gaseous which are dangerous and harmful too. But, the most troublesome are the liquid and solid wastes. A lot of harmonized effort must be put into the reduction of wastes at every stages of production, collection, transportation, treatment and at last disposal in order to achieve the results. The scenario of Municipal waste management (MWM) is significantly different between developed and developing nations because developing nations are lacking of proper collection, transportation, recycling and disposal mechanism of waste. So, MWM has become very critical and receives priority due to progressive concern related to environmental degradation and sustainability.[1] In present days it is very compulsory for every country to search the beneficial methods to overcome the problem of waste and issues related with it. Each and every product that are used and thrown away at open places contains very dangerous chemicals which can affect human health than ever before as more than 60,000 chemicals found entry in daily use consumer products.[2] Hence, this paper conclude how microorganisms act as agent of municipal solid waste management, by keeping in view their various roles in the environment, waste generation and management in India.

Role of microorganisms in the environment:

The activities of mankind in environment involve a lot of chemical process in the conversion of natural resources in the environment into other usable forms for consumption. In the process of creating products, pollution is also created by man. As a result, the most authentic solution to the generation of wastes in the environment is such that will conveniently put them back into the environment. That method involves the use of microorganisms (usually yeasts, bacteria, or fungi). These microorganisms or their products are integrated to yield desired industrial products such as bioleaching, biodetergen, biotreatment of wastes, biofiltrations, biomass fuel production, biomonitoring, and so on. Also, microorganisms are important to humans and the environment, as they play important role in the carbon and nitrogen cycles, and also in recycling other organisms' dead remains and waste products through decomposition. Microorganisms also have a vital place assymbionts. Some common examples of the uses of microorganisms in the environment are discussed below:

Sewage treatment:

The majority of all decaying waste treatment methods depend on a large number of microorganisms to decay organic waste matters that are not disposed to sedimentation or flotation. Anaerobic microorganisms are also used to lessen solid waste producing methane gas and a sterile mineralized residue. In potable water treatment, one method that employs, a complex layer is the slow sand filter composed of a wide range of microorganisms to remove all particulate material from water.

Soil treatment:

The nitrogen cycle rely on the fixation of atmospheric nitrogen. Diazotrophs play a vital role in this. This can occur in the nodules of the roots of legumes that contain symbiotic bacteria of the genera Rhizobium, Mesorhizobium, Sinorhizobium, Bradyrhizobium, and Azorhizobium. Nutrients and Minerals in the soil are available to plants with the help of microorganisms that produce hormones that cause growth, stimulate the plant immune system and trigger stress responses. In general, a more diverse soil microbiome results in higher yields and less plant diseases.

Energy generation:

Microorganisms are used in biogas reactors to produce methane and also in fermentation to produce ethanol. Researchers are researching the use of algae to produce liquid fuels, and bacteria to convert various forms of farming and municipal waste into usable fuels. Bacteria with tiny wire-like appendages called nanowires can digest toxic waste to produce electricity. For example Shewanella, can grow oxygen-seeking nanowires when placed in less-oxygen environments. Researchers discovered that when the microbes' nanowires are pierce with platinum electrodes, they carry a current. If these potential can be utilized properly, they can be used in waste treatment plants to simultaneously digest waste and produce power.

Waste generation and management in India:

Municipal Solid wastes is termed as non-liquid and nongaseous products which is caused due to mankind, which is termed as useless, waste, garbage and sludge. The municipal solid waste generation completely relies on the population of that area, Industries and commercial

activities being taking done in the area. India generates about 62 million tonnes of waste each year out of which 43 million tonnes (70%) are collected of which about 12 million tonnes are treated and 31 million tonnes are dumped in landfill sites without proper treatment which is causing problem every day. Most importantly, the amount of municipal solid waste is being generated continuously to and slower is the treatment process. Municipal solid waste management in some part of India is characterized by inefficient collection methods, transportation method and proper disposal system which are causing serious challenge in this era.

S. No.	Name of the state	No. of cities	Municipal population	Municipal solid waste (t/day)	Per capita generated (kg/day)
1	Andhra pradesh	32	10,845,907	3943	0.364
2	Assam	4	878,310	196	0.223
3	Bihar	17	5,278,361	1479	0.280
4	Gujrat	21	8,443,962	3805	0.451
5	Haryana	12	2,254,353	623	0.276
6	Himachal pradesh	1	82,054	35	0.427
7	Karnatka	21	8,283,498	3118	0.376
8	Kerala	146	3,107,358	1220	0.393
9	Madhya Pradesh	23	7,225,833	2286	0.316
10	Maharashtra	27	22,727,186	8589	0.378
11	Manipur	1	198,535	40	0.201
12	Meghalaya	1	223,366	35	0.157
13	Mizoram	1	155,240	46	0.296
14	Orissa	7	1,766,021	646	0.366
15	Punjab	10	3,209,903	1001	0.312
16	Rajasthan	14	4,979,301	1768	0.355
17	Tamil Nadu	25	10,745,773	5021	0.467
18	Tripura	1	157,358	33	0.210
19	Uttar Pradesh	41	14,480,479	5515	0.381
20	West Bengal	23	13,943,445	4475	0.321
21	Chandigarh	1	504,094	200	0.397
22	Delhi	1	8,419,084	4000	0.475
23	Pondichery	1	203,065	60	0.295
		299	128,113,865	48,134	0.376

Source: Status of MSW generation, collection, treatment and disposal in class-I cities (CPCB, 2000).[3]

Municipal Waste management methods:

Municipal Waste management includes collection, transport, processing or disposal, managing and monitoring of waste materials to lessen its impact on mankind and Nature. Municipal solid waste treatment methods reduces the volume and toxicity of solid waste, changes it into a more convenient and usable form. Waste management is seen as a method of source reduction, refuse recycling, controlled combustion and controlled landfill; energy generation from waste, solid waste disposal. A large number of techniques are used in properly managing municipal solid waste, which include supervising, collecting, transporting, processing, recycling, incineration, land filling and bio composting.

Use of microorganisms in waste management:

The microbes that live in the aerobic biological treatment systems include bacteria, fungi, algae, protozoa, rotifers, and other higher animals. The growth of all types of microorganisms in a given industrial waste disposal system rely upon the chemical characteristics of the industrial waste, the environmental limitations of the particular waste system and the biochemical characteristics of the microorganisms. All of the microorganisms which grow in a given industrial waste

disposal system contribute to its over-all characteristics, both good and bad. It is important to recognize the contributions made by each type of organism to the over-all stabilization of the organic wastes.

Bacteria:

The bacteria are the smallest units in aerobic waste treatment systems. The diverse biochemical nature of bacteria makes it possible for them to digest most, organic compounds found in industrial wastes. Aerobic bacteria and facultative bacteria are found in all aerobic waste treatment process. Growth of any species relies upon its competitive ability to obtain a share of the available organic material in the environment. Bacterial predomination will normally divide itself into two major groups: the bacteria using the organic compounds in the waste, and the bacteria utilizing the lysed products of the first group of bacteria. Characteristics of the bacteria, the most important characteristic is their ability to flocculate. All aerobic biological waste treatment process relies upon the flocculation of the microorganisms and their separation from the liquid phase for complete.

Fungi:

Fungi play vital role in the maintaining balance of organic wastes. Like the bacteria, the fungi can digest all organic compound found in industrial wastes. The fungi have the potential ability to predominate over the bacteria but they do not except under unfavorable environmental conditions. The filamentous fungi predominate over the bacteria of low oxygen concentration, low pH, and at low nitrogen. Low oxygen tension results from a low oxygen supply or from a high organic load which causes the demand to exceed its supply. Most of the fungi grow at pH 4 to 5 in comparisons to this some bacteria grow well in competition. But fungi require less nitrogen than bacteria per unit mass of protoplasm, fungi produces more active masses of protoplasm from the wastes than are the bacteria in the waste that lack nitrogen. Bacteria in average produces approximately 10 to 12% nitrogen but fungi range from 5 to 6% nitrogen. Under normal environmental conditions fungi will be present and will aid in the stabilization of the organic matter. But the fungi are of secondary importance and will not predominate. The degradation of plastic by *Bacillus subtilis* is analyzed using liquid culture method and observed that *Bacillus subtilis* degrade plastic in about thirty day or more time period.[4] The different microbes have capability to degrade types of plastic material in a different time span and in different environment. The microbial associated biodegradation has been accepted as a safe clearance of plastic waste but investigation is still ongoing for its enhanced efficiency. Microorganisms have the natural capability to degrade plastic in natural condition. But microbes able to decompose polythene are limited to 17 and 09 genera of bacterium and fungi.

Algae:

The algae are the third form of biological plants which play a part in the over-all stabilization of organic wastes. Since the algae obtain their energy that synthesis from sunlight, there is no requirement to metabolize the organic compounds in comparison like the bacteria and the fungi. To form protoplasm the algae primarily uses the inorganic constituents of the wastes, for example, ammonia, carbon dioxide, phosphate, magnesium, potassium, iron, calcium, sulfate, sodium and other ions. It is possible to have algae and the bacteria predominate together since they do not utilize the same waste components. During protoplasm synthesis the algae release oxygen which is taken by the bacteria to bring about complete aerobic stabilization of the organic matter. In the absence of sunlight algae is needed to obtain the energy required to stay alive from the metabolism of organic matter in the same manner as bacteria and fungi. This organic matter normally comes from stored food within the cell but in some algae it comes from the organic components in the wastes.

Viruses:

These are substances assembled from the biopolymers, which are efficient of multiplying and producing new virus particles in living prokaryotic or eukaryotic cells. Viruses are important as it is removed, retained or destroyed during water and wastewater treatment, bacteriophages can contaminate and degrade the bacterial cultures in the environment.

Protozoa:

The protozoa are the easiest animals found in waste disposal systems. The role that the protozoa play in stabilizing organic wastes has only recently been clarified by combining a study of pure culture protozoa with the natural observations in various biological treatment systems. This study showed that rather than being the primary mechanism of purification, the protozoa were responsible for reducing the number of free-swimming

bacteria, thus aiding in producing a clarified effluent. The succession of protozoa had long been observed in biological waste disposal systems.

Microbial waste management:

Generally, solid waste can broadly be categorized into biodegradable and non-biodegradable. The biodegradables (bio wastes) are those solid wastes generated, which could be decomposed by micro organisms and does not constitute major sources of pollution for a long period of time. They include paper products and wastes of plant origin, wastes of animal origin (faecal matter, carcass, droppings, and poultry waste products). These groups of solid waste even though they are easily degraded by microorganism in minimal time, give off offensive odour and constitute nuisance to the aesthetic environment more than the non-biodegradable solid wastes. On the other hand, non-biodegradable solid wastes are not degradable by microorganisms. This implies that other means of treatment such as incineration, landfill, and recycling are employed as ways of disposing them.

Municipal solid waste management:

Municipal solid waste management reduces or lessens its effect on the mankind and leads to increase in economic development of country and maintain the sustenance of life. Composting is used as biological solid waste treatment method which is the controlled aerobic decomposition of organic waste materials with the help of microorganisms. Composting is a method in which organic waste matte are decomposed and then recycled as compost for use in agriculture and landscaping applications. The most common composting techniques include static pile composting& vermin-composting.

Recent advances in microbial waste management:

The review of recent researcher activity which is applied in microbes to both environmental management and biotechniques is informed by the polluting effects on the world by soil erosion, harmful fertilizers and pesticides, and the improper treatment of sewage these harmful processes have caused in serious environmental and social problems around the world. Notably, biological methods are being used in sustainable environmental and waste management techniques. Some of the techniques are bioremediation, biostimulation, bioaugmentation, phytoremediation, and so on.

CONCLUSION

Waste is any material, which is of no use to anyone. Humans with nearly all activities produce waste. The major component of municipal solid waste is organic fraction, mostly from domestic, agricultural and industrial sources. There are many different methods of municipal waste management. These include physical, chemical and biological methods. Conventional waste management practices usually involve one negative consequence or the other. This necessitated the search for and development of biological methods, with the use of microorganisms which produce environmental-friendly results. Biological methods include the use of microorganisms such as bacteria, fungi, algae, virus and protozoa in process like composting, activated sludge, trickling filters and oxidation ponds. Recent scientific progress in applying microorganisms to environmental management includes hybrid techniques that combine microbial methods with physical and chemical ones; these include landfills, anaerobic digestion and vermiculture technology.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

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Proximate composition of some underutilized edible wild plants mainly consumed by Munda tribe of Jharkhand

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Abstract-Underutilized edible wild herbaceous plants mainly consumed by Munda tribes of Jharkhand namely *Senna tora, Commelina benghalensis, Alternanthera paronychioides* were analyzed for their proximate composition. Proximate composition was determined by using standard methods of Association of Official Analytical Chemists (AOAC). The result showed that proximate composition of the selected wild edibles ranged between; moisture (9.56±0.2 - 11.15±0.13%), ash content (9.82±0.12 - 11.56±0.14%), crude protein (6.12± 0.11 - 6.52±0.18%), crude fiber (4.78±0.1 - 6.15±0.2%), fat (1.16±0.08 - 1.91±0.11%), carbohydrate (69.33±0.58 - 73.06±0.76%), caloric value (317.1±0.61 -328.38±1.2 kcal/100g). The varied proportion of nutrient is found in this wild edibles which helps to obtain a scientific data baseline for their wider acceptance as additional, nutritional supplementary source of food.

Key words: Underutilized edible wild, Proximate composition, Munda tribe, Jharkhand

INTRODUCTION

Jharkhand (The land of forest) is primarily known as tribal state with total tribal population of 70,89,068 which is about 28% of state population. There are 32 tribal communities inhabiting here in which Munda, Oraon, Ho, Ashur are in majority. In Jharkhand most of the rural inhabitants depend on wild edible plants to meet their additional food requirements. The wild edible plants have traditionally occupied an important position in the socio-cultural, spiritual and health arena of rural and tribal lives in Jharkhand.

The wild plants from forest provides many essential nutrients which helps to improve both the physical and mental well-being of rural people. In developing countries, the wild plants have been a main source of food and medicine for tribal people. It is estimated that 80% of the forest dwellers depend on forests for 25-50% of their annual food requirements.[1] They are inexpensive, easy to cook and rich sources of macro and micro nutrients.[2,3]

Proximate and nutritional analysis of edible fruit and vegetables play a crucial role in assessing their nutritional significance.[4] Since the selected wild edible herbaceous plants are underutilized, information on their nutritional and medicinal properties are lacking. As there are not many published studies on nutritional and chemical composition of these wild edible varieties, so there is need to initiate the awareness for its wider acceptability among mass urban population. In this research work underutilized edible wild herbaceous plants used by Munda tribe namely *Senna tora, Commelina benghalensis and Alternanthera*

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paronychioides were analyzed for proximate composition using standard AOAC method.

MATERIALS & METHODS

Sample collection and preparation:

Four underutilized edible wild herbaceous plants used by Munda tribe namely *Senna tora, Commelina benghalensis and Alternanthera paronychioides* were collected from different localities of Ranchi (Jharkhand).

After collection, the leaves were removed from the branches and were sorted out to get the healthy and disease free edible parts. The leaves were washed properly and dried them under shade at 25 $\pm$ 2°C so as to prevent the decomposition of chemical compounds present in them. The dried plant sample were powdered by just crushing them, sieved through 2 mm mesh sieve and stored in air tight container at room temperature for further investigation.

Methods

Nutritional composition of these dried leafy plant samples were analyzed by using standard method of Indian Standard and Association of Official Analytical Chemists (AOAC)[5] methods. The moisture content was determined by oven dehydration at 103°C for 6 hours. The ash content of the sample was determined by keeping the sample in silica crucible dish and ignited them in muffle furnace at 525 $\pm$ 25 °C until greyish white ash results, then it is cooled in a desiccators and weighed the incinerated residue, according to Association of Official Analytical Chemists (AOAC)[5] method. The total protein content was estimated by Micro-Kjeldahl method.[5]

Crude fat or oil were extracted by Soxhlet extraction with petroleum ether for 16h.[5] Crude fibre were determined by acid- base digestion method.[6] Total carbohydrate content was estimated by difference method as described by James (1995)[7]. Add moisture, fat, protein and ash content and deduct the value from 100 to get carbohydrate content by difference. Caloric value (Kcal/ 100g) were estimated by using the method described by FAO (2003)[8] i.e., {4× (% of protein)} + {9×(% of fat)} + {4×(% of carbohydrate)}.

RESULT & DISCUSSION

Some selected species of underutilized edible wild plant showed variant proportions of proximate composition as mentioned in the Table 1.

Moisture Content

Moisture in vegetables is a good source of water and is necessary as it is considered that around 20% of the total water consumption must come from food moisture.[9] On dry basis the moisture content of the underutilized edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 9.56 %, 10.29 % and 11.18 % respectively. The value of moisture content was comparable to other commonly cultivated species such as *Spinacia oleracea, Coriandrum sativum* and *Brassica oleraceae* were 14.51 %, 10.49 % and 24.00 % respectively.[10]

Ash Content

From the biochemical point of view, ash content plays an important role as it contains important nutritional elements mainly minerals which is helpful in the physiological function of the body. The ash content of the underutilized edible wild herbaceous species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 8.82%, 13.85% and 16.84 % respectively. The ash value were more or less same, compared with the commonly cultivated species such as *Spinacia oleracea, Coriandrum sativum* and *Brassica oleraceae* were 22.50 %, 20.07% & 8.21% respectively.[10]

Crude Fat Content

Fats are a type of lipids which is stored in the adipose tissue and under the skin of animals. It is mainly used as an energy storage molecule in the body as 1gram fat gives 9 Kcal energy. The fat content of the selected edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 1.18%, 1.93% and 1.18% respectively. The crude fat content of the selected wild species were lower as compared to 7.12% in *Spinacia oleraceae*[11] and are same when compared with other wild edibles like *Ipomoea aquatica* (1.2%), *Lippia alba* (1.39%), *Cayratia trifolia* (1.61%)[12]. Due to the general low level of crude fat in vegetable leaves and its high level of total unsaturated fatty acid, their consumption in large amount would be beneficial to individual suffering from overweight or obesity and this would constitute a good dietary habits.

Crude Fibre Content

Fibre is the indigestible part of plant food which is important for our digestive health. It can improve cholesterol and blood sugar level and helps in preventing

some diseases such as diabetes, heart diseases and colorectal cancer. Crude fibre content of the underutilized edible wild herbaceous species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 4.78%, 5.71% and 6.18% respectively. Fiber content was lower compared to *Spinacia oleracea, Coriandrum sativum* and *Brassica oleraceae* which were 22.50 %, 20.07%, and 8.21% respectively.[10]

Crude Protein Content

Crude protein content of the selected underutilized edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 6.38 %, 6.57 % and 6.12 % respectively. The value obtained was lower than the value of the commonly cultivated species such as *Spinacia oleracea, Coriandrum sativum* and *Brassica oleraceae* were 17.29 %, 18.36 % & 9.59 % respectively.[10] However, the protein content of these edible wild species was higher compared to other wild edibleslike, 2.11 % in *Amaranthus viridus*, 2.98 % in *Chenopodium murale* leaves and 2.76 % in *Nasturitium officinale.*[13]

Total Carbohydrate Content

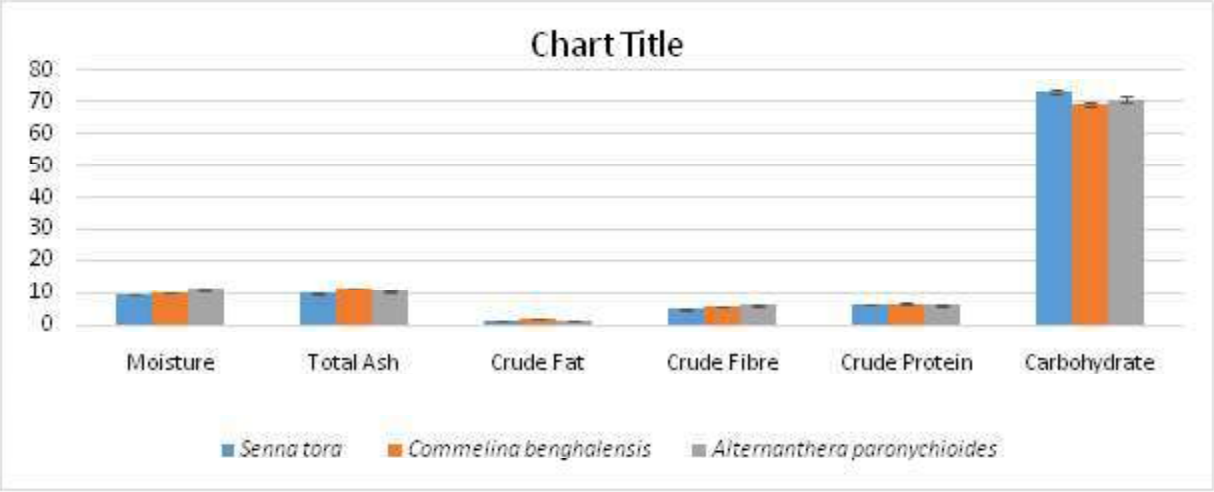
Carbohydrate is the primary source of energy in the body. The 1 g of carbohydrates yield 4 kcal energy. The carbohydrate content of the selected underutilized edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 74.06 %, 67.37 % and 64.68 % respectively. It was higher compared to other commonly cultivated species such as *Spinacia oleracea* (*44.56 %*), 49. 64 % in *Coriandrum sativum* and 56.18 % in *Brassica oleraceae.*[10]

Energy (Caloric Value) kcal/100g

The overall calculated energy values of the selected edible wild species *Senna tora, Commelina benghalensis* and *Alternanthera paronychioides* were 332.38 kcal/100g, 313.09 kcal/100g and 293.82 kcal/ 100g respectively. These energy values are similar to *Brassica oleracea* i.e. 319.80 kcal/100g[14] and to *Spinacia oleracea* i.e. 306.35 kcal/100g[11].

Table 1- Proximate composition of some underutilized edible wild herbaceous plants used by Munda tribe (on dry weight basis)

Sample name	Moisture (%)	Total Ash (%)	Crude fat (%)	Crude fibre (%)	Crude Protein (%)	Carbohydrate (%)	Caloric value (kcal/100g
Senna tora	9.56±0.2	9.82±0.12	1.18±0.06	4.78±0.1	6.38±0.13	73.06±0.76	328.38±1.2
Commelina benghalensis	10.1±0.19	11.56±0.14	1.91±0.11	5.7±0.18	6.52±0.18	69.33±0.58	320.59±1.4
Alternanthera paronychioides	11.15±0.13	10.84±0.14	1.16±0.08	6.15±0.2	6.12± 0.11	70.61±0.43	317.1±0.61



Graph 1- Graphical representation of proximate composition of some underutilized edible wild herbaceous plants used by Munda tribe (on dry weight basis)

CONCLUSION

These underutilized wild edibles possess significant amount of carbohydrate, protein, dietary fiber and calories and may be recommended as nourishment to people suffering from malnutrition. Therefore, the maintenance of good health, vitality and longevity by the rural tribal may be due to consumption of these nutritious wild edibles.

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The larval survival of selected fresh water ornamental fishes (*Carassius auratus* and *Trichopodus trichopterus*)

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Abstract-Ornamental fish breeding is an emerging sector and can be a world scale opportunity area for fish farmers in India, especially in rural population. Larval rearing is one of the most essential criteria for any successful aquaculture. In developing countries fish seed production and larval survival is most unsuccessful due to poor nutrition of these graceful living jewels. Proper statistical analysis estimation has been done which support the management of both the rearing environment and feeding regime in order to assure productive growth and survival rate of larvae of selected ornamental fishes (*Carassius auratus* and *Trichopodus trichopterus*). Information regarding age specific food preferences of fish on different food items is deficient and survivability is also less which needs research trust for augmenting larval survival and it also heavily depends on the supply of suitable feed which should also be cost effective. The experiment was conducted to investigate the effect of alternative formulative feed on larval survival, a control diet (CD), different types of Spirulina diet (S_1, S_2 and S_3) and experimental diets (DEF, DCF, DSF and SPF). The experiment was carried out for 20 days and the larval survival rate was calculated of two selected ornamental fishes (*Carassius auratus* and *Trichopodus trichopterus*). All the five different feeds were readily accepted from the beginning of feeding and was found to be high with SPF whereas low with CD in both the larvae of *Carassius auratus* and *Trichopodus trichopterus*. In the present investigation attempt has been made to formulate cost effective, high quality, high yielding, protein and lipid rich nutritious feed for gaining high survival and growth rate in the larvae of *Carassius auratus* and *Trichopodus trichopterus* in a healthy aquatic environment.

Key words: Ornamental, emerging sector, feeding regime, aquatic environment, formulated feed.

INTRODUCTION

India is the third largest producer of ornamental fish in the world and the fishery sector is a major foreign exchange earner. Ornamental fish keeping is emerging as one of the most popular hobbies across the world next to photography.[1]

With the increase demand for ornamental fishes especially in USA, Europe and Japan and many countries in Asia including India have started capturing and culturing beautifully colored ornamental fishes, the true living jewels. Watching them swimming gracefully in the aquarium is a pure sense of joy and mental pleasure to many enthusiastic. Larval rearing is one of the essential criteria for any successful aquaculture. In developing countries, fish seed production and larval survival is mostly unsuccessful due to poor nutrition. There are a number of factors related to food and feeding which directly affect larval survival. They are the duration of development from the embryo stage to the time of the first feeding responses occur. The preferred food species

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and its abundance and distributions. The behavioral relation between the larva and its prey, the success of feeding responses, the swimming ability of larvae in search of food and the required food ration for growth and metabolic expenditure.

The small mouth gap of the larvae at first hatching makes feeding over the first few days crucial. Fish larvae are the smallest self-supporting vertebrates and in order to increase their chances of survival, they need to complete their morpho-functional systems to escape predictions and to obtain food. Food preferences of fish on different food item is deficient and the survivability is also very less which needs research thrust for augmenting larval survival and it also heavily depends on the supply of suitable life feed.[2]

When yolk sac is exhausted, fish larvae will enter into exogenous feeding and must successfully establish the transition of exogenous nutrition or suffer progressive starvation.[3] The rearing of larvae to the fry stage is most critical in the cycle of fish seed production in hatcheries and its rearing under controlled hatchery condition requires the development of specific culture techniques.[4] In aquaculture survival and production of early-stage larvae is still low.

Growth and survival data are powerful tools for understanding the effects of both live and manufactured diets on first feeding fish larvae.[5] The selection of these live feed organisms is based on many factors such as nutritional requirements of the cultured larvae, the size of mouth gap, development of digestive tract of the cultured larvae, nutritional value, availability of live food and suitability of production. Successful rearing of fish larvae is partially dependent upon the proper availability of lipid, protein, carbohydrate, vitamins and minerals through the diet. Deficiencies of these nutrients result in poor growth, low feed efficiency anemia and high mortality. The factors affecting larval survival can be divided into biotic and abiotic factors.[6] Abiotic factors like water temperature, salinity and light can influence larval survival by directly affecting developmental rates.[7]



Carassius auratus



Trichopodus trichopterus

LITERATURE REVIEW

Ornamental fish farming is becoming a luxurious hobby. Fish feeds contribute to more than one half of the variable operating cost in the aquaculture sector. Many studies on the effects of nutrition on growth in edible fishes have well documented. Paulet (2003)[8] studied the effect of diet and feeding rate on growth, morphological development and behaviors of larval and juvenile god fish, *C.auratus*. Amornsakun1 *et al.* (2004)[9] investigated the period of yolk absorption, the onset of first feeding mouth development and starvation in larval *Siamese gourami*, *T.pectoralis*. Caroline and Okpokwasil (2012)[10] determined the effect of PH on hatching success and larval survival of African cat fish, *C. gariepinus*. Allain *et al.* (2007)[11] reported the influences of environment on the survival of anchovy, *Engraulis encrasicolus* larvae in the Bay of Biscay. Sahoo *et al.* (2006)[12] evaluated the effect of different feeds of animal origin in composition on growth and survival of *Wallago attu* larvae.

MATERIALS & METHODS

Fish Feed Preparation:

To estimate the growth and survival of larvae, five different types of food were formulated for two different species of ornamental of fishes, *C.auratus* and *T.trichopterus* given below:

1. Control Feed- C (Feed 1)
2. Dried Earth worm Feed - DEF (Feed 2)
3. Dried Chironomid Feed - DCE (Feed 3)
4. Dried Spirulina Feed - DSF (Feed 4)
5. Specially Prepared Feed - SPF (Feed 5).

Newly hatched larvae of *C.auratus* and *T. trichopterus* were utilized for the present study. The experiment was conducted with the 5th day of hatched larvae and terminated when more than two-thirds of larvae in a group completed their metamorphosis to juvenile phase. Gentle aeration was supplied in the experimental tanks and increased with the growth of larvae. Daily water

exchange rate of 20% was carried out during the experimental period. During larval rearing, dead larvae were counted daily and the waste was siphoned off every day to avoid any means of stress.



Larvae of *T.trichopterus*



Larvae of *C.auratus*

New hatched larvae were randomly assigned into eight groups with 100 larvae in each group. To investigate the effect of alternative feed on larval survival, a control diet, different types of spirulina diets (S_1, S_2 and S_3) and experimental diets (DEF, DCF, DSF and SPF) were given to the larvae during the experimental period. The experiment was carried out for 20 days and the larval survival of *C.auratus* and *T.trichopterus* was calculated using the following formula:

Larval Survival Rate (%) = Final Larvae Number*100/Initial Larvae Number

Statistical Analysis: One way ANOVA without replication was carried out using Microsoft Excel. During the experiment, the water quality parameters (Temperature: 27-30°C; dissolved oxygen: 4.5-5.5 mg/l; PH: 7.5-8.2; hardness: 120-140 mg/l; nitrite: 0.02-0.05 mg/l and ammonia: 0.060.08 mg/l) were estimated and maintained.

RESULT

The effect of four different feeds on larval survival of selected experimental fishes *C.auratus* and *T.trichopterus* was analysed.

Effect of Spirulina diet on larval survival of selected ornamental fishes:

Experimental diets	Larval survival (%)	
	C.auratus	*T.trichopterus*
C	37.5±1.52	45.8±0.96
S_1	85.5±3.47	85.5±2.34
S_2	89.2±2.93	90.5±3.68
S_3	98.5±3.56	99.6±2.78

The larva of *C.auratus* fed with S_3 diet exhibit the highest survival rate (99± 3.56%) when compared to larvae fed with other diets with lowest (40.5±1.52%) with control diet. Similarly maximum larval survival (99.6±2.78%) was observed in *T.trichopterus* fed with S_3 diet and the lowest (45.8±0.96%) with control diet.



Effect of different levels of Spirulina on larval survival in *C.auratus*



Effect of different levels of Spirulina on larval survival in *T.trichopterus*

One way ANOVA for larval survival of selected ornamental fishes

Source of variation	*SS*	*df*	*MS*	*F*	*P-value*
Between Groups	0.96	1	0.96	0.020007	0.894355
Within Groups	191.93	4	47.983		
Total	192.893	5			

P < 0.05 statistically significant

One way ANOVA showed a significant variation ($P<0.05$) existed in survival rate between the larvae of *C.auratus* and *T.trichopterus* fed with different types of experimental feed.

Effect of experimental feed on larval survival of selected ornamental fishes.

All the five different feeds were readily accepted from the beginning of feeding in *C.auratus* and *T.trichopterus*. The larval survival was found to be high (97.32±2.95%) in *C.auratus* fed with SPF whereas the larvae fed with control diet exhibited a low survive rate (40.52±1.35%). In *T.trichopterus* maximum survival rate was observed in larvae fed with SPF diet (98.37±3.46%) followed by DSF (88.91±2.98%), DCF (86.34±3.56%), DEF (85.76±2.87%) and minimum in control diet (45.64± 0.098%).

Effect of experimental feed on larval survival of selected ornamental fishes

Experimental feed	Larval survival (%)	
	C.auratus	*T.trichopterus*
C	40.52±1.35	45.64±0.98
DEF	86.35±3.25	85.76±2.87
DCF	84.59±2.64	86.34±3.56
DSF	82.84±3.17	88.91±2.98
SPF	97.32±2.95	98.37±3.46

± SD from triplicate groups of fish

Effect of different types of experimental feed on larval survival of ornamental fishes

Source of Variation	SS	df	MS	F	P-value
Between Groups	8.5698	1	8.5698	0.223406	0.65316
Within Groups	230.1582	6	38.3597		
Total	238.728	7			

$P < 0.05$ is statistically significant

CONCLUSION

The management of both the rearing environment and feeding regime are the most important aspects of the larval survival. In our present study, five different types of experimental feed were formulated in relation to the mouth size of the larvae. Lipid of earthworm has nutritional and medicinal values, and it is believed that the dissolved substances in its lipid create a good taste to flood which can be useful in aquatics.[13] Spirulina is considered to be with rich source of protein, vitamins, minerals, essential amino acids and fatty acids (gamma linolenic acid) and antioxidant pigments such as carotenoids.[14] In addition, it is effective as an immunomodulator.[15]

Improvement in larval rearing techniques for ornamental fish larvae is considered as an urgent need as there is a demand for ornamental fish seed supply in our country. With the live feed, the larval survival of *C.auratus* and *T.trichopterus* can be improved by enriching with formulated feed. This provides positive impact on economy of fish industry for formulating low-cost feed for the ornamental fishes so that optimum sized fingerlings can be produced to make its culture a variable commercial activity.

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Arsenic induced renal toxicity and its correlation with high lipid peroxidation in swiss albino mice

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Abstract- Around the world, millions of people get exposed to a high level of heavy metals through their drinking water. Among these metals, Arsenic is one of the most potent toxic agent and prevalent in the environment in its inorganic form i.e. Sodium Arsenite ($NaAsO_2$). Sodium Arsenite is observed to possess high reactivity with thiol groups. The present study was conducted to investigate the effects of low doses of Sodium Arsenite on kidney and its correlations with oxidative stress in Swiss albino mice. Mice were dissected for histological & lipid per oxidation evaluation after 4th, 6th and 8th weeks. Significant rise in lipid per oxidation was observed in Sodium Arsenite dosed Swiss albino mice. Mild to severe types of necrosis and degenerative changes in the kidney via histological studies were noticed. Mononuclear cell infiltration and vacuolization in the tubules resulted in reduced glomerulus space. It could be safely concluded that degenerative changes in kidney tubular epithelium may be attributed to Arsenic. Metabolites of Arsenic which are excreted from kidneys may also cause cellular damage leading to kidney dysfunction. These findings in mice model might be useful for a better understanding of the toxic effects of Arsenic, to develop an effective ameliorative treatment against Arsenic mediated effects on human health.

Key words: Arsenic, Swiss Albino Mice, Kidney, histology and LPO.

INTRODUCTION

Heavy metals are hazardous compounds which a major concern to human health alleviating the quality of life. These heavy metals are potentially toxic to the multicellular life due to high accumulation in the body.[1] Many studies have shown that heavy metals are widely present in the environment and cause deleterious effects on human health.[2-8] Among these heavy metals, Arsenic is one of the most 20 hazardous substances, is known to cause several diseases like carcinogenicity, diabetes, genotoxicity, hypertension, peripheral neuropathy, hyperkeratosis, cardiovascular diseases etc.[9]

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Arsenic (As) is an element with atomic number 33 and relative atomic mass 74.92. Arsenic can exist with many minerals, usually in conjunction with Sulphur and metals and also as a pure elemental crystal. It was first documented by Albertus Magnus in 1250, the Arsenic is a metalloid. Arsenic is present in the environment and humans, all over the world are exposed low to high amounts, mostly through food, water and air. The presence of high levels of arsenic in groundwater, the main source of drinking water in many countries around the world, has drawn the attention of the scientific community. Millions are exposed to high levels of inorganic arsenic through drinking water. It has become a major public health problem in many countries in South and East Asia which is a great burden on water supply authorities. The

Environment Protection Agency lowered the permissible level of Arsenic in drinking water in the USA in 2001 from 50 ppb to 10 ppb. The most affected areas in the world are Bangladesh and West Bengal, India. In 42 districts in southern Bangladesh and nine adjacent districts in West Bengal, 79.9 million and 42.7 million people respectively are exposed to groundwater Arsenic concentrations that are above the World Health Organization maximum permissible limit of 50 ug/l. Prolonged ingestion of water contaminated with Arsenic may result in the manifestations of toxicity in all systems of the body as subsequently discussed. Symptoms and diseases engendered by Arsenic include dermal lesions, anaemia, and an increased risk for cardiovascular disease, liver damage, diabetes and cancer. Arsenic induced cancer targets such organs as the lungs, skin, bladder, liver and kidney.[10] The most serious concern is the potential of Arsenic to act as a carcinogen. The complex chemistry of the metals makes them hard to completely understand their mechanisms, but in general, metals such as arsenic can affect multiple aspects of cellular function including proliferation, apoptosis, differentiation and cell transformation.

The toxicity of Arsenic and its derivatives depends chiefly on the oxidation state and chemical composition. Inorganic Arsenic is far more toxic than organic Arsenic as it readily reacts with sulfhydryl groups of various enzymes and proteins in a biological system. Thus, altering multiple cellular pathways including the promotion of apoptosis, inhibition of DNA repair and increasing oxidative stress ultimately leads to Arsenic mediated adverse health effects.[11,12]

It was demonstrated that ROS formed during oxidative stress can initiate lipid peroxidation (LPO) and oxidizes proteins to its inactive state and causes DNA strands to break, damaging normal cellular function.[13] It has been reported that Arsenic induced ROS enhances lipid peroxidation and cellular damage in liver and renal tissues. Ineffective elimination of ROS has been shown to induce oxidative stress and damage to various organs such as liver, lungs, kidney and spleen. Moreover, due to increased oxidative stress macromolecules such as lipid, protein and DNA may become vulnerable to its toxicity leading to disruption of cellular structural integrity and capacity.[14] Although there is proof that arsenic toxicity is associated with the induction of oxidative stress in vital organs like kidney through overproduction of reactive oxygen spices (ROS) and inhibition of antioxidant enzymes activity.[15] As per Varner *et al.* (1998)[16], the vital role of the kidney is xenobiotics metabolism. Since blood and kidney are susceptible to oxidative stress and xenobiotics metabolic disturbance. Thus, kidney damage due to oxidative stress further provokes the kidney dysfunction which may result in additional toxic metabolites load to the renal system and may induce more injury to the kidney due to high oxidative stress.

The kidney is one of the toxics target organ due to its capacity to extract and concentrate toxic substances by highly specialized cells and also due to its large blood flow.[17,18] It has been reported that the kidney is a primary organ for the excretion of metabolites and hence appears to be one of the main targets of Arsenic.[19] The toxicity of As in the kidney has been demonstrated in the human population and animals through renal pathology and functional changes.[20]

In addition, the mechanism of kidney damage by Arsenic has been investigated in numerous studies.[21] Oxidative stress is one of the major factors for kidney damage. LPO is an autocatalytic mechanics leading to oxidative destruction of cellular membranes and ultimately cell death.[22] Among the suggested mechanism, oxidative stress is one of the best-accepted theories.[23]

As above mentioned, the kidney is the organ for excretion of unwanted and toxic substances, it is more vulnerable to Arsenic toxicity. Moreover, oxidative stress is one of the major factors for nephrotoxicity. Hence, the present study has been designed to examine the impact of Arsenic on the kidney by evaluating histological damages in renal tissue and its correlation with lipid peroxidation parameter.

MATERIALS & METHODS

Experiments were carried out on Swiss Albino Mice. The animals were procured from the laboratory animal resources section of Mahavir Cancer Institute & Research Centre, Patna, India (CPCSEA Regd. No. 1129/bc/07/CPCSEA, dated 13/02/2008). The research work was approved by the IAEC (Institutional Animal Ethics Committee) with IAEC No. IAEC/2012/12/04.The experimental animal (Swiss albino mice) (n=20) selected for the study were 8 to 10 weeks old and their weight was measured as 28 ± 2 grams. Mice were kept in the polypropylene cages with paddy rusk at room temperature

25 ± 2ºC and humidity 50±5% in a controlled light (12 hrs light and 12 hrs dark). Animals were maintained in ideal condition as per animal ethical guidelines. After a week of acclimatization to the environment, the mice were separated into four groups.

The drug was procured from the pharmacy of Mahavir Cancer Institute & Research Centre, Patna. Sodium Arsenite @ 2mg/kg body weight was administrated orally to the experimental mice at the interval of 4 weeks, 6 weeks and 8 weeks during the designed protocol for the study.

The mice were segregated into four groups with 5 mice (n=5) in each group. Group A was kept as control and served with equal volume of distilled water by Gavage method. Group B,C & D were treated with Sodium Arsenite orally @2mg/kg body weight for 8 weeks.

Body weight:

Body weight of each group of mice were measured before and after the administration of Sodium Arsenite. Each group of mice were sacrificed after 4 weeks, 6 weeks, and 8 weeks of administration with Sodium Arsenite and kidney were dissected.

Lipid Peroxidation:

Blood serum was obtained by centrifuging the blood collected from each group at 3000 rpm for 10 minutes and store at -20ºC. The lipid peroxidation level was estimated through formation Thio Barbituric Acid Reactive Substance (TBARS) by standard method.[24]

Histological analysis:

The kidney tissues were fixed in formalin (10%) and 4-5 µm thick sections were sliced. The sections were further stained with Hematoxyline and Eosin and observed under light microscope.

RESULT

The mice were sacrificed in order to collect the tissue samples for the analyses of histological examinations and blood sample for lipid peroxidation evaluation. The body weights of the mice (control and treated) were recorded before the dissection. The mean ±SEM of the body weights of mice in control group (Group A) was found to be 27.80±1.056g. On the other hand, the mean ±SEM of body weights of mice treated for 4 weeks (Group B), 6 weeks (Group C) and 8 weeks (Group D) were calculated as 29±1.012g, 30.30±0.9028g and 31.80±0.9566g respectively (Table 1). The result clearly indicates that body increased with the increase in the duration of treatment with sodium arsenite (Figure 1).

Table 1: The body weight, organ weight of kidney and LPO parameters of control and Sodium Arsenite treated of swiss mice. (Values are expressed as Mean± SEM of p<0.05)

TEST	CONTROL (N=5) (Group A)	SODIUM ARSENITE @2mg/kgb. w		
		4th week(N=5) (Group B)	6th week(N=5) (Group C)	8th week(N=5) (Group D)
Body weight (g)	27.80±1.056	29.00± 1.012	30.30± 0.9028	31.80± 0.9566
LPO(nmol/ml)	27.26±1.138	35.38±1.824	37.12 ±1.840	44.01±1.177

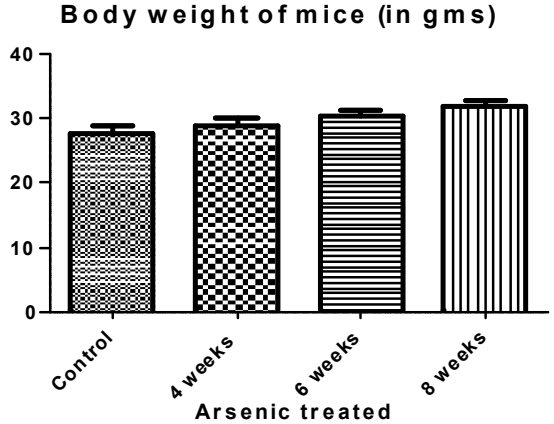


Figure 1: Comparison of body weight (mean±SEM) between control and arsenic treated mice of 4 weeks, 6 weeks and 8 weeks. *Significantly different from control group at p<0.05. p -value was performed by using one way ANOVA

The LPO levels in the blood serum collected from control and treated groups were analyzed. The mean±SEM of LPO level in the control group (Group A) was observed to be 27.26±1.138 nmol/ml. In contrast, the mean±SEMs of LPO levels in group B, group C and group D were recorded to be 35.38±1.824 nmol/ml, 37.12±1.840nmol/ml, 44.01±1.177nmol/ml respectively (table 1). The LPO level in control was observed to be lowest as compared to group B, C and D, which reflects that longer duration of arsenic treatment induces higher oxidative stress level (figure 2).

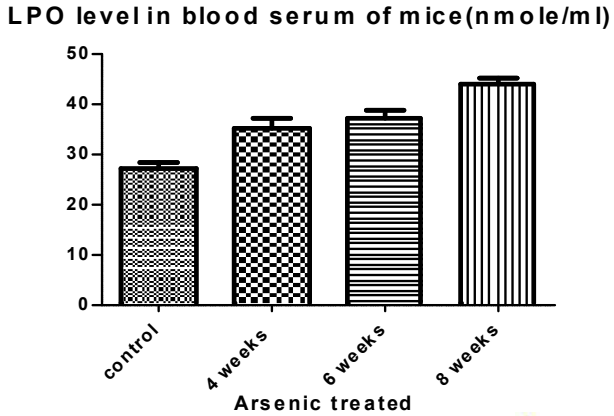


Figure 2: Comparison of LPO levels(mean ± SEM) between control and arsenic treated mice of 4 weeks, 6 weeks and 8 weeks. *Significantly different from control group at p<0.05. p-Value was performed by using one way ANOVA.

Histology:

To understand the arsenic induced tissues alterations, we have performed histopathology of organs (kidney). We have observed deformed glomeruli, clustered nuclei, diminished glands, weakening periplasm in arsenic treated mice kidney tissue (fig 4-6). In contrast, tissue architecture, glomeruli were well structured in control of kidney tissue (fig 3). It shows prominent to necrosis in tissue (fig 6). The degree of biochemical parameters can predict the level of histological damage in the kidney. The LPO level was found to be significantly increased only in as exposed group as compared to the control group (fig 2).

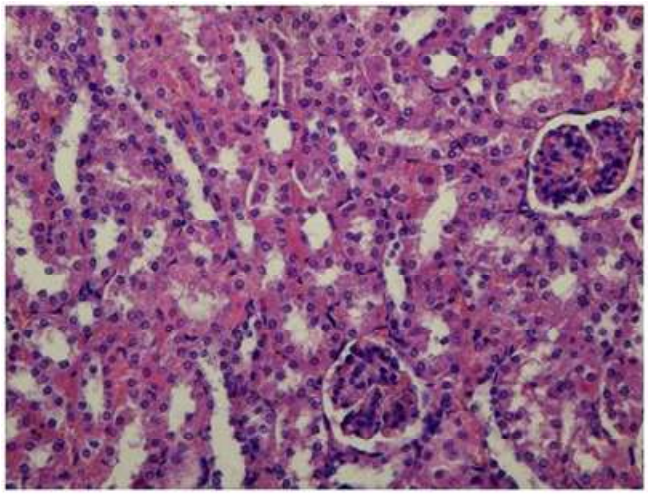


Figure 3: Photomicrograph of a normal kidney showing well - structured glomeruli, proper nuclei arrangement, well-shaped glands H&E (x100).

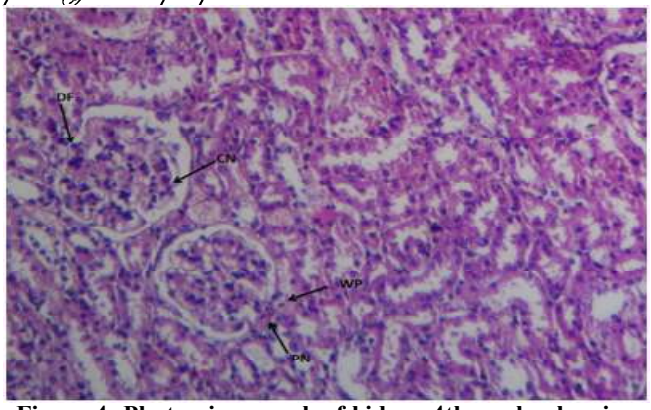


Figure 4: Photomicrograph of kidney 4th weeks showing deformed glomeruli, pervasive nuclei (PN), deformed glands (DF), clustered nuclei (CN), weakening of periplasm (WP), H&E (X100)

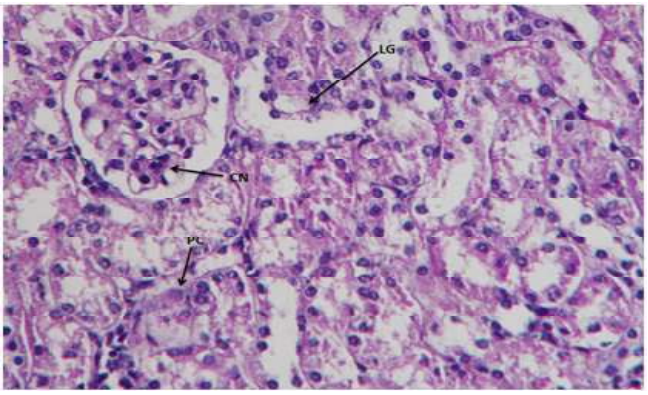


Figure 5: Photomicrograph of kidney 6th weeks showing unstructured glomeruli, irregular arrangement of nuclei also clustered nuclei (CN), loss of glands (LG), pervasiveness of cytoplasm (PC) (X100).

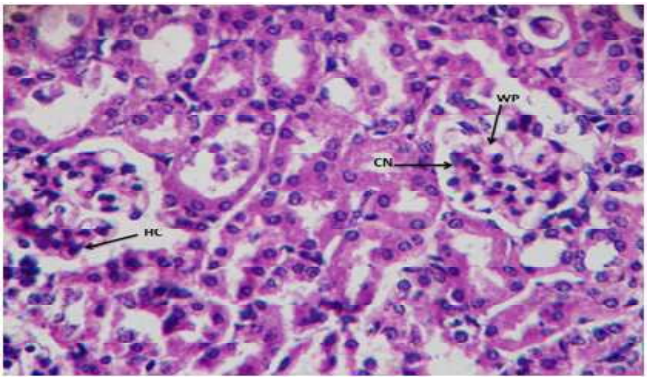


Figure 6: Photomicrograph of kidney 8th weeks showing necrosis, nuclei in the glands are spread out and dispersed (Shows low mitotic count - apoptosis may be implicated), clustered nuclei (CN) may also be seen somewhere, glomeruli completely diminished, hyperchromatins (HC) can also be seen in glomeruli region, cytoplasm is seen but not clearly defined as nuclei are absent, weak periplasm (WP), H & E (X100).

DISCUSSION

The purpose of this study was to correlate the high lipid per oxidation with histopathological alterations induced by sodium arsenite in the kidney of mice. Sodium arsenite contributes to the increased lipid per oxidation and altered histological appearances in the experimental mice. An earlier report suggests, Arsenic is an exogenously oxidative inducer, which can trigger the production of reactive oxygen species (ROS), causing damage to DNA, lipids and proteins.[14] Because the kidney is rich in phospholipids, and play a vital role in xenobiotic metabolism, is highly vulnerable to oxidative damage.[16]

In the present study, increased LPO levels in mice serum were found highly significant in sodium arsenite group (mean value 35.38, 37.12 and 44.01 nmol/ml for 4 weeks, 6 weeks and 8 weeks respectively) as compared to the control group (27.26 nmol/ml). Increased oxidative stress in tissue due to arsenic exposure is seemed to be the major cause of arsenic induced toxicity in mice. Santra *et al.* (2000)[25] has also reported oxidative stress in an experimental animal model due to arsenic exposure. Arsenic mediated oxidative stress can be indicated to induce the organ degeneration during the exposure. During the excretion process by the kidney, a significant amount of Arsenic inflows along with blood and other metabolites which further might interact with cellular protein or intercellular lipid or DNA molecules and thereby produce free radicals. This process may be responsible for the architectural changes of the kidney.[26] We have observed the degenerative changes, a cluster of nuclei, glomerulus diminished, the pervasiveness of cytoplasm; in kidney tissue in arsenic treated mice (figure 4-6). These damaging changes were observed as reflected action of toxic metabolites of Sodium arsenite. Microscopic changes were more pronounced in treated mice suffering from mild to high necrosis, indicating the toxic effect of chemical agents in the present study. Singh *et al.* (2011)[27] has also been reported renal failure, tubular necrosis due to arsenic toxicity. Apart from renal disorders, the toxic effect of arsenic has also been associated with hepatic disorders in an animal model study.[28]

It has been reported that exposure to arsenic is associated with metabolic disorders, hypertrophy of adrenal glands[29] and anaemia[30]. Exposure of sodium arsenite (2 mg/kg b.wt) to mice for the definite time intervals results in a significant increase of LPO levels in their serum.

CONCLUSION

Increased LPO leads to degeneration of lipids in the bilayer lipid layer of the plasma membrane which results in weak periplasm and hence nuclei inconsistently and irregularly present. Effect of LPO is applied in all the slides where irregularly spaced nuclei are observed i.e. 4th & 8th week treated slides.

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Effect of automobile pollution on Chlorophyll content of road side herbs along NH-31 from Katihar to Purnea (Bihar)

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Abstract- Automobile exhaust contains a large no. of air pollutants such as carbon monoxide, unburnt hydrocarbons, Sulphur and nitrogen oxide, tetraethyl lead and tetramethyl lead and several other lead compounds. These air pollutants adversely affect road side herbs by decreasing their chlorophyll content. In the present study, chlorophyll content of 10 herbs (*Catharanthus roseus*, *Ageratum conyzoides*, *Tridax procumbens*, *Acalypha indica*, *Phyllanthus nururi*, *Parthenium hysterophorus*, *Croton sparsiflorus*, *Euphorbia hirta*, *Emilia sonchifolia*, *Phyla noduliflora*) was estimated in road side plants and countryside plants i.e. one km away from the road side in summer season. The chlorophyll contents of road side herbs were less than those of countryside herbs.

Key words: Automobile exhausts, carbon monoxide, unburnt hydrocarbon, sulphur & nitrogen oxides, lead compounds

INTRODUCTION

Air pollution in India is quite a serious issue with the major sources being fuel wood and biomass burning, fuel adulteration, vehicle emission and traffic congestion. In autumn and winter months, large scale crop residue burning in agriculture fields - a low cost alternative to mechanical tilling - is a major source of smoke, smog and particulate pollution. India has low per capita emissions of greenhouse gases but the country as a whole is the third largest after China and the United States. Transport is the dominant source of emissions of Sulphur dioxide (SO_2), 57% of oxides of Nitrogen (NOx), 80% of lead (pb) 81% of Carbon monoxide (CO) and 75% of hydrocarbon (HC) emissions.[1]

A number of countries have targeted vehicles and associated sectors (such as fuel) to overcome the problem of air pollution. Notable successful initiatives are: conversion of public transport from diesel to CNG, switching Vikrams (tuk-tuks) from diesel to electricity, shifting from leaded to unleaded gasoline in many countries etc. Still the pollution problem in urban cities is continued. The Air (Prevention and Control of Pollution) Act was passed in 1981 to regulate air pollution. Automobile exhaust emissions contribute significant air pollution.[2-4]

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MATERIAL & METHOD

The study area was selected NH-31 from Katihar to Purnea, a total distance of 30km. 10 common herbs- *Catharanthus roseus*, *Ageratum conyzoides*, *Tridax procumbens*, *Acalypha indica*, *Phyllanthus nururi*, *Parthenium hysterophorus*, *Croton sparsiflorus*, *Euphorbia hirta*, *Emilia sonchifolia*, *Phyla noduliflora* were selected for study. All selected herbs were collected in summer season. Herbs one km. away from road side were also collected.

Estimation of Chlorophyll:

Chlorophyll content of leaves from each plant was estimated in laboratory. 250 mg of fresh leaves were cut into small pieces and transferred in 25 ml 80% acetone. It was incubated in dark for 24 hrs. absorbance at the extract was taken at 645 nm and 663 nm with spectrophotometer. Chlorophyll was estimated by the formula.[5]

$$\text{Total Chlorophyll}\left(\frac{\text{mg}}{\text{gm}}\text{Fw}\right) = \frac{20.2(A645) + 8.02(A663)}{1000X\ Fw} X\ V$$

A_{645} = Absorbance at 645 nm
A_{663} = Absorbance at 663 nm
V = Volume of Chlorophyll extract
Fw = Weight of fresh leaves

RESULT

In the present study, 10 herbs belonging to family Euphorbiaceae, 4 from family Compositae, 1 from Apocyanaceae and 1 from Verbinaceae were collected from road side of NH-31 (Katihar to Purnea) in summer season. Same herbs were also collected from one km away of road side. Chlorophyll pigment was estimated from each herb.

It was observed that the chlorophyll content in each herb of road side decreased in comparison to herbs from one km away of road side. The result is tabulated in table 1.

The chlorophyll content was highest in *Phyla noduliflora* (19.162) of family Verbinaceae followed by *Acalypha indica* of family Euphorbiaceae and minimum in *Ageratum conyzoides* of family Compositae (1.245).

CONCLUSION

Heavy traffic passes through NH-31 (Katihar to Purnea). A large no. of truck, bus, car, auto, tractor and several other heavy vehicles continuously passes through this road. The pollutant released from these vehicles adversely affect herbs present at the road side. These pollutants causes decrease in chlorophyll content of herbaceous plants. In summer season, the atmosphere remain dry due to which pollutants remain within air for a long time. The vehicular pollutants mostly contain carbon monoxide, unburnt hydrocarbons, Sulphur and nitrogen oxide, tetraethyl lead and tetramethyl lead and several other lead compounds. These pollutants are absent in countryside so the chlorophyll content of the countryside herbs is higher than those of road side herbs.

Table 1- Comparison of Chlorophyll content in road side plants and plants one km away from road side

S. No.	Name of Plant	Family	Chl. Content of road side plant	Chl. Content of plant one km away from road side
1	*Catharanthus roseus*	Apocyanaceae	5.423	6.432
2	*Ageratum conyzoides*	Compositae	0.984	1.245
3	*Tridax procumbens*	Compositae	7.563	8.236
4	*Acalypha indica*	Euphorbiaceae	13.945	15.678
5	*Phyllanthus nururi*	Euphorbiaceae	10.645	12.421
6	*Parthenium hysterophorus*	Compositae	4.213	5.641
7	*Croton sparsiflorus*	Euphorbiaceae	8.456	9.898
8	*Euphorbia hirta*	Euphorbiaceae	4.321	5.814
9	*Emilia sonchifolia*	Compositae	8.974	10.573
10	*Phyla noduliflora*	Verbinaceae	17.961	19.162

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Comparative study of moisture content of ethnomedicinal plants used in Ranchi district of Jharkhand for curing chronic kidney injury

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Abstract- Jharkhand's Ranchi district has been a storehouse of vast natural resources, including vegetational abundance. The people that dwell in the area have their own traditions and taboos. Chronic kidney disease is the one of the major problem in India and all over the world. Some common plants like *Coriandrum sativum* L. and *Moringa oleifera* Lam. are used in this problem. The study was conducted to investigate moisture content in leaves of these two plants.

Key words: Moisture content, ethnomedicinal plants, chronic kidney disease, tribal communities.

INTRODUCTION

Throughout the history of human society, man and plant have had a very close relationship.[1] Ranchi is the capital of Jharkhand. Traditional treatments for various diseases are mostly plants used by the tribals of Ranchi district. Chronic Kidney Disease (CKD) is one of those diseases. Chronic kidney disease is a gradual loss over a period of months to years of kidney function. It is widely believed that once there has been sufficient initial damage, no matter what the cause, renal failure inevitably progresses.[2] The CKD may be associated with coexistent risk factors, such as hypertension, proteinuria, glycemia, anemia, and disturbances of lipid or mineral metabolism.[3]

The comparative study of moisture content of some plants, which are used in chronic kidney injury, was done. *Coriandrum sativum* L. is a medicinal and aromatic plant from the Apiaceae family that is found all over the world[4] and *Moringa oleifera* Lam. is a plant that belongs to the Moringaceae family and is considered under used. The plant is also known as drumstick, sahjan or sohanjana in India.[5] Both plants are used to cure chronic kidney injury. Indigestion, worm infections, rheumatism, loss of appetite, convulsions, sleeplessness, anxiety, and joint pain have all been treated using *Coriandrum sativum* L.[6] *Moringa oleifera* Lam. leaves are the most often used, and they have been shown to be effective in CKD as well as a number of other chronic conditions, such as hyper cholesterolemia, diabetes, hypertension, insulin resistance, non-alcoholic liver disease, and general inflammation.[7]

The moisture level of harvested plants is often high, ranging between 60 and 80 percent. If the moisture content is not considerably reduced, dangerous biological processes can mature. As a result, the beneficial active

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ingredient in herbs can be eliminated, and the medications' outward qualities can deteriorate.[8] The amount of moisture in a sample is expressed as a percentage of the sample's initial weight. The difference between initial weight of fresh leaves and final weight of same leaves after drying is the moisture content. The moisture content of each plant was estimated as a percentage, which differed from one to the next.

MATERIALS & METHODS

In study of moisture content, plants of two different family were harvested during early morning from the different areas of Ranchi district. The major blocks for collection of plants in Ranchi district has been shown in Fig. 1. Plants species were identified with the help of Botany of Bihar and Orissa.[9,10] The collected leaves were properly cleaned twice in running tap water to remove any dirt or surface contaminants that may have been present. The leaves were washed and then dried in a cold area before being weighted in the weighting machine. The leaves were then placed in a hot air oven at 55°C for half an hour. This activity was repeated 3-4 times. The dried leaves were then weighed. The difference between initial weight of the green leaves and final weight after drying was the moisture content. Percentage of moisture content has been calculated with this formula:

$$\text{Moisture content} = \frac{\text{Initial weight of leaves} - \text{final weight after drying}}{\text{Initial weight of leaves}} x\ 100$$

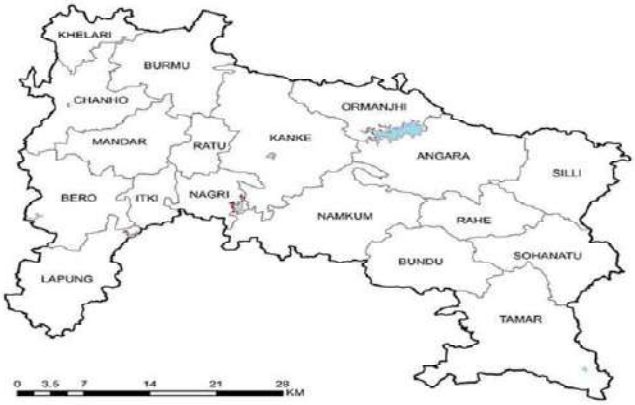


Fig. 1- Different blocks of Ranchi district for collection of plants.

RESULT & DISCUSSION

The comparative study of moisture content of two ethnomedicinal plants viz. *Coriandrum sativum* L. and *Moringa oleifera* Lam. has been done. *Coriandrum sativum* L. has a high moisture level, while *Moringa oleifera* Lam. has low moisture content as shown in table 1 & 2. The comparative study of weight of dried leaves (in gm) for *Coriandrum sativum* L. and *Moringa oleifera* Lam have been shown in Fig.2. It is found that *Moringa oleifera* Lam. has relatively high residue after drying. The *Moringa oleifera* Lam. plants have a lot of therapeutic properties as it has low average moisture content (69.91%) as shown in Fig.3. Their regular use can help prevent chronic renal disease. The *Moringa oleifera* Lam. plants can be used in our daily life to improving the filtration rate of kidney.

Table 1: Moisture content of *Coriandrum sativum* L.

Material	Number of observation	Weight of fresh leaves (in gm)	Weight of dried leaves (in gm)	Difference (in gm)	% of Moisture content
Fresh leaves of *Coriandrum sativum* L.	1.	10	2.02	7.98	79.8%
	2.	10	1.96	8.04	80.4%
	3.	10	1.96	8.04	80.4%
	4.	10	1.94	8.06	80.6%
	5.	10	1.94	8.06	80.6%
	6.	10	1.98	8.02	80.2%
	7.	10	1.88	8.12	81.2%
	8.	10	1.82	8.18	81.8%
	9.	10	1.86	8.14	81.4%
	10.	10	1.82	8.18	81.8%
Total					**808.2%**

Average Moisture content = 80.82%

Table 2: Moisture content of *Moringa oleifera* Lam.

Material	Number of observation	Weight of fresh leaves (in gm)	Weight of dried leaves (in gm)	Difference (in gm)	% of Moisture content
Fresh leaves of *Moringa oleifera* Lam.	1.	10	2.90	7.10	71.0%
	2.	10	2.92	7.08	70.8%
	3.	10	3.06	6.94	69.4%
	4.	10	2.86	7.14	71.4%
	5.	10	2.98	7.02	70.2%
	6.	10	2.96	7.04	70.4%
	7.	10	2.88	7.12	71.2%
	8.	10	3.22	6.78	67.8%
	9.	10	3.10	6.90	69.0%
	10.	10	3.21	6.79	67.9%
Total					**699.1%**

Average Moisture content = 69.91%

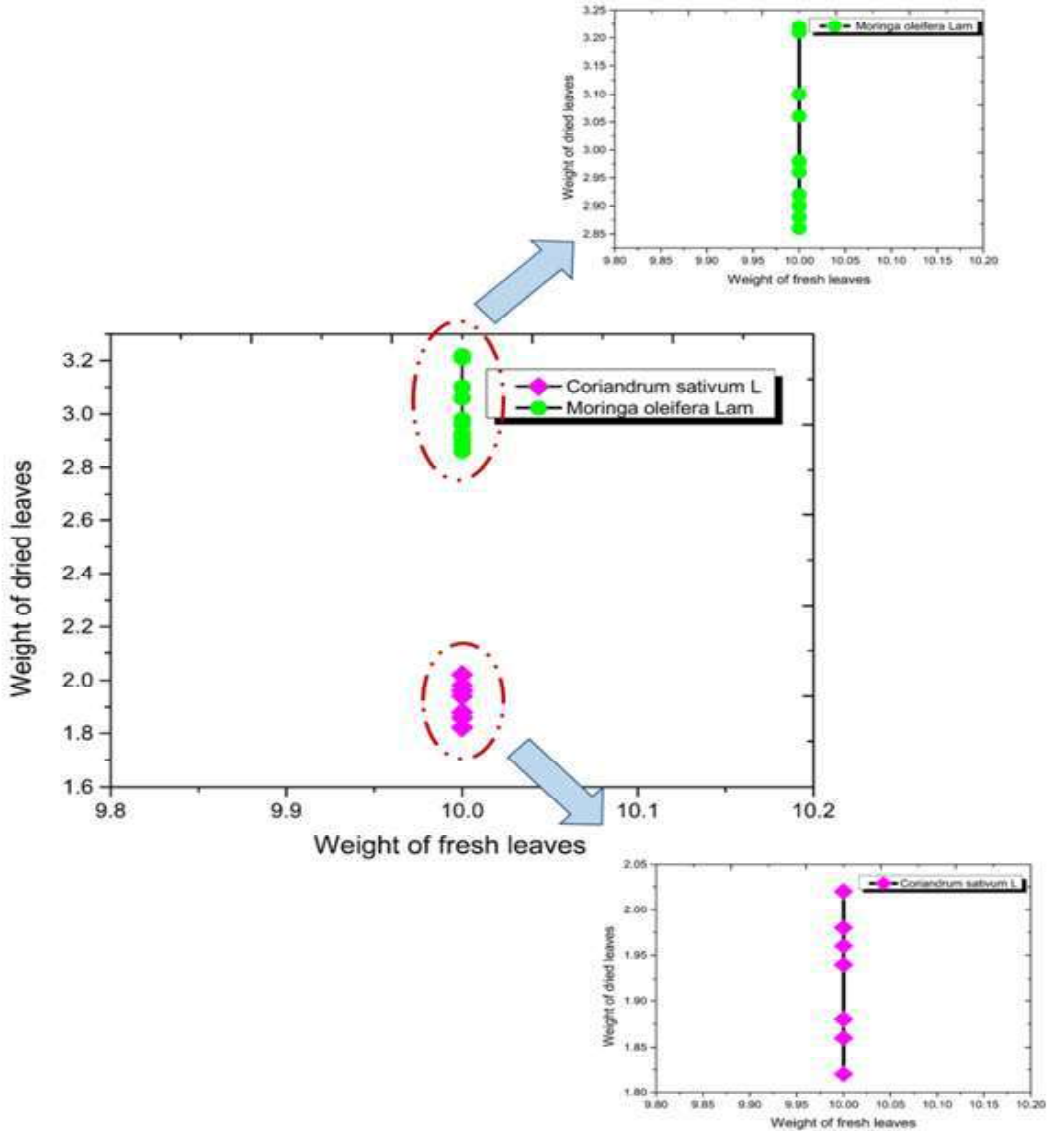


Fig. 2: Comparison of weight of dried leaves (in gm) for *Coriandrum sativum* L. and *Moringa oleifera* Lam. (Initial weight is taken 10 gm)

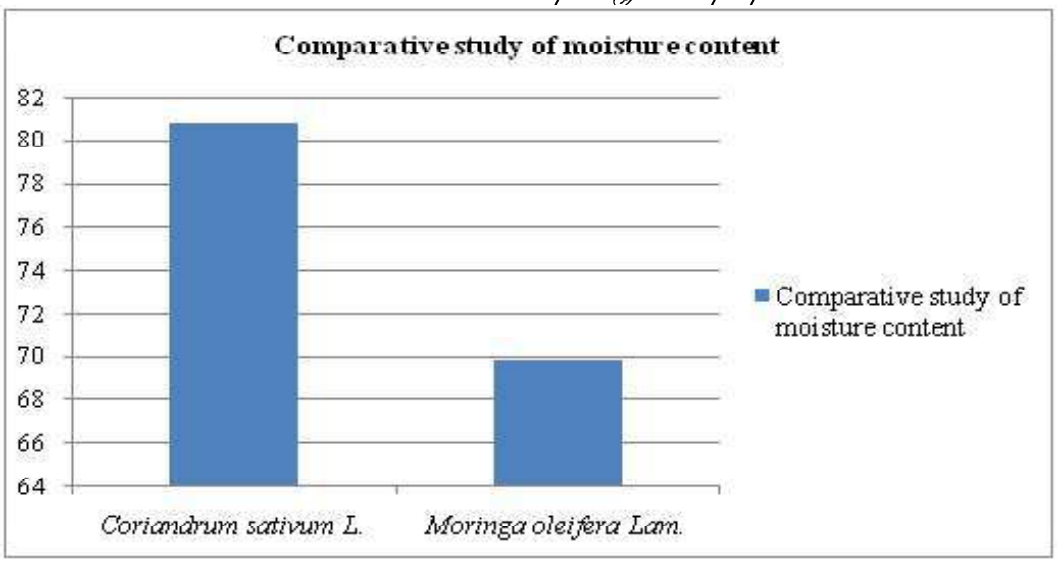


Fig. 3: Average moisture content of *Coriandrum sativum* L. and *Moringa oleifera* Lam.

CONCLUSION

The present study shows the comparative analysis of moisture content of two ethnomedicinal plants viz. *Coriandrum sativum* L. and *Moringa oleifera* Lam. It is observed that *Moringa oleifera* Lam. has relatively high residue and low average moisture content, i.e. 69.91%. This signifies that *Moringa oleifera* Lam. plants can preferably use to help for preventing chronic renal disease and other severe kidney injuries.

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Haematological alterations induced by cyclophosphamide administration in albino rats

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Abstract- In spite of having broad therapeutic efficacy against different types of cancer and immune related conditions, cyclophosphamide has been reported to exert toxic effects on the healthy cells with high proliferation rate. In this study, an effort has been made to investigate probable toxic effects of cyclophosphamide treatment on the functioning of bone marrow, expressed in terms of haematological changes. Experimental animals, male albino rats with average body weight of 150- 160 g were randomly divided into two groups (n= 5 in each). Vehicle control group (I) rats were treated with distilled water, whereas CYP- treated group (II) rats were intraperitoneally administered with single dose of cyclophosphamide (200 mg/kg body weight) on day 1 of the experiment. On 7th day, blood collected from rats of each group were analysed for various haematological parameters using automated haematology analyser. Data obtained were statistically analysed using Student's *t*- test. Single dose administration of cyclophosphamide resulted in a significant reduction in RBC count ($p<0.01$), haemoglobin ($p<0.001$), total WBC count ($p<0.001$), platelets ($p<0.001$), percent count of lymphocytes, PCV, MCV, MCH and MCHC as compared to non- treated group (I) rats ($p<0.01$). There was a decrease in percent count of neutrophil in rats of group II, but the change was not significant as compared to group I. Cyclophosphamide induced reduction in all types of blood cells and red cell indices can be attributed to its inhibiting effect on bone marrow activity, resulting in pancytopenia, erythrocytopenia, anaemia, panleucopenia and thrombocytopenia. So, monitoring of haematological parameters is advised for patients undergoing chemotherapy.

Key words: Cyclophosphamide, Bone marrow, Haematology, Pancytopenia, Thrombocytopenia

INTRODUCTION

Chemotherapy is a standard modality of cancer treatment that uses chemical agents or drugs to destroy cancer cells or inhibit their growth and spread.[1] In many cases, chemotherapy is combined with other treatment regimens, such as, radiation therapy and surgery to increase the effectiveness of cancer treatment. The antineoplastic drugs used in chemotherapy are used either singly or in combination with other drugs with different mechanism of action, resistance and side- effects.[2] Most of the anticancer drugs currently being used in chemotherapy are cytotoxic to normal cells at their therapeutically effective dosage.[1] Being non- selective in nature, chemotherapeutic drugs destroy rapidly dividing normal cells; such as, digestive tract, cells in bone marrow and hair follicles, leading to mucositis, alopecia, myelosuppression and immunosuppression.[2]

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Cyclophosphamide (N, N-bis (2-chloroethyl) tetrahydro-2H-1, 3,2- oxazaphosphorin-2-amine 2-oxide), an alkylating agent of Nitrogen Mustard family, has been known for its therapeutic efficacy against different types of cancer and immune related conditions, such as, multiple sclerosis, rheumatoid arthritis and systemic lupus erythematosus and organ transplantation. Cyclo phosphamide is a pro- drug which is activated through its conversion into two metabolites namely Phosphoramide mustard and acrolein. Although phosphoramide mustard is responsible for its anti- tumor activity, acrolein leads to oxidative damage to cellular macromolecules such as lipids, proteins and nucleic acids.[3,4] However, cyclo phosphamide exerts its anticancer activity by interfering with DNA synthesis and division of cancer cells, its non-selectivity makes high proliferating cells of bone marrow susceptible to the damage.[3] In the present study, an effort has been made to investigate the effects of single dose administration of cyclophosphamide on various haematological parameters in male albino rats.

MATERIALS & METHODS

Experimental animals

Male albino rats with an average body weight of 150-160 g were purchased from Jaz Scientific Ranchi, Jharkhand, India. Rats were maintained under controlled and hygienic environmental conditions (22- 24°C temperature and 12/12 h light/ dark cycle) with provision of ad libitum access to standard pellet diet and tap water. Animals were maintained following the provisions of institutional ethical committee of Ranchi University, Ranchi.

Experimental drug

Cyclophosphamide (Endoxan- N; 200 mg) was purchased from Azad Pharma, Ranchi, Jharkhand, India. The white crystalline powder of cyclophosphamide was dissolved in distilled water immediately before use.

Experimental design

After one week of acclimatization, rats were randomly divided into two group (n= 5 in each). Rats in group I served as vehicle control and were administered with distilled water throughout the study. Rats in group II (CYP- treated) were intraperitoneally administered with cyclophosphamide at a dose of 200 mg/kg body weight on day 1 of the experiment. Rats were maintained for 6 days. On 7th day, rats from each group were anaesthetized and blood was collected via cardiac puncture and stored in EDTA- coated vials. The dosing schedule of experimental drug to rats was based on previously published reports of Samdershi *et al.* (2019).[5]

Heamatological examination

Red Blood Cell (RBC) count, Haemoglobin (Hb) content, Packed Cell Volume (PCV), platelets, total White Blood Cell (WBC), differential percent of lymphocytes and neutrophil along with red cell indices such as, Mean Corpuscular Volume (MCV), Mean Corpuscular Haemoglobin (MCH) and Mean Corpuscular Haemoglobin Concentration (MCHC) were analysed in the blood sample collected from rats of vehicle control and CYP- treated group using automated haematology analyser.

Statistical analysis

The data obtained were expressed as mean ± Standard Deviation (SD). The difference between means of control and treatment group was analysed using Student's *t*- test and p value of ≤ 0.05 was considered as statistically significant.

RESULTS

Table 1 and Fig. 1 showed the effects of administration of cyclophosphamide on various haematological parameters in albino rats. Single dose administration of cyclophosphamide at the dose of 200 mg/kg body weight significantly reduced the count of RBC (49.58 %), PCV (56.26 %), haemoglobin content (57.25 %) and platelet count (68.26 %) and when compared to vehicle control group rats ($p<0.01$ and 0.001). Study of leucogram revealed a significant decline of 77.69 % in total WBC count of group II rats when compared to group I ($p<0.001$). Additionally, a significant decrease of 23.68 % was observed in % count of lymphocyte in rats of group II, treated with single dose of cyclophosphamide, when compared to non- treated group rats ($p<0.01$). There was a decrease of 20.27 % in % count of neutrophil of CYP-treated rats, but the decrease was not significant as compared to group I rats. Administration of cyclophosphamide caused a reduction in red cell indices such as, MCV (12.52 %), MCH (22.02 %) and MCHC (15.36 %), as compared to vehicle control group rats ($p<0.01$).

Table 1: Effects of single dose intraperitoneal administration of cyclophosphamide (200 mg/kg body weight) on haematological parameters in male albino rats

S. No.	Parameters	Groups		*t*-value	% change (+) Increase/ (-) Decrease
		Group I Vehicle control (Distilled water)	Group II CYP- treated (200 mg/kg body weight; i.p.)		
1.	RBC ($\times 10^6/mm^3$)	6.03 ± 1.12	3.04 ± 0.64*	4.635	- 49.58 %
2.	Haemoglobin (g/dl)	12.47 ± 2.20	5.33 ± 1.14**	5.754	- 57.25 %
3.	WBC ($\times 10^3/mm^3$)	5.38 ± 0.69	1.20 ± 0.20**	11.887	- 77.69 %
4.	Platelets ($\times 10^5/mm^3$)	4.60 ± 1.06	1.46 ± 0.51**	5.317	- 68.26 %
5.	Lymphocyte (%)	72.5 ± 3.50	55.33 ± 7.50*	4.145	- 23.68 %
6.	Neutrophil (%)	25.50 ± 6.73	20.33 ± 1.53	1.499	- 20.27 %
7.	PCV (vol %)	31.85 ± 7.95	13.93 ± 1.05*	4.468	- 56.26 %
8.	MCV (fL)	52.70 ± 2.46	46.10 ± 2.91*	3.459	- 12.52 %
9.	MCH (pg)	21.48 ± 2.12	16.75 ± 1.06*	3.981	- 22.02 %
10.	MCHC (g/dl)	39.70 ± 3.00	33.60 ± 0.62*	3.981	- 15.36 %

Values were mean ± SD (n= 5) * p<0.01, ** p<0.001 when compared to Group I.

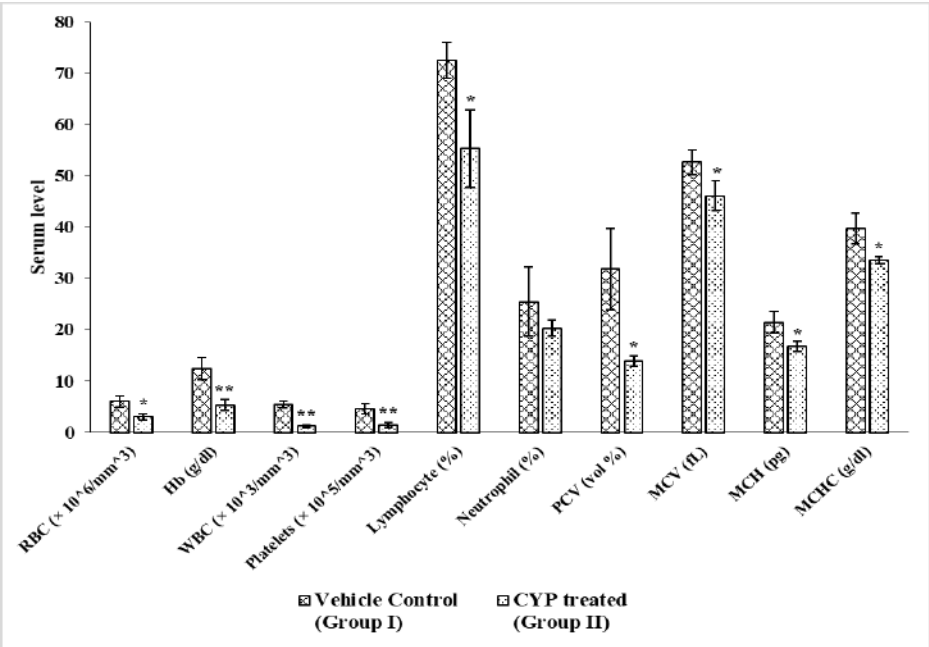


Fig. 1- Histograms showing alterations in various haematological parameters following cyclophosphamide administration in male albino rats

DISCUSSION

In the current study, single dose administration of cyclophosphamide (200 mg/kg body weight; i.p.) resulted in a significant decrease in RBC count ($p<0.01$), PCV ($p<0.01$) and haemoglobin content ($p<0.001$) of rats as compared to vehicle control group, indicating a state of erythrocytopenia and anaemia. This finding was in agreement with Shruthi *et al*., (2017)[6] who reported a significantly reduced value of RBC and haemoglobin in Swiss albino mice intraperitoneally administered with two chemotherapeutic drugs, cyclophosphamide (50 mg/kg) and cisplatin (10 mg/kg), advocating for their haematosuppressive effects on the individuals. Cyclophosphamide induced significant decline in red cell indices, such as, MCV, MCH and MCHC reported in the current study (Table 1, Fig 1) was supported by Bhalchandra *et al*., (2018)[7] who documented a significant decline in level of cell indices as well as RBC count, PCV and haemoglobin level after administration of cisplatin, a cytotoxic agent used to treat a variety of cancers, at the dose of 7 mg/kg/day for 15 days in male Wistar albino rats. Toxic effects of cisplatin treatment on RBC and haemoglobin might be due to suppression of activity of haematopoietic tissues, impaired erythropoiesis and heme biosynthesis, accelerated destruction of RBCs because of altered membrane permeability, increased mechanical fragility, and/or defective iron metabolism and haemorrhage.[7] Additionally, administration of chemotherapy has been reported to reduce erythropoietin mRNA synthesis in the liver and kidneys.[8] A lower value of MCV, MCH and MCHC in the group II rats indicated a reduction in size of RBCs, amount of haemoglobin per RBC and amount of haemoglobin per unit volume following administration of cyclophosphamide, suggesting for a clinical condition of microcytic hypochromic anaemia caused due to chemotherapy.[7,9,10]

Current study also revealed a toxic effect of cyclophosphamide on leucogram of rats. A significant decrease in total WBC count ($p<0.001$) and percent count of lymphocyte ($p<0.01$) advocated for immunosuppressive activity of the experimental drug (Table 1). The leucopenic, especially lymphopenic activity of chemotherapeutic drugs has been reported in earlier studies also.[6,7,11] Reduction in total number of WBCs might be due to increase in occurrence of infections and inflammation during chemotherapy treatment.[7] In contrary to the decline in neutrophil level of rats treated with cyclophosphamide, as reported in present study, a state of neutrophilia expressed in terms of increased level of neutrophil has been reported after administration of chemotherapeutic drugs.[11-14] Increase in neutrophil count might be attributed to the marginalisation of phagocytic cells, showing improvement in defensive responses under normal circumstances.[12] Cyclophosphamide induced decline in neutrophil count was also in line with study made by Ilamkar *et al*. (2020)[10] in methotrexate (1.75 mg/kg body weight biweekly orally) treated mice, but they reported an increment in lymphocyte percent following treatment with the drug. A significant decrease in the total WBC count and differential count of lymphocyte and neutrophil in rats of CYP- treated group might be attributed to the high dose (200 mg/kg body weight) and longer duration (6 days) of exposure to the drug. Alterations in differential count of WBC can also be related with the immune status of the individual receiving the treatment.

Our study also reported a significant decline in platelet count in CYP- treated rats (Table 1). Apart from regulation of haemostasis and thrombosis, platelets have been reported to be secondarily involved in maintenance of innate immunity, tumor growth and extravasations in the vessels.[15] A significant decline in platelet count following chemotherapy advocated for a condition of thrombocytopenia and weakened immunity in the individuals.[6,7,13,16] Bhalchandra *et al*. (2018)[7] explained that the decrease in platelet count following cisplatin treatment might be due to its bone marrow inhibiting activity resulting in decreased production of platelet or increased consumption of platelets or due to increased platelet aggregation.

Based on the findings of present study and corroborations made with previous reports, a haematotoxic effect of cyclophosphamide treatment can be recorded. As all types of blood cells are produced by haematopoietic cells present in bone marrow, a condition of pancytopenia (reduction in all types of blood cells) induced by cyclophosphamide exhibited its suppressive effects on the activity of bone marrow. Atrophic changes in the histological architecture of spleen tissues and loss of cellularity in white pulp region can provide evidences for cyclophosphamide induced perturbations in haematological parameters.[5] Wondimneh *et al*., (2021)[17]

reported a significant decrement of haematological profiles in post- chemotherapy compared to pre-chemotherapy sessions during cancer treatment. Chemotherapy induced myelosuppression as well as immunosuppression can be attributed to the loss of stem cells and inability of bone marrow to regenerate new blood cells following drug administration, mediated via induction of oxidative stress.[4,7,13,18]

CONCLUSIONS

Bone marrow suppression leading to a decline in level of haematological parameters following chemotherapy adversely affects the immune status of the individual undergoing treatment. It not only affects the effectiveness of the treatment but also renders the individual susceptible for infections from different pathogens, thereby deteriorating the quality of life and reducing the longevity. So, a better understanding of the immunosuppressive effects of the chemotherapeutic drugs can be helpful in management of its dose and duration of administration in the individual undergoing therapy and in mitigation of its adverse consequences.

List of Abbreviations

CYP- Cyclophosphamide, DNA- Deoxyribonucleic acid, EDTA- Ethylenediamine tetraacetic acid, Hb- Haemoglobin, MCH- Mean Corpuscular Haemoglobin, MCHC- Mean Corpuscular Haemoglobin Concentration, MCV- Mean Corpuscular Volume, PCV- Packed Cell Volume, RBC- Red Blood Cells, SD- Standard Deviation, WBC- White Blood Cells.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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Excessive use of urea is responsible for nitrogen pollution in India: A systematic review

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Abstract- Urea is the main source of artificial fertilizer in crops in India. Fertilizer sources of emissions to all the processes that cause it to reach the environment and cause pollution. It also includes nitrogen minimization measures which suggest that this can be largely prevented by reducing the use at the source. As per the review of literature, it accounts for 84 percent of the total nitrogen fertilizer produced in the country. The study found that nitrogen fertilizers in India reduced soil carbon by 28 percent of the original amount, while nitrogen, phosphorus, potassium, and organic manure combined reduced farm carbon by 2.9 percent has decreased. The impact of nitrogen pollution on the Himalayan forests and mountains has not been extensively studied. In its lower part lies the Gangetic plain, which is probably the global epicenter of nitrogen air pollution. Excess nitrogen in the atmosphere can produce pollutants such as ammonia and ozone, which can impair the ability to breathe, limit visibility and alter plant growth. When excess nitrogen comes back to earth from the atmosphere, it can harm the health of forests, soils, and waterways.

Key words: Nitrogen pollution, Urea in Agriculture, Chemical fertilizer.

INTRODUCTION

Nitrogen is essential for plants as it restricts nutrients. Plants prepare food from it through chlorophyll and protein synthesis.[1] Naturally, these essential nutrients are present in the soil through diazotroph bacteria. They are present in the roots of plants like pulses. But this natural presence was not sufficient after the Industrial Revolution as it was unable to provide food to the growing population.[2] For this reason, artificial fertilizers were invented to supplement the naturally present nitrogen.[3]

Urea was used throughout the country after the Green Revolution (1965-66) to make the country strong in the field of agriculture. In the initial years, farmers were not duly informed about this area in many areas of the country.[4] Rather, sacks of urea were quietly dumped in the fields at night, when this did not work out, through the sarpanch of the village, the farmers were urged to use urea in their respective fields. From 1966-1967, the government urged all the farmers to use urea but it seemed to us that the loss due to farming would be converted into profit by the use of urea.[5] It happened the same. We were happy with the increase in yield. Although initially, were also afraid that the use of urea might ruin the crop. For the first time in 1967, and when the harvest was ready, we could not believe our eyes. Because for the first time we're seeing that we got three times more wheat yield.[6]

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Around 1975, it had become a common practice by farmers in the country to apply three to five times the amount of urea in their respective fields. It is also true that their production was also increasing at the same rate.[7] But there was no program far and wide to give any official information from the government to the farmers in this regard. As a result, the farmer was continuously increasing the use of urea in his fields. Between 1985 and 1995, by adding 125 kg of urea per acre, the yield increased ten to 12 times.[8] From here we started descending. Yields started decreasing in agriculture while we kept increasing the amount of urea year after year. In 1997 we put 175 kg of urea and found what? That's just ten times.[9] That is, now due to increasing urea, the yield of the fields was not increasing but was decreasing.

Yields have been declining steadily since 1995. Indiscriminate use of urea for forty years has left the fields barren.[10] After all, once again the farmers started putting manure of cow dung, etc. in their fields. This did not increase the yield but also reduced the yield. But by 2010, even cow dung manure was not easily available.[11,12] Today's situation is such that farmers are not deterred even by putting 225 to 250 kg of urea in one acre of land. The reason is that in any case, the 12 times the yield of 15-20 years ago is what is needed. Now all the efforts are proving to be in vain.[13]

Ironically, a single farmer has used more than 6,610 kg of urea in five decades. Whereas till now 195.88 million tonnes of urea have been used in the entire country.[14] Now the side effects of nitrogen have started coming to the fore. Nitrogen pollution has been assessed for the first time in India by more than 120 scientists. The objective of this study is to assess the impact of urea use on the nitrogen cycle.

METHODS

This paper has earned the adjective systematic if it is based on a formulated question, identifies relevant studies, appraises their quality, and summarizes the evidence by use of the explicit methodology. It is the explicit and systematic approach that distinguishes systematic reviews from traditional reviews and commentaries. Whenever we use the term *review* in this paper it will mean a *systematic review*. The process is given below:

Step: 1	Framing questions for a review
Step: 2	Identifying relevant work
Step: 3	Assessing the quality of studies
Step: 4	Summarizing the evidence
Step: 5	Interpreting the findings

Inclusion/exclusion criteria: Inclusion criteria for refereed study samples included original research articles published in a research journal. Other articles types such as conference abstracts, communications, commentaries, editorials, brief reports, position, and hypothesis-generating statements were excluded. Non-refereed publications were also excluded. A flowchart of the study screening and selection is presented in Fig. 1.

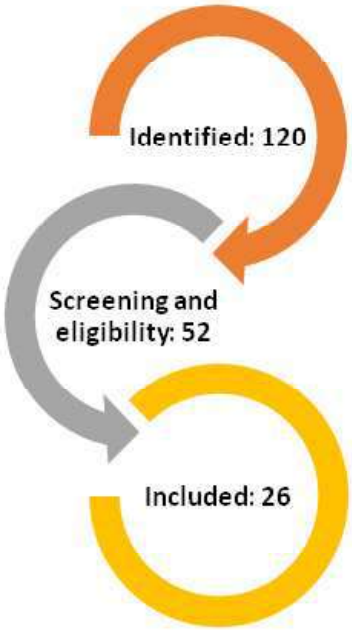


Figure 1: Process of included research articles

After applying searching strategies by using keywords Nitrogen pollution, Urea in Agriculture, and chemical fertilizer and Nitrogen pollution in India, we have found 120 articles. The 52 articles remained after screening for the eligibility check; overall 26 research articles were included in the study.

RESULT

According to Hua *et al.,* (2020)[1] by reducing the use of nitrogen fertilizers and recycling organic manure, Indian farmers have a better chance of producing safe food grains. This will also benefit the government will also save a huge amount of money. Carbon and nitrogen-

saving practices also need a lot of attention at this time. Doing so will help in better nitrogen management and improve soil structure. At the same time, drought resilience, soil erosion, and other climate-related hazards can be reduced." According to him, by adopting these methods, India can lead the world in the field of global nitrogen management. India is already working in this direction through ING, which has published the assessment and leads the South Asian chapter in the International Nitrogen Initiative.[15]

How is India's nitrogen pollution different from the world's?

The contribution of nitrogen pollution sources may vary from country to country. Sources can be from agriculture, domestic and municipal sewage to fossil fuels, automobiles, industries, and stubble fires.[16] The biggest man-made source of nitrogen pollution is unused farm fertilizers, whether organic or chemical in origin. Precision agriculture is unaffordable in India and labor cost is high for the precise use of fertilizers. Therefore, farmers use more fertilizers. But crops are not able to make full use of fertilizers, so they contribute to nitrogen pollution. Most of the pollution comes from cereals, while in other countries such as China, horticulture crops and cash crops contribute the most.[17] In Europe, it is more polluting than fodder, cattle, and animal husbandry. Africa has a negative contribution. This means that due to the non-availability of chemical fertilizers, in sufficient quantity, farmers are exploiting the nitrogen available in the soil, which is reducing the quality of the soil.[18]

According to Brunekreef (2020)[19] "The annual limit for nitrogen was set at 44 teragrams (Tg). He said that the use of nitrogen mainly as a fertilizer is about 150 teragrams per year, which is three times more than the limit of the earth. Artificial nitrogen through fertilizers has exceeded Biological Nitrogen Fixation (BNF) globally. DLN Rao, a member of the Indian Institute of Soil Science and contributing to the assessment, says, "The global ratio of BNF to fertilizer will be around 1.4:1.[20]

Problems due to nitrogen pollution

Reduction in yield:

Yields have been declining steadily since 1995. Indiscriminate use of urea for forty years has left the fields barren. After all, once again the farmers started putting manure of cow dung, etc. in their fields. This did not increase the yield but also reduced the yield. But by 2010, even cow dung manure was not easily available.[21] Today's situation is such that farmers are not deterred even by putting 225 to 250 kg of urea in one acre of land. The reason is that in any case, the 12 times the yield of 15-20 years ago is what is needed. Now all the efforts are proving to be in vain. Recalling the old times, Poddar says that 50 years ago today, the Prime Minister of our country was chirping us to use urea.[22] Today, after five decades when farms are about to be completely exhausted, the present Prime Minister is now appealing to us (in Mann Ki Baat aired on December 26, 2017) to minimize the use of urea.[23] Not only this, but now the government is also reducing the subsidy on urea. 72,438 crore subsidy was earmarked for urea in the 2015-16 budget, which was reduced to Rs 70,000 crore during 2016-17. Poddar gave an estimate and said that in the last 50 years; we used about 6,610 kg of urea in one acre of land and got about 5,337 quintals of yield (wheat and rice).

Nitrogen pollution is dangerous for children:

Nitrates in drinking water can cause a blue baby syndrome in babies up to six months old.[24] The amount of oxygen in the blood of children suffering from this disease decreases because nitrates interfere with the effect of hemoglobin. Hemoglobin is the carrier of oxygen. This can lead to frequent diarrhea in children. It also impedes respiration. It also increases high blood pressure and blood pressure in school children.[25]

Potassium deficiency in food:

Excessive use of urea reduces the amount of potassium in food grains. Along with controlling high blood pressure, potassium also keeps the heart-healthy. He says that the rapid increase in the number of blood pressure and heart patients across the country is a result of the Green Revolution.[26] He says that this revolution increased the production of wheat and paddy beyond expectation but the pulses sector was badly affected by it. Whereas, in the country it is a major source of protein. Due to its deficiency, malnutrition has become a serious problem in Bihar, Jharkhand, Chhattisgarh, Madhya Pradesh, and Odisha. The number of tuberculosis patients is also increasing due to malnutrition.[27]

Warming effects of nitrogen pollution:

Nitrous oxide is released in large amounts each year and has a positive radiative forcing, i.e. its molecules store thermal radiation from the sun, thus contributing directly to the warming of the atmosphere. Per unit of weight,

nitrous oxide is a more powerful GHG than CO_2. Galloway *et al.,* (2008)[28] calculate that over 100 years, nitrous oxide has a global warming potential 296 times larger than an equal mass of CO_2. Estimates vary as to how much nitrous oxide is released by the EU each year.[13] The United Nations Framework Convention on Climate Change estimates 819 gigagrams (Gg) per year, but this figure does not include emissions from EU Rivers, continental shelf, and estuaries, nor indirect emissions following nitrogen deposition and leaching. These indirect emissions of nitrous oxide have themselves been estimated at 71.1 gigagrams (Gg) and 34.6 Gg per year.[24]

Reactive nitrogen and its impact on methane from livestock Livestock:

Such as cattle emit methane from the methanogenic microbes that live in their digestive systems. With globally increasing demands for meat and dairy produce, methane emissions from livestock have been increasing.[26] Different livestock systems and animals emit different quantities of methane. The rearing of pigs, poultry, and dairy cattle emit the most methane, as they require large amounts of reactive nitrogen used as fertilizer to grow feed for the animals. These intensive forms of livestock production emit more methane than the rearing of cattle and sheep for meat production. Cattle and sheep tend to graze grass, and although pastures are often fertilized, this is not as nitrogen intensive as growing food crops, such as maize and soy, and feed does not need to be transported.[29]

Reactive nitrogen, tropospheric ozone, and carbon sequestration:

Nitrogen contributes to the formation of tropospheric ozone. This affects the carbon cycle, as tropospheric ozone impedes photosynthesis and thus reduces carbon sequestration. Galloway (2008)[28] raises this possibility and draws attention to uncertainties in how tropical ecosystems will respond to rising reactive nitrogen inputs, as most research has been undertaken in northern latitudes.[30] This is an important question given that tropical ecosystems are expected to receive 'the most dramatic increases in reactive nitrogen in the future, and are already under pressure from a warming climate.[23]

Measures to reduce:

Based on its analysis, SCN recommends that India formulate such policies to increase its NUE. By doing this, the impact of nitrogen pollution on the environment, human health, and climate change can be reduced .[31] This can be significantly reduced by controlling the use of nitrogen. This will reduce the burden of subsidies given by the Government of India on fertilizers.[8]

After the liberalization of the economy in 1991, the government took control of other fertilizers, but urea, which contains the most nitrogen, is still being controlled completely. In other words, the government controls its production, import, and distribution.[32] The government also gives a subsidy to the farmers to buy urea. It is cheaper than other fertilizers, so farmers use it freely. The result is that the soil is deprived of other nutrients and this harms its health.[9]

The valuation found that based on the expected selling price of US$78.8 per tonne of nitrogen fertilizer, the cash subsidy amounts to around US$7 billion. This puts a heavy burden on the taxpayers of the country. Scientists estimate that India uses nitrogen worth about US$ 10 billion every year.[33] Whereas the impact of nitrogen pollution on health, ecology, and climate is as much as US$75 billion per year. They have also found that if the NUE is improved by 20 percent, there will be a profit of about US$ 170 billion every year globally.

CONCLUSION

Food production in India is not growing in the same proportion as fertilizers are being consumed. The contribution of nitrogen to grain production has decreased over the past two decades. Irrespective of other factors, increasing the use of fertilizers will only be seen in the form of low productivity of crops. The environment will also have to pay a price for this. Irrigated areas where many crops are grown in a year need a logical balance on the unbalanced use of nitrogen fertilizers. Ensure that only as much fertilizer is used in all areas as is necessary, even if the use of fertilizers has to be increased slightly in monsoon-dependent lands. Most importantly, need to curb non-agricultural sources of nitrogen which are increasing pollution at a very fast rate. The Government of India has inadvertently taken steps in the right direction by promoting neem-coated urea. This can reduce the crisis. Neem-coated urea releases nitrogen at a slow rate and gives the plants time to absorb it. It makes maximum use of nitrogen. Looking at the size of India, it can be said that Indian emissions are much less than America and China, but still, it contributes a lot to the world.

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Influence of low light and dark treatment on callus survival of *Elaeocarpus ganitrus*

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Abstract- An attempt has been made to analyze the effects of low light conditions on callus cultures of *E. ganitrus* on full strength MS medium for long time preservation. For the study dark and light conditions were taken on the basis of time differences out of which low light conditions for 16h and dark conditions for 8h were selected to recapitalize the callus of *Elaeocarpus ganitrus* when supplemented with plant growth regulators and different kinds of antioxidants in growth medium.

Key words: *Elaeocarpus ganitrus*, Callus color, Dark treatment, Low light condition

INTRODUCTION

For efficient callus and shoot initiation, light plays crucial role which make it as one of the most important factor in an outcome of any tissue culture study.[1] The etiolation and de-etiolation mechanisms are important for knowing about the accession of photosynthetic capability at the situation of biogenesis of chloroplasts and response of plant with respect to light.[2] The influence of dark and light conditions was analyzed on callus cultures of *Citrus reticulata.* Under both kinds of treatment explants culture shown variations in the response. Under the dark and light incubation treatment it was observed that genetic variation regarding callus induction exists.[3]

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In a study it was observed that regeneration of shoot is light independent and no etiolation character was observed in the regenerated shoots in dark, except they were lacking chlorophyll which was further developed when 16 h light exposure was given.[4] To analyze the effects under various spectra of light rose plants were grown.[5]

In the tissue culture seedlings of *Cunninghamia lanceolata* the composite LED light effects on the antioxidant capacity and root growth and development was reported.[6] The tissue culture study on indirect regeneration of *Lilium ledebourii* Bioss was reported, in which all the stages were investigated under the dark conditions for the first time.[7]

For micropropagation study using nodal segments as explants of *Elaeocarpus sphaericus*, the cultures were

kept under 8h dark and 16h light conditions (using 1200 lux, 40 watt cool white fluorescent tubes).[8] To maintain the shoot cultures of *E. sphaericus* for *in vitro* propagation 40-50µmol $m^{-2}s^{-1}$ of irradiance was used for 16h light condition.[9] The calli of *E. sphaericus*[10] and *E. tuberculatus*[11] were developed and well maintained using *in vitro* tissue culture protocol which had shown positive outcome.

MATERIALS & METHODS

The callus samples were developed from immature fruits of *Elaeocarpus ganitrus* on MS medium earlier[12] in Plant tissue culture laboratory of School of Biological Engineering and Life Sciences, Shobhit Institute of Engineering and Technology, Modipuram, Meerut (UP) India were procured for the investigation purpose. The callus samples were sub-cultured on full strength MS[13] medium in conical flasks of 250 ml (Borosil), containing 100 ml of medium. The procedure of growth media preparation was similar to that which was applied earlier for callus initiation from immature fruit of *E. ganitrus* explants material.[12]

Sample treatment

The dark treatment was provided to the cultures for 10 days (Fig.2A). After 10 days of treatment the cultures were maintained on the growth medium and placed under the single tube light of 40 watt, 6 inches away on the rack separating the open area using black colored paper of A4 Size (Fig.2B).

During the initial first week after the sub-culturing of the callus, the cultures were maintained at a range of light and dark conditions depending on time variations in hours. The callus cultures treated under complete dark conditions (24 h) were taken as control.

Statistical Analysis

The experiments were performed with ten technical and three biological replicates. The analysis of variance was performed (one way ANOVA) for significant difference (Pd" 0.05) in between the control and treated experiments. The asterisk mark used for denoting significant difference.

RESULTS & DISCUSSIONS

It was observed that callus growth was very slow when the dark conditions were provided to the cultures. When complete dark conditions for 10 days were provided, the color of the callus turned from light green to dark green and then to white and then light yellow and then to dark yellow to yellowish brown (Fig.2A). It was observed that new callus tissues were started appearing during the 1st week after first sub-culturing which was performed just after dark treatment and further in later stages of treatment i.e., from 2nd to 4th week with similar low light and dark conditions (16 h /8 h) started becoming light green (Fig.2B). From 4th week to 6th week of exposure callus was completely appearing light green to dark green in color.



Fig 1: Trees of *Elaeocarpus ganitrus* located at SIET campus, Meerut (UP) India

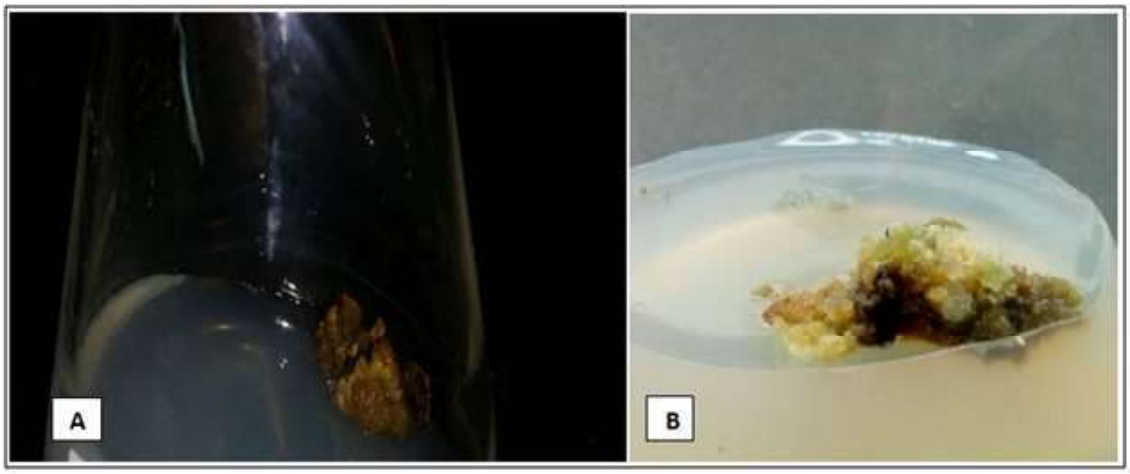


Fig 2: Callus culture under dark conditions of 10 days [A] and Callus culture re-multiplied during 21st day to 42nd day under treatment of 16h low light/ 8 h dark conditions [B].

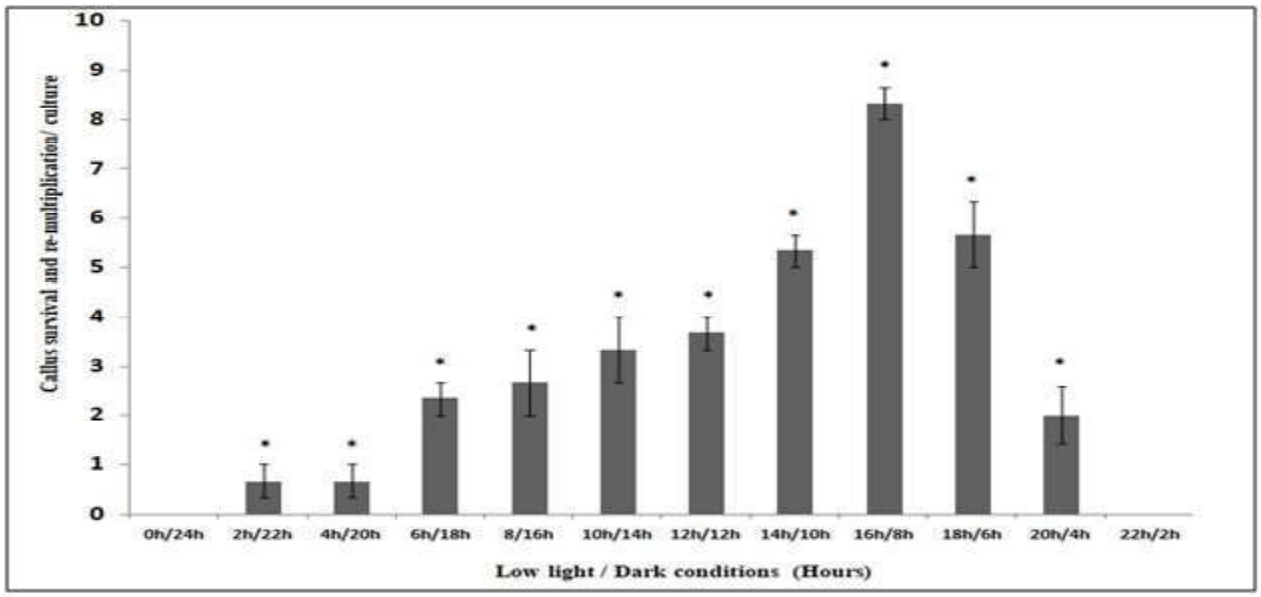


Fig 3: Callus cultures under 24 hours dark conditions were taken as control and under the treatment of light and dark conditions it was observed that low light for 16 h and 8 h dark environment provided better results.

Depending on the physical condition of callus the second sub-culturing may be performed in the 5th week otherwise it should be performed during the 6th week. The entire process may be repeated again keeping the same parameters to maintain the *in vitro* callus cultures. It was noted that low light was helpful for callus multiplication, and complete dark incubation for slow multiplication which was observed during the callus study of *E.ganitrus.*[14]

However, low light and dark environment based on a range of time variations in hours has shown the survival and re-multiplications of cultures (Fig.3). The use conical flasks for callus studies were previously reported.[14] The light and dark effects were studied for initiation and regeneration of callus for *Nicotiana tabaccum.*[15] During propagation of *Haworthia* through tissue culture for better organogenesis, best union of plant growth regulators and intensity of light was determined.[1] The effects of TDZ and dark conditions for callus formation from rose leaf explants were analyzed.[16]

CONCLUSION

The callus re-multiplications were observed higher under 16h low light and 8h dark environment. By using this protocol the number of sub-culturing for callus cultures of *Elaeocarpus ganitrus* may be reduced and

cultures can be preserved for long time period. Because the cultures can be remain available for a long time period several other investigations can be taken into consideration.

Many more investigations based on effects of light and dark conditions on *in vitro* cultures yet needed. The study may be helpful for investigating the effects of light and dark conditions on *in vitro* callus cultures survival and re capitalization for other endangered tree species.

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COMPETING INTEREST

Declared none

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Studies on the biodiversity of some families of butterflies (Lepidoptera: Rhopalocera) at Kanke, Ranchi, Jharkhand, India

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Abstract- Present study was an attempt to investigate the biodiversity of the butterflies (Lepidoptera: Rhopalocera) at Kanke, Ranchi, Jharkhand in the year 2019 to 2021. The seasonal collections were made at the interval of 15 days. The objective of this work was to understand the status of species richness and diversity of the butterflies, mainly during the COVID-19 lockdown. Total 702 individuals of butterflies belonging to 21 species under the three families Nymphalidae, Pieridae and Papillionidae with their host plants were reported. Nymphalidae was the most dominant family. The diversity was calculated using Shannon Weiner Index (H). The calculated values of Shannon Weiner diversity index (H) was found to be highest (3.012) during the year 2021. These values indicate that the area of study contains good butterfly diversity and it might play a role in the biodiversity conservation.

Key words: Lepidoptera: Rhopalocera, Nymphalidae, Pieridae, Papillionidae, Shannon Weiner Index, Kanke, Ranchi

INTRODUCTION

Most abundant animal types belong to the class Insecta of phylum Arthropoda.[1] In this class, the order Lepidoptera is considered to be the second largest and highly diverse[2]. It comprises butterflies and moths.[3-5] Butterflies are studied more than any other insect because they are easily accessible and can be captured and identified. Moths play a role as pest on many economically important plants.[6] Both of them serve as pollinators. So, they have great economic importance.[7] Studies on the animals have been performed[8-10] mainly for its abundance, species richness, relationship with vegetation and abiotic factors. They show behaviour of mimicry, migration and seasonality that make them different from the others. Butterflies are the most attractive creatures in the world of insects due to their diverse and beautiful colour. So they have aesthetic value, as well. The diverse habitat and climate of India is responsible for the different colour, size, and species of butterflies. They are recognized as a potent ecological indicator because of their sensitivity towards changes in habitat, climate, and weather[11-13] and also as an important component of food chain. They can be good study material for the analysis of genetics, coevolution and insect - plant relationship.[14] As the abiotic factors like ecological factors and host plants directly influence the larva and adult of them and so influence their diversity.

Aim and objective of this work was to collect the butterflies from study areas of Ranchi to record the ecological factors at the same place, to calculate the diversity of the collected butterflies and then correlate with the seasonal variations and host plants, because

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butterflies directly depend on food plants.[15] The study of the diversity of butterflies mainly took place during the COVID-19 period when the anthropogenic disturbances were less, because of the lock down.[16] Morrison-Godfrey (1950)[17] reported about 121 species and 134 sub-species from Ranchi and Hazaribag. Rapid urbanization and destruction of natural habitat for the commercial and residential purpose at Ranchi can be a potential threat to the biodiversity of butterflies. The present study was designed to study the effect of ecological factors and host plants on the population dynamics and diversity of the insects. The collection work was made during the years 2019-2021.

MATERIALS & METHODS

Collection, Identification and Preservation of samples:–

Collection of butterflies from Kanke, Ranchi in the state of Jharkhand was done. Its geo-coordinate is 23° 26' 4.3764'' N (latitude) and 85° 19' 14.5272'' E (longitude). The collection was accomplished through flying net technique. One individual of each species was preserved for reference and rest were released. Counting was made through quadrate method. For the preservation, butterflies were killed using ethyl acetate in an airtight killing jar. Stretching of wings and proper pinning was done as per prescribed format. Identification was done with the help of available literature, (https://www.ifoundbutterflies.org)[18] and by comparing with the zoological keys and reference collection available at the Zoological Survey of India and National Centre for Biological Science[19]. After a few days of spreading, the dried specimens were shifted from the spreading board to the wooden box by removing the pins and preserving them with the labelling. The collection work was divided into three seasons – Pre-monsoon (January to April), Monsoon (May to August) and Post-monsoon (September to December).

Tabulation of Data: -

Data analysis:- It was completed with the help of MS-Excel and PAST software. Shannon-Weiner Diversity Index (1948)[20] and Simpson's Dominance Index

Shannon Weiner Species Diversity Index

$(H') = \Sigma Pi.lnPi$ $< H = -\sum[(p_i) \times \ln(p_i)] >$

Where, Pi = species diversity index.

N = total number of species.

Table 1: Yearly report of the environmental factors like ambient temperature, relative humidity and rainfall at Kanke site. The ambient temperature was between 22°C to more than 24°C while relative humidity was calculated between 76% to more than 77 % and the rainfall was 3mm to more than 5mm.

Year	Ambient temp.(˚C)	R.H. (%)	Rainfall (mm)
2019	24.75 ± 5.15	76.04 ± 3.13	5.64 ± 18.75
2020	22.10 ± 6.08	77.14 ± 5.08	4.38 ± 13.36
2021	24.61 ± 5.04	77.29 ± 1.75	3.19 ± 12.80

Plant diversity at the site: A rich diversity of plants belonging to trees, shrubs and herbs was found in the collection area of Kanke. Agro-horticulture areas were also found nearby to the site. *Asclepias curassavica* (Kaura-Dodi), *Calotropis gigantea* (Safed Ak), *C. Procera* (Madar, Ak), *Cryptolepis dubia* (Karanta) *Bambusa bambos* (bans), *Andropogon* (beard grass),*Cymbopogon* (Lemon grass), *Apluda, Eleusine* (grass), *Ficus reliogiosa* (Peepul), *Oryza sativa* (Paddy, Rice), *Panicum maximum* (Guinea Grass), *Zea* (maize), *Sinningia speciosa* (Gloxinia), *Hygrophila auriculata* (gokanta), *Nelsonia canescens* (blue pussy leaf), *Ricinus communis* (Castor Seed Plant), *Tragia cannabina* (Ground creeper), *Tragia involucrate* (Indian Stinging Nettle), *Barleria* (vajradanti), *Celtis timorensis* (stinkwood) *Ficus benghalensis* (Banyan), *Ficus glomerata* (Counry fig, Goolar), *Ficus indica* (cactus pear) *Holarrhena antidysenterica* (Kura, Kurchi, Dudhi), *Ichnocarpus frutescents* (Kali Dudhi), *Nerium oleander* (Oleander), *Bombax ceiba* (red silk cotton tree) *Leguminosae* (Beans), *Hibiscus rosasinensis* (china rose), *Jasminum officinale* (jasmine) and others.

RESULTS & DISCUSSION

Total 702 butterflies' sightings of 21 species were recorded during the collection period. The species compositions with their taxonomic position, family wise presented in Table 2.

Species Composition

The present study covers a smaller area. Among three families, butterflies of family Nymphalidae were prevalent than two other families. During the compilation done in the year 2019-21 in Kanke, Ranchi, 13 species of butterflies belonging to Nymphalidae family, four species each of Pieridae, and Papillionidae family were found (Table-1). The thirteen species of Nymphalidae family were *Danaus chrysippus chrysippus (*NYM001*), Melanitis phedima bela* (NYM002) *Junonia almana*

(NYM003), *Callerebia scandao pima* (NYM004), *Ariadne merione merione* (NYM005), *Melanitis phedima varaha* (NYM006), *Junonia atlites atlites* (NYM007) *Rohana parisatis parisatis* (NYM008) *Danaus genutia genutia* (NYM009) *Euploea core core* (NYM010), *Neptis hylas nicobarica* (NYM011), *Lasippa viraja viraja* (NYM012), and *Acraea terpsicore* (NYM013). Four species of butterflies of the family Pieridae were *Eurema andersonii* (PIE001), *Delias eucharis* (PIE002), *Catopsilia pomona* (PIE003), and *Belenois aurota aurota* (PIE004). Four species of butterflies of the family Papillionidae were *Graphium macareus indicus* (PAP001), *Papilio polytes* (PAP002), *Papili odemolus demolus* (PAP003), and *Pachliopta aristolochiae aristolochiae* (PAP004). As far as butterflies are concerned, 61.9% belonged to Nymphalidae followed by 19.05% of both Pieridae and Papillionidae families. The similar trend was reported by Verma (2009)[21] and Joshi & Arya (2007)[22].

Maximum types of butterflies were belonging to the most common family Nymphalidae. Joshi & Arya (2007)[22] has reported that its dominance was found equally in both higher and lower altitude areas. Verma (2009)[21] has also reported presence of butterflies (NYM001, NYM003, NYM005, NYM007, NYM010, NYM011, PIE002, PIE003, and PAP002) at Dalma Wildlife Sanctuary, Jharkhand found in consonance with the results of author.

Table 2: - List of butterflies collected from Kanke, Ranchi during 2019 – 2021.

Sl. no	Name Of The Butterfly Species	Sample Code	Common Name	Family
1	*Danaus chrysippus chrysippus* (Linn.)	NYM001	Plain Tiger	Nymphalidae
2	*Melanitis phedima bela* (Carmer)	NYM002	Dark Evening Brown	Nymphalidae
3	*Junonia almana* (Linn.)	NYM003	PeacockPansy	Nymphalidae
4	*Callerebia scandao pima* (Kollar)	NYM004	Pallid Argus	Nymphalidae
5	*Ariadne merione merione* (Carmer)	NYM005	Dakhan Common Caster	Nymphalidae
6	*Melanitis phedima varaha* (Carmer)	NYM006	Dark Evening Brown	Nymphalidae
7	*Junonia atlites atlites* (Linn.)	NYM007	Grey Pansy	Nymphalidae
8	*Rohana parisatis parisatis* (Westwood)	NYM008	Black Prince	Nymphalidae
9	*Danaus genutia genutia* (Carmer)	NYM009	Oriental Plain Tiger	Nymphalidae
10	*Euploea core core* (Carmer)	NYM010	Indian Common Crow	Nymphalidae
11	*Neptis hylas nicobarica* (Linn.)	NYM011	Common Sailor	Nymphalidae
12	*Lasippa viraja viraja* (Moore)	NYM012	Yellow jack Sailor	Nymphalidae
13	*Acraea terpsicore* (Linn.)	NYM013	Indian Tawny caster	Nymphalidae
14	*Eurema andersonii*(Moore)	PIE001	One Spot Grass Yellow	Pieridae
15	*Delias eucharis*(Drury)	PIE002	Common Jezebel	Pieridae
16	*Catopsilia pomona* (Fabricius)	PIE003	Lemon Emigrant	Pieridae
17	*Belenois aurota aurota*(Fabricius)	PIE004	Pioneer White	Pieridae
18	*Graphium macareus indicus*(Godart)	PAP001	Lessor Zebra	Papillionidae
19	*Papilio polytes* (Linn.)	PAP002	Indian Common Mormon	Papillionidae
20	*Papilio demolus demolus*(Linn.)	PAP003	Oriental Lime swallowtail	Papillionidae
21	*Pachliopta aristolochiae aristolochiae* (Fabricius)	PAP004	Common Rose	Papillionidae

Table 3: A chart containing year wise collection list of individual species and their families and the Shannon Weiner diversity index values.

Year	Family	Total number of species	Total number of individuals	Diversity index (H´)
2019	Nymphalidae	13	87	
	Pieridae	04	40	
	Papillionidae	04	26	
	Total (A)	**21**	**153**	2.940
2020	Nymphalidae	13	190	
	Pieridae	04	51	
	Papillionidae	04	33	
	Total (B)	**21**	**274**	2.993
2021	Nymphalidae	13	195	
	Pieridae	04	54	
	Papillionidae	04	33	
	Total (C)	**21**	**282**	3.012
		Grand Total (A+B+C)	**709**	

Species Diversity

Shannon-Weiner diversity index of the collected butterflies were calculated year wise. In the year 2019, 2020 and 2021 it was 2.940, 2.993, and 3.012 respectively (Table 3). The results show that the biodiversity of collected samples increased yearly. It was highest in the year 2021 while lowest in 2019. This trend is reported as a new finding by the author because it was not similar to the diversity index cited by Verma (2009)[21].

Species Richness

In the year 2019, *Ariadne merione merione* (NYM005) and *Belenois aurota aurota* (PIE004) were abundantly collected while in the year 2020, *Ariadne merione merione* (NYM005) was abundant. But in the year 2021, *Danaus chrysippus chrysippus* (NYM001), *Melanitis phedima bela* (NYM002), *Ariadne merione merione* (NYM005), *Neptis hylas nicobarica* (NYM011), *Belenois aurota aurota* (PIE004) were abundant (Table 3). It indicated that *Ariadne merione merione* (NYM005) was abundantly found in all three years while in 2021 the dominance of some other species was also reported.

The result of species richness of the Ranchi in and around was found in inconservence with the results of Morse *et al.* (1988)[23], Ahmad (2017)[24], Kumari *et al.* (2021)[25].

CONCLUSION

Total 702 butterflies' sightings of 21 species were tabulated during the collection period. The diversity was found to be good in all three years. COVID-19 lockdown may have helped their diversity in year 2020 and 2021 because of less fabricated disturbances. Further researches are needed to develop more correlation with the situation and diversity as well as species richness of butterflies in Kanke, Ranchi, Jharkhand.

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LIMITATION

The study area was small so, results may not be significantly similar in a bigger collection area. It depends on further researches.

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Qualitative analysis of catalase production by the gut bacteria of *Labeo rohita*

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Abstract- Enzyme producing bacteria has become popular among the researchers due to its remarkable participation in the field of textile, food and beverage. Apart from producing beneficial enzymes, bacteria are easy to handle. Persistent and cost-efficient production is possible as well as epigenetic modification can also be performed conveniently. Hydrogen peroxide is a highly reactive chemical compound produced by nearly all organisms that live in the presence of oxygen. This chemical is a Reactive Oxygen Species (ROS) produced during numerous metabolic processes of living cell and can perpetrate oxidative damage to cellular components. To cope up with its harmful effect catalase enzyme plays a crucial role. It converts hydrogen peroxide into water and oxygen and, thus, protects the cells from oxidative damage. In the present investigation an attempt has been made to isolate the catalase producing bacteria from the gut of fish *Labeo rohita*. It is a freshwater herbivore fish commonly found in Indian freshwater ponds, rivers and streams. Bacteria was isolated on NA media and purified by streaking. The bacteria were identified as Gram-positive *Bacillus* species. Catalase producing ability of bacteria was identified in vitro by using the chemical hydrogen peroxide and a positive test result was observed. The optimum temperature for bacterial growth was also identified which was about 40°C. In fabric industry, catalase is used for the removal of excess hydrogen peroxide from fabric. This enzyme is mostly used along with other enzymes in food processing industries. Catalase is extensively used for food preservation. It is also used for elimination of oxygen from wine before bottling. Contemplating the beneficial effects of catalase, the isolated bacteria from the gut of *Labeo rohita* can be used for mass production of this enzyme in a cost-efficient manner.

Key words: Hydrogen peroxide, Catalase, *Labeo rohita,* Reactive oxygen species (ROS), Oxidative damage

INTRODUCTION

Existence of living organisms is a great example of coordination between physiological as well as molecular components. With the advancement of science and technology a meticulous microscopic world came into existence that is microbiology, which is the scientific study of microorganisms. Bacteria, which is one of the class of microorganisms are considered to be cosmopolitan in distribution.[1,2] Not only in the environment, but they are also reported to be found inside the body of living organisms[3] in which some bacteria are harmful but some are found suitable or beneficial.[4]

Gut bacteria of fish have dragged the attention of many researchers and thus various researches have been done in this field. The micro flora of the digestive tract represents a pivotal and diversified enzymatic potential. Ray *et al.* (2012)[5] reported that some fish species accomplish their need of intestinal enzyme from the bacteria inhabiting the gut. Apart from helping in digestion of food, these bacteria also help in the protection of its host body from other pathogens by competitive exclusion.[6]

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Hydrogen peroxide (H_2O_2) is a highly reactive chemical compound widely found in living cells and has been reported to cause oxidative damage to the cell in many ways.[7] This hydrogen peroxide (H_2O_2) can be neutralized by an enzyme called catalase. This catalase enzyme is reported to be produced by some bacteria which are termed as catalase positive. The catalase enzyme which is produced by the bacteria has various industrial significances.[8] In the present investigation an attempt has been made to extract catalase positive bacteria from the gut of freshwater fish *Labeo rohita.*

MATERIALS & METHODS

Healthy *Labeo rohita* fish was bought from the fish farmers training centre, Shalimar, Dhurwa, Ranchi and brought to the laboratory for further investigation. Fish was starved for 24 hrs in order to clear its digestive tract.[9]

Collection of bacteria

Using ethanol, outer surface of fish was washed properly. The ventral portion of fish body surface was opened and gut was removed. Removed gut was washed thrice with distilled water. About 1g of intestine was homogenized in 9ml of distilled water and this was followed by centrifugation. 1ml of cell free supernatant (CFS) was diluted serially using serial dilution method[10] upto 6th dilution. 1ml of 6th dilution was inoculated on NA media plates and incubated at 37°C for 24 hours. Using streak plate technique, a randomly picked bacterial colony was purified.[11]

Identification and characterization of bacteria

Using Bergey's manual of determinative bacteriology[12], the isolated bacteria was identified carefully and further preceded for Gram staining.

Screening of catalyst production

A loop full of isolated bacteria was placed on a clean glass slide. Then 1-2 drops of hydrogen peroxide (H_2O_2) were poured on the bacteria and result was observed.

Effect of temperature on bacteria

Temperature has a significant role in the bacterial growth and metabolism. In the present investigation a loop full of isolated bacteria was inoculated in 50 ml of nutrient broth and incubated at different temperatures in separate test tubes for 24 hours. The temperatures taken were 25°C, 35°C, 40°C, 50°C and 60°C.

After the incubation, each sample was serially diluted upto 6th dilution. 1ml of sample from each test tube, that was kept at different temperatures was taken and spread over NA media plate and incubated for 24 hours. Bacterial colonies were then counted and colony forming unit (CFU) per ml of sample was calculated.

RESULTS

Isolated bacteria from the gut of *Labeo rohita* were analysed. The morphological analysis showed cream coloured, large, circular bacterial colonies. [Fig. 1] The edge of the colonies was also found to be smooth. The isolated bacteria were Gram positive and a *Bacillus* species.

Catalase activity of the isolated bacteria was tested using hydrogen peroxide (H_2O_2). A positive test result was identified due to the occurrence of brisk effervescence when bacteria came in contact with hydrogen peroxide H_2O_2. [Fig. 2]

Survival of the isolated bacteria was also tested at different temperatures. The selected range of temperature was between 25°C to 60°C. After incubation of bacteria at different temperatures, a maximum bacterial growth was observed at the temperature of 40°C, which means that 40°C is the optimum temperature for the isolated bacteria. On the other hand, deflection of temperature from 40°C showed fall in bacterial growth. [Fig. 3]



Fig. 1- Bacterial growth on NA media



Fig. 2- Positive catalase test result, Showing brisk effervescence when H_2O_2 was poured on loop full of bacteria

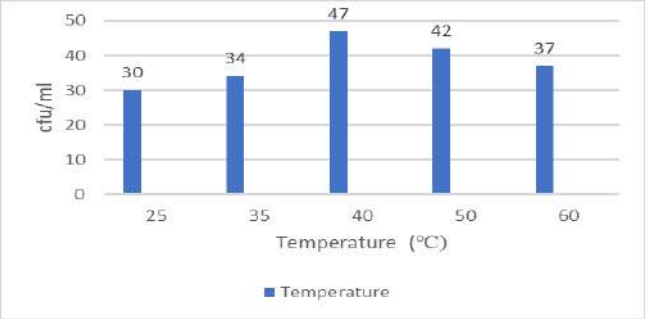


Fig. 3- Effect of temperature on bacterial growth

DISCUSSION

Enzymological and growth studies have revealed the enzymes producing ability of many bacteria. These enzymes are responsible for digestion and providing immunity to its host. Recent studies have also put a light on the capability of bacteria for utilizing Reactive Oxygen Species (ROS), like hydrogen peroxide (H_2O_2), and for its signalling and survival.[13] Reactive Oxygen Species (ROS) are a group of molecules that are produced in the cell through metabolism of oxygen. Endogenous Reactive Oxygen Species (ROS) such as hydrogen peroxide (H_2O_2) have been recognized as a destructive molecule.[14]

Watt *et al.* (2004)[15] have documented that hydrogen peroxide (H_2O_2) causes toxic effect in many ways, some of which are lipid peroxidation and corrosive damage. However, in 1970s, a small collection of studies reported that exogenously added hydrogen peroxide (H_2O_2) could mimic the signalling activity of some hormones.[16] Gough and Cotter (2011)[14] have also reported that hydrogen peroxide (H_2O_2) can modulate enzyme activities by altering its working mechanism which is known as oxidative modulation.

Catalase is an enzyme that is also found in biological systems. It actually catalyses the hydrogen peroxide (H_2O_2) decomposition which ultimately converts it into non-toxic compounds i.e. water and oxygen. It is a well known enzyme for protecting cell from oxidative damage.[17]

$$2H_2O_2 \xrightarrow{\text{Catalase enzyme}} 2H_2O + \underset{\text{Air bubbles}}{O_2\uparrow\uparrow}$$

Whitten bury (1964)[18], stated that bacteria like *Bacillus*, *Leuconostoc*, *Streptococci* and *Pediococci* have the capability of producing catalase enzyme. This shows resemblance with the result obtained in present investigation in which the isolated bacteria was found to be a *Bacillus* species which is also catalase positive.

So far, the reaction velocity is concerned; it has extremely high turnover number which means one molecule of catalase can convert millions of hydrogen peroxide (H_2O_2) molecule into water and oxygen per second.[19,20] This shows resemblance with the present experiment in which brisk effervescence was observed which is the indication of high reaction rate of hydrogen peroxide (H_2O_2) and catalase.

Catalase enzyme has a great role in protecting the cells and other biological components from damage. However, Zamocky *et al.* (2008)[21] opined that only selective organisms have catalase producing ability. In some studies, it has also been reported that catalase activity is acid resistant, which means catalase activity will not get affected by stomach acid if the bacteria is fed as probiotic.[18,22]

On the other hand, Kirchman *et al.* (2005)[23] opined that temperature plays a significant role in the growth and metabolism of bacteria. Membré *et al.* (2005)[24] added that effect of temperature can vary from species to species of microbes. On examining the effect of temperature on the isolated bacteria, it was observed that 40°C was the optimum temperature for its growth. So far, the commercial aspect is concerned, catalase enzyme has a significant role in textile industries as an agent for removing excess hydrogen peroxide from fabric. Catalase is also enormously used for food preservation. It is also in use for elimination of oxygen from wine before bottling.[25,26]

Considering the multiple benefits of this catalase enzyme and due to the easy handling of bacteria, the isolated bacteria can be used for mass production of enzyme catalase in cost efficient manner.

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Blood biochemical profile in carbohydrate metabolism in desi rabbit of Madhepura district of Bihar

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Abstract- The desi rabbit of Madhepura district was selected for biochemical study of the blood parameters. It was found that the total protein content of honey treated rabbits was found to be increased significantly as compared to control, suggesting anabolic effect of honey in rabbit. However no effect on sex was observed. There is highly significant effect of honey treatment is event as increased level of blood glucose was found which provide energy and non-significant effect on sex was observed. There is significant effect of honey on total cholesterol content among group's increases but non-significant effect on sex was observed. In total lipid content show non-significant effects on group but have significant effect on sex. There is non-significant effect of honey on serum phospholipids among group and sex.

Key words: Desi Rabbit, Biochemical blood parameters, carbohydrate metabolism

INTRODUCTION

India is a vast country. The rapid growth of the human population results in continued widening up the gap between demand and supply of the required quantity and quality of food. The problem of malnutrition resulting from food shortage is bound to grow more and more acute with the increase in the human population, to bridge the wide gap between large requirements and low animal protein availability. It is essential to improve and multiply all meat-producing animals in the country.

The rearing of domestic rabbits is an established micro livestock industry in many countries. In many European and Asiatic countries, rabbits are domesticated for meat. Broiler breeds of rabbits have also been introduced in India to explore its avenue as an alternative source of animal protein. The physiological responses of animals are almost crucial in a better understanding of environmental stress and adaptability, which will significantly enhance managerial capability.

Rabbit farming is a widespread livelihood practice across the country. This livestock industry is becoming popular in Bihar also due to its ease of business. Rabbits are best known for being prolific; these herbivores efficiently convert fodder to food. The motive behind meat production is to convert plant proteins of little use to people as food into high-value animal protein. Inefficient production systems, rabbits can turn 20 per cent of the proteins they eat into edible meat. Comparable figures for other species are 22 to 23 per cent for broiler chickens, 16 to 18 per cent for pigs and 8 to 12 per cent for beef.

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Rabbits are reared for their meat, but their hair and skin are also important. Their meat is rich in protein and can be part of a good diet. They have very little fat and are almost cholesterol-free. They are palatable, easily digestible, and consumed in any weather condition, making them a good source of nutrition.

The rabbit meat becomes more prevalent among people requiring less fat, cholesterol, and sodium. So it is not harmful to the heart patient. The cholesterol content of Rabbit meat is lower than other meat; it is suggested because of the inter-relationship between cholesterol, saturated and polyunsaturated fatty acid and caloric value.

The magnitude and trend of rabbit mortality in various seasons and age groups are vital for commercial rabbit farming. Proper nutrition is one of the essential aspects of broiler rabbit production. The protein and energy content of the diet plays a vital role in rabbit nutrition.

The information on biochemical blood profiles for cholesterol, lipid and protein in desi rabbits of Bihar is lacking. The present studies were made to see the effect on these parameters supplemental with antistress or immunomodulatory agent like honey on Rabbit, which will give an additional guide line for future planning or general health status of the Rabbit. Honey is a natural sweetish substance prepared by honey-bee from the nectar of flower contains majority (70-75%) of simple sugars as glucose and fructose, water (18-20%) small amount of sucrose (3-4%) and trace (1-2%) of vitamins, mineral, enzymes, acids and unknown factors including antibacterial effects. In addition, it serves as an instantaneous source of energy, vitamins and trace mineral.

Biochemical characterization of Rabbits in India or tribal belt will help better understand Rabbit in relation to growth, meat, fur and wool quality. In perspective, the present study was undertaken with 'active to assess the gross biochemical constituents including the enzymes of serum in control and honey treatment group and to characterize genetically in desi rabbits available in Madhepura through biochemical polymorphism study'.

MATERIALS & METHODS

Desi rabbits six each of either sex, were divided into two groups. One is honey treated group and the other is the control group. Group II had given 5% simple sugar level in drinking water. All the Rabbit were kept under uniform management practices till the end of the experiment. Blood samples were collected at the end of the investigation directly from Rabbit's heart with or without anticoagulants into a sterile test tube. Serum was harvested from whole blood and kept refrigerated until analyzed. Biochemical parameters were studied with the help of separated serum. Estimation of total blood glucose: The Nelson Somogyi method (1945) estimated blood glucose. 0.1 ml of blood in a test tube was taken, and 9.5 ml of zinc sulphate solution was added and mixed by rotation. Then alkaline copper reagent was mixed by tapping the top of the tube. It was covered with a marble. It was placed upright in a boiling water bath for 5 minutes. Then 1 ml arsenomolybdate coloring reagent was added and mixed, and then it was diluted to 10 ml with distilled water. A blank and standard was run simultaneously. Then absorbance was recorded at 540 nm in photoelectric colorimeter model AE-11 Japan.

Estimation of serum cholesterol: Total serum cholesterol in blood was estimated by method of Zlatkis, *et.al*. (1953)[1]. 0.1 ml serum sample was taken in a test tube. Glacial acetic acid 6.0 ml and coloring reagent 0.4 ml were added. The contents were mixed well. Simultaneously, 0.1 ml of water as blank and 0.1 ml of standard cholesterol solution 2 mg /ml were run similarly treated. Then tubes were allowed to cool at room temperature. The optical density was recorded at 540 nm in photoelectric colorimeter model AE-11 Japan.

Estimation of serum lipid: The total serum lipid estimation was done by using the method of Mandal (2002)[2]. 1 ml of serum was taken with 19 ml of chloroform:methanol (2:1 v/v) mixed well and allowed to stand overnight. The extract was filtered by using Watsman no.1, filter paper in 50 ml of conical flask. Filter paper with residue was cut into pieces and again transferred into 19 ml chloroform kept overnight. This procedure was repeated 3 times. The filtrate was combined with previous extract.

The extract was evaporated at 45°C - 50°C in Sand bath for breaking the volume of original lipid extract in chloroform:methanol:water (64:32:4 v/v) and evaporated to dryness in Sand bath at 45°C - 50°C. This step was repeated thrice and dried residue was dissolved in chloroform:methanol (2:1 v/v) and transferred to separating funnel for removing nonlipid impurities. The

lipid extract was layered with 1/ 5th volume of normal saline (0.9% NaCl) and mixed several times by gentle inversion. It was allowed to stand for 6 to 8 hours at room temperature. The lower chloroform layer was collected, evaporated to dryness in sand bath at 45°C-50°C and was repeated thrice. The extract was dissolved in known volume of chloroform. The total lipid extract were determined aluminium/stainless steel planchets and dried at 40°C - 50°C in an oven until constant weight were reached. Increased weight gives total serum lipid. Estimation of Phospholipids. The total phospholipid in the serum was estimated by method of Post and Sen (1967)[3]. An aliquot 0.1 of lipid extract was taken into micro kjeldahl flask and 0.4 ml of 60% perchlororic acid solution was added. It was digested directly on a sand bath for 20 minutes. Glass beads (2-3) were added to each flask to avoid bumping during digestion. The 0.1 ml of digested solution was taken in a tube and was added to 8 ml water. It was then mixed with 2.0 ml coloring reagent (1 part of 10% ascorbic acid and 6 part of 0.42%) was added. It was incubated in heated water bath at 37°C for 1 hour, after mixing absorbance was taken at 660 nm in colorimeter AE model Japan against H_2O. Which was expressed as Mg lecithin/ml blood data collected, different traits were analyzed as per the standard statistic techniques.[4-6]

RESULTS & DISCUSSION

Total serum protein content of treatment and control group of Rabbits are presented in Table 1. The analysis of variance of this data in presented in Table - 2. The result showed significant difference ($P < 0.01$) in total serum protein content between treatment and control group. The significantly higher value in treatment group was observed but non-significant differ between sexes were find such differences in protein content of serum was clearly evidenced.

The value of total blood glucose level of control and treatment groups of rabbits are summarized in Table - 3. The analysis of variance of this data is presented in Table-4. The finding indicated significant ($P < 0.01$) difference between groups. But no any changes between sexes were observed.

The total serum cholesterol levels of different groups of rabbit are summarized in Table-5. The analysis of variance of this data presented in Table - 6. The finding showed significant differences ($P < 0.01$) in total serum cholesterol level among treatment and control group and non significant between sexes. However increased trend was observed.

The total lipid level in serum of treatment and control groups are presented in Table - 7. The analysis of variance of this data presented in Table - 8. The presented findings showed no significant difference in total lipid level between control and treatment group. But significant ($P < 0.01$) difference found between male and female. Male showed higher value than female, such observation are clearly visible.

The total phospholipid content in serum is presented in Table 9. The analysis of variance of this data presented in Table - 10. The present findings showed non-significant difference between control and treatment group also did not showed difference between sexes. The decreased levels of total phospholipd are clearly visible.

Table-1: Showing the values* of total serum protein (gm %) of different groups of rabbit.

Group No.	Sex	Total serum protein	
		Sex	Group
Control	Male	5.44±0.134	5.15±0.184
	Female	4.82±0.167	
Treatment	Male	6.58±0.134	6.41±0.106
	Female	6.24±0.167	

* Value (Mean ± SE) bearing the same superscript did not differ significantly.

Table-2: Analysis of variance of the data presented in table 1 showing the effect of treatment and sex on total serum protein content in rabbit.

Source of Variation	d.f.	S.S.	M.S.S.	F
Between group	1	4.78	4.78	29.87**
Between Sex	1	0.75	0.75	4.7^{NS}
Error	9	1.45	0.16	

**- Significantly at 1% level ($P < 0.01$)
NS- Non significant

Table-3: Showing the values* of total blood glucose (mg %) of different groups of rabbit.

Group No.	Sex	Total serum protein	
		Sex	Group
Control	Male	139.3±7.86	142.38±10.89
	Female	145.4±4.24	
Treatment	Male	206.03±7.86	193.91±7.65
	Female	181.40±4.24s	

*Value (Mean ± SE) bearing the same superscript did not differ significantly.

Table-4: Analysis of variance of the data presented in table 3 showing the effect of treatment and sex on total glucose content in rabbit.

Source of Variation	d.f.	S.S.	M.S.S.	F
Between group	1	7967.05	7967.05	11.78**
Between Sex	1	248.43	248.43	0.36^{NS}
Error	9	6085.19	676.13	

**- Significantly at 1% level (P < 0.01)
NS- Non significant

Table-5: Showing the values* of total serum cholesterol (gm %) of different groups of rabbit.

Group No.	Sex	Total serum protein	
		Sex	Group
Control	Male	48.50±1.96	46.08±4.54
	Female	43.83±1.23	
Treatment	Male	65.16±1.96	59.75±2.6
	Female	54.33±1.23	

* Value (Mean ± SE) bearing the same superscript did not differ significantly.

Table-6: Analysis of variance of the data presented in table 5 showing the effect of treatment and sex on total serum cholesterol content in rabbit.

Source of Variation	d.f.	S.S.	M.S.S.	F
Between group	1	572.28	572.28	6.81**
Between Sex	1	233.12	233.12	2.77^{NS}
Error	9	755.51	83.94	

**- Significantly at 1% level (P < 0.01)
NS- Non significant

The total protein content of honey treated rabbits was found to be increased significantly as compared to control, suggesting anabolic effect of honey in rabbit. However no effect on sex was observed.

There is highly significant effect of honey treatment is event as increased level of blood glucose was found which provide energy and non-significant effect on sex was observed.

There is significant effect of honey on total cholesterol content among group's increases but non-significant effect on sex was observed. In total lipid content show non-significant effects on group but have significant effect on sex. There is non-significant effect of honey on serum phospholipid among group and sex. The anabolic effect of honey evident from present result is on protein, lipid and carbohydrate metabolism.

There are significant differences in the level of lactate dehydrogenase and succinic dehydrogenase among groups. LDH shows significant and SDH shows non-significant effect on sex, these observations indicating variation on glycolytic pathway. On the other hand effect of treatment as found on Kreb's cycle which provide the major source of energy to the animals.

The effect of treatment on ALT & AST were non-significant indicating no influence on protein atabolism only ALT value slightly increase between groups.

The effect of treatment evident on ATPase activity as well as alkaline phosphatase such inding suggested significant effect of honey on energy transfer process in rabbit. There is no significant differences were observed in biochemical haemoglobin polymorphism which suggested further details studies using different breeds.

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Change in the fundamental niche of *Mesocyclops* species by the action of *Brachionus* species at Ranchi Lake

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Abstract- Changes in the niche structure of a cultural eutrophic lake at Ranchi were studied in relation to the population structure of three zooplankters *Mesocyclops leuckartii, Brachionus qudridentatus and Brachionus angularis*. Twenty two parameters were used to define the broad environmental framework of the lake. These parameters create an n-dimensional hypervolume within which the plankton selects the optimum combination of variables that creates its fundamental niche. The fundamental niche of *Mesocyclops* is defined by three parameters: food resource, temperature, and alkalinity. However, the same food resource is used by other zooplankton and thus compresses the fundamental niche to a realized niche. The parameters defining the fundamental niche are in turn are controlled by a variety of other subsystems. The result showed that the *Brachionus* species caused a change in the fundamental niche of *Mesocyclops* species.

Key words: Hypervolume, temperature, alkalinity, niche, correlation

INTRODUCTION

Rotifers of the Subclass Monogononta (Phylum Rotifera) are common metazoans in the plankton of freshwater environment.[1,2] Although small (usually 50–500 µm), they often are extremely abundant and important in the transfer of energy from algae, microbes, and detritus to higher trophic levels in community food webs.[3]

Rotifer life cycles often include several types of females and eggs.[4] Females sequentially produce single eggs throughout their reproductive life. Most rotifers are oviparous for instance in *Brachionus* eggs are attached to the mother until they hatch[5] and are released into the environment, or deposited onto a substratum.

An early, long – held view of the rotifer life cycle was a seasonal progression of events in which resting eggs hatch, populations develop via female parthenogenesis as long as environmental conditions are favourable, and then bisexual reproduction with resting-egg production occurs when conditions turn unfavourable at the end of the growing season.

The objective of the present study was to identify the changes in the niche structure of *Mesocyclops* species which shrinks its fundamental niche to a realised niche by the action of another species *Brachionus* at Ranchi Lake, Ranchi.

MATERIALS & METHODS

Plankton samples were collected once each month and used for species identification and counting on a Sedgwick rafter cell at 10x magnification. Twenty two parameters were used to define the broad environmental framework of the lake.[6] An n-dimensional hyper volume is created by these parameters. The plankton selects the

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optimum combination of variables among these which create its fundamental niche. The estimation of chlorophyll was done by using spectrophotometer at 630nm, 645nm and 665nm.[7] The measurement of water temperature, salinity, conductivity and alaklinity[8] was done every month.

RESULTS & DISCUSSION

Results are presented in Table 1 and Figures 1 to 9. Figure 1 (a) and 1 (b) depict the *Brachionus* species with attached eggs. Bisexual reproduction is clearly evident in *Brachionus* species (Figure 2).

At Ranchi Lake three zooplankters were in abundance that is *Mesocyclops leuckartii, Brachionus qudridentatus and Brachionus angularis* (Figure 3). Organisms are primarily dependent on their food source. *Mesocyclops* is omnivorous feeding on algae, insect larvae and dead organic matter.[9,10] However, their primary dependence is on available algae.

Figure 4 represents changes in the phytoplankton population with respect to food availability at Ranchi Lake during the study. Another aspect which affects the plankton is competition from *Brachionus* which in the present case is *Brachionus angularis*. Populations of *Brachionus angularis* were quite common in Ranchi lake and had a fundamental niche defined by a temperature of 15-25ºC; a salinity of 9-40 (parts per thousand %); conductivity of 5mS cm^{-1} and food resource. The average life span of *Brachionus* species is 8-10 days and produces two eggs at a time. The maximum individuals were found in October at a temperature of 24.7°C and minimum in March at a temperature of 26.4°C (Figure 4).

The basic energy source of the zooplankton is the phytoplankton population specifically the various species of algae. The phytoplankton population at Ranchi Lake showed two peaks, one in the month of April – May, and the other in the month of August – September (Figure 5), it exhibited fluctuations with nutrient availability as the nutrients were released through decomposition of organic matter (Figure 6).

The phytoplankton population reached its peak in April-May and decreased rapidly in June due to unfavorable environmental conditions. It was estimated by chlorophyll method. The phytoplankton population was calculated from the concentration of chlorophyll and a significant negative correlation (r = -0.98) was noted between the organic matter and the phytoplankton population (Figure 7).

Brachionus competes for food resource to a certain extent but it has an alternative food source in the form of dead particulate organic matter. However, *Mesocyclops* can change its food preference from, algae to predation on living zooplankton, including *Brachionus* and dead organic matter.

One of the competitors of *Mesocyclops* with respect to food resource is *Brachionus*. A significant negative correlation (r = - 0.82) was noted between the population of *Mesocyclops* and *Brachionus* (Figure 8). *Brachionus* species shrinks the fundamental niche of *Mesocyclops* species to a realized niche (Figure 9).

$\Delta \log$ Mesocyclops = - 0.5481Δ log Brachionus – 0.0278 (r = - 0.82)

Brachionus uses the same food resource as *Mesocyclops* and competes with the latter for food. This competition shrinks the fundamental niche of *Mesocyclops* species to a realized niche (Figure 9 & 10).

Table 1: The parameters and their range

Parameters	Range
Temperature	15 – 25 °C
Salinity	9 – 40 ‰
Conductivity	5 mS cm^{-1}
Alkalinity	200 – 250 mgL^{-1}
Chlorophyll	12.55 μgL^{-1}

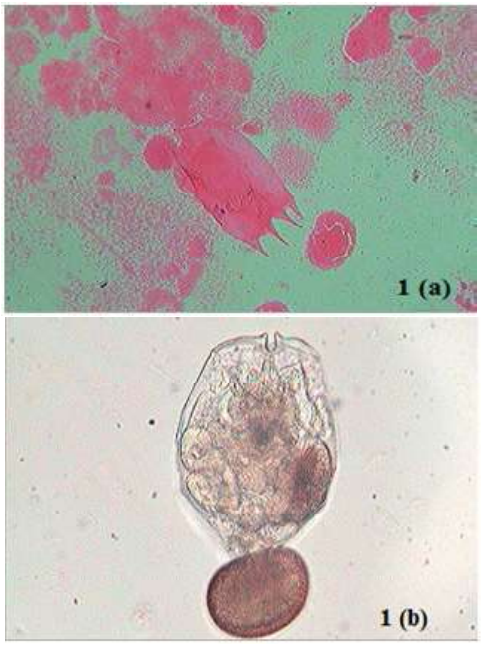


Fig 1: Eggs are attached to the female of *Brachionus angularis*

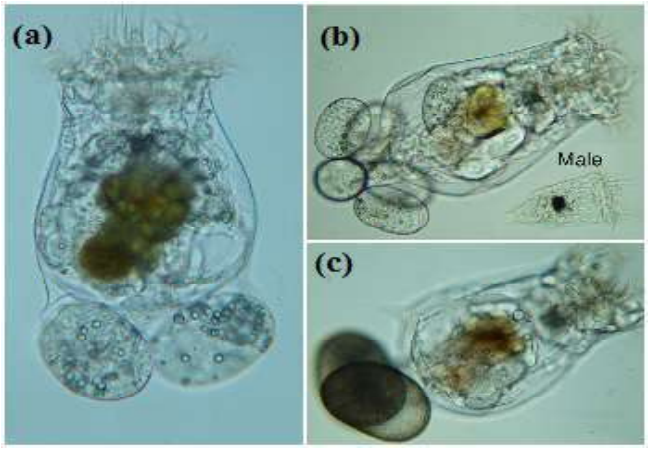


Fig. 2: Life-cycle stages of *Brachionus calyciflorus*. (a) Female carrying two diploid eggs developing parthenogenetically into females. (b) Female carrying five haploid eggs developing parthenogenetically into haploid males, and one hatched male. (c) Fertilised female carrying two encysted diapausing embryos or resting eggs.

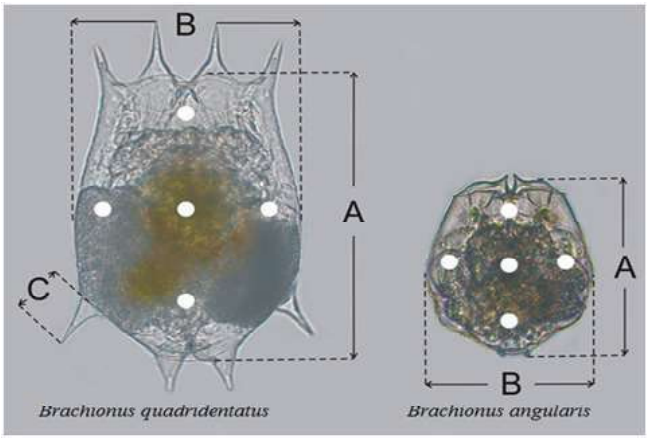


Fig. 3: Biometric characters of *Brachionus* species (A) body length; (B) body width; (C) length of posterolateral spine

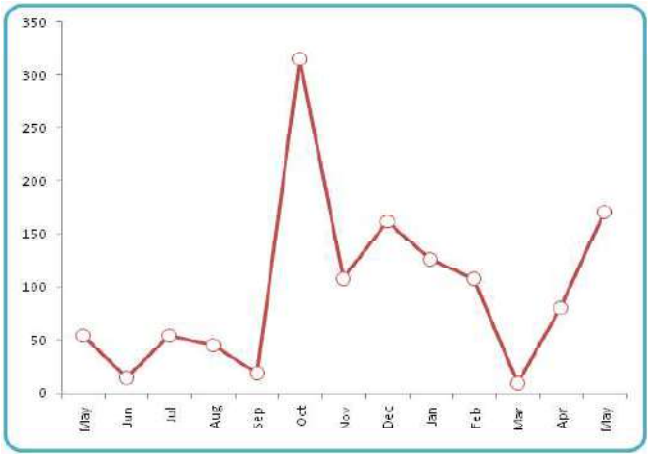


Fig. 4: The abundance of zooplankton population with respect to the temperature at Ranchi Lake

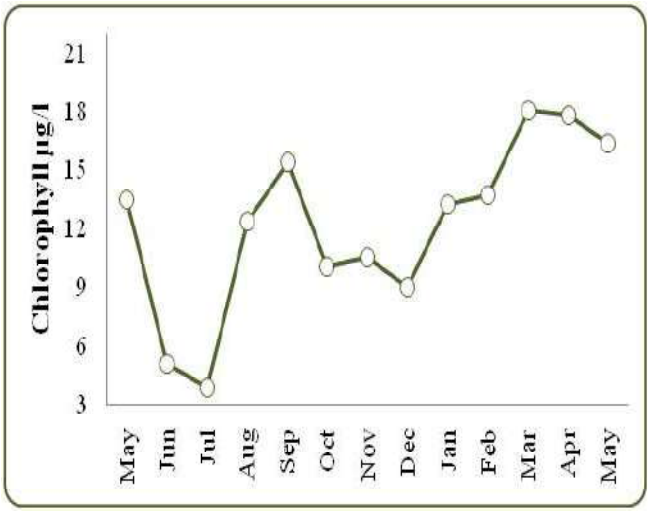


Fig. 5: The abundance of phytoplankton population with respect to the food availability at Ranchi Lake

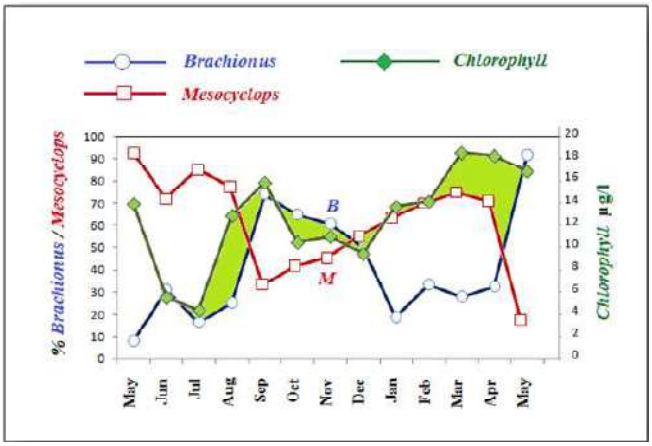


Fig. 6: The fluctuations of *Mesocyclops* and *Brachionus* at Ranchi lake

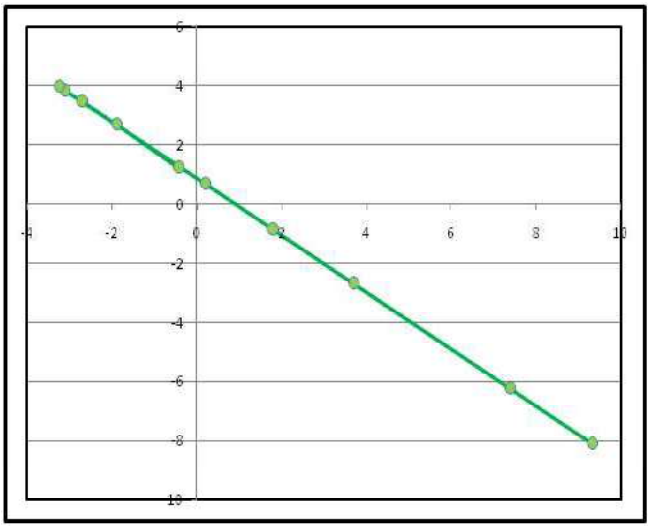


Fig. 7: showing a negative correlation between chlorophyll and dead organic matter

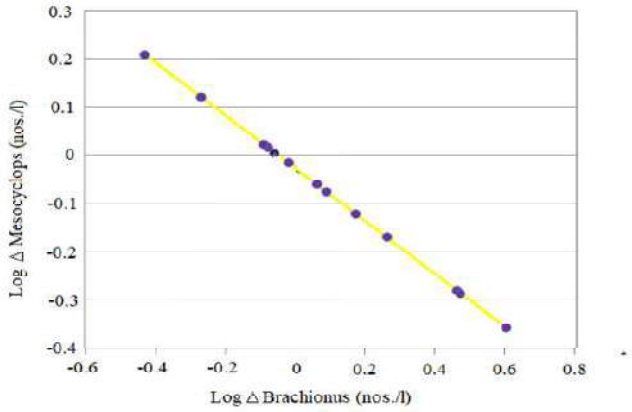


Fig. 8: Double logarithmic plot showing the negative correlation between *Mesocyclops* and *Brachionus*

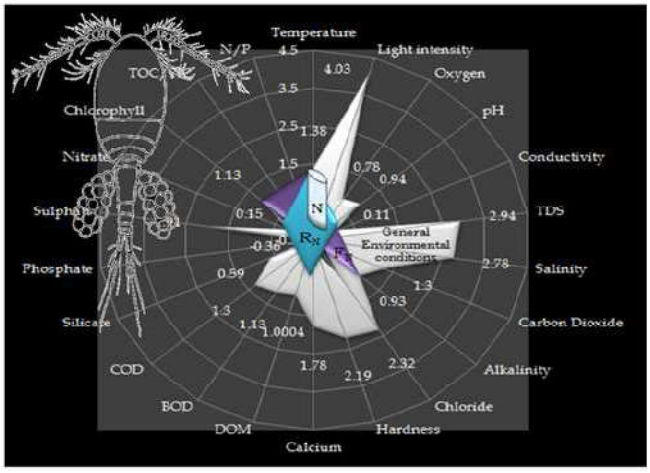


Fig. 9 The general environmental conditions; fundamental niche (FN); Realized niche (RN) and the *Mesocyclops* population (N)

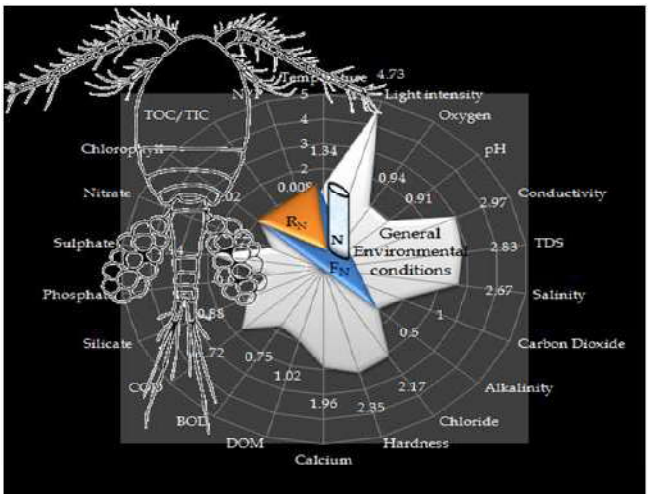


Fig. 10: The general environmental conditions; fundamental niche (FN); Realized niche (RN) and the Mesocyclops population (N) in the month of November

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Effect of phytohormones (2,4-D & KN) in callus induction of *Lawsonia inermis* L.

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Received : 18th December, 2021 ; Revised : 19th January, 2022

Abstract-Different concentration of phytohormones affected callus formation of *Lawsonia inermis* L. explant was cultured in Ms medium supplemented with different concentration of 2, 4-D and KN. Cultures were kept on 25±2ºC temperature and 16 hr photoperiod while callus was observed on different concentration of auxin and cytokinien in combination. Most suitable medium of callus formation from shoot tips was that 2, 4-D (2.5mg/L) + KN (0.5mg/L).

Key words: *Lawsonia inermis*, auxin, cytokinien, callus

INTRODUCTION

Lawsonia inermis is glabrous branched shrub of small tree (2 to 6 mts in height). *Lawsonia inermis* L. belongs to family lythraceae is much branched shrub and small tree that grows in Middle East Africa. Leaves are small, opposite, entire margin elliptical to broadly lanceolate sub-sessile, about 1.5 to 5 cm long, 0.5 to 2 cm wide greenish brown to dull green, petiole short and glabrous acute or obtuse apex with tapering base. *L. inermis* contains lawsone, a red orange pigment, chemically, the molecule known as lawsone in 2- hydroxyl-1, 4- napthoquinone.[1] Henna flowers have four sepals and 2 mm calyx tube , with 3 mm spread lobes. Its petals are ovate, with white or red stamens found in pairs on the rim of the calyx tube. The ovary is four called, 5 mm long, and erect. Henna fruits are small, brownish capsules, 4-8 mm in diameter, with 32 – 49 seeds per fruit and open irregularly into four splits.[2] The seeds are blue black angular and small.[3] It produces the most dye when grown in temperatures between 35 and 45 degree Centigrade.[4]

Medicinal properties include the cure of renal lithiases, jaundice, wound healing, prevent skin inflammation. The bark is traditionally used in treatment of jaundice and enlargement of the Spleen; renal calculus, leprosy and obstinate skin diseases.

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MATERIAL & METHODS

In the present analysis shoot tips were used for experiments. The segments of shoot tip were inoculated in Ms medium supplemented with different phyto hormones and combinations.

Culture media

Ms medium with varying concentration of phytohormones was employed for callus initiation.

Culture condition

The cultures were maintained and controlled environment at 25±2ºC with 16 hours photoperiod. All experiments were conducted under sterile conditions.

RESULTS & DISCUSSION

In *Lawsonia inermis* L. inductions of callus was achieved by supplementing low concentration of auxin and cytokinin. In Ms+2,4-D (2.0mg/L) + KN (0.5mg/L) 65%, Ms+2,4-D (5.0mg/L) +KN (0.5mg/L) 71% Ms+2,4-D (2.5mg/L) + KN (0.5mg/L). 90% callus was observed.

After 4-5 weeks when callus was transferred. Some works reported that young stem of *Lawsonia inermis* were responsive in Ms media supplemented with NAA also.[5] Callus which shows stable characteristics under specific condition after subculture through many successive passages, is a suitable material for cytodifferentiation. The advantage of using such callus is that it is composed of a fairly homogeneous mass of cells and can be proliferated in large amounts under known culture condition. The vascular strands in Syringa callus derived from the cambial region of the stem or grafted apices of shoots.[6]

Table 1- Different concentration of 2, 4-D (0.5 to 10 mg/l), kinetin (0.5mg/l) in Ms Medium callus induction was maximum

Ms+2, 4-D (0.5 to 10mg/L) + KN (0.5mg/L)	% culture showing callus induction				
	1st	2nd	3rd	4th	5th
2,4-D (0.5mg/l) + KN (0.5mg/l)	00	07	10	15	20
2,4-D (1.0.mg/l) + KN (0.5mg/l)	00	08	12	18	22
2,4-D (2.0mg/l) + KN (0.5mg/l)	00	09	17	37	65
2,4-D (2.5mg/l) + KN (0.5mg/l)	00	20	38	59	90
2,4-D (5.0mg/l) + KN (0.5mg/l)	00	15	17	33	71
2,4-D (7.5mg/l) + KN (0.5mg/l)	00	13	15	20	23
2,4-D (10.mg/l) + KN (0.5mg/l)	00	10	10	15	21

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Destruction of *Solanum lycopersicum* (L), by the pest *Helicoverpa armigera* Hubner, (Lepidoptera:Noctuidae) (tomato fruit worm) a case study at Ramgarh, Jharkhand, India

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Received : 20th December, 2021 ; Revised : 21th January, 2022

Abstract- The plant *Solanum lycopersicum* (L) 'Tomato' is cultivated and commonly used in domesticated agro business, consumed as raw and in cooked form. The plant naturally contains tannins and cyanide precursors protect it from pests, predators and fungal attack. The serious pest *Helicoverpa armigera* Hubner (Tomato fruit worm) destructs the crop and causes the loss of its agribusinesses. It is a polyphagus pest infests the green plant *Solanum lycopersicum* (L) and other cultivated vegetables. High reproductive potential makes it the serious pest causing significant economic damage. High dispersal capacity of the pest makes it cosmopolitan. Activity of *Helicoverpa armigera* was maximum during February to October and completed 03 to 04 over lapping generations in a year. The caterpillars of the pest damage the standing crops at night. During day time they hide in cracks and crevices in soil. The larva attains its maximum size in 17 to 30 days and enters into cracks or crevices and forms their earthen chamber and pupates underground. The pupal period varies from 13 to 25 days and the moth emerges out at night. The life cycle gets completed in 35 to 65 days. The life cycle was studied during the year 2018-2020 at Ramgarh. And it was found that the life-cycle of the experimental pest *Helicoverpa armigera* Hubner (Lepidoptera:Noctuidae) was very simple. The larvae after 3rd to 4th instar were photo negative, hidden in the soil during day time. In extreme hot climatic season the pest larvae and pupa hide under 20 to 30 mm in cracks and crevices in the soil. There was an observation that the moth migrates towards cold region of hills during extreme hot climate. Destruction of host plant by pest was 50% to 80%. Prevention from the pest was very difficult. The farmers were advised to control the pest by spraying the chemical pesticides as lindane 02% or carbaryl 0.1% or endrin 0.1% etc. Various natural enemies like moles, grubs, wasps, predaceous and parasitic flies destroy the larvae of the pest and were natural control of the pest population.

Key words: pest, cholesterol, quality, health supplement, photo negative, food, larvae, natural control, chemical control, endrin.

INTRODUCTION

Tomato fruit worm or tomato fruit borer is genetically the *Helicoverpa armigera* Hubner is a polyphagus pest. The morphology of larva on tomato plant is quite different than the pest larva found on other host plants it was lighter and smaller in size. It was one of the serious pests to destruct the various types of cultivated and ornamental crops, damage the crops as well as the financial capacity of farmers of Ramgarh. It was observed that the experimental vegetable *Solanum lycopersicum* (L.) is an important agribusiness product of Ramgarh and the farmers cultivate it on commercial level. The pest *Helicoverpa armigera* Hubner was a serious pest damage this commercial crop significantly, sometimes it damages the standing crop up to 80%. It was also observed that

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Legumes and dry bean have been preferred by host whereas Corn and Sorghum was the least preferred crop of the experimental pest. The attack of the pest starts in the month of February, March, when the flowers of the host plants become visible. The larvae of the pest consume the crop, the tender leaves and the flowers were consumed by small larvae and the developed larvae feed the berry. The first instar larvae feed on the chorion and then crawl individually in search of food if no suitable food is found they will consume unhatched eggs and other less active larvae, but in normal stages cannibalism was not observed. The pest *Helicoverpa armigera* Hubner having global distribution and serious infestation of cultivated plant causes significantly economic loss in agricultural products. The experimental host plant produces one of the most important vegetable for agro-industries consumed by the people around the globe. It is rich in essential vitamins, fibres and lacks in cholesterol, fat. It is the main ingredient of most of the delicious dishes. It also offers a variety of health benefits and apply as a sauces, salad and drinks.

The pest *Helicoverpa armigera* Hubner, cause severe damage to different host plants including vegetables, field crops, weeds and turf grasses. It infests a number of agricultural plants very fast and destroy 50 to 80 percent of crop population, sometimes in absence of host plant it consumed non host plant for their survival. Considerable works had been done by several workers on the life history of *Helicoverpa armigera* Hubner, On different crops but none of the researcher studied the life history of the pest *Helicoverpa armigera* Hubner, (Lepidoptera : noctuidae) on the vegetable *Solanum lycopersicum* (L) at Ramgarh. The present author deals with the destruction of *Solanum lycopersicum* (L) by the pest *Helicoverpa armigera* Hubner, (Lepidoptera: noctuidae), tomato fruit worm. A case study at Ramgarh, Jharkhand, India.

Ramgarh is a district town located at latitude: 23.38^{o}N and longitude: 85.34^{o}E, situated at 40 km South-East from Ranchi. The temperature is moderate with more humidity. This good climate is favourable for the growth and development of animal and plant population. The forest of Ramgarh plays a considerable role for the study of diversity in animals and plants. The experimental pest successfully completed 03 to 04 overlapping generation in a year from February to October. The adults emerges after sunset, the female mates several times and lays 80 to 120 eggs per day in cluster or in single and it continues for 04 to 15 days and up to 800 eggs. They lay eggs at night on various suitable places like under surface of leaves of host plant, grasses, other plants, and the moist soil near the host plant. The tiny caterpillar emerges after the incubation of 04 to 10 days. Neonates start to feed their chorine and crawl individually in search of food to tender leaves and flowers of the host plant and then to fruits, after 03 to 04 instar the larvae avoid day light and become nocturnal feeder and remain in the soil in day time. In 17 to 30 days the larvae develop completely, and then it enters 25 to 35 mm deep in cracks or crevices in the soil and pupates there. Pupal period last in 15 to 25 days, then moth emerges out during night. 03 to 04 over lapping generations were observed in a year.

It had been observed that the moth is migratory in nature and can move a long distance with the help of flow of wind. These peculiar characters of pest makes hurdle in study of lifecycle. Prevention from pest was very difficult and costlier. Farmers were advised to apply chemical pesticides as 2% lindane or carbaryl 0.1% or endrin 0.1% after the sowing of crop. Field should be often flooded moderately. Some natural enemies were observing to control the pest population.

MATERIALS & METHODS

Standard methodology was applied for the destruction of *Solanum lycopersicum* (L) by the pest *Helicoverpa armigera,* Hubner, (Lepidoptera: noctuidae), a case study at Ramgarh. The life history of the experimental pest and the nature of damaging the host plant were studied in the crop field as well as in the laboratory. The life history was studied during the year 2018-2020. Ten healthy selected plants of *Solanum lycopersicum* (L) were completely and carefully covered by a small mesh mosquito net separately. One pair (one male and one female) of experimental pest were introduced on the plant covered by mosquito net and observation was taken daily. One pair (one male and one female) of the experimental pest were kept in the cage of 20 cm X 20 cm X 30 cm. Space and fresh, soft leaves of host plant were provided for egg laying and hatching, this was food for caterpillar as well. In a separate cage some leaves of other plant were also provided to larvae to observe their feeding interest and survival rate. The data were recorded for further study.

Methods of prevention of crop and control of pests by chemical and biological methods were applied, observed and recorded.

OBSERVATION

Solanum lycopersicum (L) is itself a tastier vegetable, applied to improve the taste of food. It is consumed as raw and cooked form is the main ingredient of many dishes. The experimental pest *Helicoverpa armigera* Hubner (Lepidoptera: noctuidae) is a major economic pest destruct the vegetable *Solanum lycopersicum* (L), also infest many floral plants, migratory in nature makes it cosmopolitan in distribution, and difficult in protection. It travels more than 1000 km with the help of wind flow infest a wide range of vegetation. It was observed that the experimental plant was not the favourable host of *Helicoverpa armigera* Hubner, development of its larvae were retarded on this plant. As a result the growth was ill and passive. Ramgarh is a beautiful industrial town surrounded by hills, covered with green and dense forest and small agricultural field. It is one of the best places to study the ecological biodiversity, a niche of variety of plants and animals. The climatic condition of Ramgarh is suitable for the vegetable and crop cultivation. Therefore, a variety of species of animal and plants increase population comfortably. The pest *Helicoverpa armigera* Hubner, was also observed here, it is a polyphagous pest which generally infest vegetables as well as the rabbi crops. The experimental crop *Solanum lycopersicum* (L) is an important cash crop; common farmers of Ramgarh cultivate it as their capacity. The experimental pest *Helicoverpa armigera* Hubner, is a serious pest and damages 50 to 80 percent of the crop production.

The present author expresses and deals with Destruction of *Solanum lycopersicum* (L) by the pest *Helicoverpa armigera* Hubner (Lepidoptera: noctuidae), a case study, at Ramgarh (Jharkhand, India). The vegetable crop *Solanum lycopersicum* (L) is a cash crop the people like the taste and farmers cultivated by a vested interest. Infestation of crop by the experimental pest ceases the financial progress of the farmers.

Morphology of the adult pest: The experimental adult moth was large in size, 14 to 18 mm long with wingspan 35 to 40 mm. tapered thorax; fine hairs on antennae, colour of fore wings of male were greenish-grey, hind wings were pale- brown a pale patch was observed near the centre of dark region as indicated by red circles. The fore wings of female moth were orange-brown in colour.

Morphology of eggs: The freshly laid eggs were white in colour, later the colour turns to green. The eggs were 0.43 to 0.60mm high and 0.50 to 0.55mm wide, spherical in shape, generally laid on the lower part of the foliage or on the moist soil as single or in cluster of 3 to 5.

Morphology of the larvae: The newly hatched tiny larvae fed on the chorine and crawl individually in search of food. They were uniformly yellowish-white in colour later change to reddish –brown and dark spotted body grown up to 07mm. Full grown larvae having pale-brown head, after 07 developmental stages some of the larvae moult to 8th stage the full grown larva developed up to 30 mm.

Morphology of the pupa: The last instars of full grown larva (7th or 8th) will borrow and enters in the soil up to 25 to 35 mm below and form an earthen chamber and get in pupation stage. The size of pupa was 10 to 13 mm, it was barrel shaped and yellowish-brown in colour later it becomes dark- brown.

Life cycle of pest: The moth *Helicoverpa armigera* Hubner (Lepidoptera: noctuidae) was a serious polyphagous pest. It is a leading species causing extensive damaged in vegetables, ornamental plants, the commercial plants, field crops, weeds etc. It was a nocturnal pest; adult moths became active at the dusk. Copulation of adults took place since late night to the early morning sometimes the copula remain upto three days. After 01 to 02 days of incubation the female laid eggs in the morning on the lower surface of leaf of the experimental host plant *Solanum lycopersicum* (L) or on moist ground around the host plant singly or in a mass of 03 to 05 eggs. The process of oviposition was continued for 04 to 10 days depended on climate and availability of food. The total number of eggs laid by a female was 80 to 120, sometimes up to 800.

After the incubation of 04 to 10 days the tiny caterpillar were hatched out from egg at night. The neonates started feeding their chorion and grow rapidly, then crawl individually in search of food. In the absence of suitable food the neonates consume unhealthy eggs and other less active larvae. Up to 3rd sometimes 4th instar the larvae remain on foliage or on flowers and attempts on

the berry, then it became photonegative in day time it enter in the soil, the crack or in the crevices and come out during night. After seven to eight instars the larva reached to full grown stage, it would be 30 mm long maximum. The mature caterpillar enters in to the soil up to 30 mm below and get in rest and stop feeding it release reddish secretion before burrowing in to the soil. Later with the duration of 13 to 25 days the moth emerges out from pupa during night. The life cycles were completed in 35 to 65 days. Three to four generations had been observed in a year.

Table 1. Tabular representation of life-cycle of pest *Helicoverpa armigera* Hubner on host plant *Solanum lycopersicum* (L)

Name of the month	No. of eggs laid/day	Incubation period of eggs in days	Larvae mature in days	Pupa mature in days	Life-cycle completed in day
February - March	80-100	05-07	20-25	15-20	40-52
April-June	100-120	04-06	18-22	13-18	35-46
July-September	90-120	05-08	17-22	15-18	37-48
October-November	80-120	05-08	17-20	18-22	40-50
December-January	70-90	07-10	20-30	20-25	47-65

SUGGESSION FOR CONTROL OF PESTS:

- Farmers were advised to adopt the clean cultivation.
- The caterpillars can be trapped using hand- pick method applied in small area.
- Field should be often flooded moderately.
- Regular raking up of soil the cultivated field help to destroy the pupa.
- The application of 02 % lindane, 0.1% carbaryl, or 0.1% endrin after the sowing of crop can control the pest.

DISCUSSION

Solanum lycopersicum (L) is a widely accepted vegetarian food, consumed in raw and cooked form also by the people of globe. It is cultivated by the common farmers of Ramgarh for agribusinesses and for their own consumption. This crop is destructed by a serious pest tomato fruit worm *Helicoverpa armigera* Hubner, it is a polyphagus serious insect pest. The experimental plant *Solanum lycopersicum* (L) is a sub optional host plant of the experimental pest. Development of larvae were retarded on this plant, it takes more time to mature. On other host the larva fully matures after 6th to 7th instar but on tomato it mature on 8th instar. *Solanum lycopersicum* (L), (tomato) is commonly called by the local people as bilaiti consumed by the population around the globe. It is rich in fibres, various essential vitamins, free from fat, cholesterol, very low caloriesuse in various traditional medicines, control the blood sugar, weight- loss, and improve immunity. The high nutritional values, high contents of minerals, low in calories, make it a good health supplement.[1,2] The experimental pest *Helicoverpa armigera* Hubner, was a serious pest commonly known as cotton bollworm. It attracts towards the flowering plants. Adult moths make pairs remain in copula for few hours to three days, after the incubation of 04 to 10 days oviposition starts. The neonates will feed on chorion and then crawl individually in search of food. If suitable food is not found they will consume unhatched eggs and other less active larvae. Subsequent generation may cause economic damage with small larvae feeding on leaves and flowers and large larvae primarily feeding on fruit. Larvae were the only destructive stage. The larvae of 03rd to 04th instar were very sensitive to light, they hide in daytime in soil and come out at night. The period of pupal diapause varies and highly dependent on temperature and photoperiod. The life cycle is directly influenced by light, temperature, humidity, pH. etc.[3-8] The pest declines the production of berry, their size, number and the taste of the crops the fruit became unfit for human consumption and unmarketable. The pest not only declines the production of the crop but also declines the financial status of the farmer.[9,10] The town Ramgarh is situated at very prime location. It is surrounded by wide National Highway and good connectivity of railway which makes Ramgarh good and prime market for vegetable sale.[7,9,10] The life cycle of this moth was very simple. It was observed that the larvae were very active during night and the full mature larvae enter the soil, cracks or the crevices, form an

earthen chamber and pupate there. The adult emerged out in the morning and took part in reproduction at night. The life span may increase depending upon availability of food and favourable temperature.[8] It has been observed that the moth migrated to hilly cold region during extreme hot season, with the help of wind current sometimes they rich up to 1000 km.

It was observed that it is one of the most expensive pest. Prevention and control was a difficult phenomenon. The spray of chemical pesticides like 02 % lindane, or carbaryl 0.1% or endrin 0.1% kill the eggs, larvae and adults on the surface of the host plant.[11,12] Some natural enemies as moles, grubs, wasps, predaceous and parasitic flies destroy the larvae and control the pest population. Viral control of pest by (HvAV-3i) also apply.

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In-vitro study of antifungal activity of nickel (II) complex with phenylbiguanide and p-phenylenedibiguanide ligand against *Aspergillus versicolor* & *Aspergillus niger*

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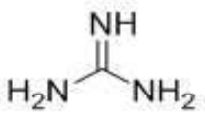
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Abstract- In last two decades, it has been observed that many of the antimicrobial drugs became non-functional any more. In other words, one can say all of them are now inefficient to control microbial growth. In the 2nd wave of COVID – 19, many of the patients have been found infected with different fungi after defeating corona disease. This has resulted in taking out different organs in order to save the life of the patients. In this research, attention has been dragged towards biochemical activity of some complexes of biguanide derivatives to check its action towards antifungal activity of two common fungi of ascomycetes family. In the present research work, it was checked whether after complexation with transition metal nickel (II) do it has such properties or not. It was found that the complexes have a positive response against the growth of fungi *Aspergillus versicolor* and *Aspergillus niger*. Antifungal activity has been compared in PDA (Potato Dextrose Agar) and SDA (Sabourand and Dextrose Agar) media by disc dilution test method & mycelium growth inhibition was calculated. It was found that p-phenyledibiguanide ligand complex was more effective than phenylbiguanide complex in both media.

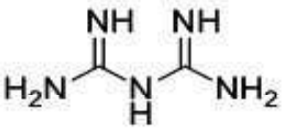
Key words: ***Aspergillus versicolor, Aspergillus niger*, PDA, SDA, ascomycetes, bidentate**

INTRODUCTION

Biguanide[1,2] is a class of drug which is used to reduce the sugar level in blood. This drug is given to diabetes type-2. Its oral intake reduces the sugar absorption by intestine and prevents liver to convert fats & protein into glucose. Also it stimulates the body to be more sensitive towards insulin release by pancreas. Biguanide as the name predict it to be diguanide i.e. formed by condensation of two guanidine molecules.



guanidine



biguanide

Biguanide[2] have been found to act as ligand and form numerous complexes with divalent and trivalent metal ions. Each biguanide moiety gets bonded to metal ion by one covalent bonding and one coordinate bonding. One of the imino hydrogen gets replaced by covalent bonding with metal ion and one amino N gets connected by

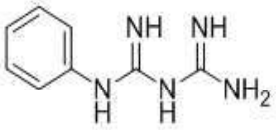
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coordinate bond. Biguanide and its substituted derivatives have been synthesized by different researchers and scientists and complexed with transition metals. Its physical and chemical properties were studied by different scholars. Beside these, nowadays their spectral studies are also been done but still biochemical studies are still lacking in the field of above said ligand complexes. Hence, in this research work attention have been dragged towards antifungal activity of nickel (II) ion complexed with phenylbiguanide & p-phenylenedibiguanide ligands. After complexation both the complexes were treated against fungus- *Aspergillus versicolor* and *Aspergillus niger*. It was found that both the complexes showed positive result towards reduction in the fungal growth.

MATERIALS & METHOD

In order to do the research work, the mentioned ligands were prepared in the form of phenylbiguanide hydrochloride and p-phenylenedibiguanide sulphate. After the preparation of the ligands, complexation was done by reacting with the metal salt. The complexes so obtained were confirmed by different analysis such as elemental analysis, spectral analysis and magnetic properties.

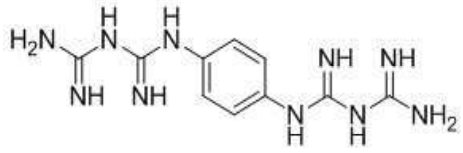
Phenylbiguanide hydrochloride (m.pt.=244-247°C)[3-6] - It was prepared by refluxing aniline hydrochloride with dicyandiamine by dissolving both with ethyl alcohol in round bottom flask on water bath for 2 hrs. On cooling shining crystals get deposited which was filtered and recrystallized with hot water.



Phenylbiguanide

p-phenylene dibiguanide[7] ($C_{10}N_{10}H_{16}$•$2H_2SO_4$)

It was prepared by refluxing a mixture of p-phenylenediamine, dicyandiamide with 32% hydrochloric acid and water for 2 hours. Refluxing was done in round bottom flask on water bath. After cooling the resulting refluxed solution was treated with alcohol. The solution so obtained was coloured and so boiled with activated charcoal to eradicate the colouration. The solution was then filtered in beaker and dil. sulphuric acid was added. The filtrate was then kept for obtaining crystals of the ligand. After 2-3 days crystals was deposited in the beaker. It was filtered and washed first with cold water and then alcohol & dried in air.



p-Phenylenedibiguanide

Nickel phenylbiguanidiumchloride or Bis (phenylbiguanidium) nickel (II) chloride[8] $[Ni(PhBigH)_2]Cl_2.\ 2.5H_2O$

It was prepared by action of phenylbiguanide hydroxide on a solution of nickel chloride. As a result, golden yellow crystals of the above said complex was obtained. The precipitate was filtered. The crystals were purified by recrystallisation with absolute alcohol and dried in air.

p-phenylenedibiguanidiumnickel (II)sulphate[9-10] $[Ni(C_{10}N_{10}H_{14})]SO_4.2H_2O$

It was prepared by mixing ammonical solution of p-phenylenedibiguanide sulphate with aqueous solution of nickel sulphate solution. As a result, a brick red flocculent precipitate of p-phenylenedibiguanidumnickel (II) sulphate was obtained. The red precipitate so obtained was filtered and washed with water and dried in air. The complex was sparingly soluble in hot water and insoluble in alcohol.

Antifungal activity[11-13]

Antifungal activity was done by inoculating the above mentioned fungus to PDA (Potato Dextrose Agar) and SDA (Sabourand and Dextrose Agar) medium. The medium was autoclaved before use and the ratio used for complex solution and medium was1:10. Further inhibition of mycelium was calculated by disc dilution test method. PDA and SDA were prepared as follows:

PDA medium preparation[14-15]- 200gms of potato tubers were peeled off and chopped into small pieces. The pieces were boiled with 100ml water in a beaker for 20 minutes and filtered with muslin cloth. The filterate so obtained is called potato extract. 20gm of dextrose, 15gm agar and 2gm peptone was then added to the extract and heated gently. The solution was then made to 1litre. pH of the solution is maintained to 5.6 by 1N HCl or 1N NaOH solution and stored in Erlenmeyer flask.

SDA medium preparation[14-15] - In this case, premix was dissolved in distilled water and boiled. The filterate so obtained was filtered and diluted to 1litre and heated gently for half an hour to make uniform solution. pH was maintained to 5.6 and stored in Erlenmeyer flask by putting glass wool on its mouth.

After preparation of the media, pure culture of the fungal spores are prepared and isolated by Warcup method.[15] On successive inoculation in new petridish with the medium, pure culture of a particular fungal spore had been obtained. After getting separated, the fungus was identified under microscope and testing was done in UV chamber of the Chemistry Department of Magadh Mahila College, Patna University.

RESULTS & DISCUSSION

Table 1- Percentage composition of elements and ions

Complexes	Formulae	Percent composition of elements and ions				
			Ni	N	Cl^-	H_2O
Bis-(phenylbiguanidium) nickel(II) chloride	$[Ni(PhBigH)_2]Cl_2.2.5H_2O$	By calculation	11.02	26.50	13.45	8.52
		Found	11.08	26.30	13.20	8.43
p-phenylenedibiguanidium nickel(II)sulphate	$[Ni(C_{10}N_{10}H_{14})]SO_4.2H_2O$		Ni	N	SO_4^{-2}	H_2O
		By calculation	12.57	29.98	20.50	7.70
		Found	12.30	29.84	20.50	5.68

The nickel absorption spectra of Ni (II) complex, Bis(phenyldibiguanidium) nickel (II) chloride (A), display one broad distinct band at 240nm & 220nm attributed to the charge transfer and $^1A_{1g} \longrightarrow {}^1E_g$ transition. The diamagnetism and UV electronic absorption pattern suggested square planer geometry of nickel (II) complexes.

The electronic absorption of Ni (II) complexes display one band (distinct) at 470nm for [Ni(p-phenylenedibiguanide)]sulphate. This medium band due to $A_{1g} \longrightarrow B_{1g}$ transition in square planer field. The strong absorption band located at 270nm attributed to the charge transfer and $^1A_{1g} \longrightarrow {}^1E_g$ transition. The diamagnetic and uv electronic absorption pattern suggested square planar geometry of Ni (II) complexes.

Table 2. Percentage inhibition of growth of fungus *Aspergillus versicolor* at indicated dose

Complex	Medium	Concentration in µg/ml	Percentage inhibition of fungus
Bis-(phenylbi guanidium) nickel(II)chloride Complex A	PDA	400	92.31
	PDA	200	78.73
	PDA	100	50.72
	SDA	400	88.50
	SDA	200	70.58
	SDA	100	47.41
p-phenylenedibi guanidium nickel(II)sulphate Complex B	PDA	400	98.51
	PDA	200	81.20
	PDA	100	59.51
	SDA	400	95.20
	SDA	200	79.25
	SDA	100	56.31

Table 3. Percentage inhibition of growth of fungus *Aspergillus niger* at indicated dose

Complex	Medium	Concentration in µg/ml	Percentage inhibition of fungus
Bis-(phenylbi guanidium) nickel(II)chloride Complex A	PDA	400	84.01
	PDA	200	65.72
	PDA	100	44.36
	SDA	400	82.09
	SDA	200	62.37
	SDA	100	42.27
p-phenylenedibi guanidium nickel(II)sulphate Complex B	PDA	400	95.80
	PDA	200	70.25
	PDA	100	50.73
	SDA	400	84.11
	SDA	200	69.59
	SDA	100	48.45

When the data mentioned in table ii and iii were compared, it was concluded that when concentration of the complex taken was 400µg/ml, minimum inhibitory concentration (MIC) in PDA and SDA media were as follows:

Table 4. Percentage inhibition of growth of fungal growth

Complex	Concentration (µg/ml)	Fungus	Percentage inhibition of fungal growth	
			PDA medium	SDA medium
A	400	*A.versicolor*	92.31	88.50
B	400	*A.versicolor*	98.51	95.20
A	400	*A. niger*	84.01	82.09
B	400	*A. niger*	95.80	84.11

From the data, it was found that in PDA (potato dextrose agar) medium, the complex B was more effective to control the fungal growth in *A.versicolor* than *A. niger.* At 400 µg/ml concentration in the case of SDA (Sabourand

Dextrose Agar) medium also, complex B was more effective to control fungal growth of *A. versicolor* than *A.niger*.

For both the fungi, *A. versicolor* and *A. niger*, when MIC was compared, it was found that under 400µg/ml concentration in PDA & SDA media, complex B was more effective in controlling the fungal growth with 98.51% & 95.20% and 95.80 % & 84.11% inhibition respectively.

CONCLUSION

From the result and discussion as described above, it is concluded that for complex B i.e. nickel complexed with p-phenylenedibiguanide ligand controls the mycelium growth more efficiently than complex A i.e. nickel (II) complexed with phenylbiguanide ligand for both the fungi- *A.versicolor* and *A.niger*.

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Chemical composition of raw milk from Cow, Buffalo and Goat of Madhepura district (Bihar)

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Received : 22nd December, 2021 ; Revised : 23rd January, 2022

Abstract- Milk is considered as a good source of Protein, Fat and Minerals. In the present work, study of chemical composition of milk from Buffalo, Cow and Goat was performed. Milk was collected from 5 dairy firms of Madhepura district and chemical constituents-Protein, Fat, Lactose, Carbohydrate and minerals were measured in the laboratory. It was observed that Protein, Fat and Lactose content were highest in Buffalo milk and lowest in Goat milk. The content of these biochemical substances in Cow milk was in between Buffalo and Goat milk. Similarly, the mineral content i.e. Calcium, Potassium, Phosphorus, Iron, Zinc etc. were also maximum in Buffalo milk and the value of these minerals varies in Goat and Cow milk. The Calcium and Potassium were in more quantity in Goat milk in comparison to the Cow milk but the Iron content was higher in Cow milk in comparison to the Goat milk. On the other hand, Zinc content was higher in Goat milk.

Key words: Protein, Fat, Lactose, Carbohydrate, Minerals

INTRODUCTION

Milk is considered as a nearly complete food since it is a good source of Protein, Fat and major minerals. Also milk and milk products are main constituents of the daily diet, especially for vulnerable groups such as infants, school age children and old age.[1] Several studies have reported the distribution and occurrence of the essential components in various animals' milks.[2,3] Milk is an excellent source of Protein, Fat, Vitamin A, Vitamin D, Vitamin B2, B12, Niacin and minerals like Calcium, Potassium, Zinc, Iron, Copper, Phosphorus, etc. Major source of milk for human consumption comes from Buffalo, Cow and Goat. Cow and buffalo milk is preferred by adults while Goat milk is preferred for infants. Buffalo milk is high in fat and solid content in comparison to Cow and Goat milk.

The value of goat milk in human nutrition has so far received very little attention despite its medicinal need for infants.[4,5] Goat milk protein and fat have many significant differences in their composition from the milk of other mammalian species, especially in relative proportion of their genetic polymorphism.[6] Goat milk have shorter rennet coagulation time, less resistant to heat treatment, curd firmness is weaker and cheese yield are less which might explain significance differences to cow and other milk in digestion by infants, and patients which traditionally have been explained by the homogenised nature of goat milk fat.[4,5]

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MATERIALS & METHODS

Cow and Buffalo milk samples were collected in one litre sterilized bottles from 5 dairy firms of Madhepura district and Goat milk were collected from 5 houses of the same area. Cow samples were labelled as CM1, CM2, CM3, CM4, CM5; Buffalo milk was labelled as BM1, BM2, BM3, BM4, BM5 and Goat milk were labelled as GM1, GM2, GM3, GM4, GM5. Milk samples were immediately brought to the laboratory in ice box and stored in refrigerator at 4°C for chemical analysis.

Chemical Analysis: Chemical constituents Protein, Fat, Lactose, Carbohydrate and Ash were determined according to AOAC (1990)[7]. Mineral constituents Calcium, Potassium, Magnesium, Zinc, Copper, Iron, Phosphorus, Manganese, and Cadmium were measured in Ash using atomic absorption spectrophotometer. Phosphorus was estimated colorimetrically in Ash.

RESULT

The chemical composition of milk of Cow, Buffalo and Goat shows major differences in Protein, Fat, Total solid, Lactose contents and Mineral contents. The maximum protein in Cow was observed 31.9gm/kg, in Buffalo 46.2gm/kg and only 30.9gm/kg in Goat. Similarly the fat content was also higher in Buffalo (75.9gm/kg) and minimum in Goat (39.1gm/kg) while in Cow, the maximum Fat content of Cow was recorded as 41.4gm/kg. The Lactose content of Buffalo was also maximum (51.8gm/kg) and minimum in Goat (41.1gm/kg) while in Cow, it was 42.1gm/kg i.e. in between Buffalo and Goat. The mineral content, Calcium, Potassium, Phosphorus, Iron and Zinc were maximum in Buffalo (Ca=1630.1, K=1470.3, P=949.6, Fe=0.986, Zn-4.372)mg/kg while in Goat and Cow, mineral nutrition in some cases were observed variable. The Calcium and Potash content was higher in Goat (Ca=1192.5, K=1964.1)mg/kg and in Cow the Calcium content was (Ca=1156.1, K=1659.8)mg/kg. Zinc content was also higher in Goat milk (3.8mg/kg) in comparison to the Cow milk (3.152mg/kg) but the Phosphate and Iron content were higher in Cow (P=1109.9, Fe=0.689)mg/kg but in Goat the Phosphate and Iron content were (P=938.8, Fe=0.55)mg/kg. The result was tabulated in Table no. 01 to 06.

Table 1- Chemical Composition of Cow Milk (g/kg)

S. No.	Sample No.	Protein	Fat	Total Solid	Lactose	Carbohydrate	Ash
1	CM1	31.5	40.3	121.3	42.1	52.8	6.5
2	CM2	31.2	40.6	120.5	41.8	52.1	6.2
3	CM3	30.8	41.4	119.7	40.9	51.9	6.3
4	CM4	31.7	40.9	122.4	41.3	53.1	6.6
5	CM5	31.9	41.2	123.2	42.5	52.4	6.9

Table 2- Chemical Composition of Buffalo Milk (g/kg)

S. No.	Sample No.	Protein	Fat	Total Solid	Lactose	Carbohydrate	Ash
1	BM1	45.3	75.02	157.5	51.05	45.4	7.6
2	BM2	44.9	74.8	156.8	50.7	45.1	7.4
3	BM3	44.5	74.3	155.9	50.4	44.8	7.1
4	BM4	45.8	75.3	158.1	51.5	45.9	7.7
5	BM5	46.2	75.9	158.8	51.8	46.2	7.9

Table 3- Chemical Composition of Goat Milk (g/kg)

S. No.	Sample No.	Protein	Fat	Total Solid	Lactose	Carbohydrate	Ash
1	GM1	30.4	38.5	116.5	40.4	43.1	6.3
2	GM2	30.1	38.1	115.9	40.1	42.8	6.1
3	GM3	29.8	37.8	115.4	39.8	42.5	5.9
4	GM4	30.6	38.7	116.9	40.8	43.5	6.6
5	GM5	30.9	39.1	117.3	41.1	43.8	6.8

Table 4- Metal content of Cow milk (mg/kg)

S. No.	Sample No.	Ca	P	Fe	Zn	K	Mg	Cu	Mn	Cd
1	CM1	1153.4	1108.3	0.683	3.146	1655.5	134.2	0.142	0.056	0.086
2	CM2	1151.7	1107.6	0.681	3.142	1653.8	133.8	0.140	0.055	0.085
3	CM3	1149.5	1106.8	0.680	3.139	1651.1	133.3	0.139	0.054	0.083
4	CM4	1154.7	1109.1	0.687	3.149	1657.6	134.7	0.145	0.057	0.088
5	CM5	1156.1	1109.9	0.689	3.152	1659.8	135.3	0.147	0.059	0.090

Table 5- Metal content of Buffalo milk (mg/kg)

S. No.	Sample No.	Ca	P	Fe	Zn	K	Mg	Cu	Mn	Cd
1	BM1	1625.3	948.2	0.980	4.366	1467.1	284.2	0.212	0.076	0.118
2	BM2	1623.9	947.4	0.976	4.361	1465.6	283.7	.0210	0.075	0.116
3	BM3	1621.1	946.1	0.974	4.357	1463.2	283.3	0.208	0.073	0.113
4	BM4	1627.8	948.9	0.983	4.369	1468.7	284.8	0.215	0.078	0.121
5	BM5	1630.1	949.6	0.986	4.372	1470.3	285.6	0.219	0.080	0.125

Table 6- Metal content of Goat milk (mg/kg)

S. No.	Sample No.	Ca	P	Fe	Zn	K	Mg	Cu	Mn	Cd
1	GM1	1188.7	936.5	0.50	3.2	1958.7	136.4	0.261	0.046	0.078
2	GM2	1186.5	935.2	0.48	3.0	1955.3	135.9	0.258	0.044	0.076
3	GM3	1183.2	934.3	0.47	2.7	1951.4	135.2	0.256	0.041	0.075
4	GM4	1190.3	937.4	0.53	3.5	1960.5	136.8	0.265	0.049	0.080
5	GM5	1192.5	938.8	0.55	3.8	1964.1	137.3	0.268	0.052	0.081

CONCLUSION

Milk is considered as complete food and a good source of Protein, Fat and major minerals. To access the chemical composition of Cow, Buffalo and Goat milk present work was undertaken. It was observed that Protein, Fat and Lactose content were highest in Buffalo milk and lowest in Goat milk. The content of these biochemical substances in Cow milk was in between Buffalo and Goat milk. Similarly, the mineral content i.e. Calcium, Potassium, Phosphorus, Iron, Zinc etc. were also maximum in Buffalo milk and the value of these minerals varies in Goat and Cow milk. The Calcium and Potassium were in more quantity in Goat milk in comparison to the Cow milk but the Iron content was higher in Cow milk in comparison to the Goat milk. On the other hand, Zinc content was higher in Goat milk.

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Acute toxicity study and long term effect of daidzein on the thyroid gland of male Wistar rat

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Abstract- Daidzein is a plant extracted natural exogenous estrogen which can mimic the role of endogenous female hormone estrogen. It is a naturally occurring compound found exclusively in soybeans, moong beans, lentin, clover and other legumes. Daidzein was used for estimation of toxicity (LD_{50}) for short- term duration and its adverse effect on thyroid gland and its hormones in Wistar rat. Lethal $dose_{50}$ was estimated by log dose/ probit regression method. Five sets containing 10 rats each were taken and serially diluted doses were given orally with the help of intragastric gavage tube. The doses were 0, 50,100,150 and 300 mg/kg body weight (BW) for daidzein. The rats were observed for 4 days and then mortality was noted for further calculation. LD_{50} of the daidzein was found to be 150 mg/kg BW. $1/10^{th}$ of LD_{50} dose was introduced for sub-chronic 90 days treatment and its effect on thyroid hormones was observed after 90 continuous day's exposure. Saline water and ethanol (9:1v/v) were used as vehicle. The vehicle and daidzein solution were given intragastrically (0.5 ml/15 mg/kg BW) once a day for 90 continuous days. Serum TSH concentration was observed to be significantly ($p<0.001$) higher in daidzein treated animals in comparison to the control and sham - control animals. The levels of T_3 and T_4 were significantly lowered in daidzein treated rats. This result shows that the isoflavone, daidzein has adverse effect on thyroid gland and its function in Wistar rat. Thus, it can be suggested that daidzein administration affects the hormonal status of this animal.

Key words: Daidzein, LD_{50}, Toxicity, thyroid gland

INTRODUCTION

Daidzein is a natural endocrine disruptor belonging to class flavonoids and sub-class isoflavones that may interfere with the endogenous endocrine system and produce adverse developmental, reproductive and immune effects in animals. Structurally, daidzein closely resembles 17β-estradiol, and it binds to the estrogen receptors (ERs), the stronger affinity being for the ERβ isoform.[1] Thus they have potential to mimic and block sex hormones, adrenal function and even affect the thyroid gland function. Mimicking as a natural selective ER modulator daidzein exerts its estrogenic agonist and antagonist action as a tissue and dose dependent manner.[2] They are ubiquitously found in fruits, vegetables, grains, nuts, mostly abundant in soy beans and other legumes.[3,4] Approachability of soybean throughout popularized consumption in recent years due to their potential curative, preventative and nutritive value.[5,6] In addition to their positive effects, isoflavones have been shown to cause direct and indirect harmful effects on thyroid function of animals. Thyroid hormones regulate several metabolic processes and are known to play an important role in development and maturation of central nervous system, regulation of bone metabolism, cardiovascular system and

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normal growth in mammals. Studies on the relationship between soybean intake and thyroid function[7-10] suggest that consumption of soybeans and soy isoflavone daidzein, is goitrogenic and alters thyroid function. Estrogens exert several effects on thyroid follicular cells. They modulate their cell cycle regulation, proliferation and function thus potentially furnishing to the pathogenesis of thyroid hyperplasia and thyroid cancer. These effects are mediated by the binding to estrogen receptors.[11,12] However, the effects of soy isoflavones on thyroid function are controversial.[7-10] Several experimental studies have shown that administration of isoflavones at doses commonly found in human diets affects the thyroid gland function. Flavonoid rich diets have the potential to beneficial as well as adverse effects in humans.[13] This is related to hormonal activity, which is therefore relevant for the evaluation of associated risk to humans. Very few works are notable on the toxicity of LD_{50} of daidzein and its effects on mammals.

Therefore, the present study was aimed at determining the lethal dose toxicity and effect of daidzein after 90 days exposure on thyroid gland in Wistar rat.

RESEARCH DESIGN

The experiment was performed according to the guidelines accepted by the Ethics Committee, Ranchi University, Ranchi for investigations on animals.

Daidzein (7,4 dihydroxy isoflavones) is a naturally occurring exogenous endocrine disruptor present in a number of plants, mainly in soybeans and soy derived products. The toxicological study has been done on rats which are related to human beings and easy to handle in laboratory. Adult Wistar rats were selected from inbreed colony. Healthy adult rats of almost equal weight (130-150 g) body weight and size irrespective of sexes were selected randomly. The rats were housed in stainless -steel cages (435× 290×150 mm) and maintained in a room fully exposed to natural photoperiod, temperature and humidity. They were provided with food (soy free rat chow) and water *ad libitum*.

DETERMINATION OF LD_{50}

The LD_{50} was determined by log-dose/ probit regression live method.[14] Five sets containing 10 rats each were taken and serially diluted doses were given orally with the help of gavage tube. The doses administered were 0, 50, 100, 150 and 300 mg/kg BW for daidzein. The rats were observed for 4 days and then mortality was noted for further calculation. A graph has been plotted between empirical probit and log dose and then LD_{50} has been calculated with the help of regression live and computerized calculation.

SUB-CHRONIC TREATMENT

LD_{50} of the daidzein has been estimated (150 mg/kg bw) and $1/10^{th}$ of LD_{50} dose was introduced for sub-chronic 90 days treatment. Normal saline and ethanol (9:1v/v) were used as vehicle. All administrations were made intragastrically (0.5 ml/15mg/kg BW) once a day between 0900hrs-1000hrs for 90 continuous days. The animals were anaesthisized (di-ethyl ether) and their blood serum was collected and stored (-80°C) until analysis.

ANALYSIS

Total tri-iodothyronine (T_3), thyroxine (T_4) and thyroid stimulating hormone (TSH) were measured by Chemiluminescence Immunoassay, CLIA.[15]

TABLES & GRAPHS

Table 1- Transformation of percentage to probit

%	0	1	2	3	4	5	6	7	8	9
0	---	2.67	2.95	3.12	3.25	3.36	3.45	3.52	3.59	3.66
10	3.72	3.77	3.82	3.87	3.92	3.96	4.01	4.05	4.08	4.12
20	4.10	4.19	4.23	4.26	4.29	4.33	4.36	4.39	4.42	4.45
30	4.48	4.50	4.53	4.56	4.59	4.61	4.64	4.67	4.69	4.72
40	4.75	4.77	4.80	4.82	4.85	4.87	4.90	4.92	4.95	4.97
50	5.00	5.03	5.05	5.08	5.10	5.13	5.15	5.18	5.20	5.23
60	5.25	5.28	5.31	5.33	5.36	5.39	5.41	5.44	5.47	5.50
70	5.52	5.55	5.58	5.61	5.64	5.67	5.71	5.74	5.77	5.81
80	5.84	5.88	5.92	5.95	5.99	6.04	6.08	6.13	6.18	6.23
90	6.28	6.34	6.41	6.48	6.55	6.64	6.75	6.88	7.05	7.33
---	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
99	7.33	7.37	7.41	7.46	7.51	7.58	7.65	7.75	7.88	8.09

Table 2- Influence of daidzein toxicity on male Wistar rat

Sl. No.	Concentration (mg/kg BW)	Concentration (ppm)	Number of animals		Percentage mortality	Log concentration	
			Exposed	Dead			
1.	300	6810	10	10	100	3.833	8.09
2.	150	1690	10	5	50	3.227	5.00
3.	100	750	10	3	30	2.875	4.48
4.	50	180	10	2	20	2.255	4.16
5.	0	0	10	-	-	-	-

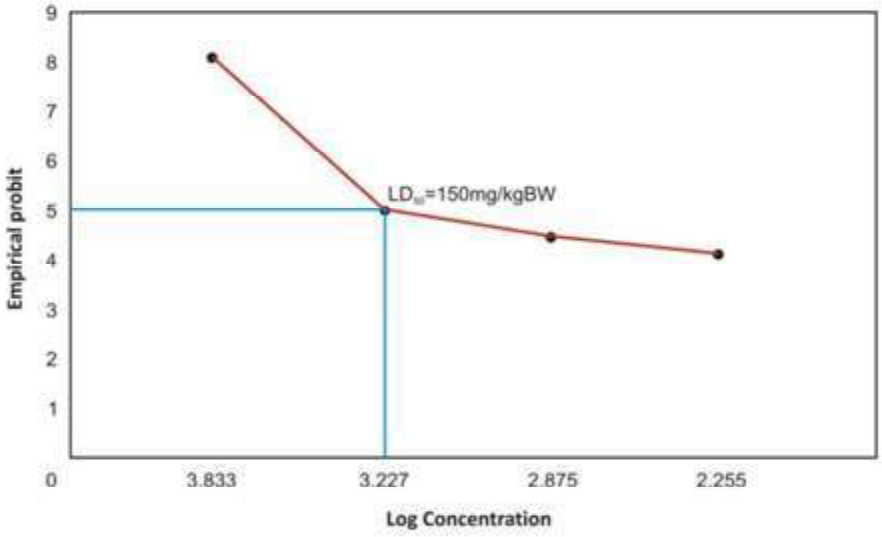


Figure 1- Plot of log concentration versus empirical probit from table 2 for calculation of LD_{50} of daidzein administered intragastrically.

Table 3- The influence of daidzein on serum thyroid hormones in mature male rats

Blood serum	Control	Sham-control	Treated
Thyroid stimulating hormone (TSH) µ/ml	0.85±0.02	0.86±0.01	2.83±0.02*
Triiodothryonine (T_3) ng/dl	132.0±0.04	131.9±0.05	77.0±0.05*
Thyroxine (T_4) µg/dl	7.30±0.04	7.31±0.08	4.60±0.06*

***Daidzein was dissolved in saline water: ethanol mixture (9:1 v/v; 0.5 ml/150g BW) and was administered intragastrically for 90 days. Values are means ± SEM for rats.**

RESULTS

In the present study the lethality of daidzein and mortality rate of rats were determined by introducing the daidzein orally. Daidzein in different concentrations/doses ranging from 50 mg/kg BW to 300 mg/kg BW were used. The animals exposed to different concentrations of daidzein, showed no mortality at 0 mg/kg BW groups, 20

percent mortality at 50 mg/kg BW, 30 percent mortality at 100 mg/kg BW, 50 percent mortality at 150mg/kg BW and 100 percent mortality at 300 mg/kg BW observed. The data was computed according to Probit Analysis Method[14] and LD_{50} value was determined.

Results for the effect of intragastric administration of daidzein at a dose of 15 mg/kg BW, for 90 continuous days on the serum levels of T_3, T_4 and TSH are presented in Table 3. It was observed that daidzein administration affected the hormonal status of the animal. Serum TSH concentration was significantly ($P<0.001$) higher in daidzein treated rat in comparison to the control and sham-control animals. The levels of serum T_3 and T_4 were lowered ($P<0.001$) in daidzein treated rats.

DISCUSSION

In the present investigation, the obtained LD_{50} value is 150 mg/kg BW.

The rats introduced to different concentration of daidzein, showed 50 percent mortality at 150 mg/kg BW. The probit mortality to the Wistar rat was calculated from percent mortality. When the probit mortality was plotted against log concentrations of daidzein, a straight line was obtained. The LD_{50} value obtained from this straight line graph is 150 mg/kg BW. The computation of percent mortality against different log concentration of the isoflavones yielded a sigmoid curve the LD_{50} value obtained from sigmoid curve was 150 mg/kg BW for 96 hours. The graphical representation of percent mortality versus log concentration and probit mortality versus log concentration of daidzein showed a typical sigmoid curve and straight line respectively which are in agreement with the principle of probit analysis.[14]

In the present study it was demonstrated that daidzein at the dose 15 mg/kg BW administered for ninety continuous days significantly increased the blood thyroid stimulating hormone (TSH) level in adult male rats. This was accompanied by a corresponding decrease in T_3 and T_4 concentrations. These present results provide evidence that intragastrically administered daidzein is able to influence TSH secretion from the pituitary gland. This may result in hormonal and metabolic disturbances. It has been reported that a subcutaneous injection of 10 mg/kg of daidzein can disturb the pituitary- thyroid axis, causing hypothyroidism in orchidectomized middle - aged rats.[16] Isoflavones such as daidzein and genistein have the ability to change thyroid cell events. Isoflavones can interfere with activity of an enzyme called thyroid peroxidase (TPO). TPO helps in attaching iodine to the amino acid tyrosine. This iodine tyrosine combination forms the basis for production of thyroid hormone. Daidzein and genistein act as suicide substrates for TPO by covalently binding to the active site. Thus, inhibition of TPO leads to a reduction of thyroid hormone levels, with a subsequent increment of TSH release. That, in turn, provides a strong growth stimulus to the thyroid gland. However, daidzein also affects the metabolism of thyroid hormones and iodide re-utilization by inhibition of sulfotransferase enzymes.[17]

In the present study the levels of thyroid stimulating hormone (TSH) in blood were significantly increased after treatment with isoflavone, daidzein. T_3 (tri-iodothyronine) and T_4 (thyroxine) levels showed an opposite effect and were lowered in treated animals as compared to the control and sham- control animals. Ferreira *et al.* (2002)[18] demonstrated that flavonoid daidzein inhibited 5'-iodothyronine deiodinase. In vivo experiments with the synthetic flavonoid EMD 21388, inhibits thyroid hormone binding to plasma transthyretin, showing a reduction of T3 content in tissues that express type II 5'-iodothyronine deiodinase.

In humans, Brahmbhatt *et al.* (2000)[19] showed that 530 children aged 6-15 years living in iodine-deficient areas of India and consuming large amount of isoflavones were found to be goitrous. This enormous rate of thyroid enlargement led to conclude, by these authors, that goiter was due to the combined effect of iodine deficiency and excess flavonoid in their diet.

Results obtained in our studying clearly indicate that daidzein significantly affects thyroid hormone levels in blood serum of the rat and suggest its involvement in thyroid action. Future research in this direction will provide a complete understanding of the isoflavone daidzein in thyroid function.

CONCLUSION

In the present study, the obtained LD_{50} value is 150 mg/kg BW. There is a shortage of information about the effect of pure isoflavones, such as daidzein, on thyroid safety in mammals. The results obtained in this experiment prove that 90 days sub-chronic intragastric administration of daidzein may change thyroid hormone function.

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Evaluation of cytotoxic and genotoxic effects of Butyl Hydroxylated Anisole (BHA) on somatic cells of *Allium cepa* L.

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Abstract- Food preservatives are intentionally added in small quantities during processing of food to improve the taste quality and shelf-life of the food. Therefore, we are constantly exposed to different food preservatives in daily life that may or may not show adverse effects. Evaluating the cyto-genotoxicity of food preservatives is necessary. Dose dependent and Time dependent effect of Butyl hydroxylated anisole (BHA), commonly used food preservative was done at two concentrations, 0.01mg/L and 0.04mg/L. The mitotic index had been calculated for the control and treated cells. The mitotic indices at 24hrs were 71.67±1.527 for control and 69.33±2.081 and 35.33±1.527, respectively. The mitotic indices at 48hrs were 54.67±2.081 and 27.67±2.081, respectively. The best results were observed in 72 hrs where the mitotic indices were 48.33±2.88 and 18.33±2.88 at 0.4mg/L. Chromosomal abnormalities like multinucleated cells, laggard and anaphase bridges were observed. The effect of preservative BHA on the somatic cells of *Allium cepa* was cyto-genotoxic.

Key words: Butyl hydroxylated anisole (BHA), onion, mitotic index, genotoxic, chromosomal aberrations.

INTRODUCTION

The genotoxic effect of various chemicals used as food preservatives, cosmetics and drugs has been well studied. Food preservatives are compounds that improve the quality and shelf life of food.[1] Natural food preservatives like salt and sugar were utilized to extend the shelf life of the food by improving their texture, taste, acidity/alkalinity, and consistency.[2] The food industry has changed dramatically in recent years, and the most challenging task is to increase global food quality, which is attained by using chemicals to enhance food flavour, colour and quality. To keep the veggies and fruits fresh for several days, various colouring compounds and waxes were and still being utilised. These compounds have never been used as food ingredient or ingested directly but added as adjuvants.[3] Adeyemo & Farinmade (2013)[4] reported that the present and upcoming generations are on the higher risk for exposure to these chemicals.

Butyl Hydroxylated Anisole (BHA), chemical mixture of 3-tert-butyl-4-hydroxyanisole and 2-tert-butyl-4-hydroxyanisole molecular formula $C_{11}H_{16}O_2$ has large scale uses in various food industries.[5] It is also the most widely used food antioxidants owing to its low-cost, world-wide availability, stable at high temperature and high performance. It is widely used in cosmetics, food

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packaging, oils, margarines, fat-containing products, rubber industry and petroleum products. At high doses, it causes cancer in rats, mice and hamsters, but it does this exclusively in the forestomach, an organ that humans don't have. In the low proportions used in food preservatives, however, many researchers consider it perfectly safe, especially due to absence of forestomach in humans. In an experiment, the higher dose of BHA was given to the Japanese house musk shrew, lacking fore-stomach, all the experimental model organisms died because of gastrointestinal hemorrhage. Whereas, the lower and mild dosages had effect on 50-60% models showing adenomatous hyperplasia of the lung.[6] There were no reports on BHA causing genotoxic or any adverse effects on human beings, but during current situation various reports are found showing the genotoxic effects.[7] BHA had been also reported as carcinogens by International Agency for Research on Cancer (IARC). The possibility that BHA may be a carcinogenic at normal human dietary levels, which are much lower than the experimental tumor-inducing levels, has also been questioned on the grounds that it is not mutagenic in a variety of test systems.

Genotoxic studies were performed to evaluate the rate of mutation which enhanced due to food preservation addition by food companies using *Allium cepa.*

Allium cepa has been chosen as a model to study the genotoxicity of BHA to test and monitor environmental pollution indicator. It has total number of 16 chromosomes only, thus studies involving chromosomal aberrations can be efficiently done during mitotic stages. The change in chromosomal morphology is due to the genetic changes induced through using food preservatives or any other harmful chemicals. *Allium cepa* test was first performed by Levan in 1938[8] and it was observed that there is similarity with the mammalian test model. Many of the chemical assays are first performed on onion root tips and then further elaboration of work is done using the mammalian system.[9,10] DNA alterations are known to be the early signs of damage in the organisms. The outcomes of these assays provide significant results having relationship with genotoxicity appearance in the mammalian body and more pounced with the children exposed perinatally.[11] In the present study the effect of BHA on the growth of onion root tips and effect on mitotic index and chromosome damage was analyzed.

MATERIALS & METHODS

The fresh bulbs of *Allium cepa*, of medium sized were taken from the local market. Butylated Hydroxy Anisole (BHA), chemical formula $C_{11}H_{16}O_2$, molecular weight 180.24 g/mol, procured from LOBA CHEMIE. For preparation of Carnoy's fixative chemicals taken from SRL was glacial acetic acid and ethyl alcohol in 1:3 ratios. For the hydrolysis of cells HCl was purchased from Thomas Bakers and the stain acetocarmine was from LOBA CHEMIE.

Raising fresh tips and treatment with BHA

The experimental set up was designed with slight modification of method given by Fiskesj (1988)[12]. The first two days onion root tip were grown on fresh water and then followed by treatment with BHA for 24 hours, 48 hours and 72 hours, respectively. To obtain fresh root tips, the dried roots were removed using sharp blade without damaging the meristematic part. The onion bulb was then placed on the coupling jar filled with water in a way that meristematic part of bulb is immersed in water. After 48 hours 2-3 cm long fresh onion root tips emerged out, the bulbs were then transferred at three different concentration (0, 0.01 mg/L and 0.04 mg/L) of BHA and after 24hrs 5-8 root tips were excised and fixed in Carnoy's fixative after 48hrs again 5-8 tips were excised and fixed in Carnoy's fixative and after 72 hours of treatment the left-over roots were excised and fixed.[12] After 24 hours treatment with Carnoy's fixative the tips were transferred to 70% alcohol at 4°C.[13]

Squash preparation of treated root tips

Stored onion root tips were placed at room temperature then the translucent area at the tip of roots (2-3 mm) was excised on the slide and treated with 0.1 N HCl for 2-3 minutes in case the tip is hard. This step is not required if tip cells separate with slight pressing with thumb. Then 2-3 drops of lukewarm acetocarmine stain were placed above the root tip for approximately 2 minutes. The root tip was covered with coverslip and with the help of blotting sheet the excess stained was removed. The slide was placed under 2-3 layers of blotting sheets and pressed using thumb, the onion root tip cells spread on the slide and further spreading of the cells is done by gentle heating and tapping. For each sample 3 slides were prepared and observed under Nikon Eclipse E200 with digital camera microscope at 40X and 100X magnification and 3 pictures were taken randomly from each slide for

the calculation of mitotic index and study of chromosomal morphology.

Mitotic index was calculated for three different concentrations of BHA using the formula:

$$Mitotic\ index\ MI = \frac{number\ of\ dividing\ cells}{total\ number\ of\ cells} x\ 100$$

Different types of aberrations were observed as an indicator of the specificity in the mutagenic action of food preservatives.

RESULT & DISCUSSION

The dose-dependent reduction in mean root length has been observed at higher concentrations significantly differ from the control but it may not be significant within the same group. In control, the roots of the onion bulbs were growing well hydroponically and morphologically it appeared long, healthy, shiny, fast growing and dense roots whereas treated root tips were less dense, slow growing and lesser number of root tips (Fig 1).

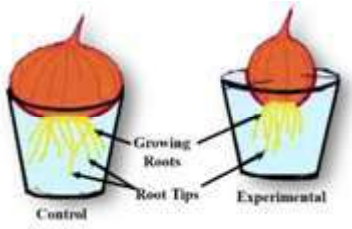


Figure 1- Experimental Set-up

Doses between 0.04 mg/L for 72 hrs showed significant reduction in growth when compared to control but insignificant when compared with each other whereas concentration of 0.04 mg/L was observed to be insignificant. The calculated mitotic indices have been tabulated for control and the treated cells (Table 1).

The BHA at recommended dose (0.01 mg/L) and higher dose (0.04 mg/L) in comparison with the control setup reveals very prominent chromosomal aberrations at recommended dosage but at higher doses most of the cells arrested at prophase stage. The mitotic index was high in controlled setup and gradually it decreased with increase in dosage as well as time of the treatment (Figure 2). The observation indicated the inhibitory effect of BHA on the onion mitotic cells both in terms of division and chromosomal aberrations. The chromosomal aberrations observed were chromosomal bridges, breaks, laggards and multipolar nuclei (Figure 3).

Table 1: The mitotic index MI of onion root tips treated with three different concentrations of BHA treated for 24, 48 and 72 hours.

Concentration of BHA	Mitotic Index (%)		
	24 hours	48 hours	72 hours
Control	71.67±1.527	67.00±1.732	82.33±2.081
0.01 mg/L	69.33±2.081	54.67±2.081	48.33±2.88
0.04 mg/L	35.33±1.527	27.67±2.081	18.33±2.88

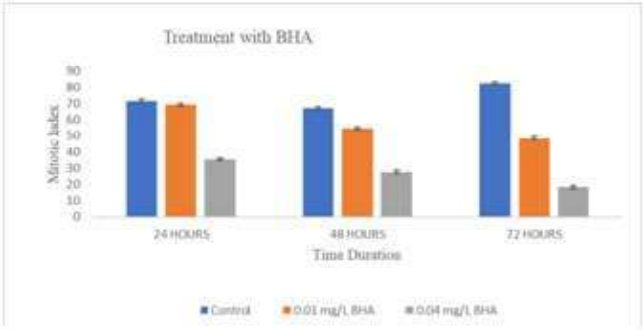


Figure 2- Bar chart showing the mitotic index of root tip cells treated with Butyl hydroxylated anisole (BHA) for different time duration

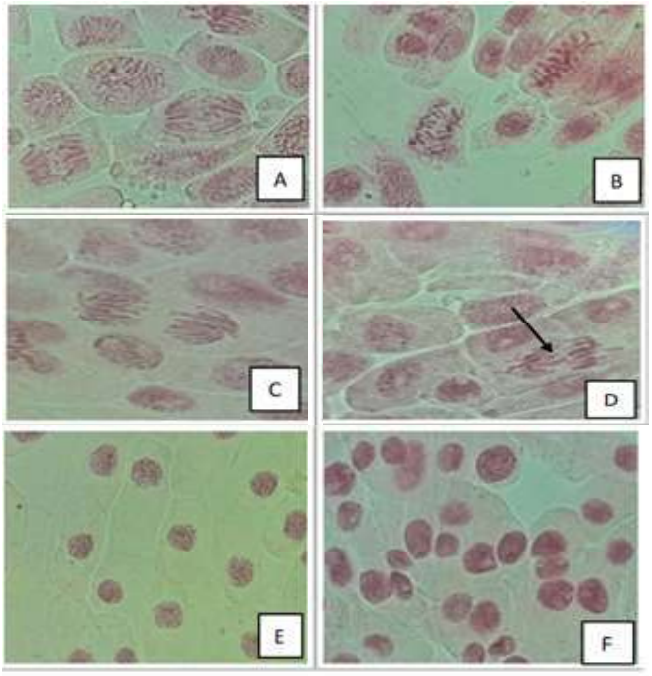


Figure 3- The mitotic cell of onion root tip (A) The untreated root tips showing normal dividing cells after 72 hrs (B) The root tips treated with 0.01 mg/L BHA for 24 hrs showing metaphase and anaphase (C) The root tip from the same onion bulb as shown in B at 48 hrs showing less dividing cells (D) One of the anaphase cell showing laggards (E& F) The root tip cell treated with 0.04 mg/L BHA after 24 and 72 respectively showing complete absence of dividing cell or at the interphase.

Mitotic division is a mandatory process of living cell and involves highly controlled steps.[14] The complete cell division is very precisely regulated in the presence of essential nutrients and other factors. Any change in condition results in halting and/or alterations in the cell division. Food preservatives which have been added in the food to increase quality, texture, taste and colour directly or indirectly causes alteration in the cell division cycle which results into chromosomal aberrations. First-tier bioassay are always easier, inexpensive and reliable in nature by involving plant system. The result of plant root tip system and mammalian culture shows positive correlation when treated with the same chemicals hence plant root tips assays are preferred.[15] The experiment results of onion root tips using various food preservatives shows warning signals which should be marked as potential health poisons.[16] Various experiments have been conducted using variety of food preservatives has been found to be genotoxic in nature.[17-19] The experiment performed by Kumar and Srivastava in 2011[17] showed inhibitory effect of the sunset yellow colour and boric acid on *Trigonella foenum-graecum* root tips. Similar results were obtained while working with Monosodium Glutamate at higher concentration affecting the cell growth, chromosomal aberrations and change in the root tips colour. Sodium metabisulphite and Potassium metabisulphite tested on the root tips of *Vicia faba* resulted decrease of mitotic index and chromosomal abnormalities.

Length of growing root tips has also been considered as perfect parameter as there is direct interaction of dividing cells with the changed environment as well as nutrients which causes significant modification in root length.[20,21] The elongation zone of the root tips grows due to expansion of cells and cellular differentiation.[22] During growth of elongation zone there is substantial increase in water uptake, sugar synthesis, plasma membrane elasticity, tonoplast elasticity and nitrogen metabolism. Any disturbance during cell division causes halting of lipid synthesis and increase in deposition of toxins which causes reduction in cell wall expansibility, loss of homeostatic regulations, enhanced cell necrosis, root growth inhibition and cellular toxicity.[23] Even shortening and decaying of roots has been reported to be evidence for cytotoxicity.[24]

Turkoglu (2007)[25] also reported that BHA exposure leads to inhibition of MI and DNA synthesis and the effect is more pronounced at higher concentrations. C-mitosis is the resultant of inhibition of spindle formation observed under the treatment of colchicines.[26] Colchicine acts as spindle poison with weak toxic effects and causes aneuploidy.[27] Failure in chromosome separation during anaphase leads to multipolarity and unequal translocation while anaphase bridge formation occurs due to fusion and breakdown of chromosomes and chromatids. Spindle formation disturbance results into multipolarity which mostly occurs when cells are treated with BHA.[10,15] Stickiness in chromosomes is further aggravated at higher chemical dosage which might be due to change in chromosomal proteins resulting in toxic effect ultimately causing cell death. Treatment with high concentration of BHA retards the movement of chromosome to opposite poles resulting in formation of laggards. Nitrites at lower concentration results in abnormalities while at higher concentrations and exposure for more duration leads to the swollen root tip cell and restriction of chromosomes at prophase. Chromosome breakage was also frequently reported in all groups which might involves an involvement of DNA distortion. BHA treated cells shows binucleated and lobulated nucleus which is an indicative of death process. Recommended dose of BHA shows low abnormalities because of the antioxidative properties and has been used to prevent rancidity in edible fats and oils and in fat containing foods. Kahl and Kappus (1993)[28] reported that BHA have less or no toxic effects but abnormalities at higher concentrations show genotoxic effects.[29-31]

CONCLUSION

The experimental outcome suggests that *Allium cepa* assay is significant and can be used for genotoxicity studies. On the basis of chromosomal aberrations like laggards, bridges and multipolar nuclei, BHA can be considered as toxic chemical. As BHA is commonly used for various purposes, it should be considered as major concern related with the health of organisms. For gaining proper knowledge about the mode of action of BHA more experimental studies should be conducted and till the time the basis of action is understood it must be used in limited doses.

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Ethnyl estradiol administration affects serum estrogen level of male *Clarias batrachus.*

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Abstract- Dose dependent effect of ethnyl estradiol was studied on the serum estrogen level of male *Clarias batrachus*. Fishes weighing 150- 200 g were used for the study. They were divided into four different groups. Experimental groups were fed with ethnyl estadiol (5µg, 10µg and 20µg Kg^{-1} feed) in the diet supplemented through premarin (oral contraceptive pill). Control group fishes were fed normal fish diet. After treatment for 30 consecutive days blood was collected and serum estrogen assayed by Enzyme Immunoassay. Results showed significant increase in blood estrogen levels in ethnyl estradiol treated fishes. Serum estrogen values were 0.1 µg/ml (control), 0.6 µg/ml (5 µg ethnyl estradiol), 1.0 µg/ml (10 µg ethnyl estradiol), and 1.3 µg/ml (20 µg ethnyl estradiol). A dose dependent effect was evident. The present results indicate that ethnyl estradiol has feminization effect on the male fish and may cause ovarian follicle development. At juvenile stage exogenous administration of ethnyl estradiol may result in gonadal malfunctions and suppress testicular development leading to sterilization or feminization. This may cause reproductive failure and result in reduced fish population in aquatic environment.

Key words: Ethnyl estradiol, *Clarias batrachus*, Students't'-test, ANOVA.

INTRODUCTION

Ethnyl estradiol is a synthetic estrogen widely used in oral contraceptive pills. Generally oral contraceptive pills contain 0.02 to 0.1 mg (20-100 µg) of ethnyl estradiol.[1] Ethnyl estradiol is partially metabolized and removed by urine and drains out to the sewage treatment plants and other aquatic environments. In surface water and sewage treatment plants, ethnyl estradiol frequently occurs in concentrations ranging between 1ng/l to 200-300 ng/l.[2,3] It has been reported that in response to low concentrations of ethnyl estradiol the fish synthesizes vitellogenin.[4,5] Aquatic organisms are highly threatened due to the harmful effects of endocrine disrupting chemicals, EDC.[6] Ethnyl estradiol is an EDC which causes vitellogenin induction in males, intersex gonads and reduced fertility in fish populations by reduction in the growth of gonads.[7] Many studies have documented that estrogens cause feminizing effects on masculine traits, like spermatogenesis.[8] It also affects the male secondary sexual characters as well as the male reproductive behaviour.[9] Reports show that ethnyl estradiol reduces the reproductive capacity in the Japanese medaka (*Oryzias latipes*), Zebra fish (*Danio rerio*) and in the rainbow trout (*Oncorhynchus mykiss*).[10,11] Ethnyl estradiol significantly reduces the reproduction of fishes at concentration of 10 ng/l. The exposure of estrogen during the period of fetal development, leads to irreversible changes in the tissue organization and this causes effects on fertility and

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behavior of the animal.[12,13] In fish, the effects of ethnyl estradiol on gonadal development, fertility and reproduction have been observed in laboratory experiments and also in wild fish.[14,15]

In Japanese medaka (*Oryzias latipes*), the developmental exposure to ethnyl estradiol during the first generation caused reduction in fertility and survival of embryo during the next two and three generations.[16] Estrogens are receptor mediated sex hormones. Ethnyl estradiol is a synthetic hormone that hinders reproductive function in humans.[17] The mode of its action is mediated via estrogen receptors. The mode of action of ethnyl estradiol in aquatic organisms is also expected to be via estrogen receptors, just like in mammalian systems.[17]

The Indian catfish magur (*Clarias batrachus*) is very hardy in nature[18] and can survive in adverse conditions. Considering the fact that the wastewater effluent and the rivers receiving the effluent have synthetic steroid hormones and their metabolites[19] and that most of the work has centered on fishes belonging to temperate zones the present study was aimed at observing the effects of ethnyl estradiol on the serum estrogen level in a tropical zone fish, *Clarias batrachus.*

MATERIALS & METHODS

Experimental animal: The study was performed on the males of the catfish, *Clarias batrachus*. Healthy males of *Clarias batrachus* weighing 150- 200g (14 ± 0.1 cm-18 ± 0.3 cm) were purchased from local breeders in the month of August. The fishes were transported in oxygenated bag and brought to the laboratory. The fishes were kept in glass aquaria under normal aquatic conditions at a temperature of 28±5°C. The fish were acclimatized for a period of 3 days.

Treatment and selection of dose: A toxicity bioassay was conducted to determine the LD_{50} for ethnyl estradiol according to the standard method opted by earlier worker.[20] LD_{50} of ethnyl estradiol was found to be 55µg kg^{-1} feed. The fish diet with increasing dose of ethnyl estradiol supplemented through premarin (Oral contraceptive pill), were provided daily to the fishes of different groups.

Experimental Design: The fishes were divided randomly into four different groups. Stocking density maintained for the experiment was 10 fish per aquarium. They were maintained in 4 separate glass aquaria. Control group fishes were fed normal fish diet. Experimental groups were fed with diets containing 5 µg, 10 µg and 20 µg of ethnyl estradiol per kg feed supplemented through premarin. The experiment was conducted for a period of 30 days. All the aquaria were aerated with air blower. The water exchange was done on daily basis.

The experiment was performed according to guidelines accepted by the "Local Animal Ethical Committee" for investigation on animals.

Biochemical Study: After treatment for thirty consecutive days, the blood was collected by puncturing the caudal vein using a medical syringe, which was previously rinsed with 2.7% EDTA solution and shaken gently in order to prevent hemolysis of blood. After collection of blood sample from fishes of four different groups, serum estrogen was assayed by Enzyme Immunoassay.[21]

Data Analysis: The data obtained was analyzed by Students't'-test and analysis of variance, ANOVA.[22]

RESULT

Results are presented in Table 1 and Figure 1. A dose dependent stimulatory effect of ethnyl estradiol was observed on the serum estrogen level of the fish. Fishes treated with ethnyl estradiol indicated significant ($p < 0.005$) increase in the serum estrogen level as compared to the control group fishes. ANOVA indicated a significant variation in serum estrogen level with respect to ethnyl estradiol dose ($F=23.41$, $p< 0.001$).

DISCUSSION

Results of the present study revealed that the level of serum estrogen is affected by exposure to synthetic hormone ethnyl estradiol. A dose dependent effect of ethnyl estradiol was evident. This finding is similar to the earlier reports in zebrafish where ethnyl estradiol alters sex differentiation and may even lead to sex reversal in males.[23] Exposure of male fishes to a high dose of 20 µg kg^{-1} feed of ethnyl estradiol resulted in significantly higher serum estrogen level (1.45 ± 0.06 ng/ml) as compared to that of control fishes (0.15 ± 0.02 ng/ml). The present results indicate that the exogenous administration of ethnyl estradiol on juvenile fish may cause gonadal malfunction and suppress the testicular development. It may cause development of ovarian follicle leading to sterilization or feminization.[24] Numerous effects of ethnyl estradiol on fish have been reported from all over the world such as deformities in the larvae of fish from North sea.[25]

Table-1 Serum estrogen levels (ng/ml) in male *Clarias batrachus* after treatment with different doses (µg kg^{-1} feed) of ethnyl estradiol supplemented through premarin (oral contraceptive pill) for thirty consecutive days

Dose of ethnyl estradiol	Serum estrogen
Control	0.15 ± 0.02
5 µg kg^{-1} feed	0.60 ± 0.01[a]
10 µg kg^{-1} feed	1.02 ± 0.04[a]
20 µg kg^{-1} feed	1.45 ± 0.06[a]

Data are expressed as mean ± SE for 10 fishes. Significance of difference from control: a, $p < 0.005$.

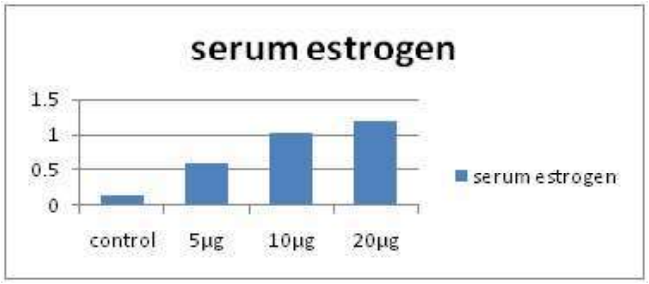


Figure 1- Dose of ethnyl estradiol (µg kg^{-1} feed) and the level of serum estrogen (ng/ml). Data indicated significant ($p< 0.005$) increase in the serum estrogen levels in fishes treated with ethnyl estradiol through premarin at the end of 30 days of experiment.

In United Kingdom, feminization was observed in male fish near outlets of sewage treatment.[26] A significant reversal of the steroidal system has been reported in carps from the Ebro River.[27] However, reports relating to fishes inhabiting tropical waters are lacking. The present results are new in the case of the freshwater Indian catfish, *Clarias batrachus*.

Ethnyl estradiol is the main active ingredient in contraceptive pill and it is excreted by women which then reaches the sewage treatment plants.[28] In women pharmaceutical estrogens, in oral contraceptives add upto the normal endogenous estrogen estradiol. It is possible that if the human fetus in utero is exposed to high doses of ethnyl estradiol, it may produce adverse developmental effects since in certain cases pregnancies do occur even after the use of oral contraceptives.[29] Ethnyl estradiol is a highly effective estrogen and it induces vitellogenin in male rainbow trout.[30] Intersex of gonads has been observed due to exposure of ethnyl estradiol in fathead minnow[31] and Japanese medaka.[32] Exposure to estrogen causes disrupted gonadal development and presence of intersex.[31,33] There are reports of arrest in development of testis and presence of ova- testis having primary stage oocyte after ethnyl estradiol exposure. In fish, this gender – bending effect has been identified. These results are in agreement with those of our present study with *Clarias batrachus* which exhibited a negative impact of ethnyl estradiol on its testes as evidenced by increase in serum estrogen levels. In the present work *Clarias batrachus* was exposed to ethnyl estradiol for duration of thirty days and this led to a significant change in the serum estrogen level of the fish. It may be opined that ethnyl estradiol exposure for a longer duration may cause adverse effects on the gonads of this fish. This requires further investigation. In tilapia (*Oreochromis niloticus*) it has been shown that exposure to high concentrations of ethnyl estradiol during the early life has long term consequences on growth, and gonads.[34] In the same study it has been reported that ethnyl estradiol exposure to tilapia influences serum levels of IGF-I and its expression in the liver and gonads. Review of literature suggests that in the aquatic toxicity base set fishes are the only taxa which have estrogen receptors.[35] Thus, ethnyl estradiol has a mode of action that binds to the estrogen receptors.

The present results clearly showed an elevation in serum estrogen level of *Clarias batrachus* after exposure to ethnyl estradiol for a period of thirty days. It is suggested that this compound affects the reproduction of the fish. Further studies are required to elucidate the complete role of ethnyl estradiol in the sexual differentiation as well as gonad development in this fish.

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Analytical studies on the effect of cestode parasites on some commercially important freshwater fishes of Madhepura region

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Abstract- Fish production plays an important role in the economy of a country. They are rich source of protein diet. There are many factors which effects the production of fishes, parasites infection is one of the main reasons for increasing mortality of fish. The present study was performed to find out the effect of parasites on some economically important freshwater fishes with special reference to cestodes (tapeworms), as two more types of parasites were there other than cestodes which can badly affect the fish production such as trematode (flatworms) and nematode (roundworms). A total of 160 commercially important fishes belonging to eight families were taken for this study. 20 each from the following species *Catla catla, Labeo rohita, Channa punctatus, Clarias batrachus, Cirrhinus mrigala, Heteropneustes fossilis, Mystus vittatus, Wallago attu* were taken. Extreme infestations were observed in the fishes. The production of fishes were decreased due to the infestation of cestode parasites. Very few work has been done in this topic till date, especially studies on cestode parasites. It is an attempt to find out those parasites which are particularly affecting these commercially important fishes.

Key words: Cestodes, Nematodes, Trematodes, fishes, infestation.

INTRODUCTION

Fishes are considered as good source of income and government of many countries promote fish farming and provides financial support to their citizens for the production. As fishes are very rich source of protein diet, many doctors also prescribe their patient to consume fish instead of red meat. Fish meat contains ample amount omega-3 fatty acids, vitamin D, vitamin B2, phosphorus, calcium, iron, magnesium, iodine, zinc and potassium.[1] It contains high quality protein with low fat, which keeps our heart and brain healthy. Fishes provide around one-sixth of worlds total protein.[2] Some studies prove that if people consume fish at least once a week, they are less likely to die of heart disease.[3] Fish have had a role in culture for centuries, religious symbols, serving as deities and many more.

Although much smaller than the host, the parasite can cause significant damage to its host. So, parasites are not passive organisms, merely 'riding' with their hosts; To be classified as a parasite, a normal organism must meet criteria that include the fact that its presence has a negative effect on its host.[4] Fish serve as host for parasites as they very diverse. Some parasites usually use fish as their intermediate host. They were found mainly in the intestine of fishes.

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MATERIALS & METHODS

Collection of the specimens

For the current study 160 species of commercially important fishes were taken, 20 species each of *Catla catla* (F. Hamilton, 1822), *Labeo rohita* (F. Hamilton, 1822), *Channa punctatus* (Bloch, 1793), *Clarias batrachus* (Linnaeus, 1758), *Cirrhinus mrigala* (Hamilton, 1822), *Heteropneustes fossilis* (Bloch, 1794), *Mystus vittatus* (Bloch, 1794), *Wallago attu* (Bloch & Schneider, 1801). Fishes were purchased from the local waterbodies with the help of fisherman. We have tried to purchase the fish of equal shape and size. Fishes were brought to the laboratory. Total length and weight of the fishes were measured in the laboratory. Their fins and gills were separated and were dissected from the ventral side. The body cavity, spleen, stomach, kidney, liver, intestine etc. all were searched for the presence of parasites. The gastrointestinal tract, the rectum to the esophagus were opened longitudinally and examined carefully.[5]

Table 1. Showing cestode parasites in different host along wwith their habitat

Sl. No.	Host	Cestode Parasites	Habitat
1.	*Catla catla* F. Hamilton, 1822	*Ligula intestinalis L.*	Intestine
2.	*Heteropneustes fossilis* Bloch, 1794	*Crecentovitus biloculus* Murhar 1963	Intestine
3.	*Heteropneustes fossilis* Bloch, 1794	*Djombangia penetrans* Bovien, 1926	Intestine
4.	*Heteropneustes fossilis* Bloch, 1794	*Procamallanus spiculogubernaculus* Agarwal, 1958	Intestine
5.	*Channa punctatus* Bloch, 1793	*Senga raoi* Majid *et al.* 1984	Intestine
6.	*Channa punctatus* Bloch, 1793	*Senga lucknowensis* Johri, 1956	Intestine
7.	*Channa punctatus* Bloch, 1793	*Gangesia paithanensis* Kadam *et al.* 1983	Intestine
8.	*Channa punctatus* Bloch, 1793	*Polyoncobothrium srivastavai* Pande *et al.* 2006	Intestine
9.	*Clarias batrachus* Linnaeus, 1758	*Lytocestus clariae* Tandon, Chakravarty and Das, 2005	Intestine
10.	*Clarias batrachus* Linnaeus, 1758	*Lytocestus longicollis* Ramadevi, 1973	Intestine
11.	*Clarias batrachus* Linnaeus, 1758	*Lytocestus birmanicus* Lynsdale, 1956	Intestine
12.	*Clarias batrachus* Linnaeus, 1758	*Djombangia penetrans* Bovien, 1926	Intestine
13.	*Cirrhinus mrigala* Hamilton, 1822	*Djombangia penetrans* Bovien, 1926	Intestine
14.	*Mystus vittatus* Bloch, 1794	*Lytocestus indicus* (Moghe, 1925) Woodland, 1926	Intestine
15.	*Wallago attu* Bloch & Schneider, 1801	*Gangesia punjabensis*	Intestine
16.	*Wallago attu* Bloch & Schneider, 1801	*Gangesia bengalensis*	Intestine
17.	*Wallago attu* Bloch & Schneider, 1801	*Gangesia vachai*	Intestine
18.	*Labeo rohita* F. Hamilton, 1822	No cestodes	

Fixation

Parasites were removed from the intestine of the infected species of fishes. The parasites collected were kept for relaxing, upon being fully relaxed parasite specimens were fixed in the fixatives prescribed for different helminthic group i.e. parasites were washed in 0.6% saline solution and then fixed in 5% formalin solution. After fixation parasites were identified, for this purpose they were stained in Alum carmine, dehydrated in glacial acidic acid, then after cleared in methyl salicylate and finally mounted in canada balsam.

RESULTS

Parasites were present in 88% of the fish specimens examined. In which *Clarias batrachus* and *Channa punctatus* were infected with most varieties of parasites, whereas *Catla catla, Cirrhinus mrigala, Mystus vittatus* were infected with only one species of cestode parasites, *Wallago attu* and *Heteropneustes fossilis* were infected with moderate number of species. The cestode infection were higher in *Clarias batrachus, Wallago attu, Channa punctatus, Heteropneustes fossilis* whereas minimum in *Catla catla, Cirrhinus mrigala, Mystus vittatus*. No cestodes were found in *Labeo rohita* probable reason for this may be, they are herbivores.

One of the species of cestode parasites i.e. *Djombangia penetrans* was common in *Clarias batrachus, Cirrhinus mrigala & Heteropneustes fossilis*.

Many researchers have studied about cestode parasite of freshwater fishes throughout the world.[6-10] In India also many workers have worked on the cestode parasitic of freshwater fishes.[11-15]

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Synthesis, spectral characterization and antimicrobial activity of Chromium (III) complex with 2-Amino phenol

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Abstract- A series of 2-aminophenols including methoxy group were synthesized using appropriate synthetic route. The structures of the compound were determined with spectroscopic techniques and analytical methods. IR spectra of the compounds including and not including hydrogen bonding were compared. The compounds 2 and 4 show the characteristic UV bands attributed to the NH-forms. On the other hand, antibacterial and antifungal activities of the compounds were investigated. Most of the compounds show selective activity toward *Staphylococcus epidermidis* and *Candida albicans*.

Key words: *Staphylococcus epidermidis, Candida albicans*, Chromium, UV bands

INTRODUCTION

2-Amino phenol is an organic compound having chemical formula C_6H_7NO with molar mass of 109.13g/mole. Along with its isomer 4-Amino phenol, it is an amphoteric molecule and a reducing agent.[1] It is useful reagent for the synthesis of dyes and hetrocyclic compound. Reflecting its slight hydrophilic character, white powder is moderately soluble in alcohol and can be recrystallized from hot water. 2-Amino phenol becomes quite popular in coordination chemistry because of its Schiff bases, which generate a rich variety of complexes.[2] Recently, the oxidation of many organic compounds with ditertiary butyl chromate has become a subject of great interest and many research works has been published in this present paper, we have reported the synthesis and study of some Chromium (III) complexes of 2-Amino phenol by oxidizing it with TBC to explore the versatility of the oxidizing agent.

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MATERIALS & METHODS

Preparation of reaction mixture

Substrate/oxidant ratio 1:1-1gm CrO_3 dissolved in 10 ml of tert butyl alcohol to prepare tert butyl chromate and 1gm 2-Amino phenol dissolved in 10 ml THF.

Substrate/oxidant ratio 1:2-1gm CrO_3 dissolved in 10 ml of tert butyl alcohol to prepare tert butyl chromate and 2.4gm 2-Amino phenol dissolved in 10 ml THF.

Substrate/oxidant ratio 1:3-1gm CrO_3 dissolved in 10 ml of tert butyl alcohol to prepare tert butyl chromate and 3.16gm 2-Amino phenol dissolved in 10 ml THF.

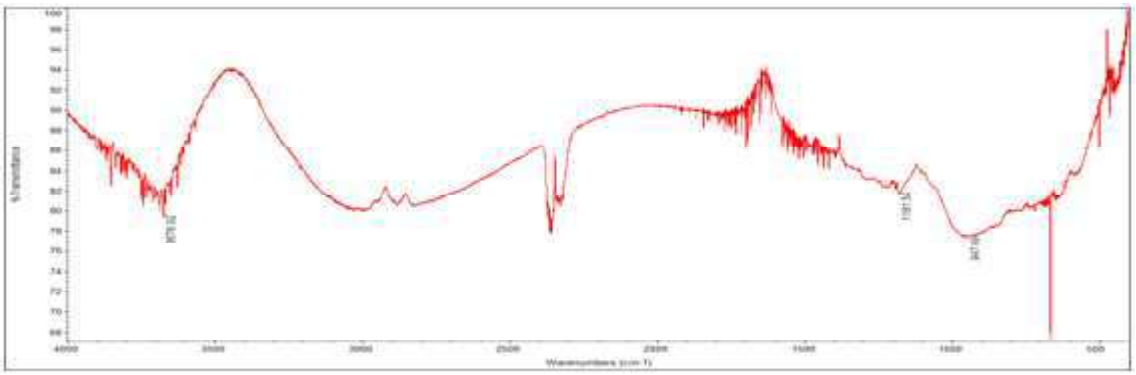
Procedure

2-Amino phenol was found to be freely soluble in selected solvents namely THF and 1,4-dioxane. Oxidation of this substrate with TBC was performed in two sets, each for the two solvents.[3] A solution of the oxidant TBC was prepared separately in situ from Chromium trioxide and tertiary butanol, varying the substrate/oxidant molar ratio as 1:1, 1:2 and 1:3 for each set. The two solution were mixed together to prepare a homogeneous reaction

mixture, stirred continuously for a considerable time using a magnetic stirrer and any change in consistency was noted. The mixture was heated in microwave oven.[4] Reaction were found to be exothermic, in general, except for those in 1,4-dioxane, in which they were endothermic as indicated by a drop in temperature of the reaction mixture. The six products formed in each of these cases were washed with acetone, dried and weighed.

Spectral character

Absence of characteristic C-H stretching frequency for an aldehyde group suggests that the amino phenol aldehyde group has been oxidized to a carboxylic group. This is further confirmed by the appearance of relevant peaks near 761.02 cm^{-1} for (C-O) stretching vibrations of a carboxylic group. The characteristic peaks for aromatic (C-H) stretching, aromatic (C-C) and (C=C) in-ring stretching. The presence of Cr-O and Cr=O bonds in these complexes is indicated by the (Cr-0) stretching frequencies ranging from 478.65-454.10 cm^{-1} and (Cr=0) peaks near 761.02, 667.55 cm^{1}. The broad and strong absorption bands appearing from 1161.69 947.69 cm^{1} shows the presence of the H-Bonded (O-H) group, intensity and peak broadening varying greatly with variation in substrate/ oxidant ratio. This indicates the difference in the extent of coordination of the (O-H) oxygen atom with the Cr atom.[5] The lowest (O-H) stretching frequency is centred at 3341.10 cm^{-1} in the compound labelled AP-101, while the maximum value recorded is for AP-111 at 3676 cm^{-1}. A strong (C-O) carbonyl absorption is visible near 1369.46 cm^{-1}. However, the vibrations near 1574.87-1556.97 cm^{-1}, 1488.19-1393 cm^{-1} and 1283.43-1203.11 cm^{-1}, respectively, indicate that the aromatic ring in amino phenol is intact and has not undergone any decomposition/ cleavage during the course of oxidation. The presence of (C-O) stretching vibrations for the methoxy group near 667.56 cm^{-1} in the IR spectrum of these complexes is indicative of the oxygen atom of the other group of Amino phenol being uncoordinated. This is supported by the weak alkyl (C-H) stretching vibrations appearing near 1147.07 cm^{-1} for each of these compounds. Other weak absorption bands for bending and rocking vibrations of (C-H) group were recorded near 686 cm^{1} and 478.65-454.10 $cm^{-1,}$ respectively.[6] A comparative spectral analysis for the six oxidation products of Aminophenol shows a considerable shift in the relative intensities and peak positions of these vibrational frequencies, which may be attributed to a difference in the percentage composition of Cr metal and variable extent of its coordination with the ligand. The strong peaks at 3341.22 cm^{1} and 2373 cm^{-1} are probably due to the impurities present in the KBr. A very weak peak near 3676.92 cm^{1} is characteristic of CO, absorption.



Antimicrobial Activity

This section will describe the antibacterial activities of Cr (III) complexes as well as their ligands.[1-3,7] It has also been proposed that chelation decreased the polarity of the metal ion mainly due to partial sharing of its positive charge with a donor group throughout the entire chelate ring system. This chelation process, therefore, enhanced the lipophilic nature of the central metal ion, which in turn facilitated its permeation through the lipid layer of the membrane, thereby allowing the metal complex to more easily cross the bacterial membrane, increasing the complexes' activity. In addition, many other factors, such as solubility, dipole moment, conductivity that influenced by a metal ion, could contribute to the antibacterial activity of the complex. On chelation, the delocalisation of pi-electrons over the whole chelate ring increases the liposolubility of the complexes. The hydrocarbon tail

serves as a lipophilic group to move the compound through the cell's semi-permeable membrane and block metal-binding sites within microorganism enzymes. The latter was attributed to the fact that complexes may deactivate different cellular enzymes, which play a vital role in the microorganism metabolic pathways. It was also suggested that the toxicant's ultimate action was the denaturation of one or more cell proteins, impairing normal cellular processes as a consequence. Other factors that also increased the antimicrobial activity were the solubility, conductivity and length of bond between metal and ligand.

A comparative study of Schiff base and its complex indicate that the metal complexes exhibit higher antibacterial activity than the free ligands.[4,8] This is probably due to the greater lipophilic nature of the complex. Such increased activity of the metal chelets can be explained on the basis of overtones concept and chelation theory. According to overtones concept of cell permeability the lipid membrane that surrounds the cell favours the passage of only lipid soluble materials due to which liposolubility is an important factor which controls the antimicrobial activity. On chelation the polarity of the metal ion will be reduced to a greater extent due to the overlap of the ligand orbital and partial sharing of positive charge of metal ion with donor groups.

RESULT & DISCUSSION

The expected broad peak for O-H stretching (H-bonded) at 3294.94 cm^{-1}, bidendate carboxylic acid functioning as ligand at 1489.05.cm, C-O stretching at 1230.58 cm^{-1}, O-C stretching at 1369.46 cm^{-1} and M-O stretching at 680.66 cm^{-1} are present in the curve. Similarly, the loss pattern is just what we expect for the formulation as shown. We observe that the FTIR curves (table -3) of sample R3 contain almost all the peaks which are expected for its formulation. We observe that the FTIR curves contain almost all the peaks which are expected for its Formulation $Cr_2O.CH_3COOH.CH_3COCOOH.2H_2O$. It is supported by the peaks at 3298.28cm^{-1} (O-H stretching), 964.41cm (C-O-H stretching) 1489.05 cm^{-1} (bidentate carbxylate ligand), 844.82 cm^{-1} (-COOH co–ordinated), 964.41.cm-1 (C-O stretching) etc. Also the loss pattern in TGA-DTA curve supports the proposed formulation. The first stage loss between the temperature ranges 31.98-140^oC corresponds to the fragmentation of CH_3COOH. The second stage loss of lattice water. The substrate might have been co- ordinated through two COOH groups, as the loss is taking at higher temperature.

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Effects of global warming & climate change- Opinions, thoughts and remedial strategies

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Abstract- Squandering of precious natural resources, burning of fossil fuels & release of CO_2 in the atmosphere acting as glass panel not allowing the heat reradiation in the outer space, suicidal pollution of rivers, lakes & oceans, unplanned construction activities, and Dam construction have caused immense destruction to biosphere resulting into global warming & climate change. The impact of climate change on the loss of biodiversity (the latter is critical in bringing resilience impacting fertility) is so great that time is not far when world would face an ecological collapse & if the worst came to the worst, the extinction of life. It is true that fossil fuel (coal), power the world economy yet if the consumption of coal, oil, gas continues unabated the environmental problem won't stop. The increasing population needs, forced humans to indulge in deforestation, killing wild animals, exploit resources, growth of industries, excessive use of pesticides, insecticide, fertilizes, use of coal for energy all have destroyed environment immensely. That an increase of temperature of 1.5 Celsius over pre - industrial level would substantially threaten life as the world would see heat wave, heavy rainfall, extreme weather conditions, shortage of water, reduced farm output, bleaching of corals, rise of sea level & many more events. The author have incorporated thoughts and opinions of different authors at one platform, discussed the consequences and have suggested certain remedial measures.

Key words: Global warming & climate change, consequences, thoughts & opinions remedial measures.

INTRODUCTION

The earth's biosphere is a complex system in which events and crisis occur which is indicated by non-linear unpredictable changes. The increasing CO_2 levels leads to melting of ice caps and increase in sea levels. This in turn would disrupt life of people from Islands of Maldives, Marshall Islands, Nauru and Tuvalu and other coastal low lying areas, their histories, traditions and cultures will be lost forever.

Climate change impacts insects and pests. The decline of a parasitoid, wasps and flies playing an important role in biological pest control has been observed at various places like Nepal (decline of natural enemies of pests) United States (decline of nearly 40% grassland bird index during 1968 and 2014). In addition to these extreme weather events cause disruption to species distribution and yields. Cold and windy days in spring disrupts pollination, rising temperature can shift the range of fishes impacting fisheries. Briefly understanding all these changes, the present authors have concentrated on what are the present scenario of global warming, its consequences and possible remedial measures. The present article includes thoughts and opinions of different authors who wrote on global warming.

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MATERIALS & METHODS

Different research papers and informations have been consulted and incorporated.

Present scenario

Pollution at glance

Oxygen in the atmosphere is an absolute necessity to live comfortably but nature has been exploited for acquiring comforts thus air has become polluted. In Delhi itself "the number of four wheelers have exceeded one crore. These vehicles emit CO_2, CO, SO_2, nitrogen hydrocarbons, etc in the air of urban areas, is now polluted.[1]

Industrialization has played a great role in polluting the air, smoke coming out of chimneys contain lead, mercury, copper, cadmium, arsenic & asbestos etc. Beside these, it also emits CO_2, CO, SO_2, H_2S hydrogen fluoride. Petroleum refinery too releases SO_2 & nitrogen oxide.

In many places, burning of solid biomass in poorly ventilated spaces add to environmental pollution. Nearly 1.3 million premature deaths in India have been reported by WHO – 2018. It also leads to approximately 30% of ambient air pollution in India. "One of the major constituents of house hold air pollution is PM 2.5 which affects local & regional climate change".[2]

The present authors opine that in order to reduce such pollution it would be worth to provide liquefied petroleum gas cylinders to each & every house hold. Ujjwala Yojana of the Govt. of India is an effort towards this venture.

Climate change resulting from growing green house gas (GHG) emissions from human (anthropogenic) activities is the prime cause of global warming and if the CHG emission is not controlled, the temperature of the earth is bound to increase. The affect of temperature increase can have potentially harmful effect on our ecosystems, biodiversity health, behaviour & livelihood, also many other events can happen.

Green house gas CO_2 is released in the atmosphere due to burning of fossil fuels (coal, oil, natural gas) etc or via domestic cooking. Apart from emissions from transport, industrial activity, changes in land use, agriculture (including live stock) & waste management are also responsible. Other short lived climate pollutants include black carbon, methane and ground level ozone, which along with other pollutants, particulate matters from the same sectors combine to aggravate air quality & cause climate change.[3]

Rajendran (2019)[4] observed that India's emission of CO_2 are low at 1.8 tonnes per capita, compared to the world average of 4.2 tonnes possibly due to reduction in the consumption of petroleum products and the decline in Cement production.

CONSEQUENCES

Health Hazards:-

In areas covered by snow such as Arctic & Himalayan region, the deposition of Black carbon onto ice & snow darkens the surface, increasing the absorption of solar radiation.[5] Higher temperature raises the level of ozone and other pollutants in the air which causes cardiovascular diseases.

In all probability it raises the level of nitrogen oxide (NOX), ozone (O_3), black carbon (BC), particulate matters, dust & floating pollen. These are pollutants to which people are exposed.

Dr. Nayar, HOD Respiratory Medicare Allergy and Sleep disorders at BLK Super - specialty Hospital opines that particles less than 2.5 can directly enter our respiratory system and can lead to breathlessness, cough, fever and even choking. Our nervous system gets affected & can experience headache, nausea or vomiting. It also affects heart.

Ozone

Ozone is a ubiquitous urban pollutant (Dobson unit is used for measuring the density of ozone; one Dobson unit = a layer 0.01 mm thin layer at 0-1 ATM (Atmospheric Pressure). It has been found that CO_2 is the main culprit of thinning of ozone layer. In human, inhalation of ozone in amounts low as 0.1 to 0.2 ppm is sufficient to cause toxicity characterized by edema, airways hyper – responsiveness, alveolar epithelium damage & impaired host defense.[6]

Acute inhalation of ozone is associated with an inflammatory response characterized by the accumulation of macrophages at the site of injury. Thus cells along with resident alveolar epithelial cells get activated & release cytotoxic & pro – inflammatory mediators.

Ozone & Agriculture

Surface ozone is generated by chemical reactions between primary pollutants such as oxides of nitrogen and volatile organic compounds in the presence of sunlight.

O_3 toxicity in plants has been studied by Elstner *et al*. (1991)[7]; Batini *et al*. (1995)[8]; Pitcher *et al*. (1996)[9]; Polle *et al*. (1995)[10]; Melhorn *et al*. (1994)[11]; Menser (1964)[12]; Castillo *et al*. (1987)[13]; Chameides (1989)[14]; Heritt *et al*. (1990)[15]; Kangasjarvi *et al*. (1994)[16]; Anttonen *et al*. (1996)[17]; Schraudner *et al*. (1996)[18]; Sandermann (1996)[19].

Like any other gas, O_3 enters the plant leaves through stomata as part of normal atmospheric gas exchange, goes in apoplast, it reacts quickly with biomolecules, water located there. Some of it, may be converted into secondary ROS (Reactive Oxygen Species) such as superoxide anion radicals ($\dot{o}_2$), hydroxy radicals ($\dot{o}H$), H_2O_2 and singlet oxygen – Hoigne *et al*. (1975)[20]; Koppenol (1982)[21]; Grimes *et al*. (1983)[22]; Byvoel *et al*. (1975)[23]; Kanofsky *et al*. (1995)[24]; Extremely higher dose of O_3 permeate to the underlying plasma membrane. O_3 reacts with apoplastic bimolecules & generate toxic species (lipid peroxidation products) that initiates damage to plasma membrane lipids & proteins resulting into decreased photosynthesis, electrolyte leakage and accelerated senescence.

Heat wave / heat stress - labour hours lost

In India 40 million heat wave exposure cases have been reported compared to 2012 there by the country lost nearly 75,000 million hours of labour in 2017 (equivalent to a year's work for 7% of working population) compared to about 43,000 million hours in 2000, an increase of more than 30 million labour hours lost in less than two decades.

Forest Fires

Dry and deciduous forests in the borderlands of Chattishgarh, Maharashtra & Telangana are affected by fires every year. Parts of western Himalayas such as Uttarakhand & Himachal Pradesh experienced severe forest fires. "Forest fires season" is time after winters when there are natural changes of forests catching fire due to rise in temperature.

Forest fires were also witnessed in Uttarakhand (Pauri, Garhwal, Rudra prayag, Nainital, Chamoli, Udham Singh Nagar, Bageshwar, Champawat & Pittoragarh. All this was reported by Roy (2018)[1]. According to the state forest department there have been 989 fire incidents between Oct. 1, 2020 & April 4, 2021 in Uttarakhand, destroying around 1,297.43 hectares of rich biodiverse forests. Since 2000 forest fires have affected over 48,000 hectares in the said state- report state forest. The causative factors are anthropogenic, carelessness & due to rise in temperature less of 'rain in winter etc.

REMEDIAL MEASURE

In order to combat increase in temperature strong international co-operation is needed to stop / minimize the causative factors to a level which does not destroy the environment. Following are advisable

1. Shift to low or zero emission
2. Electrifying transport
3. Development of green infra - structure
4. Increasing the energy efficiency.

Chandra Bhusan Deputy Director General of Centre for Science and Environment (CES) opines that "through it will be very difficult in the current global economic system to limit warming to 1.5°C, it is not impossible. This will require acting on all fronts to rapidly reduce emission by 2030 without an active participation of the United States, this will be impossible".

The present authors suggest that "plants with more CO2 absorbance capacity can be developed by 'gene editing technology' (CRISPR technology - i.e. clustered regularly interspersed short palindromic repeats".

The idea may appears to be fantastic yet an effort in the direction needs consideration.

There are trees available which imbibe air pollution which can be planted as for needs. Plants have to be selected on the basis of the types of pollutants, their intercity, location and easy availability & suitability of the climate conditions. Hydrogen is considered a source of clean energy. As a fuel it can run electric motos, or burned in internal combustion engine & has a number of industrial uses. It can be compressed or converted to liquid, easily stored.

Normally, hydrogen is produced through stream methane reforming and coal gasification. This raised the fuel's carbon foot print, despite it burns cleanly. The traditional catalysts through produce hydrogen from water yet it relies on expensive platinum group elements raising the cost and therefore large scale splitting becomes rather impractical.

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Isolation and screening of potassium solubilizing soil bacteria from Rhizosphere of paddy plant

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Abstract- Potassium solubilizing bacteria were isolated from the rhizosphere of the Paddy plant. Altogether seven isolates were collected from the 15 villages of Madhepura district. All isolates were cultured in Aleksandrow medium and colonies were further sub-cultured in nutrient medium. Morphological and biochemical test were performed for colony morphology, gram-staining, Citrate-utilization test, Voges-Proskauer test, Catalase test, Oxidase test and Methyl test were performed. For the determination of solubilizinig efficiency and solubilisation index bacteria were cultured in Aleksandrow medium, Halozone and colony diameter were measured and Solubilizing Efficiency (SE) and Solubilization Index (SI) were calculated. The highest value was observed in isolate no. K_1 (SE=11.33, SI=4.581) and the minimum for isolate no. K_4 (SE=5.66, SI=2.698).

Key words: Solubilization Index, Solubilizing Efficiency, Citrate test, Catalase test, Oxidase test

INTRODUCTION

Potassium is an essential macronutrient for plants. Among the three essential nutrients required by the plant one of them is Potassium. It plays a vital role in enzyme activation, osmotic regulation, opening and closing of stomata, starch synthesis, Protein synthesis and water nutrient transport. Soil contain large amount of potassium, however most of the Potassium is unavailable for plant uptake. Application of chemical fertiliser is one of the sources to fulfil potassium requirement of plants but chemical fertilizers have a considerable negative effect on soil. The eco-friendly agriculture system has emerged as an important thrust area globally for long-term soil environmental sustainability and to minimise the environmental pollution associated with extensive use of chemical fertiliser to conserve our existing resources and to minimise environmental pollution hazards.[1-5] Hence, the production and management of bio fertiliser containing KSB can be an effective alternative to chemical fertiliser. It is well-known that potassium solubilizing bacteria *Bacillus subtilis*, *Bacillus mucilaginous*, *Bacillus megaterium*, *Paenibacillus macerans*, *Paenibacillus agglomerans*, *Pseudomonas aureofaciens*, *Pseudomonas fluorescens*, *Agrobacterium radiobacter*, *Azospirillum brasilense*, *Azospirillum lipoferum*, *Azotobacter chroococcum* etc. can solubilize Potassium which become available to Plants.

These soil bacteria are commonly known as KSB (Potassium Solubilizing Bacteria). The diversity and ability of KBS depends upon soil and climatic condition. KSB are able to dissolve silicate minerals and release K through the production of organic and inorganic acids,

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Acidolyses, Polysaccharides, Complexolysis, Chelation and exchange reactions.

MATERIAL & METHOD

The soil samples were collected from rhizosphere of Paddy plants from different fields. Each sample consisted of 200 g of soil. Samples were collected in septic bags and transported to lab for further processing. Details of soil sample and culture media used are mentioned in Table no. 01.

Isolation of Potassium solubilizing bacteria:

1 gm of soil sample was mixed thoroughly in 10 ML autoclaved distilled water. Serial dilution was made up to 10^{-6}. Diluted soil samples were placed on Aleksandrow medium. The plates were incubated at 30°C for three days. Colonies exhibiting clear zones were selected as potassium solubilizer. Selected colonies were placed on nutrient agar medium for obtaining pure culture. The pure culture was again inoculated in Aleksandrow medium to confirm their K solubilizing efficiency.

Estimation of Solubilisation index and Solubilizing efficiency:

Qualitative analysis of K solubilizing activity of various isolates was calculated by measuring halozone and colony diameter on Aleksandrow medium. Halozone and colony diameter was measured using Calipers.

Calculation of Solubilization Efficiency (SE) and Solubilization Index (SI) were calculated by following formula:

$$SE = \frac{\text{Halozone} - \text{Colony diameter}}{2}$$

$$SI = \frac{\text{Halozone}}{\text{Colony diameter}}$$

RESULT

A total of 15 soil samples were collected from different Paddy fields of following villages of Madhepura district: Rampatti, Chousa, Budhma, Murliganj, Barbana. Details of soil sample, Date of collection, Culture medium and Isolate code are described in Table no. 01. All together seven isolates were selected as K solubilizing bacteria for morphological and biochemical characterisation. All isolates were examined for colony morphology, gram-staining, Methyl red test, Voges-Proskaurtest, Nitrate reaction test, Catalase test and Citrate test. The morphological characters are tabulated in Table no. 2. Biochemical characters are tabulated in Table no. 3. The value of Potassium solubilizing efficiency and Potassium solubilization index are mentioned in Table no. 4.

Table 1- Details of K-solubilizing Bacteria isolated from different soil samples

S. No.	Soil Sample	Place	Date	Culture medium	Isolate code
1	S_1	Rampatti	03-08-20	NA-AM	K_6
2	S_2	Barbana	05-08-20	NA-AM	K_2
3	S_3	Chousa	09-08-20	NA-AM	K_4
4	S_4	Rampatti	12-08-20	NA-AM	K_1
5	S_5	Rampatti	17-08-20	NA-AM	K_6
6	S_6	Budhma	22-08-20	NA-AM	K_5
7	S_7	Chousa	28-08-20	NA-AM	K_3
8	S_8	Barbana	05-09-20	NA-AM	K_1
9	S_9	Murliganj	08-09-20	NA-AM	K_4
10	S_{10}	Chousa	12-09-20	NA-AM	K_7
11	S_{11}	Barbana	18-09-20	NA-AM	K_4
12	S_{12}	Budhma	22-09-20	NA-AM	K_6
13	S_{13}	Murliganj	24-09-20	NA-AM	K_7
14	S_{14}	Budhma	26-09-20	NA-AM	K_1
15	S_{15}	Murliganj	29-09-20	NA-AM	K_2

NA: Nutrient-Agar medium
AM: Aleksandrow medium

Table 2- Morphological Character

Sl. No.	Isolate	Form	Elevation	Colony size	Colony colour	Cell shape	Gram staining	Remark
1	K_1	Regular	Flat	Small	White	Oval	-ve	*Cocci*
2	K_2	Regular	Flat	Small	White	Oval	-ve	*Cocci*
3	K_3	Irregular	Flat	Small	White	Rod	+ve	*Bacillus*
4	K_4	Irregular	Flat	Small	White	Oval	-ve	*Cocci*
5	K_5	Regular	Flat	Small	White	Oval	-ve	*Cocci*
6	K_6	Regular	Flat	Small	White	Oval	-ve	*Cocci*
7	K_7	Regular	Flat	Small	White	Oval	-ve	*Cocci*

Table 3- Biochemical Character

Sl. No.	Biochemical Test	KBS Isolates						
		K_1	K_2	K_3	K_4	K_5	K_6	K_7
1	Citrate Utilization	+	+	+	+	+	+	+
2	Voges-Proskauer	+	+	+	+	+	+	+
3	Catalase	+	+	+	+	+	+	+
4	Oxidase	+	+	+	+	+	+	+
5	Methyl red	+	-	+	-	+	+	+

Table 4- K-Solubilization efficiency and K-solubilization index of Bacterial Isolates

S. No.	Isolate	Colony Size (mm)	Solubilization Zone (mm)	Efficiency (mm)	Solubilization Index
1	K_1	6.33	29.00	11.33	4.581
2	K_2	7.2	21.00	6.9	2.916
3	K_3	6.45	25.00	9.27	3.875
4	K_4	6.67	18.00	5.66	2.698
5	K_5	6.57	22.00	7.71	3.348
6	K_6	7.3	26.00	9.35	3.561
7	K_7	6.75	23.00	8.12	3.407

CONCLUSION

Potassium is one of the most essential nutrients for plant which is absorbed by plant from the soil. Potassium solubilizing bacteria fulfil the requirement of Potassium by the plants. In the present investigation, altogether 7 isolates of Potassium solubilizing bacteria were isolated from rhizosphere of Paddy plant. All isolates were tested for their morphology, biochemical test and Potassium solubilizing capacity. One isolate (K_3) was gram positive while all other were gram negative. The biochemical test for Citrate-utilization, Voges-proskauer, Catalase and Oxidase were positive for all isolates. Methyl test for isolates no. K_2 and K_4 were negative and for all other was positive. Solubilizing efficiency (SE) and Solubilization index (SI) was highest for isolate no. K1 (SE=11.33, SI=4.581). SE and SI of isolate no. K_4 was minimum (SE=5.66, SI=2.698).

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Parental Mark - Genome Imprinting

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Abstract- Genome imprinting is a vital genetic mechanism in mammals and is thought to influence the transfer of nutrients to the foetus and the newborn from the mother versus father. Imprinting defects can involve multilocus or isolated epigametic changes that may have no evident genetic cause or imprinting disruption can be traced back to alteration of cis-acting elements or transacting factors that control the establishment maintenance and erasure of germiline epigametic imprints. The study at transgene methylation has shown that methylation pattern can be in parent origin in specific manner 3'- 7' suggesting that DNA Methylation may play role in genomic imprinting.

Key words: Heredity, Mendel, Genome Imprinting

INTRODUCTION

Heredity is one of the fundamental characteristics of all life on earth. Naturally, it has been a subject of intense investigation and keen observation since early days. Mendel in the late 19th century unveiled the basic principles guiding the hereditary traits. This virtually became an axiom for the geneticists. But later investigators found some genes showing non-Mendelian behaviours. Genome imprinting is one of such exceptions. It contradicts Mendelian principle of equivalence of traits in reciprocal crosses and shows that the sex of the parents contributing the gene does matter in its expression in the offspring. It is a recent finding. It became known in 1960's after Helen Crouse's experiments on insects. But the interest in the phenomenon really took off in mid 1980s. By now it is well known in Humans triggering several genetic diseases. A large number of investigators are actively engaged in understanding its mechanism and working.

Genome imprinting is the process which imprints or marks the genes passed by males and females in a way that some genes function properly when they are donated by mother while others do this only when they come from father. The phenomenon, thereby, includes all cases where genes have differential expressions depending upon the sex of the parents who inherit it.

The imprints are actually a special kind of modification in gene structure. These arise from varied activity of genes that operate differently in males and females. The imprints are specific to the sex of the individual. Consequently, the behaviour of the alleles varies in effect in succeeding generation. The offspring's

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that receive the gene from mother behave differently from those that receive it from father. Even when parents contribute identical genes, the genes have different effects. However, the imprints are not a permanent modification in gene or chromosome. These do not persist in the next generation. They are, rather, temporary and erasable. Old imprints are erased in each generation and new ones specific to sex of the individual are made during gamete formation.

The sperm and egg cells have chromosomes with new imprints, which are then passed on to the offspring. Thereby both sexes inherit some genes with paternal imprints and some with maternal imprints. The imprints result in differential expressions of alleles during development and in adult.

The mechanism of genome imprinting is not very much clear. Several molecular geneticists consider it as a consequence of vastly different physiology and biochemistry of the gametes. The data, so far available, indicate that the imprints lie within the gene structure itself and are a product of differential Methylation.[1] Methylation occurs after the base has been incorporated into DNA during replication. It is accomplished by ubiquitous nuclear enzyme Methylases. Different genes are methylated differently. Experiments on state of Methylation clearly showed that it is directly associated with the expression of genes. It is critical at specific site or in a restricted region. Many genes are not methylated at 5' end when they are expressed, although they are Methylated at 3' end.

However, the pattern of Methylation is variable in two sexes and hence allelic expressions differ. A gene is more methylated in females than the Males. The more highly methylated a gene the less likely it is to be expressed. The pattern changes as the gene traverses through alternating sexes. The highly Methylated genes become demethylated in Mother's son, Who then makes a new mark, and pass on to its offspring. In other words, female imprints are erased in son and a now one reflecting the male pattern is put on to it. Thereby, genes are methylated differently because of their maternal and paternal origin.

The pattern is believed to be guided by alterations in the chromatin surrounding the gene(s). This implies that the genes lying in tightly wound chromatin regions or in its vicinity are more likely to be imprinted upon. Based on the observations that winding protein may alter gene expression Sapienza *et al.* (1989)[2] suggested that such proteins dictate the process of genome imprinting.

Functions of genome imprinting are also not very clearly understood. There are contrasting views. Some people believe it to be a special case of "Dominance Modification". Many traits respond to the activity of other genes that modify their expressions. The modifier gene is postulated to control imprinting act differently in males and females.

While a widely held view is that imprinted genes regulate growth during development. The opinion is based on the nuclear experiments in Mouse embryo by McGrath and Solter (1984)[3]. It was observed that imprinting process selectively turned off some genes that would act otherwise during development. It was also seen that paternal over expression increased growth and maternal over expression reduced it. It is, thereby, obvious that the differential modification depended upon whether it was maternally or paternally inherited. This view is also substantiated with the evolution of genome imprinting in Mammals alone among vertebrates. Mammalian embryo is nourished directly from maternal tissues. The genes that are expressed in embryos can influence the quantity of resources that the offsprings receive from their mother. But significant effects are not expected in oviparous taxa where the amount of yolk is predetermined.

There are other suggestions also. Moore and Haige (1991)[4] suggested that imprinting in Mammals has evolved to prevent parthenogenesis. Parthenogenetic reproduction is often presumed to have short-term benefit for females but long term disadvantage for the species.

A few are of the opinion that genome imprinting allows flexible gene expressions and sophisticated control.

The problems of genome imprinting remain open for research. For example, it has not been conclusively proved that Methylation is the primary cause for genome imprinting or it merely reflects some hidden and more fundamental biochemical event. Besides, there are also difficulties in assuming that the state of Methylation provides general means for controlling gene expression. In Drosophila melanogaster and other Dipterans there is no methylation of DNA. The other differences between active and inactive, chromatin is the same as in other species that display Methylation, Furthermore, suggestions so far made, do not satisfactorily explain why

among vertebrates mammals alone should require such mechanisms. Future investigations will answer these. The prominent workers engaged in such investigations are Rotondo *et al.* (2013)[1], Sapienza *et al.* (1989)[2], McGrath and Solter (1984)[3], Moore and Haig (1991)[4], Reik and Walter (2001)[5], Barton *et al.* (1984)[6], Herrick and Seger (1999)[7], Ferguson (2011)[8].

CONCLUSION

Sex of the parents does matter in the expression of genes in offsprings. Each parent imprints its genome in a way that some genes function properly when they are donated by mother while others do so only when they come from father. The Imprints are modification in gene structure. Their pattern is specific to sex of the individual. This results in differential expression of alleles during development and growth.

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Composition and abundance of benthic macro invertebrates in floodplain lakes (chaurs) of North Bihar, India

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Abstract- Floodplain wetlands in North Bihar, India are common features of river Gandak and Koshi river basins which have been used for a number of activities such as fisheries, agriculture and other human activities. Species composition and abundance of macro-invertebrates were studied in the floodplain lakes (chaurs). This study was conducted from July, 2019 to June, 2020. Samples were collected monthly at two floodplain lakes of Madhepura district of North Bihar. Macro-invertebrates of the lakes consist of 5 groups, viz., Oligocheata, Hirudinea, Insecta, Gastropoda and Pelecypoda. In total, 21 species of benthic macro-invertebrates were recorded, of these, 19 species were recorded at Ambhobasa chaur and 15 species at Doura chaur. The density of benthos was higher in poor-water condition at Doura chaur varied from 105 individuals/m^2 to 1922 individuals/m^2 with mean of 973.74±122.42 individuals/m^2, however lower in clean-water environment at Ambhobasa chaur ranged from 85 individuals/m^2 to 1443 individuals/m^2 with mean of 713.08±87.96 individuals/m^2. Species composition and abundance of the benthic fauna were almost same not differ much between the studied lakes, however minor differences could be due to heterogeneity of the habitat. Gastropoda was the most dominant group of zoo-benthos contributing of 42.65% and 43.18% followed by Oligochata 21.24% and 30.89%, Insecta 19.84% and 21.37%, Bivalvia 14.46% and 3.29% of total benthic fauna at Ambhobasa chaur and Doura chaur respectively. Benthic fauna abundance shows seasonal variation, higher in winter followed by summer and lower during monsoon. The results of this study will help to update information on benthic invertebrate resources of North Bihar.

Key words: Floodplain lakes, benthic macro-invertebrates, species composition, abundance

INTRODUCTION

Benthic invertebrates are those organisms that live in or on the bottom deposit of a body of water.[1] Benthic fauna are aquatic bottom-dwelling animals can be seen with the naked eye without using microscope, collectively referred to as macro-zoo benthos or benthic macro-invertebrates. Benthic invertebrates play a key role in ecological processes, such as food chain dynamics, nutrient cycling and recycling, productivity and decomposition.[2,3] Benthic invertebrates are important and integral part of aquatic food chain forms a vital link between primary producers, unavailable nutrient and energy in detritus and higher trophic levels in aquatic food web.[4,5] Most benthic fauna feed on debris that settles on bottom of a water body. Benthic fauna are of great significance in fisheries as they constitute main food source for many bottom feeder fishes. Benthic invertebrates are regarded as best biological indicator use

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for the assessment of water quality and overall aquatic ecosystem health.[6,7]

Wetlands associated with floodplain of major rivers (floodplain wetlands) are common feature of Indian landscape particularly along the Ganga and Brahmaputra river systems covering area of 0.2 million ha.[8] Not much attention has been given to study composition and abundance of benthic invertebrates existing in the floodplain wetlands of India. However, macro-invertebrates of the floodplain wetlands of this country have been studied by some workers.[3,9-12]

Many floodplain wetlands are available in North Bihar, mostly in Gandak and Koshi river basins either in form of floodplain lakes (tectonic-lakes) or ox-bow lakes (cutoff portion of river meander) or residual channels, locally called *chaurs*, *manus* and *dhars*, respectively. Floodplain wetlands cover an area of 40,000 ha,[13] where fish production is mainly depending on the natural food including benthos. Despite of their key role in aquatic ecosystems, the studies on benthic fauna inhabiting in the floodplain lakes is still meager. The main objective of the present study is to determine composition and abundance of benthic macro-invertebrate fauna in two floodplain lakes of the district Madhepura of North Bihar.

MATERIALS & METHODS

Study Area

The field work is carried out in the floodplain lakes of the district of Madhepura of North Bihar. Among the selected floodplain lakes, Ambhobasa *chaur* is located at 25°34'48" North latitudes and 86°52'28" East longitudes, however Doura *chaur* at 25°30'59" North latitudes and 86°56'32" East longitudes (Fig. 1). Studied lakes receive water during the monsoon season from flood and rainfall and contain abundant macrophytes belonging to several species.

Sampling

Sampling was initiated in July, 2019 and continued up to June, 2020. Sediment substrate samples were collected with the help of an Ekman's dredge of 15.5×15.5 cm^2. Monthly three replicate samples were taken from each floodplain lake.

Sample Preparation

Sediment samples collected were taken into a labeled polyethylene bag and transferred to a bucket and mixed with water. Then mixed water passed through a standard brass sieve No.40 (256 meshes per sq. cm.). All benthic fauna retained on sieve were transferred to specimen bottle and fixed with 10% buffered formalin for detailed examination in the laboratory.

Sample Analysis

In the laboratory, for identification, samples were taken into a white enamel tray for easy vision. The animals were then sorted out, enumerated to different taxonomic groups using brush and forceps and identified up to genus or species level using appropriate literature. [14-18] Hand lens and low power binocular microscope were used to identify benthic fauna. Benthic invertebrate was computed in terms of individual per meter square.[19]

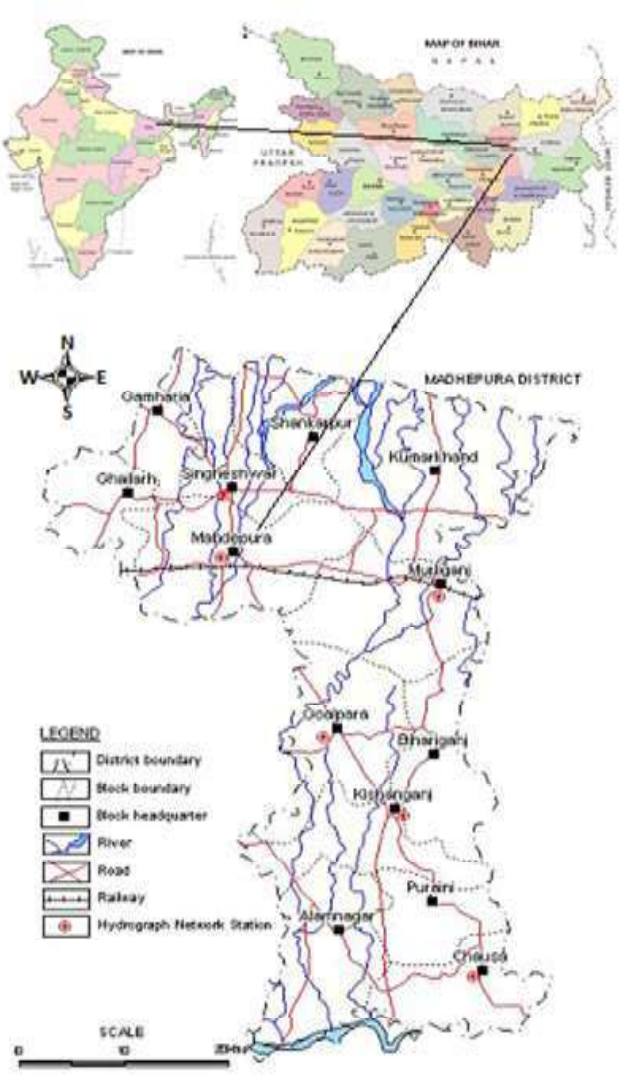


Figure 1- Location map of Madhepura district of North Bihar (India).

RESULTS & DISCUSSION

Composition and Abundance

Benthic invertebrates recorded in two floodplain lakes (*chaurs*) are presented in Table 1. Variation in the density of the total benthos is shown in Fig. 2. The studied lakes support a great diversity of benthic invertebrates. Altogether, 21 species of benthic macro-invertebrates were recorded from both lakes comprises of 5 major groups, such as Oligocheata, Hirudinea, Insecta, Gastropoda and Pelecypoda belonging to 3 major invertebrate phyla (Mollusca, Annelida and Arthropoda). In total, 19 species of benthos belonging to Oligocheta with 2 species, Hirudinea 2 species, Insecta 3 species, Pelecypoda 4 species and Gastropoda 8 species were

Table 1: List of benthic macro-invertebrates recorded in the studied wetlands.

Macro-invertebrates	Ambhobasa Badi *chaur*	Doura *chaur*
Phylum : Annelida		
Class : Oligochaeta		
Family: Naididae		
1. *Branchiodrilus semperi* (Bourne, 1890)	-	+
Family: Tubificidae		
2. *Limnodrillus hoffmeisteri* (Claparède, 1862)	+	+
3. *Branchiura sowerbyi* (Beddard, 1892)	+	+
Class : Hirudinea (Leeches)		
Family: Glossiphoniidae		
4. *Glassophonia* (Johnson, 1816)	+	+
Family: Hirudinidae		
5. *Helobdella* (Linnaeus, 1758)	+	-
Phylum : Arthropoda		
Class : Insecta		
Family: Chironomidae		
6. *Chironomus* sp.	+	+
Family: Lestidae		
7. Damselfly nymph	+	-
Family: Gomphidae		
8. Dragonfly nymph	+	+
Phylum : Mollusca		
Class : Bivalvia (Pelecypoda)		
Family: Unionidae		
9. *Lamellidens marginalis* (Lamarck, 1819)	+	+
10. *Parreysia* (*parreysia*) *favidens* (Benson, 1862)	+	-
Family: Corbiculidae		
11. *Corbicula bensoni* (Deshayes, 1854)	+	+
12. *Corbicula striatella* (Deshayes, 1854)	+	-
Class : Gastropoda		
Family: Viviparidae		
13. *Bellamya bengalensis f. typica* (Lamarck, 1822)	+	+
14. *Bellamya crassa* (Benson, 1836)	+	-
Family: Ampulariidae		
15. *Pila globossa* (Swainson, 1822)	+	+
Family: Bithyniidae		
16. *Bithynia* (*Digoniostoma*) *pulchella* (Benson, 1836)	+	+
Family: Thiaridae		
17. *Thiara* (*Melanoides*) *tuberculata* (Müller, 1774)	+	+
Family: Lymnaeidae		
18. *Lymnaea acuminata f. typica* (Lamarck, 1822)	+	+
Family Planorbidae		
19. *Gyraulus convexiusculus* (Hutton, 1849)	+	+
20. *Gyraulus euphraticus*	+	-
Family Bullinidae		
21. *Indoplanorbis exustus* (Deshayes, 1834)	-	+
Total Zoo-benthos	19	15

+ Present, - Absent

recorded at Ambhobasa *chaur*. Although,15 species belonging to Oligochata with 3 species, Hirudinea 1 species, Insecta 2 species, Pelecypoda 2 species and Gastropoda 7 species were collected at Doura *chaur* (Table 1).Number of species was the maximum in clean-water environment at Ambhobasa *chaur* than the poor-water condition at Doura *chaur*, but no significant differences were found in composition of benthic invertebrates. Previous workers have reported different diversity and density of benthos in different floodplain wetlands of India. Khan (2002)[20] reported 78 taxa of benthic fauna from ox-bow lakes, Beri Gopalpur *baor* and Sosadanga *baor* of North 24 Parganas district of West Bengal. Sugunan *et al*. (2000)[21] recorded benthic organisms from *beels* of West Bengal dominated by molluscs, insects and annelids. Sharma and Agrawal (2011)[22] reported 15 species of Gastropoda belonging to 2 orders and 6 families and 7 species of Pelecypoda belonging 2 orders and 2 families from flood plain wetlands of North Bihar. Sharma and Agrawal (2011)[22] reported 14 Gastropoda species belonging to 6 genera and 6 species of Pelecypoda belonging to 3 genera from Surha *Tal* of Ballia district of Uttar Pradesh. Patial *et al*. (2015)[23] recorded 18 species of benthos during pre-drought and 17 in post drought period in a floodplain wetland of Vaishali district of Bihar. Singh *et al*. (2019)[3] recorded 26 species of benthic macro-invertebrates belonging to 17 genera and 17 families from floodplain wetlands of Madhepura district of North Bihar. Singh and Sharma (2020)[24] recorded 29 species of benthic invertebrates from high altitude wetland, Dodi Tal of Garhwal Himalaya.

Density of benthic macro-invertebrate population ranged from 85 individuals/m^2 to 1443 individuals/m^2 with mean of 713.08±88 individuals/m^2 at Ambhobasa *chaur*, and varied from 105 individuals/m^2 and 1922 individuals/m^2 with mean of 972.74±122.42 individuals/m^2 at Doura *chaur* (Table 2). The highest density of benthos reflects eutrophic condition in the studied lakes agree with the findings of previous workers.[25] Density of benthos in both lakes were fairly same not differ much from each other, though minor difference observed could be due to heterogeneity of the environments, such as difference in water quality, sediment types and food quality and quantity.[26] Benthic macro-invertebrates showed seasonal

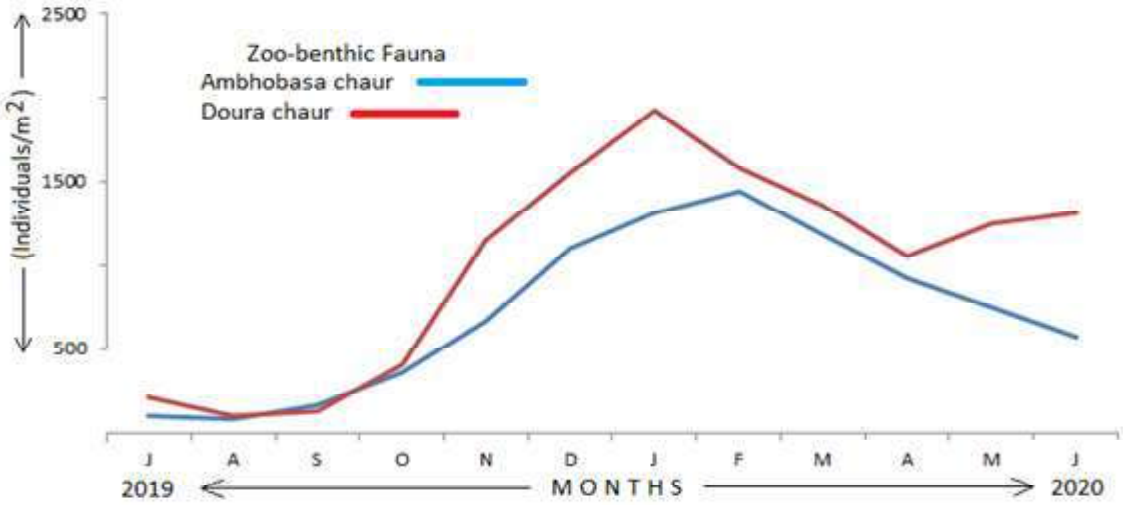


Figure 2- Richness in benthic fauna in Ambhobasa badi *chaur* and Doura *chaur*.

Table 2- Range and mean ± standard error (±SE) of benthic fauna (individual/m^2)

Macro-invertebrates	Ambhobasa badi *chaur*		Doura *chaur*	
	Min. - Max.	Mean ± SE	Min. - Max.	Mean ± SE
Oligochaeta	21 - 389	185.45 ± 22.74	21 - 691	328.73 ± 44.11
Hirudinea	21 - 43	24.23 ± 2.19	21 - 43	27.6 ± 3.36
Insecta	21 - 345	157.45 ± 21.98	21 - 455	226.77 ± 28.12
Pelecypoda	21 - 257	105.21 ± 13.27	21 - 108	45.18 ± 6.41
Gastropoda	42 - 710	310.25 ± 41.60	63 - 951	420.04 ± 54.33
Total Benthos	85 - 1443	713.08 ± 87.96	105 - 1922	972.74 ± 122.42

variation, the maximum density was observed in winter followed by summer and the minimum during monsoon season (Fig. 2). Similar trend was also observed by the earlier workers in the floodplain wetlands of this country.[3,9,20] Higher density was observed in winter could be due to decreased temperature, increased dissolved oxygen, medium to high water level and low predation pressure. The abundance and composition sharply declined during monsoon season could be attributed to dilution factors, incoming rain water and flood water dilutes summer population of benthos.

Different groups of benthic invertebrates showed different requirement to survive in the floodplain wetlands. Morphometery and size of the lake, discharge of domestic sewage, run-off from agriculture field and weedy environment brought significant alteration in water quality. The species composition and composition of benthic fauna depend on water quality of their habitat. Clean-water environment increase probabilities of survival of animals might be responsible for decreased density and increased diversity of benthos at Ambhobasa *chaur*, often dominated by pollution-sensitive benthos. However, poor-water condition might be accountable for increased density and decreased diversity at Doura *chaur*.

In terms of abundance and diversity, Oligochaeta was the second most dominant group of benthic fauna contributing 21.24% at Ambhobasa *chaur* and 30.89% at Doura *chaur* (fig. 3). The total density of Oligochaeta ranged from 21 individuals/m^2 to 389 individuals/m^2 at Ambhobasa *chaur* with mean density of 185.45±2.74 individuals/m^2, however varied from 21 individuals/m^2 to 691 individuals/m^2 at Doura *chaur* with mean of 328.73±44.11 individuals/m^2 (Table 2). The number of Oligochaeta was higher throughout the period of study, except monsoon when they were typically absent or less abundant. *Limnodrilus hoffmeisteri* is small, thin worm existing in all freshwater habitats worldwide.[27] In this study, *Limnodrillus hoffmeisteri* was most dominant oligochaete worm and along with *Branchiura sowerbyi* recorded in both studied lakes. *Branchiodrilus semperi* recorded only in Doura *chaur* (Table 1). Brinkhurst (1975)[4] reported occurrence of *Limnodrilus hoffmeisteri* with high abundance in the organically enriched aquatic environments often with *Tubifex tubifex*. Though, in this study, *Limnodrilus hoffmeisteri* were recorded with high abundance with *Branchiodrilus semperi* at Doura *chaur* where muddy-bed contain huge amount of organic matter frequently detritus. Prior to this study *Limnodrillus hoffmeisteri* has been also reported from freshwater bodies of this region.[3,28]

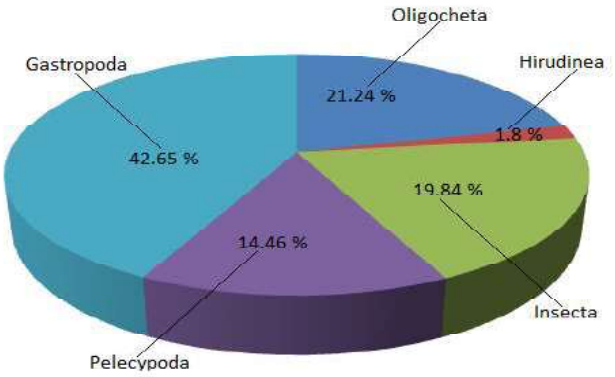


Figure 3- Relative composition of macro-invertebrates in Ambhobasa badi *chaur*.

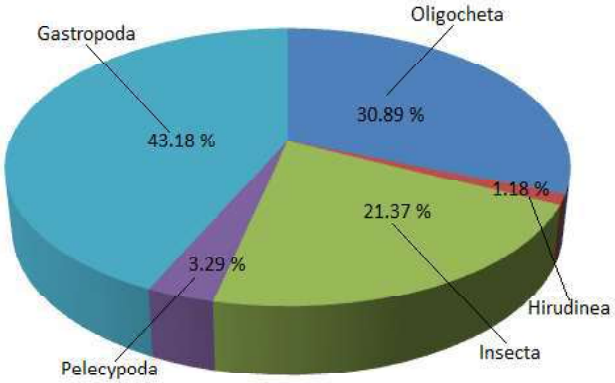


Figure 4- Relative composition of macro-invertebrates in Doura *chaur*.

In total, 2 species of Hirudinea (Leaches) were recorded from the studied lakes. Of these, *Glassophonia* was recorded in both lakes, though *Helobdella* only in Ambhobasa *chaur* (Table 1) contributing 1.8% and 1.18% of the total benthos (Fig. 3). Leaches were recorded with low abundance. Factors structuring leech community is not clearly known, however, their presence indicates organically-enriched environment.[29] The present study supports facultative nature of Hirudinea as recorded from both clean-water environments at Ambhobasa *chaur* and poor-water conditions at Doura *chaur*.

Insecta was the prominent group of benthic fauna. Benthic insects consist of one larva and two nymphs of flying insect. Density of benthic insects ranged from 21 individuals/m^2 to 345 individuals/m^2 with mean of 157.45±22 individuals/m^2 at Ambhobasa *chaur*, though,

varies from 21 individuals/m^2 to 455 individuals/m^2 with mean of 226.77±28 individuals/m^2 at Doura *chaur* (Table 2) contributing 19.84% and 21.37% of the total benthos (Fig. 3). *Chironomus* (dipteran larvae) was most dominant taxa. They were pollution-tolerant facultative organisms, thus found in both lakes.[30] Density of *Chironomus* larvae was generally high in poor-water condition at Goura *chaur* as they were well-adapted to oxygen-deficient environments[31] as their blood contains hemoglobin which traps oxygen[32]. Nymphs recorded from both lakes were juveniles of adult flying insects. Nymphs of damselfly and dragonfly were recorded in oxygen-rich water at Ambhobasa *chaur* as they need additional oxygen for their cellular metabolism.[33] Nymph of damselfly was absent in Doura *chaur* could be due to poor-water conditions. Nymphs are pollution-sensitive benthos restricted to clean-water environment,[34] their absences reflect water may be polluted. Benthic insects feed on bottom debris, planktons and other micro-organisms available in the floodplain lakes, though they themselves form leading food items of many fishes.

Pelecypoda (Bivalvia) found buried in the sediment on the lake-bed and majority is filter feeders. In total, 4 species of Pelecypoda was recorded from both the lakes (Table 1). Population density of Bivalvia ranged from 21 individuals/m^2 to 257 individuals/m^2 with mean density of 105.21±13.27 individuals/m^2 at Ambhobasa *chaur*, however varied from 21 individuals/m^2 to 108 individuals/m^2 with mean of 45.18±6.41 individuals/m^2 at Doura *chaur* (Table 2). Density of Pelecypoda was high at Ambhobasa *chaur* than at Doura *chaur*. All the 4 species of bivalves, like *Lammellidens marginalis, Parreysia* (*parreysia*) *favidens*, *Corbicula bensoni* and *Corbicula striatella* were recorded at Ambhobasa *chaur* (Table 1) contributing 14.46% of the total benthos, though only one species, *Corbicula* was recorded at Doura *chaur* constituting 3.29% (Fig. 3). Bivalvia population was maximum in summer and minimum during many months at Ambhobasa *chaur*, though the higher in winter and lowest during many months at Doura *chaur*. Ecological condition of floodplain lakes, such as the depth, altitude, temperature and physical stability of bottom substrate, all have varying degree of influence on species composition and abundance of bivalves. Occurrence of Bivalvia and their population structure allow easily receiving necessary information about water condition.[35] However, the present knowledge of this group of benthic fauna is very poor.

Gastropoda (snails) was the most dominant benthic fauna contributing major share to total molluscan fauna of both lakes. Gastropods showed a variety of shapes, in total, 9 species belonging to 7 genera and 7 families were recorded in this study. Out of these, 8 species were recorded at Ambhobasa *chaur* and 6 species at Doura *chaur* (Table 1), contributing 42.65% and 43.18% of the total benthic fauna (Fig. 3). Density of Gastropoda ranged from 42 individuals/m^2 to 710 individuals/m^2 at Ambhobasa *chaur* with mean of 310.25±41.60 individuals/m^2, however varied from 63 individuals/m^2 to 951 individuals/m^2 at Doura *chaur* with mean of 420.04±54.33 individuals/m^2 (Table 2). Maximum density of Gastropoda was recorded in winter and minimum during monsoon in both the lakes (Fig. 2). Among gastropods, *Thiara* (*Melanoides*) *tuberculata*, *Bellamya bengalensis f. typica* and *Pila globossa* were numerically most abundant at Ambhobasa *chaur* and rest was less abundant. Though, *Bellamya bengalensis f. typica* was most dominant Gastropoda followed by *Thiara* (*Melanoides*) *tuberculata* and *Bithynia* (*Digoniostoma*) *pulchella* at Doura *chaur*. Earlier workers reported Gastropoda as characteristics benthos of organically enriched environment.[36] Probably due to their facultative nature. In the present study, density of gastropods was higher in poor-water condition at Doura *chaur* than the clean-water environment at Ambhobasa *chaur* agree with the findings of earlier studies.[37,38]

CONCLUSION

Results of the present study showed that floodplain lakes of North Bihar support diverse groups of benthic fauna. Qualitative analysis of benthos reveals presence of 21 species belonging to 5 major groups, like Oligochaeta, Hirudinaria, Insecta, Pelecypoda and Gastropoda. Benthic fauna were relatively more divers at Ambhobasa *chaur* (19 species) than the Doura *chaur* (15 species). Higher density of benthos reveals eutrophic nature of the studied lakes. It was observed that species present at Ambhobasa *chaur* were absent at Doura *chaur* probably due to intolerant nature of the concerned benthic fauna. Food availability and environmental condition are most possible reason for declined number and disappearance of organisms. Benthic fauna constitutes the food of fish; if effective measures are not taken numerous species of benthos may become extinct in coming years. Information obtained during the present study can be used

for further analysis of benthic fauna in the floodplain lakes of this region.

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Erratic changes in an air breathing catfish *Heteropneustes fossilis* due to effect of stress

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Abstract- Fish and fish products provide low fat resources and are very healthy substitute of red meat. Fish industry is growing very rapidly, it provides a good source of income. Hence, it is necessary to produce healthy quality of fishes. Fishes get affected by many environmental factors such as temperature, humidity, pH etc. Proper handling of the water body is also necessary for the production. In the current study, the effect of above factors and the erratic changes in air breathing catfish *Heteropneustes fossilis* were studied. The effect of these stressors were studied for a period of two weeks. Subsequent decrease in haemoglobin level and muscle glycogen level was observed. Glycogen content was decreased by 57% when exposed to temperature stress. Haemoglobin level is in relation with erythrocytes count. These erratic changes effects the growth of *Heteropneustes fossilis* which adversely effects the production.

Key words: Erratic changes, Haemoglobin, erythrocytes, *Heteropneustes fossilis*

INTRODUCTION

Fishes are good source of income for human. The fishes are very much dependent on the natural source to feed themselves. They are part of moderate and important food supply in the world.[1] In past these products are thought to be unlimited but now it is realized that these products are finite and in knife edge with the physical chemical and biological factors of the aquatic environment. Fishes are considered as renewable source of income, but some factors such as overfishing pollution, overfishing, alteration in environment has depleted fish production.[2] Fishes are important recourses for human especially as food. Now a day's fish production has become one of the best business worldwide. Fish industry is growing rapidly day by day therefore it is necessary to produce healthy quality of fishes. One of the main obstruction in fish farming is their growth and survival which are effected by many factors. In the current paper we are discussing the erratic changes due to role of stress in the production of fishes. Changes in fish are not completely natural,[3] what precautions should be taken by those who keep fish, is very important. We have taken a number of species of catfish *Heteropneustes fossilis* for this investigation. The erratic changes in *Heteropneustes fossilis* is monitored for a period of two weeks. *Heteropneustes fossilis* are commonly called as singhi and are found in freshwater. It is mainly found in freshwater ditches, marsh lands, swamps, ponds etc. It usually breeds

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in monsoon and in constricted water. It is omnivorous in nature. This catfish emanates poison from a gland on its pectoral fin spine to deliver a extremely painful sting to humans. It can tolerate slightly brackish water. It is in great demand due to its medicinal value also.[4-6]

MATERIALS & METHODS

Study area

Madhepura is one among the thirty-eight districts of Bihar situated in 25.920°N latitude and 86.792°E longitudes. The district is 29th largest in terms of area, whereas it is at 34th place in terms of population. The maximum temperature of this district ranges from 35°C to 41°C and the minimum temperature ranges between 7°C to 9°C. The average rain fall in this area is around 1300mm. The district is blessed by a river Kosi, which is an important source of water for agriculture, household use, fish culture etc. The district got its name because it stands at the centre of Kosi ravine, due to which it is termed as Madhyapura (centrally situated) and in due course transformed as Madhepura.

Samples collection

Fifty adult live specimens of *Heteropneustes fossilis* irrespective of sex consideration were collected from the local market of Madhepura. Tanks were maintained at the normal room temperature. The fishes were brought and kept in tanks.The pH of the tanks was maintained near neutral. The pH above and below a certain threshold level is deemed dangerous for the survival of the fishes. The humidity of the lab was also optimum. A good supply of phytoplankton and zooplankton was provided in the tanks from time to time, to ensure proper nutrition to the fishes. Special care was taken while handling the live fishes as their sting can be very painful. It was seen that the tanks donot overcrowd avoiding turbidity. Turbidity usually decreases the photosynthesis rate.

Methods

The fishes were kept in tanks at normal room temperature i.e., 28°C. Simple trial and error methods were used to see the affect of stress such as erratic increase or decrease in temperature, pH,humidity, turbidity, starvation, etc. on fishes. The duration of the experiment was 2 weeks.

Temperature

The tanks were maintained at a normal temperature of 28°C for the first two days of the experiment. This allowed the fishes to get acclimatized or adapted to the new surroundings. After 2 days the temperature of the tank was increasedby 20 with the help of a blower. The fishes were observed at this stressful temperature for 24 hours. Next readings were taken at 48 hours and 96 hours respectively. The movement of fishes both upwards and downwards was seen.

pH

The pH of the tank was kept near neutral for the first two days. The pH of the tank was naturally increased by adding baking soda. The pH increased from around 7 to about 8.3. The fishes were observed for next 24, 48 and 96 hours.

The pH can also be lowered from 7.5 to 6.5 by adding white vinegar to the tank.

Humidity

The humidity on the tank was increased using a vaporizer. The vaporizer locally increases the humidity.

Turbidity

Turbidity affects the amount of sunlight that goes to the bottom of a water body. More turbid water means less sunlight that reaches the water. Less sunlight means less photosynthesis which eventually leads to less food, therefore starvation. Turbidty was increased in the tanks by increasing the number floating plants that cover the surface. The activity of the fish in response to low light was observed.

Starvation

After initial feeding of 2 days, the food provided was minimized by 50%. For the next 24 hours the fishes were observed. The next 48 hours the fishes were provided with only 25% of the food, followed by absolute starvation in the next 96 hours. The metabolic activities were observed carefully.

RESULTS & DISCUSSION

It was seen that introduction of stress in any form be it temperature, pH, humidity, turbidity or starvation leads to restlessness in the initial period. However, if the stress is continued, the fish slowly gets accustomed to the change. If, however, the changes are too erratic the fish's metabolic state is compromised and it slowly settles down at the bottom of the tank. The respiration rate slows down, slowing down all other activities requiring rapid energy as well.

Temperature

When fishes are exposed to high temperature their central nervous system recognizes the threat and gets activated. It shows physiological stress response and release catecholamines hormones whose secretion gets continuous increase with the increase of stress.[7] The secretion of cortisol hormones is not as rapid as release of catecholamines hormones. Fishes seem to tolerate small fluctuations in temperature.[8] The fish increases its metabolic activities with increase in temperature but only upto a threshold temperature. Higher temperatures increase the respiration rate, which increases the need of oxygen. It has effects on the rate of reproduction, usually decreasing it.

pH

Increase in pH up to 8.3 usually does not harm the fish productivity, but decreases it gradually. The optimum pH for fish's maximum productivity is between 6.5 to 7.4. Any increase or decrease in pH from the threshold can affect the hormonal equations, leading to loss in vigor or egg production.

Humidity

The humidity is not seen to exert any effects on the metabolic activity of the fish.

Turbidity

Initially turbidity doesnot seem to affect the fishes metabolic activity, however on the long run, it decreases the photosynthetic rate, hampering the phytoplankton and zooplankton's mass. This can lead to starvation in the absence of any addition of food manually.

Starvation

The physiological processes affect organs like liver, skeletal muscles, stomach, during starvation. Changes in the metabolic processes like breathing, oxidation states, intermediates formed, enzymatic action etc are all followed by starvation.

Initial starvation process is followed by lessened respiration rate, slow movement, and finally cessation of any bodily metabolism. Long periods of starvation usually accounts for hibernation. However in this case, repeated starvation caused loss in mass, less vigor, no reproduction and eventually death.

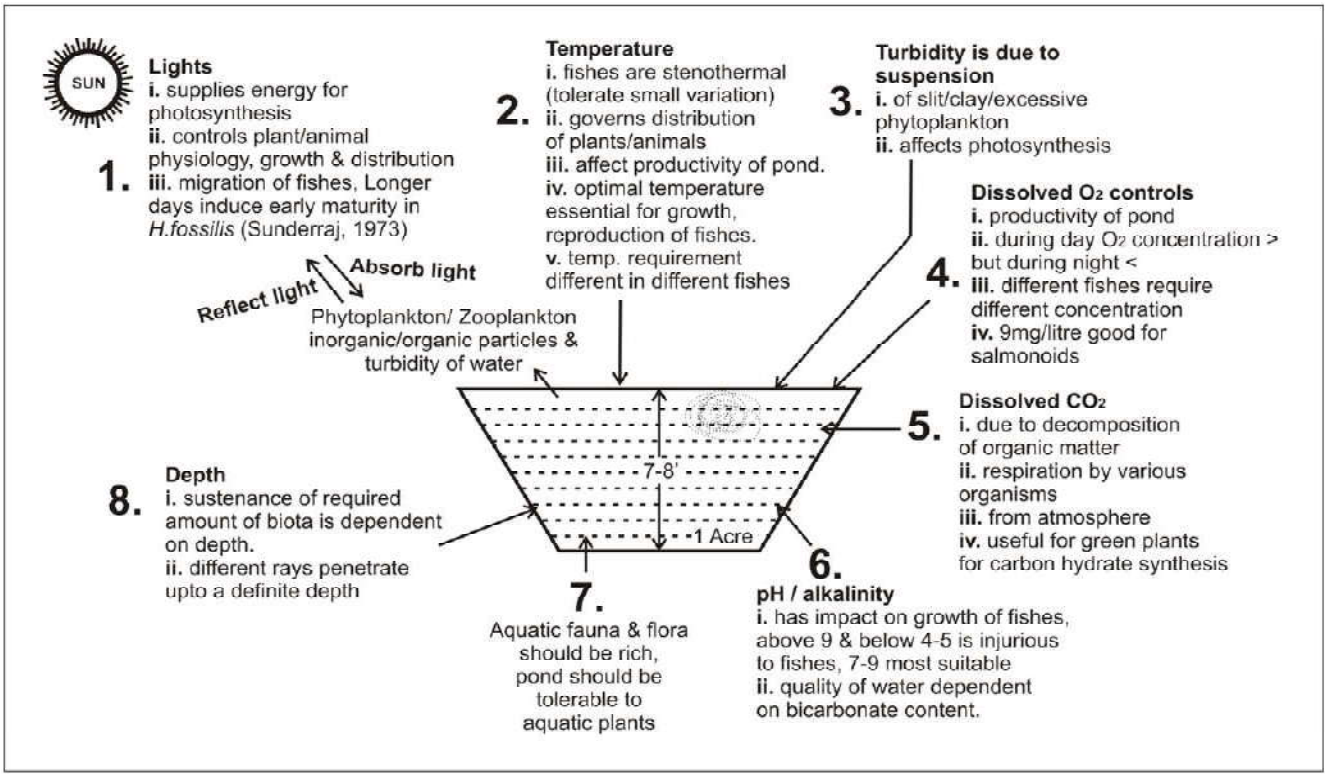


Fig. 1- Conditions of a pond is dependent upon above parameters

CONCLUSION

From the above study it was concluded that the stress can affect the production of fish. All types of stressors can cause depletion of glycogen level in muscle which adversely effect the quality. This study also concludes that fishes are good indicators of aquatic pollution. By maintaining the water quality properly one can produce a good quality of fishes.

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Survey report on National Research Centre for Orchids

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Abstract- National Research Centre for Orchids was established on 5th October, 1996 by the Indian Council of Agricultural Research (ICAR), New Delhi to organise programme on improvement in productivity quality and commercialization of orchids. Orchids are one of the high value crop cultivated for cut flower and pot plants in Sikkim. Usually, the orchids are grown on the plant in Sikkim. Orchids are one of the most sought after flower, they are expensive, elegant dainty, rare to find and possess a miraculous beauty of courses not everyone can afford to buy an orchid on their way home as a gift to often. However, even a one-time gift of an orchid signifies feelings of love romance and friendship in this project we will study history of orchids, findings of orchids, growth & conservation of orchids etc.

Key words: Orchids, NRCO, ICAR, Elegant, Miraculous, Conservation

INTRODUCTION

Orchids are mostly found in the North Easternmost tip of India, like Arunachal Pradesh, Sikkim Darjeeling, Ranipool, Kalimpong, Rangpo, Gangtok, Pakyong, Dikling, and adjoining areas. Sikkim is called 'The Land of Rising Sun' it is also fondly called 'Land of Down lit Mountains' or the 'Orchid State' for the abundance of a beautiful and rare orchid species found here. Eurasia, America and Canada are known as the home of orchids but today they are found almost in every part of the world. A French botanist Noel Bernard was the first to discover this magnificent flower, back in the year 1899, he came across seeds of a terrestrial orchids *Neottia*. He found that these seeds were germinated in some fruit which were infected by mould.

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Another discovery was by Charles Darwin, he noted a single orchid plant *Ophrys*. Charles Darwin, in a letter to Joseph Hooker, wrote, I never was more interested in any subject in my life than orchid.

Orchids constitute on order of royalty in the world of ornamental plants and they are of horticulture importance and play a very useful role in balance the forest ecosystem. Theophrastus, a student of Plato noted the round praised bulbous of one Common European Orchids and named it the Greek word for testicle. Later this name was applied not only to one genes but also to the entire Family Orchidaceae.

MATERIALS & METHODS

Stationary items, DSLR camera, Mobile Temperature G.P.S etc. NRCO at Dikling Road is a wonderful place where I met Director of NRCO, Dr. Rampal Sir, there I

visited laboratory, had taken photographs of Orchids of Poly houses and Net houses. In the NRCO the research area was divided in two parts the NRCO Office and Administrative Building, where researchers works in their lab. In this building library and conference hall are well maintained. On 5th October 1996, ICAR established NRCO on the basis of planning commission in 5 year plan institute works, as for vision for research & development activities orchids commercialization and sustainable utilization the library system.

OBSERVATION

Tissue culture lab were operating using multiplication of Orchids. Entomology section, Plant pathology section, Horticulture, Soil Science lab were well maintained. 325 different species of orchids were grown in 4 acre land for conservation purpose. Growing Orchids in polyhouse chamber, Tropical Orchid chamber, Natural Net House, is a very often process. Germination of polyhouse Rapidly Growth in Polyhouse. Rapidly increases Growth of Net house and Lower Part of the NRCO where cold climate are found Orchids are attached with the wood or in Natural condition.

DISCUSSION

There are four Veda in Sanskrit literature Rig Veda Sam Veda, Ayur Veda and Atharva Veda, written between 400 and 1600 B.C. Ayur Veda deals with use of plant drugs, it is considered as an Up Veda. The Indian orchids were brought to the notice of the world by Charak the great Indian medicine man. Charak the great Indian medicine man brings in notice of the world about the existence of orchid around 125 A.D. He described 'VANDA' and several others orchids in his book Charak Samhita which provides description of present known orchid like *Flickingeria, Malaxis, Eulophia*, species[1] having tremendous horticulture and medicinal value. It is believed that Chinese were the first to cultivate and describe orchids. They were certainly the first to describe orchids for medicinal uses.[2] In early 200 BC the Chinese Pharmacopeia "The San Nung Pen Tsao Ching" mention Dendrobium as a stringent, analgesic, anti-inflammatory substances. Reinikka reports a Chinese legend 'Shunnug' who described *Striata* and a *Dendrobium* species in his book 'Materia Medica' of the 28th Century B.C. One of the earliest nomenclature used around 937-286 BC for Orchids was that:- "The Orchids a generic name which stands for the testicles like things are basically the pair of underground tuber of many European terrestrial orchids".

In breeding programme, selection of good and healthy plant and flower by visual observation accounts to a great extent. Evidences of natural hybridization. Occurring among wild species were reported[3] in Dendrobium, Offspring's of reciprocal crosses show variations in characters like cane length and flower colour, flower size, flowering season and flower yield.[4] Selection of flower size, and flower colour are effective in Dendrobium improvement programme.[5] In *Cymbidium*, fragrance is the most important character sought after by breeders. *Cymbidium munronianum* has been used as parent in several breeding programme.[6] The higher order hybrid, Paphiopedilum F.C Puddle with six species in parentage plays a predominant role in breeding for white flower colour in the genus.[7] An extensive breeding programme in phalaenopsis using 29 wild species and 873 varieties were studied by a group of workers[8] for development of new hybrids inter:- group hybridization between Phalaenopsis type Dendrobium extends flowering season, expands the range of flower colour and shapes and increases the flowering.[9] Majority of commercially grown orchids today are hybrids derived from Arachnis, Vanda, Renanthera, Ascocentrum, Cymbidium, Catteleya, Dendrobium, Oncidium, Phalaenopsis and Paphiopedilum.[10] *Vanda sanderiana* and *Vanda coerulea* are the two important Vandal species found in the back ground of most of the Vandaceous hybrids.[11] Most orchids have two basic sets (diploid, 2x) of chromosomes. Tetraploid plants are more Fertile bearing flowers of better texture, bigger and more intense colouration.[12] Many Indian species have earned worldwide recognition in breeding program due to their inherent attractiveness coupled with their ability to transmit these character to hybrids. Some of the leading species are *Aerides multiflorum, Cymbidium devonianum, C.lourianum, C.tracyanum, C.elegans, Dendrobium aggregatum, Dendrobium chrysotoxum, Dendrobium formosum, Dendrobium nobile, Paphiopedilum venustum, Vanda coerulea* etc.[13]

CONCLUSION

Collection of orchids from their natural bodies should be made illegal unless the same is made for scientific study or for propagation and preservation at

institutional level. Plant should not be collected and natural homes of orchids should not be destroyed without the permission of the local authorities. The market of Orchids can increase throughout the World.



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Studies on bioaccumulation of heavy metals in gills and muscles of freshwater edible fish species *Clarias batrachus* and *Labeo rohita* of Bara Talab, Bundu, Jharkhand

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Abstract- Agricultural, domestic wastes and industrial discharges have contributed as source of inorganic pollutants in freshwater bodies, that are also a home for many species of edible freshwater fishes. Heavy metals Zinc and Copper are common inorganic components bioaccumulating gradually in different body parts of the fishes, particularly gills and muscles. Consumption of these fishes by people of nearby areas lead to bioaccumulation of toxic heavy metals in the body. In the present study freshwater pond Bara Talab, Bundu, Jharkhand was selected, and bioaccumulation of toxic heavy metals in gills and muscles of two freshwater edible fish *Clarias batrachus* and *Labeo rohita* were studied. Concentration of Zinc in gills of *Clarias batrachus* and *Labeo rohita* was found to be 2.69±0.8 µg/g and 1.8 ±0.6 µg/g respectively, whereas, concentration of Copper was found to be 1.28 ±0.15 µg/g and 0.48 ±0.09 µg/g respectively. Concentration of Zinc in muscles of *C. batrachus* and *L. rohita* was 1.4 ±0.5 µg/g and 0.8 ±0.04 µg/g respectively. Gills showed higher bioaccumulation of both Cu and Zn heavy metal. This result showed that due to deposition of agricultural and domestic waste in Bara Talab, Bundu, freshwater edible fishes are facing the degradation of health as a result of bioaccumulation of toxic heavy metals Zn and Cu.

Key words: Bioaccumulation, *Clarias batrachus, Labeo rohita,* Whatman's filter paper, ANOVA

INTRODUCTION

The term bioaccumulation is defined as uptake, storage, and accumulation of organic and inorganic contaminants by organisms from their environment.[1] Bioaccumulation therefore results from complex interactions between various routes of uptake, excretion, passive release, and metabolization.

For fish, the bioaccumulation process includes two routes of uptake: aqueous uptake of water-borne chemicals and dietary uptake by ingestion of contaminated food particles. Many inorganic pollutants involving toxic metals and different types of salts cause bioaccumulation, that are sourced by agriculture and domestic wastes and industrial discharges. A heavy metal can be defined as a naturally occurring element that has a high atomic weight and high density which is five times greater than that of water.[2] Due to their toxic nature and nature to accumulate in the environment, heavy metals have received a paramount attention among all the pollutants. Usually heavy metals are present in trace amounts in natural waters but many of them are toxic even at very low concentrations.[2] Heavy metals contamination may have devastating effects on the ecological balance of the recipient environment and a diversity of aquatic

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organisms. Due to their toxicity and potential bioaccumulation in many aquatic species heavy metals have gained considerable attention in the past decades. High metal concentration in gills reflects high metal concentration in dissolved phase. Zn, although is important for synthesis of important enzymes formation and stabilization of large amount of proteins, it becomes toxic when present in high concentration. Cu also causes Mense disease, Wilson disease, and kidney damage when present in high quantity.

In the present study, bioaccumulation of two heavy metals Zn and Cu in gills and muscles of two important edible freshwater fish species *Clarias batrachus* and *Labeo rohita* found in Bara Talab, Bundu, Jharkhand, was quantitatively studied.

MATERIALS & METHOD

Selection of study area-

In this project Bundu lake, was selected for study because it is widely spread out with abundant flora and fauna. It is a huge lake where freshwater fishes like *Labeo rohita* and *Clarias batrachus* inhabit, other species like *Catla catla* is also found. This was selected for the study mainly because fishes of this lake are brought to the city of Ranchi and are widely sold out. This lake is highly polluted and is vastly covered with water hyacinth. Water hyacinth degrades the water quality and is known to reduce oxygen levels in water bodies.



Fig. 1- partially decomposed plastic pollutants thrown in Bara Talab, Bundu



Fig. 2- Discharge from drains into Bara Talab, Bundu

Fish and tissue collection –

10 specimen of *Clarias batrachus* and *Labeo rohita* each were purchased from the fishermen directly from the site and were transferred to ice box and taken to the laboratory for further assessment. After complete cleaning and removal of scales, fishes were dissected to obtain their gills, accessory respiratory organs and muscles to study the bioaccumulation of heavy metals.

Digestion of tissues-

The tissues were digested following the protocol of Chindah *et al.* (2009)[3]. The tissues were dried in incubator to obtain a constant dry weight 10 ml of concentrated HCl and 1 ml of concentrated HNO_3 was taken in a beaker, 4 g of dry tissue was weighed, ground and added to the beaker gradually. 50 ml of deionized water was added. The mixture was heated in a steam bath to obtain a thick yellow liquid and allowed to cool. The mixture was then filtered using Whatman's filter paper and filtrate obtained was made upto 100 ml using distilled water. Filtrate was stored in plastic bottles and analyzed using atomic absorption spectrophotometer.

Statistical analysis-

All values were expressed as mean of 10 fishes. The difference obtained in the analysis of concentration of metals between two different species were analysed by ANOVA. The difference obtained in the metal concentration in different organs of two different species were statistically analysed by Student's t test to test the significance.

RESULT & DISCUSSION

Table 1- Comparison of concentration of Zinc in gills and muscles of *Clarias batrachus* and *Labeo rohita*

Fish species	Concentration of Zn in Gills (µg/g)	Concentration of Zn in Muscles (µg/g)
Clarias batrachus	2.69±0.8	1.4±0.5*
Labeo rohita	1.8±0.6	0.8±0.04

*=significant at 5% level

Table 2- Comparison of concentration of Copper in gills and muscles of *Clarias batrachus* and *Labeo rohita*

Fish species	Concentration of Cu in Gills (µg/g)	Concentration of Cu in Muscles (µg/g)
Clarias batrachus	1.28±0.15***	0.2±0.08*
Labeo rohita	0.48±0.09	0.08±0.008

*=significant at 5% level

In the present study (Table 1) showed the bioaccumulation of Zinc in gills and muscles of *Clarias batrachus* as 2.69 ± 0.8 µg/g and 1.4± 0.5 µg/g respectively. *Labeo rohita* showed lesser bioaccumulation of Zn in gills and muscles as 1.8±0.6 µg/g and 0.8±0.04 µg/g respectively. Zn concentration in muscles of *C. batrachus* was significantly higher than that of *L. rohita* at 5% significance level. This was in accordance with Yousafzai *et al*. (2017)[4] where they obtained higher concentration for Zn was found in gills as compared to muscles .

Table 2 showed higher bioaccumulation of Copper in *Clarias batrachus* as compared to *Labeo rohita*. Cu was found to be higher in gills of *C. batrachus* as 1.28 ±0.15 µg/g and muscles as 0.2 ±0.08 µg/g. Gills and muscles of *Labeo rohita* showed comparative lesser bioaccumulation of copper as 0.48±0.09 µg/g and 0.08±0.008 µg/g respectively. Statistical analysis showed that bioaccumulation of Cu in gills of *C. batrachus* was significantly higher than that of *L. rohita* at 0.1 % level. Also, bioaccumulation of Cu in Muscles of *C.batrachus* was significantly higher than that of *L. rohita* at 5% significance level. This is different from the findings of Pandey *et al.* (2014)[5] where the high accumulation of Cu was detected in the muscles. From the data, the storage of Cu was highest in the liver in the studied species which is in congruency with Oldewage and Marx (2000)[6] who used *Clarias gariepinus* for their study and obtained that liver had the highest Cu concentration.

According to Forstner and Prosi (1979)[7] the harmful effects of heavy metals as pollutants results from incomplete biological degradation[8]. The water bodies are contaminated by industrial discharge and agricultural wastewater discharge, possible sources of heavy metals, which are absorbed by the suspended sediments which gets precipitated to be a part of the surface sediment, which in case of certain disturbances gets released into the water column and serve as the potential secondary source of contaminants which can be a potential threat to ecosystems.[9]

Fishes are integral component of human diet,[10] because it's a good source of high-quality protein, amino acid (lysine, sulphur containing amino acids etc), iodine, calcium, trace elements (Mn, Fe, Cu, Zn, Se etc), vitamins and omega-3-polyunsaturated fatty acid.[11] Fish are the organisms that absorb dissolved or available metals[12] from the surrounding environment and therefore are considered as an excellent biomarker of metal pollution in an aquatic ecosystem. In an aquatic ecosystem fishes occupy higher tropic levels this is another reason for considering it as a biomarker of metal pollution. Since fishes are among the top consumers in an aquatic environment, the tropic transfer which is also referred to trophodynamics of elements along a food chain can result in an increase (biomagnification) or decrease (bio-dilution) or even no change in the element concentrations from lower to upper components of the food web.[13] Therefore, since they play an important role in human nutrition, fish need to be screened to ensure that unnecessarily high levels of the toxic heavy metals are not being transferred to man through them.

CONCLUSION

Both Zinc and Copper heavy metals were found to be heavily bioaccumulated in the gills and muscles of the two commonly consumed freshwater fishes of Bara Talab i.e., *Clarias batrachus* and *Labeo rohita*.

Such extent of bioaccumulation in both fish species is suspected to be because of the pollutants in the Bara Talab in response to anthropogenic activities taking place in and around it.

Acknowledging it as one of the rice source of edible fish supply in Jharkhand, the Government of Jharkhand should take major steps to protect this fresh water body from any kind of pollution, so as to better the health condition of fishes present there and thereby providing a healthier source of protein to the markets.

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An experimental investigation on growth rate of water lettuce (*Pistia stratiotes* L.)

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Abstract- *Pistia stratiotes* commonly called water lettuce is free floating aquatic herbs is the member of family areaceae. This is an alien invasive species. This grows very fast and soon covers the entire surface of the water bodies. It is often grown as an ornamental plant in the water bodies or small water tubs owing to its beautiful foliage. But due to its rapid growth it causes it causes nuisance to water bodies as because it is very difficult to control once it cover the entire surface. During the course of study of aquatic and semi aquatic angiosperms of Jharkhand it has been observed that several ponds have been covered with the mat of *Pistia stratiotes*. In order to know its pace of spread, an experiment was conducted at department of botany KKM College, Pakur to observe the multiplication. It was initiated in a water tub. Soon the entire surface of the tub was covered with *Pistia*. In the present communication number in increase of *Pistia* has been recorded on daily basis for a month and same has been discussed.

Key words: *Pistia stratiotes,* ornamental plant, nuisance, water bodies.

INTRODUCTION

Pistia, a genus of family araceae is an aquatic floating herb. Commonly called water lettuce or water cabbage. The genus "*Pistia*" is derived from the Greek word "pistos" meaning watery. The specific epithet "*stratiotes*" is derived from a Greek word that means soldier, a possible reference to the species *stratiotes* which contains an aquatic plant species with leaves shaped like swords. It is the single genus of the tribe Pistieae that shows its systematic isolation inside the family. It has only one species i.e., *Pistia stratiotes*, This was first reported from the Nile close to Lake Victoria within Africa. Which further spreaded through naturally or by anthropogenic activities to almost entire tropical and subtropical fresh waterway. Water lettuce is among the world's most dynamic freshwater aquatic plants due to the reason it is considered as highly invasive species. The invasion of this species in the environment leads to serious environmental as well as socio-economic damage to its surrounding the communities. In water with high nutrient load this plants shows luxuriant growth and presents weedy appearance.

Water lettuce is among the world's most dynamic freshwater aquatic plants and is considered as an invasive species. The species can be introduced to new areas by water dispersal, fragmentation, and hitchhiking on marine transportation or fishing equipment.

It reproduces rapidly through vegetative fragmentation with the help of stolon as well as seeds. Under optimal environmental condition in doubles its number in three week. It occurs in the surface of lakes

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and ponds. The heavy growth of this plant hampers the effective uses of water causes clogging of canals and dominates in wetland environments.

Description of the plant:

Free floating herbs having 2-10 cm long leaves, surrounded at its base by a membranous sheath, pubescent, obovate-cuneate; apex rounded and truncate. Spathes 10-12 mm long, yellowish unite, obliquely campanulate, contracted near the middle. Carpels obliquely adnate to the spadix; apex free. Ovary one loculed with numerous ovules on a marginal placentae. It flowers and fruit in the month of August - April.

Systematic position

Kingdom	-	Plantae
Subkingdom	-	Viridiplantae
Infrakingdom	-	Streptophyta
Superdivision	-	Embryophyta
Division	-	Tracheophyta
Subdivision	-	Spermatophytina
Class	-	Magnoliopsida
Superorder	-	Lilianae
Order	-	Alismatales
Family	-	Araceae
Genus	-	*Pistia* L.
Species	-	*P. stratiotes*

MATERIALS & METHOD

One pair of *Pistia stratiotes* was taken and is allowed to grow and multiply in a controlled condition. The increase in number of *Pistia* has been observed on daily basis. The date wise increase in numbers were recorded and photographs of the increased number were taken. On the basis of the same a graph has been plotted.

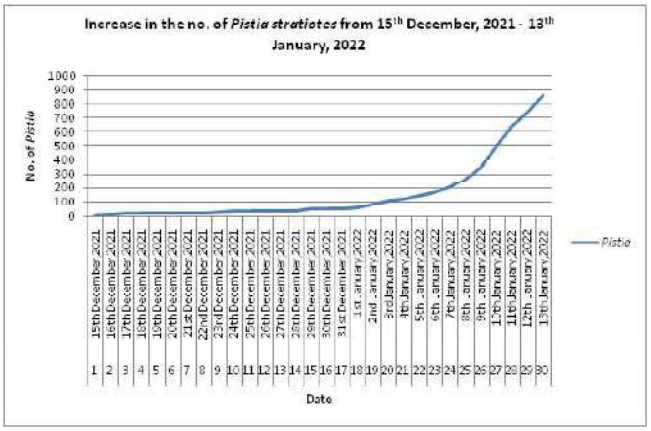
RESULT & DISCUSSION

One pair of *Pistia* was allowed to grow on a controlled condition in a pot on 15th of December, 2021, very next day i.e. on 16th of December, 2021, the number increased gradually to 11, 18, 18, 18, 18, and then 19, 19, 27, 31, 31, 32, 32, 32 further 46, 46, 49, 54, 80, 100, 120, 140, 165, 200, 255, 300, 500, 640,746 and finally 859 on 13th January, 2022.

This plant being alien invasive soon covers the entire surface, causing disturbance in reaching the sunlight for bottom feeder and alters the grazing the food chain. The huge decomposition of *Pistia* settles in the bottom and reduces the depth.

Table 1- Datewise multiplication of *Pistia stratiotes*

Sl. No.	Date	No. of *Pistia*
1.	15th December, 2021	2
2.	16th December, 2021	11
3.	17th December, 2021	18
4.	18th December, 2021	18
5.	19th December, 2021	18
6.	20th December, 2021	18
7.	21st December, 2021	19
8.	22nd December, 2021	19
9.	23rd December, 2021	27
10.	24th December, 2021	31
11.	25th December, 2021	31
12.	26th December, 2021	32
13.	27th December, 2021	32
14.	28th December, 2021	32
15.	29th December, 2021	46
16.	30th December, 2021	46
17.	31st December, 2021	49
18.	1st January, 2022	54
19.	2nd January, 2022	80
20.	3rd January, 2022	100
21.	4th January, 2022	120
22.	5th January, 2022	140
23.	6th January, 2022	165
24.	7th January, 2022	200
25.	8th January, 2022	255
26.	9th January, 2022	350
27.	10th January, 2022	500
28.	11th January, 2022	640
29.	12th January, 2022	746
30.	13th January, 2022	859



Graph 1- Showing the growth pattern of *Pistia* for one month



Fig. 1- Datewise multiplication of *Pistia stratiotes*

As far as the growth and multiplication of the *Pistia* is concerned, so far no study has been conducted, however several work on aquatic flora has been done so far, pioneer among them are as follows:-

The aquatic and semi aquatic flora of district level has been done, studies on aquatic and semi aquatic angiosperms of Lohardaga.[1] Studies on aquatic and semi aquatic angiosperms of Ranchi.[2] Study of aquatic weeds and their control in and around Jamtara district.[3] The floristic and ecological studies of aquatic angiopsperm of Jamtara district and adjoining Barakar basin.[4] Studies on floristic diversity of Jubillee Park of Jamshedpur.[5] The floristic composition and vegetation profile of aquatic angiosperms of Hazaribag and adjacent area.[6] Studies on angiospermic flora of ponds of Ranchi, Jharkhand,[7] Hydrophytes of Ranchi.[8,9] Work on floristic studies of aquatic and semi aquatic angiospermic bio-diversity of Ranchi has also been published.[10] Weeds diversity of Pakur.[11] Floristic studies of aquatic and semi aquatic angiosperms of Ratu Maharaja Pond.[12] Biodiversity of aquatic flora in Raja bandh pond of Jamtara district.[13] Aquatic biodiversity of Victoria Lake of Lohardaga.[14] Studies on the aquatic and semi aquatic angiosperms of Kanke Dam, Ranchi.[15] A survey conducted in the Dumka district reveals the occurrence of 30 species of invasive aquatic and semi aquatic medicinal plant representing 26 genera and 17 families.[16]

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Late Prof. (Dr.) Mahendra Prasad

26th January 1953 - 20th April, 2021

- Former Professor & Head, University Department of Zoology, Ranchi University, Ranchi
- Former Dean, Faculty of Science, Ranchi University, Ranchi
- Founder President MSET-ICCB, Ranchi
- Former Editor-in-Chief of Biospectra
- M.Sc. and Ph.D. in Zoology (Entomology)- From Patna University, Patna, Bihar - 1973, 1984
- Date of Joining Ranchi University Service through BPSC Recommendation - 31-08-1977
- Started organising national/international level conferences from 2008 and so far successfully organised following conferences :-
  1. ICTBR, Namkum, Ranchi, Jharkhand - 2008
  2. ICBMSL, Lalgutwa, Ranchi, Jharkhand - 2009
  3. ICBRD & NRMCU, KGVK, Ranchi, Jharkhand - 2010
  4. ICLRAD, CPRS, ICAR, Patna, Bihar - 2011
  5. RAMBSWERD, KSSRDI, Bengaluru - 2013
  6. ICBNEHRI, IINRG, Namkum, Ranchi - 2014
  7. GOSLANRUF, METAS, Ranchi - 2017
- Established five Life Scientist Memorial chairs in Zoology and four in Botany under MSET-ICCB from 2010 through which several PG Scholarships, young, senior, emerging scientist awards as well as life time achievement awards have been conferred to outstanding researchers in the field of animal and plant science research.

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